

FINITE TIME BLOWUP FOR NAVIER-STOKES

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ABSTRACT. Viscosity measures how strongly the fluid resists shear. For every positive viscosity, this paper builds a solution of the three-dimensional incompressible Navier–Stokes equations. The fluid is incompressible, so it cannot be squeezed into a smaller volume. It starts at rest. In a finite amount of time its velocity becomes unbounded, which means that its speed rises past every fixed limit. Its kinetic energy, the integral of half the squared speed over all space, stays below one fixed finite bound throughout.

CONTENTS

1.	Introduction	1
2.	Physical picture of the blowup	3
3.	Outline of the proof	6
4.	Building the leading-order flow	24
5.	Correcting the base flow at every order	45
6.	The auxiliary torus and keeping oscillating supports apart	62
7.	Using oscillations to produce and correct the residual stress	73
8.	Mean corrections with compact support	88
9.	Improving the residual and building the local field	100
10.	Compact forcing and breakdown in the whole space	116
Appendix A.	Matching radial moments and building the heat exterior	126
Appendix B.	Analytic profiles near the axis and continuing them	144
Appendix C.	Producing the admissible stress cone	157
	References	165

1 Introduction

This paper studies the three-dimensional incompressible Navier–Stokes equations. The main question is whether a solution that starts from smooth data can develop unbounded velocity in finite time while its kinetic energy stays bounded. Here smooth means that derivatives of every order exist and vary continuously. The flow constructed below starts with zero velocity and is driven by a smooth force with compact support, so the force is zero outside some bounded part of space and time.

Theorem 1.1. *For every $\nu > 0$, so for every positive viscosity, there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0, 1)$ satisfying*

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = 0, \quad (1.1)$$

where the notation says that this force is infinitely differentiable, has three components, and is zero outside a compact part of space and positive time. The condition $\text{supp } u(\cdot, t) \cup \text{supp } p(\cdot, t) \subset K$ holds for every $0 \leq t < 1$, which means the velocity and pressure are both zero outside the same compact set. They also satisfy

$$\sup_{0 \leq t < 1} \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty, \quad \limsup_{t \uparrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty.$$

The first inequality says that the square root of the total squared speed stays below one fixed bound at every time before one. The second says that, along times approaching one, the largest speed rises past every fixed bound. So there is no smooth solution (u, p) on $\mathbb{R}^3 \times [0, \infty)$ with the same force and initial datum whose kinetic energy is uniformly bounded $\sup_{t \geq 0} \frac{1}{2} \int_{\mathbb{R}^3} |u(x, t)|^2 dx < \infty$.

This proves the option labeled (C) in Fefferman’s statement of the Navier–Stokes Millennium problem [13]. Because the fields have compact support, the same construction also works on $\mathbb{T}^3 = \mathbb{R}^3 / \mathbb{Z}^3$, the three-dimensional torus obtained by identifying points in three-dimensional Euclidean space that differ by an integer vector. This proves option (D) in [13]; see Corollary 10.6.

1.1 Earlier work and how it leads here

The classical Euler equations describe flows that are incompressible and inviscid, meaning that they have no viscosity [12].

$$D_t u := \partial_t u + (u \cdot \nabla)u = -\nabla p + f, \quad \nabla \cdot u = 0. \quad (1.2)$$

The material acceleration $D_t u$ is the acceleration measured while following one moving parcel, or small moving part, of the fluid. Its velocity changes for two reasons. The flow at a fixed point can change with time, and the parcel can move to a place where the flow has a different velocity. The term $(u \cdot \nabla)u$ records the second change. It is nonlinear in u , so changing the velocity does not change this term in direct proportion; this is where most of the difficulty comes from. The first equation is Newton's second law per unit mass. It says that the parcel's acceleration equals the negative pressure gradient plus the external force. The second equation says the fluid is incompressible. Squeezing it in one direction must be matched by stretching it in another.

The Navier–Stokes equations [18, 21] in (1.1) add viscosity. The resulting viscous force resists shear, which is the sliding of neighboring parts of the fluid past one another. This force is linear in u , but it changes the flow in an important way because it smooths sharp changes in velocity. The regularity problem asks whether that smoothing always prevents a finite-time singularity in three dimensions. Here a singularity means that a solution which began smooth stops being smooth after a finite amount of time.

In 1934, Leray [16] built global finite-energy weak solutions in three dimensions. They satisfy an energy inequality, which says that the kinetic energy plus the energy lost through viscosity is no greater than the starting kinetic energy plus the work done by the force. A weak solution may lack the derivatives needed to read the equation point by point. Instead, the equation holds after it is integrated against smooth test fields. Leray's result did not settle whether smooth starting data always give a solution that stays smooth. Caffarelli, Kohn, and Nirenberg [5] studied suitable weak solutions, which satisfy an energy inequality not just for the flow as a whole, but also after a smooth cutoff limits the calculation to part of space and time. They proved that the set of space-time points where such a solution may be singular has zero one-dimensional parabolic Hausdorff measure. In this measurement, a space ball of a given radius is paired with a time interval whose length scales like that radius squared. Zero one-dimensional parabolic Hausdorff measure means that, no matter how small a positive total length is requested, the singular set can be covered by such space-time cylinders whose spatial radii add up to less than that length. This still allows isolated singularities. Escauriaza, Seregin, and Šverák [11] proved regularity for the unforced Cauchy problem, meaning the initial-value problem with no external force, when the scale-invariant $L_t^\infty L_x^3$ norm is bounded. This norm first measures the velocity in space with exponent three and then takes its largest essential value over time, so changing the value on a set of times of measure zero does not change the norm. Scale-invariant means that the standard change of space, time, and velocity that leaves the Navier–Stokes equations unchanged also leaves this norm unchanged.

Tao [22] proved finite-time blowup for an averaged version of Navier–Stokes whose nonlinear term still cancels out of the kinetic-energy balance, as it does in the original equations. For the original equations, Buckmaster and Vicol [4] used convex integration, a method that builds a solution by adding finer and finer oscillating corrections. They obtained nonuniqueness among finite-energy weak solutions, so the same data can lead to more than one such solution. Albritton, Brué, and Colombo [1] later built two distinct suitable Leray–Hopf solutions with zero starting velocity and the same force. A Leray–Hopf solution is a global finite-energy weak solution that obeys the energy inequality. Their construction uses an unstable vortex written in similarity variables, which rescale space and time around the possible singularity. Their force belongs to $L_t^1 L_x^2$, so its spatial size with exponent two is integrable in time, but the force is singular at the initial time.

Lifschitz and Hameiri, and Friedlander and Vishik [17, 14], developed methods that follow a wavevector and a velocity polarization along a background flow. The wavevector records the direction and spatial frequency of the oscillation, while the polarization records the direction in which its velocity moves. Craik and Criminalle [9] built exact waves of finite amplitude on affine background flows, whose velocity depends linearly on position up to a constant term. Their waves work because the quadratic interaction of each wave with itself cancels. Other earlier results include the centrifugal-instability criteria of Leibovich and Stewartson [15] and of Billant and Gallaire [2, 3], which test when a rotating flow is unstable, and the exact viscous shearing waves of Singh and Sridhar [19]. Daneri and Székelyhidi [10] used oscillations in Euler flows to produce a prescribed stress, so the required momentum-flux field is chosen first and the oscillations are built to match it.

Córdoba and Martínez-Zoroa [6] built finite-time singularities for the forced three-dimensional Euler equations. Their construction repeatedly amplifies vortex layers while concentrating each new layer more tightly. Córdoba, Martínez-Zoroa, and Zheng [8] extended the method to hypodissipative Navier–Stokes, where the viscous term uses a smaller positive power of the Laplacian than the usual equation does. Their force lies in a local well-posedness class, which is regular enough to give short-time existence and uniqueness. These papers developed a way to form a singularity by passing amplification from one scale to the next while keeping

the external force regular. Strain, the stretching and squeezing caused by the flow, acts on a larger scale and amplifies vorticity, the local spinning motion, on a smaller scale. The construction keeps each smaller layer's main interaction with itself from entering the leading equations, and also keeps that layer from feeding back onto the larger scales. For the incompressible porous-media equation, a different model for flow through a porous material, Córdoba and Martínez-Zoroa [7] built singularities from smooth starting data by repeatedly amplifying oscillating layers. Each higher-order approximation cancels more of the error. As a result, every spatial derivative of the source has one fixed bound that works at every time. The construction in this paper also uses dynamical amplification,

where the motion of the flow itself makes small disturbances grow. Here those growing disturbances do a different job. Oscillating pulses produce a mean momentum flux, the average transport of momentum through the fluid, and that transport provides the force missing from a background vortex as it collapses.

2 How the blowup happens

First, this section describes what the flow in Theorem 1.1 is doing. Section 3 then gives a full sketch of the proof, showing its main steps before the later sections give all the details.

For any incompressible flow u and pressure p , the external force f can always be set equal to the residual in (1.1), the result of adding up the four terms on its left-hand side. The Navier-Stokes equations then hold by construction, because the force was chosen to equal that result. The hard part is choosing a flow that blows up while this residual stays smooth. Even though any one of the four terms in the momentum residual may become unbounded, the terms must cancel one another so that their sum and all its derivatives still extend smoothly through the singular time, when the velocity becomes unbounded.

The construction uses a vortex whose leading profile is self-similar. This is the profile kept before the later corrections are added. As the vortex shrinks, its shape stays the same after distance and velocity are measured on their new scales. Its radial width, measured away from the axis, shrinks faster than its axial length, measured along the axis. In the middle of the vortex, fluid spirals inward while flowing out along the axis. As the core contracts, the speeds around and along the axis increase, making the sliding between neighboring layers stronger in the ring-shaped region around the core.

Inside the core, the equations for this first profile make its momentum terms balance. Joining this inner flow to a smooth outer flow whose speed drops with radius leaves a nonzero residual in the annulus, the ring-shaped region between them. For the background flow alone, the momentum residual becomes unbounded as $t \uparrow 1$, so it cannot be the smooth force required by the theorem. Spatially oscillating pulses are added, meaning velocity disturbances that alternate rapidly across space. Their nonlinear momentum fluxes, the transport of momentum produced by products of their velocity components, cancel the singular part of this residual. Further corrections remove the singular errors that remain, leaving a force that extends smoothly through $t = 1$.

2.1 Inside the core

The singularity is built to form at the origin at time $t = 1$. Cylindrical coordinates about the vertical axis are used.

$$x = (r \cos \theta, r \sin \theta, z),$$

Here r is the distance from the axis, θ is the angle around it, and z is height. The velocity components u_r , u_θ , and u_z give the motion toward or away from the axis, around the axis, and along it. The leading flow $u^{(0)}$ is axisymmetric, which means that its components depend on r, z, t but not on θ .

In the middle of the inner region, fluid spirals toward the axis. A dividing layer near $z = 0$ separates fluid moving upward from fluid moving downward; see Figure 1. Farther above and below this middle region, the inner profile also contains flow moving outward from the axis. Pressure drops toward the axis, providing the leading centripetal force, the inward force that bends the fluid's path around the axis. Its axial gradient is the rate at which pressure changes with height. Along the axis, this change in pressure is one of the terms balanced with transport and viscosity. Near the axis, the pressure force in the axial direction points toward the plane $z = 0$.

Flow toward the axis helps the vortex spin faster by carrying angular momentum to smaller radii. The torque on a particle is the rate at which its angular momentum changes. If no torque acts on a fluid particle, its angular momentum per unit mass ru_θ is conserved, so it does not change. Moving inward must therefore increase u_θ . In the actual core, viscosity carries angular momentum outward. The typical speed still increases because inward transport and loss through viscosity balance one another. Incompressibility keeps the incoming fluid from piling up near the axis. Flow out along the axis carries it away, so the inward motion and faster spin can continue.

To leading order, the core changes self-similarly. Its velocity profile keeps the same shape when distances and velocities are measured in their own changing scales. The scales measured away from and along the axis shrink at different rates.

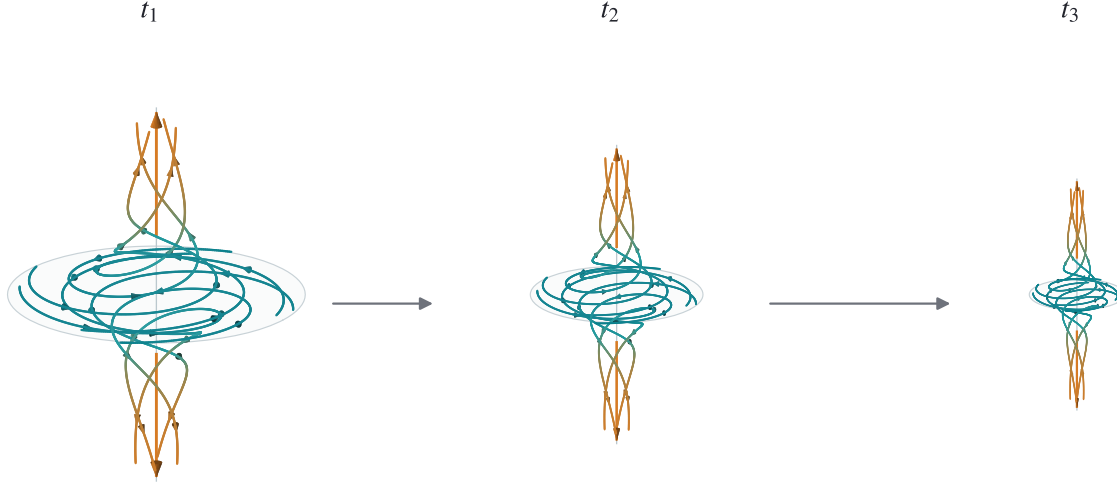


Figure 1: The middle of the inner core at three successive times $t_1 < t_2 < t_3 < 1$. Fluid in this region spirals inward. A dividing layer near $z = 0$ separates flow along the axis in one direction from flow in the other. As the singular time gets closer, the region where the flow is strongest shrinks and its speed rises. Its radius shrinks faster than its height. The drawing exaggerates this difference.

Write $\tau = 1 - t$ for the time left before the singularity. The two length scales then obey

$$\ell_r \asymp \tau^{1/2}, \quad \ell_z \asymp \tau^{1/2-h}, \quad 0 < h < 1/100,$$

The fixed number h lies between zero and one hundredth. The statement marked by $\asymp$ means that the ratio of the quantity on its left to the quantity on its right stays between two fixed positive numbers that do not depend on τ . Both lengths go to zero, but $\ell_r/\ell_z \asymp \tau^h \rightarrow 0$. The core therefore becomes a thinner and thinner column at the origin. Its volume stays between fixed positive multiples of $\tau^{3/2-h}$.

The typical sizes of the velocities in the leading flow obey

$$|u_\theta^{(0)}|, |u_z^{(0)}| \asymp \tau^{-1/2-h}, \quad |u_r^{(0)}| = O(\tau^{-1/2}).$$

These are the typical sizes across the core, not the value at every point. In particular, the velocity around the axis is zero on the axis itself. The upper-bound statement for the radial velocity means that one fixed constant times the displayed power bounds its size from above when the time left is small. The flow is smooth at every $t < 1$, but its largest speeds rise past every fixed bound inside a region shrinking toward the origin as $t \uparrow 1$. The core's kinetic energy stays between fixed positive multiples of $\tau^{1/2-3h}$, so it goes to zero even though the speeds rise.

The different velocity scales give different Reynolds numbers. For a typical speed U , distance ℓ , and viscosity ν , the Reynolds number $U\ell/\nu$ is the viscous diffusion time ℓ^2/ν divided by the travel time ℓ/U . It compares how long viscosity takes to spread motion across that distance with how long the fluid takes to travel the same distance. Using the core radius as the distance, the angular and radial Reynolds numbers are

$$\text{Re}_\theta := \frac{|u_\theta|\ell_r}{\nu} \asymp \tau^{-h} \rightarrow \infty, \quad \text{Re}_r := \frac{|u_r|\ell_r}{\nu} = O(1),$$

for each fixed $\nu > 0$.

The angular Reynolds number grows past every bound, so the fluid makes more and more turns in the time viscosity takes to spread motion across the core's radius. The radial Reynolds number stays bounded, so viscosity keeps competing with the inward radial flow.

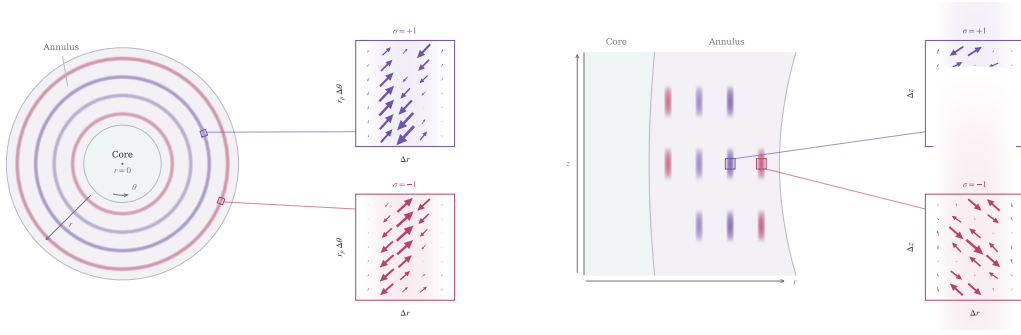


Figure 2: The shapes of the pulses and the velocities they add. *On the left*, at one fixed height, the active part of each pulse fills a complete ring. The close views show the added velocities toward or away from the axis and around it. *On the right*, a slice through the axis, where each small patch is the cross-section of a complete ring. These close views show the added velocities toward or away from the axis and along it. The colors mark the two pulse families. This is a schematic of the pulse arrangement, so it does not give the exact active positions, spacing, or aspect ratios between width and height.

The transport and diffusion rates obey

$$\frac{|u_r|}{\ell_r} = O(\tau^{-1}), \quad \frac{|u_z|}{\ell_z} \asymp \tau^{-1}, \quad \frac{\nu}{\ell_z^2} \asymp \tau^{-1}.$$

So radial diffusion, transport by the radial flow, and transport by the axial flow all remain in the leading balance. None of the three becomes too small to affect that balance as the core contracts. Diffusion along the axis is smaller. Its size compared with radial diffusion is

$$\frac{\nu/\ell_z^2}{\nu/\ell_r^2} = \frac{\ell_r^2}{\ell_z^2} \asymp \tau^{2h} \rightarrow 0.$$

The ratio goes to zero, so axial diffusion becomes negligible next to radial diffusion on these scales.

The core also has to provide enough shear at its outer edge to make the pulses there grow. Away from the middle plane, axial flow carries angular momentum from layers that rotate faster. This helps keep the angular velocity, the rate at which the fluid circles the axis, dropping steeply as radius increases. With exact reflection symmetry across $z = 0$, the flow above and below the plane would be mirror images, and this transport would disappear at the middle plane. The same symmetry would force $u_z(r, 0, t) = 0$, so there would be no radial shear of the axial velocity, no change in that velocity with radius, to take its place. The construction instead uses a slightly asymmetric axial profile, with a small upward bias and nonzero velocity at $z = 0$. The place where rotation gives little amplification is then separate from the place where the radial shear of the axial velocity vanishes. Near the middle plane, that radial shear provides the extra amplification. Farther away, rotation already provides enough.

2.2 The annulus and the pulses

At the edge of the core, the inner vortex joins an outer flow across the annulus. The leading momentum terms balance inside the core, but they no longer balance in this connecting region. The external force that would keep the background flow going becomes unbounded as $t \uparrow 1$, so it cannot be the smooth external force the construction needs.

The construction adds an oscillating correction, called a perturbation, to the velocity so that the fluid's own motion supplies the missing momentum transport. The nonlinear advection term, the part of the equation that records velocity being carried by the flow, lets this perturbation produce extra momentum fluxes. Because those fluxes change from place to place, they act as an internal force on the background. This contribution is made to cancel the singular part of the background residual, leaving only a remainder that extends smoothly through $t = 1$. These internal forces move momentum from one part of the fluid to another.

The perturbation is a sequence of pulses that oscillate across space. Each pulse is confined to a small range of radii, heights, and times, but runs all the way around a complete ring. An exponentially small external force starts each pulse, so its size falls exponentially rather than as a fixed power when the construction moves to finer scales. The background shear then makes the pulse grow. Each cylindrical component of the added velocity has average zero around the ring, but the momentum fluxes made from products of those components need not average to zero. For example, an outward radial velocity multiplied by a positive surplus in azimuthal velocity gives a positive outward flux of angular momentum. An inward radial velocity multiplied by a negative deficit in azimuthal velocity gives the same sign, so it also carries angular momentum outward. Changing both signs leaves the product unchanged. Axial momentum is moved in the same way. The total transport into or out of each region is arranged to provide the contribution missing from the background momentum balance.

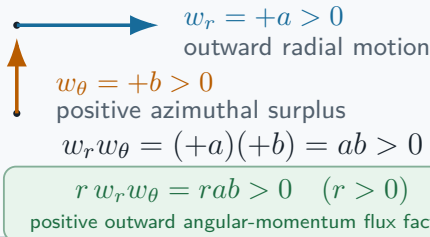
Same product, same outward flux

EXACT SIGN/PRODUCT DIAGRAM

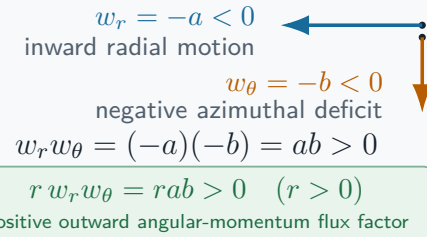
A local identity for the added pulse velocity w at one fixed cylindrical radius $r > 0$

Orientation key only $+e_r$: away from the axis $+e_\theta$: chosen positive tangent direction 

Case A: outward + surplus



Case B: inward + deficit



Sign reversal leaves the product fixed.

$$(w_r, w_\theta) \mapsto (-w_r, -w_\theta)$$

$$(-w_r)(-w_\theta) = w_r w_\theta.$$

Covariance bridge. $C_{r\theta}(w) = \langle w_r w_\theta \rangle_\theta|_Y$.

Fixed-amplitude shared cosine phase: $\langle w_r w_\theta \rangle_\theta = ab \langle \cos^2 \phi \rangle_\theta = ab/2 > 0$.
The angular frequency is a nonzero integer.

Scope and limit. Arrows show component sign and orientation only, not pulse geometry, paths, profiles, magnitudes, support, or the full three-dimensional solution. Their length and placement carry no scale or geometric information.
EE-ILL-P006-001 Source: paper p. 6; equation (7.23); Proposition 7.5

Figure 3: Both sign choices give the same positive product. The diagram shows only the signs of the radial and azimuthal velocity components at one fixed positive radius. It does not show a pulse's shape, path, size, support, or full three-dimensional motion.

The pulses grow by taking energy from the background shear. A simple rotating flow shows how. If a patch of fluid moves faster around the axis than the fluid around it, that surplus in azimuthal velocity produces outward acceleration. If angular velocity drops fast enough as the radius increases, moving outward makes the patch's surplus larger compared with its new surroundings. The outward motion and the growing surplus strengthen one another. Together, they make the pulse grow exponentially whenever the amplification is greater than viscous damping, the weakening caused by viscosity. In the full flow, shear in the axial velocity also helps the pulses grow.

The same shear also changes the pulse's pattern in space because fluid at different radii carries that pattern at different speeds. The pulses are pointed in directions that make their radial wavelength get steadily shorter. As their direction changes, the amplification becomes weaker. At the same time, the shorter radial wavelength raises the total squared wavenumber, the squared size of the vector that records the oscillation's spatial frequencies, and this raises viscous damping. The starting wavelength is chosen so that amplification is larger at first and damping becomes larger later. The tails of a pulse are the parts near its two time endpoints, where its size is exponentially small. They are small enough that the errors made by smoothly switching each pulse on and off, together with every derivative of those errors, vanish at the singular time.

Two pulse families are combined. The families have different ratios of radial angular-momentum flux to radial axial-momentum flux, so they carry different relative amounts of angular and axial momentum across

radial surfaces. The background profile and the directions of oscillation are chosen together so that the averaged quadratic velocity products, the averages of products of two velocity components, provide both components of the required leading stress, the momentum-flux field needed before the later corrections are added. Pulses appear on finer and finer spatial and time scales as the singular time approaches. Further corrections cancel the singular terms still left in the momentum residual. The resulting velocity becomes unbounded while its residual force extends smoothly through the singular time.

2.3 The outer flow and the spatial cutoff

Outside the annulus, the flow is purely azimuthal and does not change with height. Fluid only circles the axis there, and its speed drops as the radius increases. Pressure provides the inward acceleration needed for that circular motion, while viscosity spreads and changes the azimuthal velocity. This outer flow is chosen to satisfy the radial heat equation exactly, the diffusion equation with radius as its space variable, so it is called the heat exterior. Its momentum residual is zero, and no external force is needed to keep this part of the local flow going.

At every fixed radius greater than zero, the outer velocity and pressure approach smooth limits as the singular time gets closer, and the same is true after any number of derivatives. The inner construction is also smooth away from the origin. Together these facts make it possible to cut the flow off smoothly in space, leaving it unchanged near the concentrating core and making it zero outside a bounded region. The extra force created by this cutoff stays smooth through the singular time, and the cutoff does not change the concentrating core.

3 How the proof works

The physical description in Section 2 leads to an exact cancellation problem. The velocity and pressure must have a momentum residual that extends smoothly through the singular time, while the velocity itself becomes unbounded. When viscosity is set equal to one, the momentum residual is the following expression.

$$\mathcal{R}(u, p) = \partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p. \quad (3.1)$$

The estimates have to control every Cartesian space-time derivative of this residual.

First, the proof builds the axisymmetric background (u_B, p_B) from Proposition 5.5. Its residual in (5.41) is the negative divergence of a stress supported in an annulus, apart from a remainder that vanishes to every order at the singular point. The oscillating velocities produce the leading part of this stress when their quadratic products are averaged, as Proposition 7.5 shows. Corrections added one after another make the residual left over from that step decay faster, as Proposition 9.6 shows. The fields are then added together and confined to a fixed region without losing incompressibility or the leading growth of the velocity. This lets the residual extend to the smooth, compactly supported force in Theorem 1.1, by Propositions 10.1 and 9.9 and Lemma 10.3. Every estimate used below is proved for the viscous equations. Figures 6 and 7 show how the whole construction fits together.

For every $\nu > 0$, any solution with viscosity one gives a solution with viscosity ν after the following change of scale.

$$u_\nu(x, t) = \sqrt{\nu} u(x/\sqrt{\nu}, t), \quad p_\nu(x, t) = \nu p(x/\sqrt{\nu}, t), \quad f_\nu(x, t) = \sqrt{\nu} f(x/\sqrt{\nu}, t).$$

This change does not move the singular time. Equations (10.22)–(10.23) check both the equation and the way the energy changes under this scaling. The rest of this outline uses $\nu = 1$.

3.1 The concentrating leading field.

The fixed velocity profiles E, U and pressure profile Π from (4.3) put the geometry of Section 2.1 into the leading field. The similarity coordinates (X, η) from (3.2) keep track of the different rates at which the field contracts in the radial and axial directions. They also write the growth of the velocity as powers of one concentration scale. This subsection sets up that representation. The next one builds profiles whose remaining momentum residual can be supplied by the pulses.

Set

$$\tau = 1 - t, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad 0 < h < 1/100,$$

and keep h fixed while the profiles are built.

In cylindrical coordinates,

$$x = (r \cos \theta, r \sin \theta, z), \quad e_r = (\cos \theta, \sin \theta, 0), \quad e_\theta = (-\sin \theta, \cos \theta, 0), \quad e_z = (0, 0, 1).$$

The leading field $u^{(0)} = u_r^{(0)} e_r + u_\theta^{(0)} e_\theta + u_z^{(0)} e_z$ is axisymmetric, meaning that its cylindrical components do not depend on θ . Its two tangential components are the azimuthal component and the axial component. Both point along directions tangent to the cylinders on which r is constant.

Use the similarity coordinates

$$\tau = q(1 - \eta^2), \quad z = q^D \eta, \quad X = \frac{r^2}{2q}, \quad q > 0, \quad -1 < \eta < 1. \quad (3.2)$$

For $\tau > 0$, removing η from these equations leaves

$$q - z^2 q^{2h} = \tau, \quad \partial_q(q - z^2 q^{2h}) = 1 - 2h\eta^2 \geq 1 - 2h > 0 \quad \text{on } |\eta| < 1.$$

The derivative is positive, so these equations determine exactly one $q = q(z, \tau)$. They also give $q \asymp \tau + |z|^{1/D}$. Here $\asymp$ means that fixed positive factors give bounds in both directions. On any fixed compact set of similarity coordinates that stays away from $\eta = \pm 1$, the more specific comparison $q \asymp \tau$ holds. If X stays bounded, letting $q \downarrow 0$ takes the coordinates toward the singular point. The endpoints $\eta = \pm 1$, with $q > 0$, instead describe $t = 1$ away from that point, as Lemma 4.1 shows.

The azimuthal and axial profiles $E(X, \eta), U(X, \eta)$ give the leading fields through

$$u_\theta^{(0)} = q^{-A} E, \quad u_z^{(0)} = q^{-A} U, \quad ru_r^{(0)} = V_0, \quad p^{(0)} = q^{-2A} \Pi.$$

Incompressibility and regularity at the axis then fix V_0 through

$$\frac{1}{r} \partial_r(ru_r^{(0)}) + \partial_z u_z^{(0)} = 0, \quad V_0(0, \eta) = 0.$$

The leading radial pressure balance, together with the normalization at radial infinity, gives

$$\partial_r p^{(0)} = \frac{(u_\theta^{(0)})^2}{r}, \quad \Pi(X, \eta) = - \int_X^\infty \frac{E(x, \eta)^2}{2x} dx.$$

The construction makes $E/\sqrt{2X}$, U , and V_0/X smooth at $X = 0$. These are the combinations that have to be smooth for the Cartesian field to be smooth at the axis whenever $t < 1$. In particular, $u_\theta^{(0)} = 0$ on the axis, while $E(X, \eta) > 0$ when $X > 0$.

Choose $X_c > 0$ and $0 < \eta_c < 1$ so that the rectangle $[0, X_c] \times [-\eta_c, \eta_c]$ stays inside the inner region from Theorem 4.6. At time $t = 1 - \tau$, define the core by

$$\mathcal{C}_\tau = \{(r \cos \theta, r \sin \theta, z) : 0 \leq X(r, z, \tau) \leq X_c, |\eta(z, \tau)| \leq \eta_c\}.$$

Throughout this core, $q \asymp \tau$, so its radial and axial sizes satisfy

$$\ell_r \asymp \tau^{1/2}, \quad \ell_z \asymp \tau^{1/2-h}, \quad \frac{\ell_z}{\ell_r} \asymp \tau^{-h}.$$

Both sizes go to zero, but their ratio goes to infinity.

The profiles give the following sizes for the three velocity components inside the core.

$$\begin{aligned} \|u_\theta^{(0)}\|_{L^\infty(\mathcal{C}_\tau)} &\asymp \tau^{-1/2-h}, & \|u_z^{(0)}\|_{L^\infty(\mathcal{C}_\tau)} &\asymp \tau^{-1/2-h}, \\ \|u_r^{(0)}\|_{L^\infty(\mathcal{C}_\tau)} &= O(\tau^{-1/2}), & \frac{\|u_r^{(0)}\|_{L^\infty(\mathcal{C}_\tau)}}{\|u_\theta^{(0)}\|_{L^\infty(\mathcal{C}_\tau)}} &= O(\tau^h). \end{aligned}$$

The lower bound for the axial component comes from the nonzero value on the axis in Proposition B.2. For any fixed $0 < X_* < X_c$, the circle given by $z = 0$, $r = \sqrt{2X_*\tau}$ has $q = \tau$, $\eta = 0$, and on that circle

$$u_\theta^{(0)} = E(X_*, 0)\tau^{-1/2-h} \longrightarrow +\infty.$$

The core is a region of space that changes with time, while its boundary always has the same similarity coordinates. A material region consists of the same fluid particles as the velocity carries them along. Incompressibility keeps the volume of every such region fixed. Figure 4(a) shows the radial regions.

The leading field continues outside the core. Outside a fixed interval of values of X , both $U = V_0 = 0$, while

$$E(X, \eta) = c_\infty X^{-1/2-h}(1 + O(X^{-1})) \quad (X \rightarrow \infty), \quad c_\infty > 0,$$

with one uniform estimate for every η . The exact exterior formula (4.29) makes the exterior field a purely azimuthal solution of the radial heat equation for the azimuthal velocity. The final spatial cutoff in Proposition 10.1 leaves the completed velocity supported inside one fixed compact region.

At viscosity one, the local radial and axial length scales $\ell_r = q^{1/2}$, $\ell_z = q^D$ give

$$\text{Re}_\theta = q^{-A}\ell_r = q^{-h} \longrightarrow \infty, \quad \text{Re}_r = |u_r^{(0)}|\ell_r = O(1).$$

The rotation rate is larger than the rate of radial diffusion. In the axisymmetric tangential equations, radial diffusion and transport in both the radial and axial directions enter the leading balance at the following rates.

$$\frac{|u_r^{(0)}|}{\ell_r} = O(q^{-1}), \quad \frac{q^{-A}}{\ell_z} = \frac{1}{\ell_r^2} = q^{-1}.$$

Axial diffusion is smaller than radial diffusion by the factor $\ell_r^2/\ell_z^2 = q^{2h}$. This small factor is the expansion parameter that orders the corrections to the background.

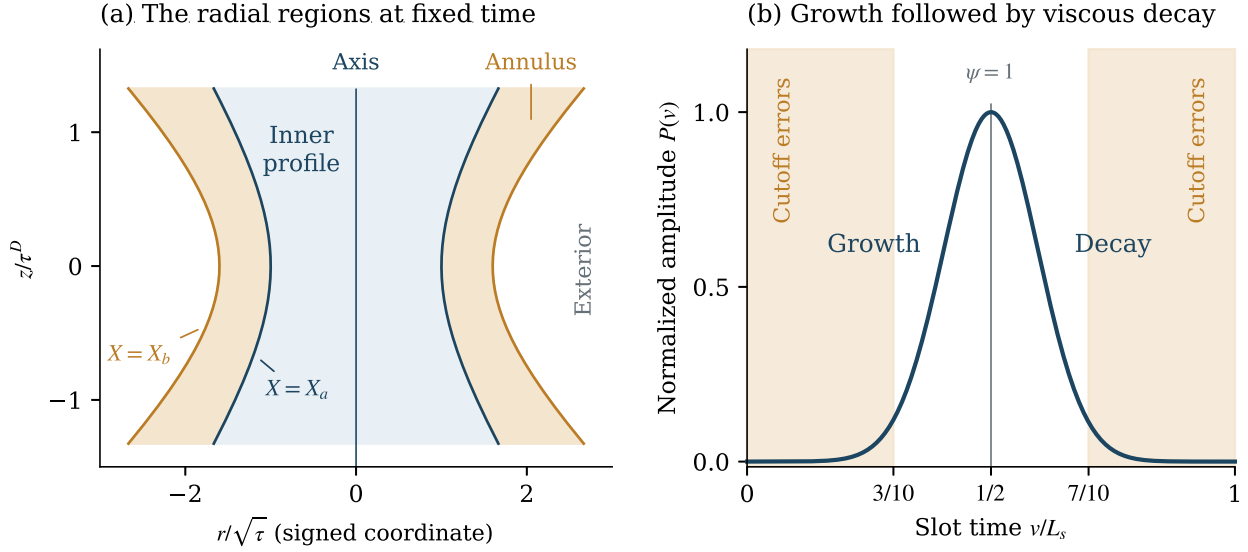


Figure 4: The concentrating geometry and the reference pulse. At one fixed time, panel (a) shows a radial-axial section. The pulse annulus lies between the two level surfaces $X = X_a$ and $X = X_b$. The horizontal and vertical axes are each measured on their own similarity scale, and both physical length scales shrink. Panel (b) shows a reference amplitude $P(v)$. It is normalized to equal one at the midpoint of a pulse interval of length L_s in v , and it first grows and then decays. The shaded parts contain the small tails made by the cutoff. The radii and pulse parameters in the picture are only representative. The proof chooses their actual values from the estimates in the sections that build the profiles and pulses.

3.2 Construction of the background stress.

Next, choose the profiles E, U so that the pulses can supply the remaining momentum transport. The radial integrals below define the resulting stress T . This stress has to vanish outside the pulse annulus, and inside the annulus it has to be a positive combination of the fluxes carried by the two wave families. Theorem 4.6(ii) gives the required support. The admissible stress cone condition from Lemma 4.5 is the condition that makes the positive representation in Proposition 7.5 possible.

Let $R_\theta^{(0)}, R_z^{(0)}$ be the tangential momentum residuals of $(u^{(0)}, p^{(0)})$. These residuals include radial viscosity and leave out axial viscosity. Construct physical stress components $T = (T_{r\theta}, T_{rz})$ that satisfy

$$R_\theta^{(0)} = -(\partial_r + 2/r)T_{r\theta}, \quad R_z^{(0)} = -(\partial_r + 1/r)T_{rz}.$$

Regularity at the axis fixes the constants of integration, giving

$$T_{r\theta}(r) = -\frac{1}{r^2} \int_0^r s^2 R_\theta^{(0)}(s) ds, \quad T_{rz}(r) = -\frac{1}{r} \int_0^r s R_z^{(0)}(s) ds.$$

Require T to be nonzero exactly in the annulus $X_a < X < X_b$, which is the same region as $\sqrt{2qX_a} < r < \sqrt{2qX_b}$, with $0 < X_a < X_b$. Make the residual zero both in the inner region and in the heat exterior, and impose the two moment identities

$$\int_0^\infty r^2 R_\theta^{(0)} dr = 0, \quad \int_0^\infty r R_z^{(0)} dr = 0.$$

These identities make the stress zero beyond the exterior radius as well as near the axis. The prescribed exterior moment identities give exactly these two zero integrals, by Lemma A.8. Matching the profiles and the five radial integrals from (4.15) keeps the exterior fields unchanged, by Lemma 4.4.

At each point of the annulus, the admissible stress cone condition lets the pulse construction choose two wave families whose covariance vectors $v_1, v_2 \in \mathbb{R}^2$ are linearly independent. Each vector has two entries. They record the azimuthal and axial momentum flux produced there by one unit of squared amplitude from that wave family in the local background. The two vectors can produce the required stress with positive weights,

$$T = c_1 v_1 + c_2 v_2, \quad c_1, c_2 > 0 \quad \text{where } T \neq 0.$$

Because both coefficients are positive, T lies inside the cone generated by v_1, v_2 . This cone is the set of all nonnegative combinations of the two vectors, and being inside it means that this representation uses both directions with strictly positive coefficients. Proposition 7.5 builds the representation from the admissible stress cone condition on the profiles. Figure 5 shows the same positive representation.

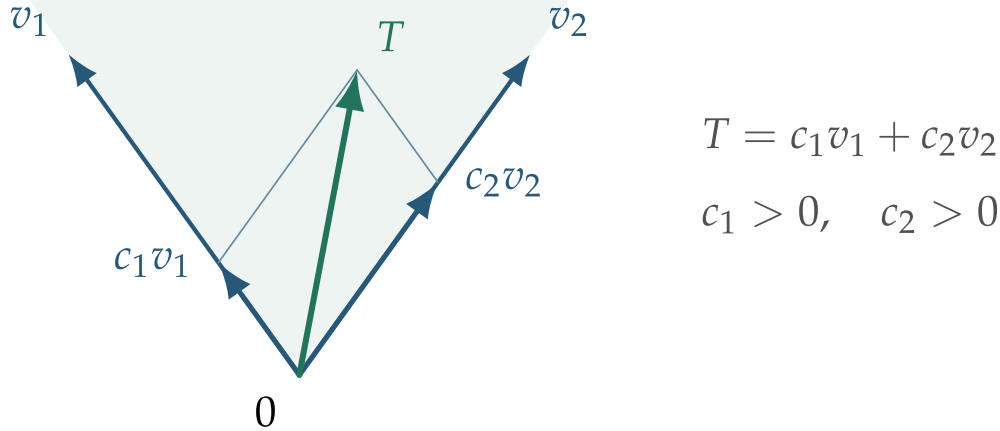


Figure 5: Positive representation of the target stress. The shaded cone is the set of all nonnegative linear combinations of the covariance directions v_1, v_2 . The target $T = c_1 v_1 + c_2 v_2$ is inside this cone because $c_1, c_2 > 0$. The dashed lines show the two weighted vectors adding to the target stress. If the two directions change by a sufficiently small amount, both coefficients stay positive.

These conditions tie the inner and exterior profiles together because both the pressure and the radial velocity depend on cumulative radial integrals, the integrals collected from the axis out to the current radius. For that reason, the pressure datum comes first. The equations are then solved near the axis, and the integrals are matched before the cone condition is imposed.

1. *Exterior and axis pressure.* Proposition A.4 builds reference profiles with a purely azimuthal tail. Their radial pressure integral (4.31) fixes $\Pi(0, \eta)$, the pressure value on the axis. Replacing this tail with the exact heat flow from Lemma A.6 changes the integral. The corrections in Proposition A.7 put back the same integral at smaller radii, so the pressure on the axis stays unchanged.
2. *Inner solution.* Using this pressure value and the specified axis data, Proposition B.2 constructs $E / \sqrt{2X}, U, \Pi$ as functions that are analytic in X, η near the axis, meaning that around each point, each function equals a convergent power series in those two variables, and satisfy (4.13). Their leading tangential residual and their stress are both zero on $X \leq X_a$.
3. *Joining and moment matching.* Corollary B.10 joins the inner profiles to the prescribed outer profiles. The correction in Proposition B.8 makes the five cumulative radial integrals agree. These are the integrals that determine the pressure, radial velocity, and stress. When both the profiles and these integrals agree at the joining radius, Lemma 4.4 gives back the prescribed exterior fields. The inner identity (4.13) and the exterior conclusion of Lemma A.8 then give $T = 0$ for $X \leq X_a$ and $X \geq X_b$.
4. *Enforcing the cone condition.* On a compact subinterval of the annulus, Proposition C.2 adds a radial oscillation with phase $N \log X$, where N is one large fixed integer. The oscillation itself has amplitude $O(N^{-1})$. Applying $X \partial_X$ removes that small factor from the derivative and produces a change of order one in the radial derivatives. This changes the leading tangential shear while changing the profile values and radial moments by only $O(N^{-1})$. The shear follows a periodic curve that satisfies the admissible stress cone condition. A separate correction, supported in the same local region, restores all five radial integrals exactly. The inner solution and the heat exterior stay unchanged, and the resulting stress has positive covariance weights everywhere in the annulus, by Proposition C.3.

The leading profiles are now fixed. Before the pulses are added, Section 5 builds corrections in the successive powers q^{2nh} , $n = 1, 2, \dots$. At each order, the coefficients already found provide the source terms in the equations for the next coefficients. This recursion takes care of the radial momentum equation, axial viscosity, and all the lower-order terms still left. Smooth cutoffs are then applied to the vector potentials and pressures before the coefficients are added together. Each cutoff equals one near the singularity, and the supports shrink from one correction to the next. Taking the curl after multiplication by the cutoff keeps the field incompressible. The resulting smooth axisymmetric background (u_B, p_B) satisfies

$$\begin{aligned}\mathcal{R}(u_B, p_B) &= -(\partial_r + 2/r)\mathcal{T}_{\text{phys},\theta}e_\theta \\ &\quad - (\partial_r + 1/r)\mathcal{T}_{\text{phys},z}e_z + \mathcal{E}_B.\end{aligned}$$

The stress $\mathcal{T}_{\text{phys}}$ is supported in the annulus, and its leading term is T . Calling a residual *flat* as $q \downarrow 0$ means that every Cartesian space-time derivative is $O(q^N)$ for every $N \geq 0$, with one uniform estimate on each bounded interval of X . Proposition 5.5 proves that the remainder $\mathcal{E}_B$ is flat in this sense.

3.3 Oscillations and momentum transport.

At this point, the background (u_B, p_B) has the required growth and the leading annular stress T that the two pulse families can produce. The next step builds divergence-free pulses, using Lemma 7.7. When their quadratic flux is averaged, its leading term supplies T , by Propositions 7.5 and 9.5. The exact increment identity below keeps this cancellation separate from the linear evolution of the pulses and from the interactions that later corrections must handle.

For a divergence-free velocity increment w and a pressure increment π , the residual obeys the exact identity

$$\mathcal{R}(u_B + w, p_B + \pi) = \mathcal{R}(u_B, p_B) + \mathcal{L}_{u_B}(w, \pi) + \nabla \cdot (w \otimes w),$$

where

$$\mathcal{L}_{u_B}(w, \pi) := \partial_t w + (u_B \cdot \nabla)w + (w \cdot \nabla)u_B - \Delta w + \nabla \pi.$$

The linear term describes how the pulses evolve in the background. The quadratic term provides the stress that cancels the leading part of $\mathcal{R}(u_B, p_B)$.

Write A_{wave} for the typical size of a pulse oscillation and ℓ_{wave} for its wavelength. At points in the pulse annulus where the profile values are fixed and nonzero, set aside the logarithmic factors and the weights that go to zero at the edge of the support. The remaining sizes are

$$\begin{aligned}A_{\text{wave}} &\asymp q^{-1/2-h/2}, & \ell_{\text{wave}} &\asymp q^{1/2+h/2}, \\ \frac{A_{\text{wave}}}{q^{-1/2-h}} &\asymp \frac{\ell_{\text{wave}}}{q^{1/2}} \asymp q^{h/2}.\end{aligned}$$

The square of the amplitude has the required stress scale.

$$A_{\text{wave}}^2 \asymp q^{-1-h} \asymp \frac{|u_\theta^{(0)}|}{q^{1/2}}, \quad \frac{A_{\text{wave}}^2}{q^{1/2}} \asymp q^{-3/2-h}.$$

The last quantity is the size of the averaged radial stress divergence. It matches the leading time derivative, transport, and radial viscosity in the tangential momentum equations.

Freeze the amplitude and the phase gradient at one point. The leading wave then has the following schematic local form.

$$w_{\text{lead}}(x) = a \cos(\tilde{\zeta} \cdot x + \varphi), \quad a \cdot \tilde{\zeta} = 0.$$

Here $a \in \mathbb{R}^3$ is the constant amplitude, $\tilde{\zeta} \in \mathbb{R}^3$ is the wavevector, and $\varphi \in \mathbb{R}$ is the phase. The displayed orthogonality condition says that the velocity oscillates in a direction perpendicular to its wavevector. Different localized pulses have to stay separate.

At the same time, the construction still needs periodic averaging. It handles both needs by introducing an extended field $\tilde{w}(r, \theta, z, t, Y)$ that depends on a separate auxiliary variable $Y \in \mathbb{T}^2$. The physical field comes from putting the fixed phase map into that variable,

$$w(r, \theta, z, t) = \tilde{w}(r, \theta, z, t, Y(r, t)),$$

where $Y(r, t)$ is defined in (6.3). The covariance is averaged before this substitution is made. The angular and auxiliary averages are

$$\langle F \rangle_\theta := \frac{1}{2\pi} \int_0^{2\pi} F d\theta, \quad \langle F \rangle_Y := \int_{\mathbb{T}^2} F dY, \quad \int_{\mathbb{T}^2} 1 dY = 1.$$

Doing the two averages one after the other is written as $\langle F \rangle := \langle \langle F \rangle_\theta \rangle_Y$. Equations (3.7)–(3.9) collect these conventions. All components are taken in the cylindrical basis. If different labeled pulses overlap in their slowly varying space–time supports, their supports in Y are chosen to be disjoint. Their products are then zero at every point, even after $Y(r, t)$ is substituted, by Lemma 6.1. Harmonics and corrections with the same label still interact with one another.

In the actual construction, the amplitudes, phases, and cutoffs all vary. First evaluate the localized vector potentials at $Y(r, t)$ to obtain $\mathcal{A}_{\text{wave}}$. Then take the physical curl,

$$w = \nabla \times \mathcal{A}_{\text{wave}}, \quad \nabla \cdot w = 0.$$

This curl makes the resulting velocity divergence-free. Every chain-rule term coming from $Y(r, t)$ is included. Differentiating the amplitudes and cutoffs also produces smaller velocity terms, and Lemma 7.7 keeps all of them.

The chosen phases have nonzero integer angular frequencies. Because of that, each wave has angular average zero, while the angular average of the square of its cosine is $1/2$. The two wave families produce the momentum-flux directions used above to choose $c_1, c_2 > 0$. Those coefficients then become the squared amplitude weights. After the physical scaling and the corrections made by taking the curl, the covariance is

$$\begin{pmatrix} \langle \tilde{w}_r \tilde{w}_\theta \rangle \\ \langle \tilde{w}_r \tilde{w}_z \rangle \end{pmatrix} = T + \text{higher-order terms}.$$

The higher-order terms include the corrections made by the curl. They are smaller by positive powers of q , with derivative bounds proved in Corollary 7.8. Taking the radial divergence of the leading covariance shown above cancels the leading negative stress divergence of the background, by Propositions 7.5 and 9.5. The extended residual still contains oscillating terms and angular means that depend on Y . In the physical equation, these terms are evaluated at $Y(r, t)$.

The same quadratic products also control the energy passed from the background to the waves. Take the homogeneous linearized model, where the linearized equation has zero right-hand side,

$$\mathcal{L}_{u_B}(w_{\text{lin}}, \pi_{\text{lin}}) = 0, \quad \nabla \cdot w_{\text{lin}} = 0.$$

If the fields decay fast enough in space, integration by parts gives

$$\frac{d}{dt} \frac{1}{2} \int |w_{\text{lin}}|^2 = - \sum_{i,j} \int (w_{\text{lin}})_i (w_{\text{lin}})_j \partial_j (u_B)_i - \int |\nabla w_{\text{lin}}|^2.$$

The first term moves energy between the background and the perturbation. The second term dissipates energy. The chosen transverse velocity amplitudes draw energy from the shear and provide the two momentum-flux directions whose positive span contains T . During each pulse, the shear increases the size of the radial component of the wavevector. Viscous damping then grows until it is larger than the amplification, so the pulse first grows and then decays. The time cutoff acts only in the exponentially small tails. The phase and amplitude equations prove these facts in Lemmas 7.1 and 7.4, and Figure 4(b) shows the resulting envelope.

The correction cycle now deals with all the linear and quadratic interactions that are still left, including the curl and cutoff errors in $\mathcal{R}(u_B + w, p_B + \pi)$.

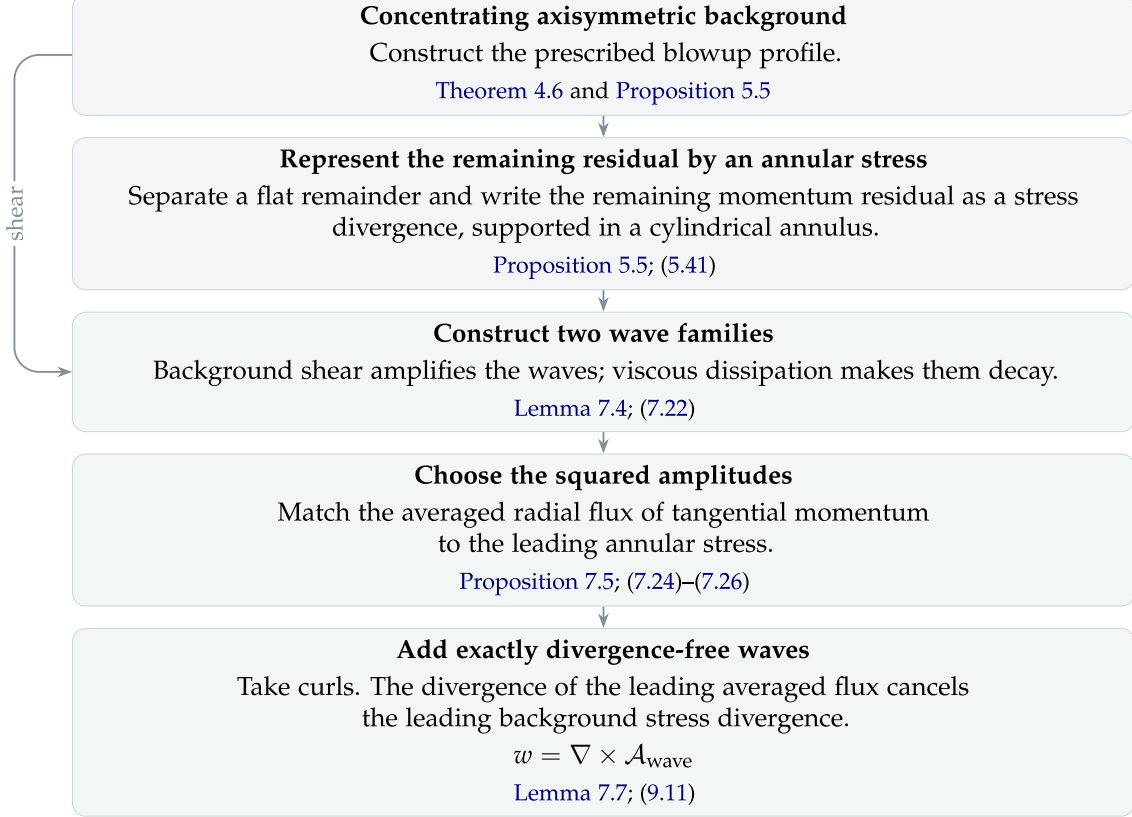


Figure 6: How the background and the first oscillatory correction are built. The averaged pulse flux cancels the divergence of the leading background stress. The cycle in Figure 7 corrects the residual that is left. The flat remainder and all curl and cutoff terms stay in the calculation.

3.4 Correcting the residual and adding up the corrections

The average wave flux is chosen to cancel the divergence of the leading stress that comes from T (Propositions 7.5 and 9.5). This leaves the remaining angular harmonics, auxiliary means, and radial integral defects, and each of these needs its own correction. The corrections always follow the same cycle. After each operation, the full residual is calculated again, as in Proposition 9.6, so any new terms created by that operation go into the next stage.

Start with $(u_B + w, p_B + \pi)$. First reconstruct the pressure, then correct the angular average of the velocity and the errors in its radial integrals. Proposition 9.5 proves that the resulting pair, $(u^{[0]}, p^{[0]})$, has all the bounds needed to begin the correction cycle. Square brackets count the number of completed cycles. The superscript (0) still marks the leading field $(u^{(0)}, p^{(0)})$.

At stage j , add a divergence-free velocity increment δu_j and a pressure increment δp_j :

$$u^{[j+1]} = u^{[j]} + \delta u_j, \quad p^{[j+1]} = p^{[j]} + \delta p_j.$$

The residual then changes by

$$\begin{aligned} \mathcal{R}(u^{[j+1]}, p^{[j+1]}) &= \mathcal{R}(u^{[j]}, p^{[j]}) + \mathcal{L}_{u^{[j]}}(\delta u_j, \delta p_j) \\ &\quad + \nabla \cdot (\delta u_j \otimes \delta u_j). \end{aligned}$$

Here $\mathcal{L}_v$ is the linear residual operator defined earlier, with v as its background. So cancelling one chosen source creates new linear remainders and new quadratic interactions. The four operations in Proposition 9.6 handle those terms in the following order.

1. The first operation solves the inhomogeneous wave-amplitude equations for the supported nonzero angular Fourier modes, the terms in the angular Fourier expansion whose integer frequency is not zero. The principal operator is the part of the linearized residual that acts on one new amplitude and its pressure at the current harmonic. It is written explicitly in (9.1), and solving its equation cancels the chosen source. The remaining linear terms and the interactions between the new pulses go into the next residual (Proposition 9.1 and Lemma 9.2).
2. The second operation builds signed changes in amplitude from the fixed positive leading amplitudes. For an amplitude change and a leading pulse, their symmetrized cross covariance adds two doubly averaged products, the leading radial component times the changed tangential component and the changed radial component times the leading tangential component. That sum gives the chosen correction to the average stress. Their interactions with earlier corrections, along with their quadratic self-interaction, remain as higher-order terms (Proposition 7.6, (9.13)).
3. The third operation corrects the angular average of the residual after its auxiliary average has been made zero. It does this by inverting the fast auxiliary-time derivative in (8.20). The axial increment comes from the vector potential in (8.14), which also gives the radial velocity that goes with it. The reconstruction leaves a remainder whose Cartesian space-time derivatives go to zero faster than every power of q ; that remainder stays in the residual.
4. The fourth operation solves the five radial moment equations in (8.25). Two of them keep the angular-momentum and axial-flux integrals in (9.10) equal to zero. The other three cancel the linear contributions to $(\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z)$. These are the errors in the radial integrals for pressure and tangential momentum defined in (8.12) and (8.15). Their nonlinear remainders decay faster, as (8.27) states. The radial integral constructions also keep the velocity and pressure corrections inside a compact support.

After each operation, the pressure is reconstructed from the updated radial equation, and its part in the axial equation is included in the new residual. Every product uses the updated fields, including the curl and cutoff corrections. The background, the leading amplitudes, and the inverse operators stay fixed for the whole induction.

The parameter σ_j in (9.8) records how fast the residual decays. For a Cartesian derivative of order m , the bound in (9.18) contains the power $q^{h\sigma_j - K_m}$. Here K_m does not depend on j . The same bound also contains logarithmic factors and the remainder that is being kept separate. The constants and logarithmic powers may depend on j, m . At the start, and after each correction cycle, the values are

$$\sigma_0 = \frac{1}{5}, \quad \sigma_{j+1} = \sigma_j + \frac{1}{10}, \quad \sigma_j = \frac{1}{5} + \frac{j}{10} \rightarrow \infty.$$

The cumulative velocity bounds in (9.9), the support conditions, and the two preserved integrals in (9.10) all continue to hold, so the cycle can be run again. Lemma 9.7 gives one common domain on which every finite stage is defined.

To add up all the corrections, first put cutoffs on their vector potentials and pressures, then take curls. Put the same cutoffs on the direct axisymmetric azimuthal increments. Each cutoff is equal to one close enough to $q = 0$, and its support gets smaller as the stage increases. The bounds in (9.17) lose a power of q that depends on the order of the derivative but not on the stage. Because of these bounds, the cutoffs can be chosen so that every differentiated tail obeys (5.35). At any point with $q > 0$, only finitely many terms in the sum are nonzero. The sum gives smooth fields $(u_{\text{loc}}, p_{\text{loc}})$ for $t < 1$. Compare their residual with the residual at any one fixed finite stage to get (9.20). The leading growth of the velocity is not changed (Proposition 9.9).

This sum gives the needed local growth of the velocity. Passing from the local construction to the whole space also needs a vector-potential representation and regularity through time one everywhere away from the origin. Proposition 9.9 proves both facts, along with the flatness of the residual. In the theorem, (u, p) means the local pair $(u_{\text{loc}}, p_{\text{loc}})$.

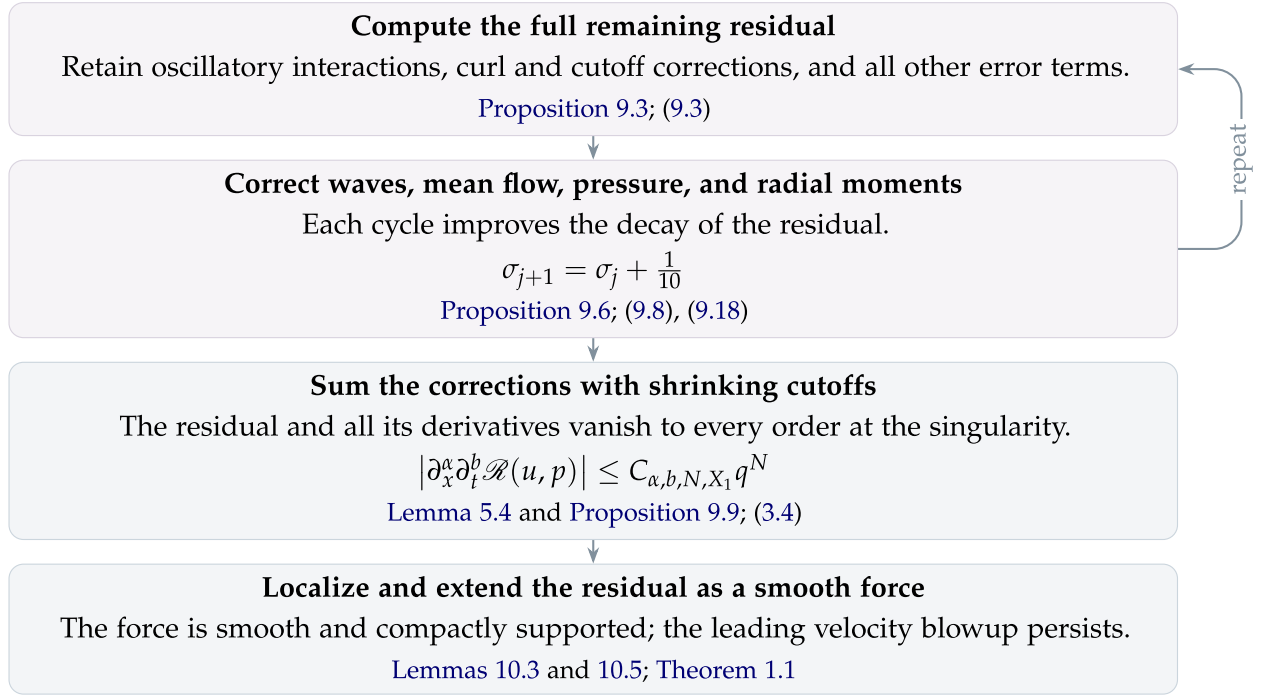


Figure 7: Correcting the residual and finishing the construction, continuing Figure 6. Each turn of the correction cycle raises the residual exponent σ_j , while every linear and nonlinear term stays in the calculation. The flatness estimate holds uniformly for $0 \leq X \leq X_1$ as $q \downarrow 0$, for each finite X_1 . The velocity stays incompressible through the whole construction.

The notation $\partial_x^\alpha \partial_t^b$ means Cartesian derivatives in space and time. Here α is a multi-index for the spatial derivatives, and $b \geq 0$ is an integer giving the number of time derivatives.

Theorem 3.1. *There are constants $0 < h < 1/100$, $q_* > 0$, $0 < X_a < X_{\text{ext}} < \infty$, and smooth fields on*

$$\Omega_* = \{(x, t) : \tau > 0, q < q_*\}$$

with the following properties.

- (i) *There is a vector potential $\mathcal{A}$ and an axisymmetric scalar $\mathcal{B}$ such that*

$$u = \text{curl } \mathcal{A} + \mathcal{B}e_\theta. \quad (3.3)$$

Both $\mathcal{A}$ and $\mathcal{B}e_\theta$ are smooth in Cartesian coordinates at the axis. The field is exactly divergence-free.

- (ii) *Every Cartesian space-time derivative of $\mathcal{A}, \mathcal{B}e_\theta, p$ stays uniformly bounded on every compact spatial set where $c \leq q \leq c' < q_*$, for each $0 < c < c'$, all the way up to the one-sided boundary $\tau = 0$. The limits of these derivatives agree with one another when another spatial or time derivative is taken.*
- (iii) *For every α, b, N and every finite X_1 ,*

$$|\partial_x^\alpha \partial_t^b \mathcal{R}(u, p)| \leq C_{\alpha,b,N,X_1} q^N \quad (0 \leq X \leq X_1, q \downarrow 0). \quad (3.4)$$

For $X \geq X_{\text{ext}}$, the residual is exactly zero, and

$$\mathcal{A} = 0, \quad \mathcal{B} = K(r, \tau), \quad p = - \int_r^\infty \frac{K(\rho, \tau)^2}{\rho} d\rho, \quad K(r, \tau) = r^{-1-2h} \mathcal{H}_{\text{ext}}(\tau/r^2). \quad (3.5)$$

Here $\mathcal{H}_{\text{ext}}$ is smooth on $[0, \infty)$, and each one of its fixed derivatives is bounded. The radial swirl heat equation obeyed by K is

$$-\partial_\tau K = \partial_r^2 K + r^{-1} \partial_r K - r^{-2} K.$$

In this statement, the second input of K is τ . Later, the notation $K(r, t)$ means the same field evaluated at $\tau = 1 - t$.

(iv) There are fixed values $X_{\text{in}} \in (0, X_a)$ and $e_0 > 0$ for which

$$u_\theta(\sqrt{2X_{\text{in}}\tau}, 0, 0, 1 - \tau) = \tau^{-A}(e_0 + O(\tau^{2h})) \quad (\tau \downarrow 0). \quad (3.6)$$

The four entries on the left give the cylindrical position and the time.

3.5 Localizing the construction and finishing the proof

The last step uses the decomposition in (3.3). Multiply the vector potential $\mathcal{A}$, the azimuthal coefficient $\mathcal{B}$, and the pressure p_{loc} by $c = \chi_x \chi_t$. The spatial cutoff χ_x is smooth and axisymmetric and has compact support $\mathcal{K}$. The time cutoff χ_t is smooth and satisfies $\chi_t = 0$ on $[0, 1 - \tau_0]$ for some $0 < \tau_0 < 1/2$. Both cutoffs are equal to one near $(0, 1)$. Since $q \asymp \tau + |z|^{1/D}$, their supports can be chosen so that $q < q_*/2$ wherever $c \neq 0$ and $t < 1$. After multiplying, take the curl and extend the fields by zero outside the support of the cutoffs:

$$u = \text{curl}(c\mathcal{A}) + c\mathcal{B}e_\theta, \quad p = cp_{\text{loc}}, \quad u = cu_{\text{loc}} + \nabla c \times \mathcal{A}.$$

The curl term and the axisymmetric azimuthal term are each divergence-free. The chosen representatives are smooth in Cartesian coordinates, and their support stays away from $q = q_*$, so extending them by zero is smooth. The fields u, p have the fixed spatial support $\mathcal{K}$, vanish throughout the stated initial time interval, and agree with the local fields near $(0, 1)$ (Proposition 10.1).

For $t < 1$, set $f = \mathcal{R}(u, p)$. When $f - c\mathcal{R}(u_{\text{loc}}, p_{\text{loc}})$ is expanded, differentiating the cutoffs gives terms such as $(\partial_t c - \Delta c)u_{\text{loc}}$ and $p_{\text{loc}} \nabla c$. The curl correction $\nabla c \times \mathcal{A}$ adds its derivatives and its advection products. The change in self-advection also contains $(c^2 - c)(u_{\text{loc}} \cdot \nabla)u_{\text{loc}}$. All these extra terms live in the parts where the cutoffs are changing, away from a neighborhood of $(0, 1)$.

Near $(0, 1)$, the cutoffs are equal to one, so f is the local residual. On $X \leq X_{\text{ext}}$, its flatness estimate (3.4), together with the exactly zero residual for larger X , gives the bound (10.9) for every Cartesian space-time derivative. In the cutoff transition regions near time one, Theorem 3.1(ii) bounds every derivative of the actual potentials and pressure because q stays positive there. In the part that remains, r stays positive while $q \rightarrow 0$. There $\mathcal{A} = 0$, and the heat bounds in (10.8), together with the normalized pressure integral in (3.5), give the limits of the derivatives. These estimates, the uniform flatness bound near the origin, and the fixed support $\mathcal{K}$ give compatible limits, uniformly on $\mathbb{R}^3$, for every derivative of f (Lemma 10.2). Lemma 10.3 realizes those limits from $t > 1$ with a force supported in $\mathcal{K} \times [0, 2]$:

$$f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3), \quad \mathcal{R}(u, p) = f \quad (0 \leq t < 1).$$

The growth path in Theorem 3.1(iv) eventually stays in the region where the cutoffs are equal to one, so the localized velocity keeps the asymptotic formula (3.6) and becomes unbounded as $t \uparrow 1$.

The concentration scales still allow the energy to stay bounded. The core $\mathcal{C}_\tau$ has volume of order $\tau^{3/2-h}$. Its leading kinetic energy E_{core} , and the integral of the squared radial derivatives of its leading velocity, D_{core} , scale as

$$E_{\text{core}} \asymp \tau^{3/2-h} \tau^{-1-2h} = \tau^{1/2-3h}, \quad D_{\text{core}} \asymp \tau^{3/2-h} \tau^{-2-2h} = \tau^{-1/2-3h}.$$

In each product, the first factor is the volume of the region, and the second is either the squared velocity or the squared radial derivative. Since $h < 1/6$,

$$\tau^{1/2-3h} \rightarrow 0, \quad \int_0^{\tau_0} \tau^{-1/2-3h} d\tau = \frac{\tau_0^{1/2-3h}}{1/2-3h} < \infty \quad (\tau_0 > 0).$$

Lemma 10.4 proves the global energy bound and the bound on integrated dissipation for the localized fields directly from their equation.

Lemma 10.5 now applies to any smooth solution with the same force and zero initial data whose kinetic energy stays uniformly bounded. That solution must agree with u on every $[0, T]$ with $T < 1$. If a global smooth solution existed, it would agree with the constructed velocity for $0 \leq t < 1$. Along the growth path, however, the constructed velocity is not bounded on any compact neighborhood of $(0, 1)$, while a global smooth solution would be. This contradiction proves Theorem 1.1.

3.6 Notation

Every choice of profile is fixed before $q \downarrow 0$. Constants may depend on those choices and on the stated order of differentiation, but not on the concentration scale. The notation $O(q^a)$ means that the quantity is bounded by Cq^a on the stated domain. Any uniformity in other parameters is stated when it is needed. The background expansion order n , the correction stage j , the dyadic band index ℓ , and the Fourier harmonic index are four different indices. The dyadic band index labels successive scales, each half the size of the preceding one, and records the scale to which the concentration parameter is comparable.

Averages. Let $F(r, \theta, z, t, Y)$ be a scalar field on the extended domain. Its auxiliary variable $Y \in \mathbb{T}^2 = \mathbb{R}^2 / \mathbb{Z}^2$ is independent of the physical coordinates. The angular average and the auxiliary average are

$$\begin{aligned} \langle F \rangle_\theta(r, z, t, Y) &:= \frac{1}{2\pi} \int_0^{2\pi} F(r, \vartheta, z, t, Y) d\vartheta, \\ \langle F \rangle_Y(r, \theta, z, t) &:= \int_{\mathbb{T}^2} F(r, \theta, z, t, Y') dY', \quad \int_{\mathbb{T}^2} 1 dY' = 1. \end{aligned} \quad (3.7)$$

The angular average varies the angle while keeping the independent auxiliary coordinate fixed. The auxiliary average varies that coordinate while keeping all the physical coordinates fixed. For vector and tensor fields, both averages act on the cylindrical components. Applying both of them, in either order, gives the fast average used for the wave fields:

$$\langle F \rangle := \langle \langle F \rangle_\theta \rangle_Y = \langle \langle F \rangle_Y \rangle_\theta = \frac{1}{2\pi} \int_{\mathbb{T}^2} \int_0^{2\pi} F(r, \vartheta, z, t, Y) d\vartheta dY. \quad (3.8)$$

These averages are taken before the field is restricted to the physical phase map in (6.3). They keep (r, z, t) fixed, or, in any one normalized chart, keep (R, Z, T) fixed. By (6.19), the normalized Haar averages agree on the absolute auxiliary torus, each band torus, and the common auxiliary torus. For the wave construction, this two-step average is the whole fast average, including its normalization. If a coefficient is already an angular mean, its part with auxiliary average zero is

$$f^\circ := f - \langle f \rangle_Y, \quad \langle f^\circ \rangle_Y = 0. \quad (3.9)$$

The profile construction also has a different auxiliary loop angle, θ' , with period 2π . Its average is $\langle f \rangle_{\theta'} = (2\pi)^{-1} \int_0^{2\pi} f(\theta') d\theta'$, as in Lemmas C.1 and 4.11. In that local profile notation, brackets with no subscript mean this loop average. The radial average is

$$\mathcal{A}_X(f)(X, \eta) = \frac{1}{X} \int_0^X f(X', \eta) dX'. \quad (3.10)$$

The auxiliary torus supplies the parameters for the oscillations. The torus in Corollary 10.6 is a different object, the spatial domain of the periodic solution.

Normalized operators and coefficient classes. Inside a dyadic chart, $Q = 2^{-\ell}$ is fixed, $q \asymp Q$, and

$$(R, Z, T) = (r/\sqrt{Q}, z/Q^D, \tau/Q), \quad \varepsilon = Q^h, \quad S_* = \ell^2.$$

The star on a spatial operator includes the factor $\sqrt{Q}$ that changes physical spatial derivatives into chart units. A physical derivative also follows the auxiliary phase map through the chain rule. With the radial operator $\mathfrak{r}$ from (6.4), set $D_r = \sqrt{Q} \mathfrak{r}$, $D_z = \sqrt{Q} \partial_z = \varepsilon \partial_Z$, and

$D_\theta = R^{-1}\partial_\theta$, as in (6.6). For a scalar f and a vector $a = (a_r, a_\theta, a_z)$ written in cylindrical components, the normalized gradient, divergence, and curl are

$$\begin{aligned}\nabla_* f &= (D_r f, D_\theta f, D_z f), \\ \operatorname{div}_* a &= (D_r + R^{-1})a_r + D_\theta a_\theta + D_z a_z, \\ \operatorname{curl}_* a &= (D_\theta a_z - D_z a_\theta, D_z a_r - D_r a_z, (D_r + R^{-1})a_\theta - D_\theta a_r).\end{aligned}\tag{3.11}$$

The separate factors that normalize the fields themselves are given in (8.11). The physical momentum residual is $\mathcal{R}$ from (3.1).

The classes $W_\alpha, M_\alpha, S_\alpha$ keep track, in normalized units, of wave amplitudes, angular mean coefficients, and radial moments. A coefficient derivative differentiates the amplitude after its oscillatory exponential has been factored out, while keeping the band, label, and harmonic fixed. At each fixed correction stage and each fixed coefficient-derivative order, ∂^I means ordinary differentiation in (R, Z, T, H) . Here H is the common auxiliary coordinate from (6.17). For moments, only (Z, T) are differentiated. Their bounds have the following forms. The first two bounds hold on $X_a < X < X_b$, and the wave bound is also restricted to its labelled rectangle:

$$\begin{aligned}f \in M_\alpha &\implies |\partial^I f| \leq C_{j,I} \varepsilon^\alpha S_*^{b_{j,I}} \zeta \delta^{-d_{j,I}}, \\ a \in W_\alpha &\implies |\partial^I a| \leq C_{j,I} \varepsilon^\alpha S_*^{b_{j,I}} \sqrt{\zeta} \delta^{-d_{j,I}} P_v \quad (0 \leq v \leq L_s), \\ F \in S_\alpha &\implies |\partial_{Z,T}^I F| \leq C_{j,I} \varepsilon^\alpha S_*^{b_{j,I}}.\end{aligned}$$

Here ζ, δ are the weights near the two radial edges from (6.23), and P_v is the pulse envelope from (7.12), used as a pointwise weight. The constants $C_{j,I} > 0$ and $b_{j,I}, d_{j,I} \geq 0$ can be different in the three rows. They may depend on the fixed stage, the derivative order, and the profile choices, but they are uniform over the band, label, copy of the rectangle, and point. Definitions 6.4 and 6.5 give the full definitions, including the support, compatibility, and smooth-extension conditions. An angular mean coefficient can still depend on the auxiliary torus.

Definition 3.2 (Regularity at the axis). A leading field is regular at the axis when its velocity and pressure extend smoothly in Cartesian coordinates across $r = 0$, for $t < 1$. For scalar auxiliary profiles, “regular at the axis” means that the profile extends smoothly in (X, η) to $X = 0$.

Definition 3.3 (Small parameters and the order in which they are chosen). For $\alpha \geq 0$ and $\beta > 0$, the notation $\alpha \ll \beta$ means that α/β must be small enough, where how small it must be can depend on all the data already fixed. In a chain of constants, choose them from right to left. A small reciprocal means that the corresponding parameter is large. When an expression depends on profile variables, the stated smallness holds uniformly on its stated domain and follows from those choices of parameters.

The next table lists the main symbols in the construction and points to their definitions. It puts the different background fields in one group. When the same letter is used more than once, the table tells the uses apart by their role or by the section where they appear. An order attached to a coefficient class refers to its normalized representative, with the weights and derivative bounds given in the definition of that class.

Table 1: Guide to the main symbols.

Symbol	Meaning and job	Definition
Physical fields and the background construction		
u, p, f, ν	Velocity, pressure, external force, and physical viscosity in the Navier–Stokes equation. The local construction takes $\nu = 1$.	(1.1); Section 3
$\mathcal{R}(u, p)$	The momentum residual at viscosity one, namely $\partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p$.	(3.1)
$u^{(0)}, p^{(0)}$	The leading axisymmetric velocity and pressure. The velocity has radial, azimuthal, and axial parts, and its swirl is $u_\theta^{(0)}$.	(4.3)
E, U, V_0, Π	The leading profiles for the swirl, the axial velocity, the radial flux $ru_r^{(0)}$, and the pressure, in that order.	(4.3), (4.7)
ϕ, F, H in the profiles	The smooth swirl profile $F = E/\sqrt{2X} = \phi/C$, and the angular-momentum profile $H = \sqrt{2X}E$. The normalization $C > 1$ is fixed.	(4.4), (4.8)
$\mathbf{m}, (M, I, J, S, C_p)$	Five cumulative radial integrals that keep the pressure, radial velocity, and stress unchanged when profile pieces are joined. The quantity C_p is the pressure increment from the axis.	(4.15); Lemma 4.4
Q_s, N_s	Normalized radial antiderivatives of the inviscid azimuthal and axial momentum residuals. They are used to build the stress vector p_s .	(4.9), (4.16), (4.11)
$\mathfrak{s} = (a, -b_s), p_s$	The normalized radial shear and the integrated inviscid-stress vector, with $\mathcal{T}_0 = F(p_s - \mathfrak{s})$. Here p_s is a stress quantity with two components.	(4.11)
t_s, v_s, P_c, J_c	The direction and strength of the shear, followed by the unnormalized components of p_s along and across that direction. Together they describe the positive covariance cone.	(4.20); Equation (4.21)
$E_n, U_n, V_n, \Pi_n, \lambda_n = 2nh$	Formal coefficient profiles for the background at order q^{2nh} . The index n counts the orders in the background expansion.	(5.1), (5.2)
u_B, p_B	The smooth corrected axisymmetric background, made by adding the coefficients with cutoffs on the vector potentials, the direct swirl terms, and the pressures.	Proposition 5.5; (5.41)
(b, V, G)	The radial, azimuthal or swirl, and axial components of the fixed background in normalized chart coordinates.	(8.1), (9.7)
$\mathcal{T}_0, T = q^{-A-1/2}\mathcal{T}_0$	The leading stress profile and the physical pair of stresses, ordered as $(r\theta, rz)$. Taking minus the cylindrical divergence of T gives the leading tangential residual, with radial viscosity kept and axial viscosity left out.	(4.11); Proposition 4.2

Continued on the next page.

Guide to the main symbols (continued).

Symbol	Meaning and job	Definition
$\mathcal{T}_{\text{phys}}, \mathcal{E}_B$	The sum of the background stresses in the annulus, and the flat residual left after its negative divergence has been removed.	(5.41); Proposition 5.5
Σ_θ, Σ_z	The fixed base stress components in chart units, given by $\Sigma_a = Q^{2A} \mathcal{T}_{\text{phys},a}$, $a = \theta, z$.	Proposition 8.1
$\mathcal{S}_n, \mathcal{A}_n$	The physical Stokes streamfunction of the background and its azimuthal vector potential $\mathcal{A}_n = (\mathcal{S}_n/r)e_\theta$. Taking the curl of that vector potential gives the radial and axial velocity.	(5.27)
$K(r, \tau), \mathcal{H}_{\text{ext}}$	The exterior swirl heat velocity and its profile, with $K = r^{-1-2h} \mathcal{H}_{\text{ext}}(\tau/r^2)$. The exterior velocity is Ke_θ .	(3.5)
$u^{[j]}, p^{[j]}$	The full fields at the finite correction stage j , including the fixed background, waves, and mean corrections. Square brackets count correction cycles.	(9.7); Definition 9.4
$Z_j = (A_j, B_j, p_j)$ in summation	The physical correction that takes stage $j-1$ to stage j , for $j \geq 1$. Its three parts are the vector potential A_j , the direct azimuthal velocity vector B_j , and the pressure increment $p_j = p^{[j]} - p^{[j-1]}$. The velocity increment is $\Delta u_j = \text{curl } A_j + B_j$.	(9.15), (9.16)
$\mathcal{A}, \mathcal{B}, u_{\text{loc}}, p_{\text{loc}}$	The summed local fields and the potential formula $u_{\text{loc}} = \text{curl } \mathcal{A} + \mathcal{B}e_\theta$. The quantity $\mathcal{B}$ is an axisymmetric scalar. Theorem 3.1 writes the local pair as (u, p) .	(3.3), (9.21)
u, p after localization	The velocity and pressure on the whole space, made by putting space and time cutoffs on the local potential formula. Their residual gives f .	(10.4), (10.5)
Coordinates, scales, and auxiliary variables		
$(r, \theta, z), t, (e_r, e_\theta, e_z)$	Physical cylindrical coordinates, time, and the unit vectors. The azimuthal and axial directions are tangent to a cylinder.	Section 3.1
τ, q	The time left, $\tau = 1 - t$, and the concentration scale $q = q(z, t)$. The equations $\tau = q(1 - \eta^2)$ and $z = q^D \eta$ determine them.	(4.1); Lemma 4.1
s, X, η	The squared radial variable $s = r^2/2$, the radial similarity variable $X = s/q$, and the axial similarity variable $\eta = z/q^D$.	(4.1)
h, A, D	The fixed small exponent h , the exponent $A = 1/2 + h$ for the growth of the tangential velocity, and the axial length exponent $D = 1/2 - h$.	(4.1), (4.3)
λ	The fixed exponent for radial decay in the reserved power-law profile, $E \propto X^{-1/2-\lambda}$.	(4.30)

Continued on the next page.

Guide to the main symbols (continued).

Symbol	Meaning and job	Definition
$\ell, Q, \varepsilon, S_*$ in charts	The dyadic band index, the fixed scale $Q = 2^{-\ell}$ of that band, the small parameter $\varepsilon = Q^h$, and the slow scale $S_* = \ell^2$. On one band, $q \asymp Q$. These chart parameters stay fixed when derivatives are taken.	(6.1)
R in profiles; (R, Z, T) in charts	In a profile, the radius is $R = \sqrt{2X} = r/\sqrt{q}$. In a dyadic chart, $(R, Z, T) = (r/\sqrt{Q}, z/Q^D, \tau/Q)$.	(6.1) and the prose after it
u_*, p_*, g_*	The normalized physical velocity, pressure, and residual or source. They are multiplied by $Q^A, Q^{2A}, Q^{2A+1/2}$, in that order. The normalization of a moment also includes its radial integration.	(8.11), (6.26)
Y, J_g, v_r, v_t, d_r	The absolute auxiliary-torus variable, its integer covering matrix, the radial and time eigendirections, and the radial phase exponent. Physical fields are evaluated at $Y = v_r r^{d_r} + v_t t \pmod{\mathbb{Z}^2}$.	(6.2), (6.3)
Y_i, H, y on auxiliary tori	The band coordinate $Y_i = J_g^i Y$, and the common coordinate $H = Y_{i_0}$ used where bands overlap. In the mean construction, that same common coordinate is called y .	(6.5), (6.17), (8.5)
$\tau, t; D_r, D_z, D_\theta, t_*$	The first two are the physical radial and time derivatives acting on extended fields before the auxiliary variable is evaluated. The remaining four are their normalized space and time versions in the wave and mean equations.	(6.4), (6.6)
$\nabla_*, \text{div}_*, \text{curl}_*$	The normalized gradient, divergence, and curl. They include the cylindrical-frame terms and the derivatives that come from following the auxiliary phase map.	(3.11)
X_a, X_b	The two edges of the active stress annulus $X_a < X < X_b$. At a fixed physical slow point, this is $\sqrt{2qX_a} < r < \sqrt{2qX_b}$.	Theorem 4.6(ii)
$\mathcal{I}_{\text{pos}}, \mathcal{I}_{\text{mean}}$	Two disjoint fixed intervals inside the active annulus. One is kept for positive-order background corrections and the other for mean-velocity corrections. Both are intervals in X .	Theorem 4.6(vi); (4.30)
$\mathcal{J}_{\text{pos}}, I_m$	The same two patches written in the coordinate $r/\sqrt{q}$. They are the images of $\mathcal{I}_{\text{pos}}$ and $\mathcal{I}_{\text{mean}}$ under $X \mapsto \sqrt{2X}$.	Lemma 5.2, Proposition 8.3 and the definition before it
$\mathcal{C}_a(\varepsilon), \mathcal{C}_b(\varepsilon)$	Fixed profile collars at the two radial edges of the annulus. Here ε is their logarithmic width and does not depend on the physical scale q .	(4.24)
Waves, amplitudes, and momentum flux		
$\gamma = (\ell, a, \sigma) \in \Gamma$	A wave label that names a band, a slow box, and one of the two wave families. This γ is not the axial part of the mean velocity listed below.	(6.8)

Continued on the next page.

Guide to the main symbols (continued).

Symbol	Meaning and job	Definition
$\eta_\gamma, K_\gamma, \mathcal{R}_\gamma$	The slow cutoff that localizes a wave, its support in physical (r, z, t) , and the rectangle that supports it on the auxiliary torus for that band.	(6.9), (6.10)
k, m, Φ, n_Φ	The carrier frequency $k = \lceil \varepsilon^{-1/2} \rceil$, a nonzero integer harmonic, the phase of the label, and the normalized spatial gradient of that phase.	(7.2), (7.3), (7.4)
t_m, π_m, f_m	The main transverse velocity amplitude, the pressure amplitude, and the prescribed source for one harmonic. The transverse velocity amplitude satisfies $n_\Phi \cdot t_m = 0$.	(7.5), (7.13)
$B, B^\ell, \mathcal{K}, \mathcal{A}_\Phi$	A 3×2 frame matrix for the moving plane $n_\Phi^\perp$, the left inverse of that frame, the matrix for background shear and rotation, and the inviscid evolution operator projected onto the moving plane.	(7.8), (7.6), (7.10)
$d, d_{\text{ref}}, \lambda(v)$ in pulses	The viscous damping coefficient $d = \varepsilon k^2 n_\Phi ^2$, its reference approximation, and the positive reference growth rate in the frame B . The two diagonal reference rates are $\pm \lambda$.	(7.5), (7.11), (7.10)
$P_v = P(v), v$ in pulses	The pulse envelope and its auxiliary time coordinate. The coordinate v moves forward at unit speed under t_* .	(7.12), (6.11), (6.12)
H, y, a_σ, W_0	The covariance matrix, the positive squared pulse weights, their square-root amplitudes, and the local leading wave before curls are taken.	Proposition 7.5; (7.24)
C_m, A_m, r_m, a_m	The coefficient of the wave potential, the oscillating vector potential, the remainder made by taking its curl, and the full harmonic velocity amplitude $a_m = t_m + r_m$.	(7.38), (7.39)
$W_0^{\text{as}} = w_0^{\text{tan}}, w$	The assembled leading wave transverse to the phase normal, and the actual wave velocity made from curls, including the curl remainders and the wave corrections added later.	(7.36), (9.7); Definition 9.4
$W_{ab} = \langle w_a w_b \rangle_\theta$	The full angular average of the wave covariance in cylindrical components. Its off-diagonal radial entries carry azimuthal and axial momentum.	(8.1), (8.3)
$\mathcal{C}(w), \mathcal{B}(w, v)$	The doubly averaged pair of radial momentum fluxes and its symmetric cross term. They are used to prescribe changes in the covariance.	(7.23)
$\Sigma, \mathcal{L}\Sigma$ in the wave correction	The prescribed signed covariance increment and its linearized correction to the main wave. The base stresses are the quantities Σ_a listed above.	(7.31), (7.34)
Mean velocity and its compatibility conditions		
(β, v, γ)	The radial, azimuthal, and axial velocity corrections that do not depend on the angle. They may still depend on the auxiliary torus.	(8.1)
p_m, p_w	The pressure correction in the angular mean and the pressure correction in the nonzero harmonics, in that order.	(8.12), (9.7)

Continued on the next page.

Guide to the main symbols (continued).

Symbol	Meaning and job	Definition
g_r, E_θ, E_z	The radial pressure source and the exact azimuthal and axial residuals of the mean. The radial mean residual is $D_r p_m - g_r$.	(8.3)
$\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z$	Three slow compatibility defects: the integral of the radial source and two integrals of momentum flux. The five-equation map corrects all three.	(8.12), (8.15), (8.25)
M_θ, M_z	The angular-momentum and axial-flux moments of the mean correction after its auxiliary variable has been averaged. Both moments must be zero.	(8.2)
$\gamma_d, \Psi_*, \Delta\gamma$	The wanted axial increment, the coefficient of its azimuthal vector potential, and the axial increment actually produced, $\Delta\gamma = \gamma_d - \mathcal{A}_1 \gamma_d$.	(8.14)
H_θ, H_z	The supported covariance targets made from the averaged tangential residuals after their weighted radial moments have been taken away.	(8.18)
Averages and inverse operators		
$\langle f \rangle_\theta, \langle f \rangle_Y, \langle f \rangle, f^\circ$	The angular average, the auxiliary Haar average, and the part with auxiliary average zero, $f - \langle f \rangle_Y$. For wave fields, brackets with no subscript mean the angular and auxiliary averages taken together. All these averages are taken before the physical phase map is evaluated.	Equations (3.7) to (3.9)
$\mathcal{A}_X(f), \mathcal{D}_X$	The first expression is the radial prefix average $X^{-1} \int_0^X f(x, \eta) dx$. The second is the logarithmic radial derivative $X \partial_X$, differentiation with respect to the logarithm of the radius while the other variable stays fixed.	(3.10), (4.2)
$\mathcal{I}, \mathcal{J}, \mathcal{I}_c$	The prefix integral that follows the phase, the full radial integral, and the compactly supported antiderivative $\mathcal{I} - \chi_m \mathcal{J}$.	(8.4), (8.5)
$\mathcal{D}_e, \mathcal{T}_e, \mathcal{A}_e$	The weighted radial derivative, its supported inverse, and the remainder from the cutoff. They satisfy $\mathcal{D}_e \mathcal{T}_e f = f - \mathcal{A}_e f$, with $e \in \{0, 1, 2\}$.	(8.6), (8.7)
$N^{-1}, N_{\text{abs}}^{-1}$	The inverse of the auxiliary-time derivative on functions whose Haar average is zero, written in chart coordinates and absolute auxiliary coordinates.	(8.19), (8.23)
Weights, coefficient classes, and correction orders		
ζ, δ, κ_s	The flat weight at the radial edges, the capped logarithmic distance from those edges, and the fixed small exponent that measures the loss from radial derivatives.	(6.23), (6.2)
W_α	Normalized wave amplitudes of order ε^α , with the radial weight $\sqrt{\zeta}$, the pulse envelope, and the stated supports.	Definition 6.5; (6.29)
M_α	Normalized angular mean coefficients of order ε^α , with the radial weight ζ . They may depend on the auxiliary variable.	Definition 6.4; (6.24)
S_α	Normalized radial moments that depend on (Z, T) and have order ε^α . Their units already include the radial integration.	(6.25), (6.26)

Continued on the next page.

Guide to the main symbols (continued).

Symbol	Meaning and job	Definition
$\mathcal{G}^{[j]}, \mathcal{F}^{[j]}$	The supported part of the residual and the flat residual kept separate from it at stage j . Flatness means that every fixed Cartesian space-time derivative goes to zero faster than every power of q .	(9.3), (9.5)
σ_j, B_j, C_j^*	The stage decay parameters are $\sigma_j = 1/5 + j/10$, $B_j = 1/2 + \sigma_j$, and $C_j^* = 1 + \sigma_j$. The last two give the orders of the wave and mean residuals as powers of ε .	(9.8); Proposition 9.6
$\chi(a_j q), a_j$	The cutoffs used to add up the physical corrections, and their increasing scale parameters. Their supports are neighborhoods of $q = 0$ that shrink as the stage increases. Each cutoff is put on a potential before its curl is taken.	Lemma 5.4; (9.21)

4 Building the leading-order flow

The concentrating flow from Section 3.1 needs an axisymmetric velocity that grows at the required rate near the origin and becomes a heat flow in the exterior. The profiles E, U, Π built here define the leading velocity and pressure $(u^{(0)}, p^{(0)})$ in (4.3). This velocity is exactly incompressible. The leading tangential momentum residual is minus the cylindrical divergence of the two-component stress with profile $\mathcal{T}_0$, defined in (4.11). The stress must be zero near the axis and in the exterior. This leaves a cylindrical annulus where the waves can act. That annulus is an interval of the scaled radius X at each fixed axial parameter η .

The direction of the stress also has to meet a condition. It must have the positive covariance representation from Proposition 7.5, as Section 3 explained. Theorem 4.6 gives profiles with this representation, with all the required regularity and support properties, and with the exact heat flow in the exterior. Two later steps deal with the momentum residual still left by these profiles. Section 5 adds higher-order axisymmetric corrections to make the background (u_B, p_B) . Section 7 then adds waves whose leading covariance gives the stress in the annulus.

First, the stress is derived from the profiles. Next come the five cumulative radial integrals in (4.15). Keeping those five integrals fixed lets profile pieces from different radial intervals be joined without changing the chosen exterior (Lemma 4.4). After that, the allowed directions of the stress are described and Theorem 4.6 is stated. Section 4.5 states the construction results needed for the theorem, and Section 4.6 proves it. Sections A to C contain the proofs of those construction results.

4.1 Similarity variables and radial integration of the residual

Start by writing $(u^{(0)}, p^{(0)})$ and their derivatives in similarity coordinates. Proposition 4.2 then recovers the tangential stress by integrating in the radial direction. The cylindrical coordinates (r, θ, z) are right-handed, with $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$. The exponents A, D are the ones from Section 3.1. Along with the relations in (3.2), set $d = 1 - \eta^2$, $L = 1 - 2h\eta^2$, and $s = r^2/2$:

$$\begin{aligned} \tau &= 1 - t, & A &= \frac{1}{2} + h, & D &= \frac{1}{2} - h, & 0 < h < \frac{1}{2}, \\ z &= q^D \eta, & \tau &= q(1 - \eta^2), & d &= 1 - \eta^2, & L &= 1 - 2h\eta^2, & s &= \frac{r^2}{2}, & X &= \frac{s}{q}. \end{aligned} \tag{4.1}$$

Every coordinate identity in this subsection is valid for $0 < h < 1/2$. Theorem 4.6 later chooses $h < 1/100$, just as Theorem 3.1 does. As Section 3.1 noted, for $\tau > 0$ the similarity relations

determine exactly one $q > |z|^{1/D}$. Here is why. At that lower endpoint, $q - z^2 q^{2h}$ is zero. It tends to infinity, and throughout this interval its derivative is $1 - 2hz^2 q^{2h-1} = L \geq 1 - 2h > 0$. This also gives $|\eta| < 1$. The endpoint values $\eta = \pm 1$ are understood as one-sided limits.

Throughout the profile construction, $\mathcal{D}_X f = X \partial_X f$ means differentiation in the logarithmic radial coordinate, with η held fixed. The next step in computing the residual of the field from Section 3.1 is to find the operators that turn time and axial derivatives into derivatives in (X, η) .

Lemma 4.1. *For any smooth profile f and any real number b ,*

$$\partial_t(q^b f) = q^{b-1} T_b f, \quad \partial_z(q^b f) = q^{b-D} Z_b f, \quad \begin{cases} T_b f = L^{-1}(-bf + D\eta f_\eta + \mathcal{D}_X f), \\ Z_b f = L^{-1}(2b\eta f + df_\eta - 2\eta \mathcal{D}_X f). \end{cases} \quad (4.2)$$

Proof. Differentiate the equation for q , then differentiate $\eta = zq^{-D}$ and $X = s/q$. This gives

$$\begin{aligned} q_t &= -L^{-1}, & \eta_t &= D\eta/(qL), & X_t &= X/(qL), \\ q_z &= 2\eta q^{1-D}/L, & \eta_z &= d/(q^D L), & X_z &= -2\eta X/(q^D L). \end{aligned}$$

The middle entry in the second row uses $L - 2D\eta^2 = d$. The formulas in the lemma now follow from the chain rule. $\square$

The leading field from Section 3.1 must be smooth across the axis. Factor out the linear radial vanishing of its azimuthal component, choose a fixed amplitude normalization $C > 1$ and a smooth scalar ϕ , and write

$$u_\theta^{(0)} = q^{-A} E, \quad u_z^{(0)} = q^{-A} U, \quad ru_r^{(0)} = V_0, \quad p^{(0)} = q^{-2A} \Pi, \quad E = C^{-1} \sqrt{2X} \phi. \quad (4.3)$$

The proof of Theorem 4.6 chooses the normalization C . As in Section 3.1, E, U, V_0, Π are functions of (X, η) . The factor $\sqrt{2X}$ makes the azimuthal velocity vanish at the axis, while the smoothness condition is put on ϕ . For the profiles built here, the following conditions give regularity at the axis in the sense of Definition 3.2.

$$E = \sqrt{2X} F, \quad V_0 = Xv_0, \quad F = \phi/C, \quad U, v_0, \Pi \in C^\infty([0, X_c] \times [-1, 1]) \quad (4.4)$$

Here $X_c > 0$. Smoothness on this closed rectangle means that every mixed X, η derivative extends continuously to the boundary from the one side that lies inside the rectangle. To see why this gives Cartesian smoothness, write $x_1 = r \cos \theta$, $x_2 = r \sin \theta$. Then $X = (x_1^2 + x_2^2)/(2q)$, and

$$\begin{aligned} u_1^{(0)} &= \frac{v_0}{2q} x_1 - q^{-A-1/2} F x_2, & u_2^{(0)} &= \frac{v_0}{2q} x_2 + q^{-A-1/2} F x_1, \\ u_3^{(0)} &= q^{-A} U, & p^{(0)} &= q^{-2A} \Pi. \end{aligned} \quad (4.5)$$

Every profile in this display is evaluated at (X, η) . The quantities q, η are smooth functions of (t, z) , and $q > 0$ whenever $t < 1$. So the coefficients in (4.5) are smooth functions of (x_1, x_2, z, t) . Notice in particular that E contains a $\sqrt{X}$ factor. At $X = 0$, smoothness is imposed on F .

Choose E, U first. Incompressibility then determines V_0 , and the leading radial momentum balance determines Π once its value on the axis has been fixed. The notation paragraph at the end of Section 3 defined the radial average of a smooth radial scalar. Its smooth extension to the axis is specified by

$$\mathcal{A}_X(f)(X, \eta) = \frac{1}{X} \int_0^X f(x, \eta) dx, \quad \mathcal{A}_X(f)(0, \eta) = f(0, \eta). \quad (4.6)$$

The incompressibility and centrifugal-pressure balances from Section 3.1 now give explicit formulas for the profiles.

$$V_0 = \frac{X}{L}(2\eta U - 2D\eta\mathcal{A}_X(U) - d\partial_\eta\mathcal{A}_X(U)), \quad \Pi_X = \frac{E^2}{2X}. \quad (4.7)$$

For the first formula, put $A + D = 1$ into $\partial_s(ru_r^{(0)}) + \partial_z u_z^{(0)} = 0$. The result is $(V_0)_X = L^{-1}(2A\eta U - dU_\eta + 2\eta XU_X)$. Integrate it and set the value on the axis to zero to get the displayed formula.

For the second formula, the coefficients of $\partial_r p^{(0)}$ and $(u_\theta^{(0)})^2/r$ are $\sqrt{2X}\Pi_X$ and $E^2/\sqrt{2X}$, respectively. Both carry the same power $q^{-2A-1/2}$. The other radial terms carry at least one extra factor q^{2h} . The same is true of axial viscosity when it is compared with radial viscosity in the tangential equations. This is why the background corrections in Section 3 use the powers q^{2nh} . These comparisons are being made with the profile coordinates fixed. Bounds for physical derivatives of the finished fields are proved separately. The full residual still includes the higher-order terms just named, and Section 5 deals with them.

After imposing incompressibility and the leading radial balance, the tangential residual still has to be written as a radial stress divergence, as Section 3 requires. Start by integrating its inviscid terms and call the resulting scalar profiles Q_s, N_s . Then add the part that comes from radial viscosity. The definitions below use only E, U, Π ; Proposition 4.2 checks that they give the required stress identity.

Take profiles that satisfy (4.7), with ϕ, U, Π smooth at the axis and $\phi > 0$, and set

$$\begin{aligned} H &= \sqrt{2X}E, & F &= \frac{E}{\sqrt{2X}}, & l &= \mathcal{D}_X \log H, \\ W &= 1 - 2D\eta\mathcal{A}_X(U) - d\partial_\eta\mathcal{A}_X(U), & H_c &= D\eta + dU. \end{aligned} \quad (4.8)$$

Here H is the profile of the angular momentum $ru_\theta^{(0)}$. The quantity F is what remains after taking the factor $\sqrt{2X}$ out of E , and l is the logarithmic radial slope of H . The two coefficients W, H_c gather the transport terms that appear in (4.14).

Define Q_s, N_s by the radial equations below, choosing the solutions that extend smoothly to $X = 0$.

$$\begin{aligned} \mathcal{D}_X Q_s + (1 + l)Q_s &= S_q, & \mathcal{D}_X N_s + N_s &= S_n, \\ S_q &= -Wl - h(1 - 2\eta U) - H_c(\log E)_\eta, \\ S_n &= -W\mathcal{D}_X U - A(1 - 2\eta U)U - H_c U_\eta - d\Pi_\eta + 4A\eta\Pi + 2\eta\mathcal{D}_X\Pi. \end{aligned} \quad (4.9)$$

These are first-order equations in the radial variable. After each equation is divided by that variable, multiplying the first by XH turns its left side into the radial derivative of the product of that factor with the first unknown. Multiplying the second by X does the same for the second unknown. That is why these expressions are called integrating factors. Before smoothness at the axis is imposed, the constants of integration may depend on η , so the solutions have the form

$$Q_s = \frac{C_Q(\eta) + \int_0^X H(x, \eta) S_q(x, \eta) dx}{XH(X, \eta)}, \quad N_s = \frac{C_N(\eta) + \int_0^X S_n(x, \eta) dx}{X}. \quad (4.10)$$

Smooth extension to the axis forces $C_Q = C_N = 0$. The reason is that $H = 2X\phi/C$ near the axis. The first constant, if nonzero, would give an X^{-2} singularity in Q_s , while a nonzero second constant would give an X^{-1} singularity in N_s . With both constants set to zero, the change of variables $x = sX$ gives

$$Q_s = \frac{1}{F(X, \eta)} \int_0^1 sF(sX, \eta) S_q(sX, \eta) ds, \quad N_s = \int_0^1 S_n(sX, \eta) ds.$$

The sources are smooth at $X = 0$, because $l = 1 + XF_X/F$ and $(\log E)_\eta = F_\eta/F$, so the formulas extend smoothly to the axis with $Q_s(0, \eta) = S_q(0, \eta)/2$ and $N_s(0, \eta) = S_n(0, \eta)$.

Put the two stress coefficients in the order $(r\theta, rz)$, and define them by

$$\begin{aligned} \mathcal{T}_0 &= F(p_s - \mathfrak{s}), \quad p_s = \left(\frac{XQ_s}{L}, \frac{XN_s}{LE} \right), \\ \mathfrak{s} &= (a, -b_s), \quad a = 1 - 2\mathcal{D}_X \log E = 2 - 2l, \quad b_s = \frac{2\mathcal{D}_X U}{E}. \end{aligned} \quad (4.11)$$

The symbol p_s now means the two-component vector of integrated inviscid terms. It is not the pressure $p^{(0)}$. The vector $\mathfrak{s} = (a, -b_s)$ records the cylindrical radial shear, with its normalization fixed by

$$\left(\partial_r u_\theta^{(0)} - \frac{u_\theta^{(0)}}{r}, \partial_r u_z^{(0)} \right) = -q^{-A-1/2} F\mathfrak{s}.$$

In (4.11), Fp_s is the inviscid part and $-F\mathfrak{s}$ is the part from radial viscosity. The calculation below shows both facts. In the notation of Section 3, the formula to prove for the physical stress is $T = q^{-A-1/2} \mathcal{T}_0$. Its components are the $r\theta$ and rz entries needed from the wave covariance, which is the tensor of averaged quadratic velocity products built in Proposition 7.5.

To check the stress identity from Section 3, separate the leading tangential terms from axial viscosity. Define the *leading tangential residuals* by

$$R_j^{(0)} = \mathcal{R}(u^{(0)}, p^{(0)}) \cdot e_j + \partial_z^2 u_j^{(0)}, \quad j \in \{\theta, z\}, \quad (4.12)$$

where e_θ, e_z are the cylindrical unit vectors and $\mathcal{R}$ is the full momentum residual defined in the introduction. Adding $\partial_z^2 u_j^{(0)}$ takes the axial-viscosity term out of the residual but leaves radial viscosity in it.

Proposition 4.2. *The residuals $R_\theta^{(0)}, R_z^{(0)}$ from (4.12) are the negatives of the cylindrical divergences $\partial_r + 2/r$ and $\partial_r + 1/r$, applied to $q^{-A-1/2} \mathcal{T}_0$. In particular, the equations that set the leading residual stress to zero are*

$$-2L \frac{X\phi_{XX} + 2\phi_X}{\phi} = S_q, \quad -2L(XU_{XX} + U_X) = S_n. \quad (4.13)$$

Proof. The angular equation takes its simplest form when written for the angular momentum $ru_\theta^{(0)} = q^{-h}H$. Equations (4.2) and (4.7) give

$$\begin{aligned} &(\partial_t + u_r^{(0)} \partial_r + u_z^{(0)} \partial_z)(q^{-h}H) \\ &= \frac{q^{-h-1}}{L} \{W\mathcal{D}_X H + H_c H_\eta + h(1 - 2\eta U)H\} = -\frac{q^{-h-1}}{L} H S_q. \end{aligned} \quad (4.14)$$

The same calculation gives $-q^{-A-1} S_n / L$ for the axial material derivative plus the pressure gradient. The integrating factors for the two radial divergences are r^2 and r . So integrating the negative inviscid residual outward from the axis gives the coefficients FXQ_s/L and $FXN_s/(LE)$. Radial viscosity contributes

$$\partial_r u_\theta^{(0)} - u_\theta^{(0)}/r = -q^{-A-1/2} Fa, \quad \partial_r u_z^{(0)} = q^{-A-1/2} Fb_s.$$

Add these viscous terms to the radially integrated inviscid residual. This gives (4.11), including its sign. If the stress is zero, then $XQ_s/L = a$, $N_s/L = -2U_X$. Put these two relations into (4.9) to get (4.13). On the axis, $l = 1 + X\phi_X/\phi$ and $(\log E)_\eta = \phi_\eta/\phi$ are smooth, so Q_s, N_s are regular on the axis in the scalar-profile sense of Definition 3.2. □

4.2 The cumulative radial integrals

The stress profile $\mathcal{T}_0$ in (4.11) depends, through the integrals in (4.10), on the profiles E, U at every smaller radius. This matters when an inner profile is joined to a prescribed outer profile, as in Section 3. The accumulated integrals must match along with the profile values. Five integrals are enough to keep the outer pressure, radial velocity, and stress unchanged, as Lemma 4.4 shows. Integrating (4.9) writes Q_s, N_s in terms of those five quantities.

$$\begin{aligned} M &= \int_0^X U dx, & I &= \int_0^X H dx, & J &= \int_0^X UH dx, \\ S &= \int_0^X (U^2 - E^2/2) dx, & C_p &= \int_0^X \frac{E^2}{2x} dx, & \Pi &= \Pi(0, \eta) + C_p. \end{aligned} \quad (4.15)$$

Each integral depends on (X, η) . Once the matching radius is fixed, each one is a scalar function of η . The quantity C_p is the pressure increment from the axis to X , while $\Pi(0, \eta)$ is the prescribed pressure value at $X = 0$.

Lemma 4.3. *For the solutions of (4.9) with $C_Q = C_N = 0$,*

$$\begin{aligned} Q_s &= -W + \frac{(1-h)I - D\eta I_\eta - dJ_\eta + 2(h-D)\eta J}{XH}, \\ N_s &= -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - dS_\eta}{X} + 4A\eta\Pi - d\Pi_\eta. \end{aligned} \quad (4.16)$$

Proof. Use $\partial_X(XW) = 1 - 2D\eta U - dU_\eta$ to integrate $-WD_XH$ and $-WD_XU$. In the angular terms, $-dU_\eta H - dUH_\eta$ becomes $-d(UH)_\eta$. For the pressure term, integrate by parts in $\Pi_X = E^2/(2X)$ to get $\int_0^X \Pi dx = XII - \int_0^X E^2/2 dx$. The source integrals are then

$$\begin{aligned} \int_0^X HS_q dx &= -XWH + (1-h)I - D\eta I_\eta - dJ_\eta + 2(h-D)\eta J, \\ \int_0^X S_n dx &= -XWU + D(M - \eta M_\eta) + 4h\eta S - dS_\eta + X(4A\eta\Pi - d\Pi_\eta). \end{aligned}$$

Divide the first line by XH and the second by X . This gives the two formulas in the lemma. $\square$

Suppose two profiles agree beyond a joining radius. If their five cumulative integrals also match at that radius, then all quantities obtained by radial integration stay the same beyond it. Their common pressure value on the axis is held fixed. If the profiles, the integrals, and their parameter derivatives change only a little, then Q_s, N_s, p_s also change only a little. This estimate does not require the radial derivatives of the two profiles to be close. The exact identity is used to join profiles, and the estimate is used to change their shear.

Lemma 4.4. *Fix $h \in (0, 1/2)$. For $i = 1, 2$, let (U_i, E_i) be smooth on $(0, \infty) \times [-1, 1]$, with $E_i > 0$. Suppose the five integrals in (4.15) are smooth functions there, including their one-sided η -derivatives at $\eta = \pm 1$. Write*

$$H_i = \sqrt{2XE_i}, \quad \mathbf{m}_i = (M_i, I_i, J_i, S_i, C_{p,i}).$$

Fix one smooth function $\Pi_{\text{ax}}(\eta)$ on $[-1, 1]$, and set

$$\Pi_i(X, \eta) = \Pi_{\text{ax}}(\eta) + C_{p,i}(X, \eta), \quad \Pi_i(0, \eta) = \Pi_{\text{ax}}(\eta), \quad i = 1, 2.$$

This gives both pressures the same constant of integration on the axis. The velocity profiles are not required to agree near the axis. Define $V_{0,i}$ by (4.7), $Q_{s,i}, N_{s,i}$ by (4.16), and $p_{s,i}, a_i, b_{s,i}, \mathcal{T}_{0,i}$ by (4.11). The formulas (4.16) used here are the ones already derived with $C_Q = C_N = 0$. For any one of these quantities f , write $\Delta f = f_2 - f_1$. There are two conclusions.

(i) Suppose that for some $X_h > 0$,

$$(U_1, E_1) = (U_2, E_2) \quad \text{for } X \geq X_h, \quad \Delta \mathbf{m}(X_h, \eta) = 0 \quad (-1 \leq \eta \leq 1).$$

Then $\mathbf{m}_1 = \mathbf{m}_2$. All the corresponding quantities $\Pi_i, V_{0,i}, Q_{s,i}, N_{s,i}, p_{s,i}, a_i, b_{s,i}, \mathcal{T}_{0,i}$ also agree whenever $X \geq X_h$ and $\eta \in [-1, 1]$.

(ii) Fix $0 < X_0 < X_1 < \infty$, an integer $k \geq 0$, and $e_{\min} > 0$. On the rectangle $\mathcal{R} = [X_0, X_1] \times [-1, 1]$, put

$$\|f\|_j := \max_{0 \leq r \leq j} \sup_{(X, \eta) \in \mathcal{R}} |\partial_\eta^r f(X, \eta)|, \quad j \geq 0,$$

and for a tuple use the largest of its component norms. If $E_i \geq e_{\min}$ on $\mathcal{R}$, then

$$\|\Delta(Q_s, N_s, p_s)\|_k \leq C_k (\|\Delta(U, E)\|_{k+1} + \|\Delta \mathbf{m}\|_{k+1}). \quad (4.17)$$

The constant C_k depends only on $k, h, X_0, X_1, e_{\min}$, the C^{k+1} -norm of Π_{ax} , and one common bound for $\|(U_i, E_i, \mathbf{m}_i)\|_{k+1}$, $i = 1, 2$. The estimate contains no radial derivative of any difference.

Proof. The definitions of the five cumulative integrals give

$$\begin{aligned} \Delta M &= \int_0^X (U_2 - U_1) dx, \\ \Delta I &= \int_0^X (H_2 - H_1) dx, \\ \Delta J &= \int_0^X (U_2 H_2 - U_1 H_1) dx, \\ \Delta S &= \int_0^X (U_2^2 - U_1^2 - (E_2^2 - E_1^2)/2) dx, \\ \Delta C_p &= \int_0^X \frac{E_2^2 - E_1^2}{2x} dx. \end{aligned} \quad (4.18)$$

The quantities on the left are evaluated at (X, η) , while each integrand on the right is evaluated at (x, η) . In part (i), the profiles U_i, E_i agree for $X \geq X_h$, so every integrand in (4.18) vanishes there. It follows that

$$\Delta \mathbf{m}(X, \eta) = \Delta \mathbf{m}(X_h, \eta) = 0 \quad (X \geq X_h, -1 \leq \eta \leq 1).$$

This is an identity between functions of η , so their η -derivatives agree too. The common pressure value on the axis gives $\Delta \Pi = \Delta C_p = 0$. Then (4.7) gives $\Delta V_0 = 0$, and (4.16) gives $\Delta Q_s = \Delta N_s = 0$. The profiles agree on a radial interval, so their radial derivatives agree there as well. The definitions in (4.11) now give equality of $p_s, a, b_s, \mathcal{T}_0$.

For part (ii), write the quantities that enter (4.16) as

$$H_i = \sqrt{2X} E_i, \quad W_i = 1 - \frac{2D\eta M_i + d\partial_\eta M_i}{X}, \quad \Pi_i = \Pi_{\text{ax}} + C_{p,i}.$$

Their differences are

$$\Delta H = \sqrt{2X} \Delta E, \quad \Delta W = -\frac{2D\eta \Delta M + d\partial_\eta \Delta M}{X}, \quad \Delta \Pi = \Delta C_p.$$

The last equality uses the common pressure value on the axis. On $\mathcal{R}$, every denominator stays away from zero, because $X \geq X_0$, $L \geq 1 - 2h > 0$, and $H_i \geq \sqrt{2X_0} e_{\min}$. For example,

$$\Delta(H^{-1}) = -\frac{\Delta H}{H_1 H_2}, \quad \Delta(E^{-1}) = -\frac{\Delta E}{E_1 E_2}.$$

Apply the product rule and these two reciprocal identities first to (4.16), then to the formula for p_s in (4.11). After no more than k derivatives with respect to η , every term contains a difference of U, E , or a difference of

a component of $\mathbf{m}$, through order at most $k + 1$. The remaining factors are bounded by the common bounds stated in the lemma. This proves (4.17), including the stated dependence of its constant. None of these formulas contains a radial derivative of an input. $\square$

Part (ii) is used for the radial modulation in Proposition C.2. The profiles U, E , the five integrals $\mathbf{m}$, and their η -derivatives stay close even when the radial shears a, b_s change a lot. The estimate controls p_s . The full stress $\mathcal{T}_0 = F(p_s - \mathfrak{s})$ also depends on those shears.

To use this comparison at a large joining radius ρ , set $\hat{U}(x, \eta) = U(\rho x, \eta)$ and $\hat{E}(x, \eta) = E(\rho x, \eta)$. Define the rest of the hatted quantities by

$$X = \rho x, \quad H = \sqrt{\rho} \hat{H}, \quad (M, I, J, S, C_p) = (\rho \hat{M}, \rho^{3/2} \hat{I}, \rho^{3/2} \hat{J}, \rho \hat{S}, \hat{C}_p). \quad (4.19)$$

Here the unhatted functions are evaluated at $(\rho x, \eta)$, while the hatted functions are evaluated at (x, η) . Put

$$\begin{aligned} \hat{\mathbf{m}} &= (\hat{M}, \hat{I}, \hat{J}, \hat{S}, \hat{C}_p), \\ (\hat{Q}_s, \hat{N}_s)(x, \eta) &= (Q_s, N_s)(\rho x, \eta), \\ \hat{p}_s(x, \eta) &= \rho^{-1} p_s(\rho x, \eta). \end{aligned}$$

Substitute these expressions into (4.16) and (4.11). The formulas for $\hat{Q}_s, \hat{N}_s, \hat{p}_s$ are then the same, with X replaced by x . On any fixed rectangle $[x_0, x_1] \times [-1, 1]$, where $0 < x_0 < x_1$, this gives

$$\|\Delta(\hat{Q}_s, \hat{N}_s, \hat{p}_s)\|_k \leq C_k(\|\Delta(\hat{U}, \hat{E})\|_{k+1} + \|\Delta \hat{\mathbf{m}}\|_{k+1}),$$

where the norms are taken on this fixed x -rectangle. The constant does not depend on ρ , provided that the positive lower bound for $\hat{E}_i$, the bounds for $\|(\hat{U}_i, \hat{E}_i, \hat{\mathbf{m}}_i)\|_{k+1}$, and the C^{k+1} -bound for the common Π_{ax} are all independent of ρ , with h fixed.

The moment corrections used below are finite linear combinations of smooth functions supported on short radial intervals. Their changes to $\mathbf{m}$ depend linearly or quadratically on the coefficients. Lemmas A.1 and A.2 prove that the moment matrices are invertible and that these finite systems have solutions. Each time those lemmas are used, the needed bounds for the moment differences and the matrix inverses are checked.

The moment conditions at infinity do a different job. The quantities M, J, S control, in that order, the radially integrated axial momentum, the axial transport of angular momentum, and the axial momentum flux including pressure. The angular momentum I is made convergent by subtracting the prescribed exterior power H_{pow} before integrating, as in (4.28). The resulting total identities make both weighted integrals of the tangential residuals vanish. Without those identities, the exterior residual can be zero while the angular and axial stress components still have tails proportional to r^{-2} and r^{-1} , respectively, because the integrated inner residual may still be nonzero. Lemma A.8 uses the total moment identities to remove these tails and prove that the stress vanishes outside.

4.3 The cone condition for the leading stress

The annular stress $\mathcal{T}_0$ has to be produced by the chosen waves with real amplitudes, so it must lie in the positive span of their covariance directions, as Proposition 7.5 proves. The next formulas put this cone condition into profile variables, and Lemma 4.5 gives equivalent inequalities. At each (X, η) , the data that matter are the radial shear $(a, -b_s)$ and the integrated inviscid term $p_s = (p_{s,1}, p_{s,2})$ from (4.11). Wherever $a > 0$, set

$$t_s = -\frac{b_s}{a}, \quad v_s = a(1 + t_s^2), \quad P_c = p_{s,1} + t_s p_{s,2}, \quad J_c = p_{s,2} - t_s p_{s,1}. \quad (4.20)$$

So the quantities P_c, J_c are the scalar products of p_s with the two orthogonal, unnormalized vectors $(1, t_s)$ and $(-t_s, 1)$, respectively. Also, $s = a(1, t_s)$. Define

$$\mathcal{U}(P_c, J_c) = P_c + \frac{J_c^2}{4} - |J_c| \sqrt{\frac{P_c - 2}{2} + \frac{J_c^2}{16}}, \quad P_c > 2, \quad v_s < \mathcal{U}(P_c, J_c). \quad (4.21)$$

When $a > 0$, the two inequalities in (4.21) are called the *relaxed cone condition*. The *admissible stress cone condition* asks for those same two inequalities and also requires $v_s > 2$. The relaxed condition is used while the profile on the axis is being joined to the outer profile. Viscous waves need the extra inequality. Once the admissible condition holds, Proposition 7.5 produces positive squared amplitudes whose wave covariances give the required stress. The next lemma gives an equivalent way to check the condition and a sufficient condition used in the outer construction of Proposition A.4.

Lemma 4.5. *Let $a > 0$, $b_s \in \mathbb{R}$, and $p_s = (p_{s,1}, p_{s,2}) \in \mathbb{R}^2$. Define t_s, v_s, P_c, J_c by (4.20). If $v_s > 2$, then the admissible stress cone condition, which means the inequalities in (4.21) together with $v_s > 2$, is equivalent to*

$$P_c > v_s, \quad (v_s - 2)J_c^2 < 2(P_c - v_s)^2. \quad (4.22)$$

Now take any nonempty compact set

$$K \subset \{(a, b_s, w) \in \mathbb{R}^3 : a > 0\}$$

such that every $(a, b_s, w) \in K$ satisfies

$$a - b_s w > 0, \quad 2b_s w + \frac{b_s^2}{a} + (a - 2)w^2 < 2,$$

There is then a number $P_K > 0$ with the following property. For every $(a, b_s, w) \in K$ and every $p_{s,1} \geq P_K$, set $p_{s,2} = w p_{s,1}$. These data satisfy the relaxed cone condition (4.21). If $v_s > 2$ as well, they satisfy the admissible stress cone condition.

Proof. Either set of conditions in the first claim gives $P_c > 2$. On this range, the polynomial $2(P_c - v)^2 - (v - 2)J_c^2$ in v has roots

$$v_{\pm} = P_c + \frac{J_c^2}{4} \pm |J_c| \sqrt{\frac{P_c - 2}{2} + \frac{J_c^2}{16}}.$$

When $P_c > 2$, the lower root satisfies $2 < v_- \leq P_c$. So on $2 < v < P_c$, the polynomial is positive exactly when $v < v_-$. Since $\mathcal{U}(P_c, J_c) = v_-$, this proves the first claim.

For the second claim, put

$$c = 1 - \frac{b_s w}{a}, \quad j = w + \frac{b_s}{a}, \quad v = a + \frac{b_s^2}{a} = v_s.$$

Then $P_c = p_{s,1}c$, $J_c = p_{s,1}j$, and

$$(v_s - 2) \left(w + \frac{b_s}{a} \right)^2 - 2 \left(1 - \frac{b_s w}{a} \right)^2 = \left(1 + \frac{b_s^2}{a^2} \right) \left((a - 2)w^2 + 2b_s w + \frac{b_s^2}{a} - 2 \right).$$

The assumptions say that $c > 0$ and $G := 2c^2 - (v - 2)j^2 > 0$ everywhere on K . Because this set is compact, there are constants $c_K, \gamma_K > 0$ and $B_K \geq 1$ such that

$$c \geq c_K, \quad G \geq \gamma_K, \quad v \leq B_K, \quad cv \leq B_K \quad \text{on } K.$$

Choose

$$P_K = \max \left\{ \frac{B_K + 2}{c_K}, \frac{8B_K}{\gamma_K} \right\}.$$

For every $p_{s,1} \geq P_K$, this choice gives $P_c \geq B_K + 2 > \max\{2, v\}$ and

$$\frac{2(P_c - v)^2 - (v - 2)J_c^2}{p_{s,1}^2} = G - \frac{4cv}{p_{s,1}} + \frac{2v^2}{p_{s,1}^2} \geq \frac{\gamma_K}{2} > 0.$$

If $v > 2$, the first claim gives the relaxed condition. If $v \leq 2$, the relaxed condition follows instead from $P_c > 2$ and $\mathcal{U}(P_c, J_c) > 2$. So the same threshold works everywhere on K , including the points where $v_s = 2$. $\square$

In stress coordinates, the cone condition is

$$\mathcal{T}_{0,\theta} + t_s \mathcal{T}_{0,z} = F(P_c - v_s), \quad \mathcal{T}_{0,z} - t_s \mathcal{T}_{0,\theta} = FJ_c. \quad (4.23)$$

Since $F > 0$, when $v_s > 2$ the admissible stress cone condition says exactly that

$$\mathcal{T}_{0,\theta} + t_s \mathcal{T}_{0,z} > 0, \quad (v_s - 2)(\mathcal{T}_{0,z} - t_s \mathcal{T}_{0,\theta})^2 < 2(\mathcal{T}_{0,\theta} + t_s \mathcal{T}_{0,z})^2.$$

These inequalities are homogeneous in $\mathcal{T}_0$, so multiplying the stress by a positive number does not change them. The stress can tend to zero at an edge of the annulus while its unit direction still satisfies the cone inequalities with a strict margin. Theorem 4.6(iii) states that boundary condition precisely. Proposition 7.5 writes every nonzero stress of this kind as a positive combination of the allowed wave covariances. Differentiating that representation gives the signed changes in amplitude used for the higher-order stress corrections in Proposition 7.6.

4.4 Properties of the leading profile

The profiles E, U must meet both the radial matching conditions of Lemma 4.4 and the stress cone inequalities of Lemma 4.5. At each edge of the annulus, the stress $\mathcal{T}_0$ must go smoothly to zero while its direction stays inside the cone. Theorem 4.6 puts fixed margins and bounds on these edge conditions. It also supplies regular behavior at the axis, the heat-flow exterior, and the moment identities needed in Sections 2.1 and 3, and it keeps aside the intervals later used for the background and mean corrections in (4.30).

The edge estimates use the names *inner collar* and *outer collar* for fixed neighborhoods of the two annular edges in profile coordinates. If $0 < X_a < X_b$ and $0 < \varepsilon < \log(X_b/X_a)$, these collars are the rectangles

$$\begin{aligned} \mathcal{C}_a(\varepsilon) &= [X_a, X_a e^\varepsilon] \times [-1, 1], \\ \mathcal{C}_b(\varepsilon) &= [X_b e^{-\varepsilon}, X_b] \times [-1, 1]. \end{aligned} \quad (4.24)$$

At a fixed η , the inner collar has $0 \leq \log(X/X_a) \leq \varepsilon$, while the outer collar has $0 \leq \log(X_b/X) \leq \varepsilon$. Their widths do not depend on η or the physical variable q . A smaller collar means the same kind of set with a smaller positive ε . If a condition is meant only for points inside the annulus, the collar is intersected with $\{X_a < X < X_b\}$.

A smooth scalar or vector g is *flat at the radial edge* $X_e \in \{X_a, X_b\}$ when every one of its radial derivatives, including the zeroth, vanishes there component by component,

$$\partial_{X_e}^j g(X_e, \eta) = 0 \quad \text{for every } j \geq 0 \text{ and } \eta \in [-1, 1].$$

The stress will be flat at both edges. Wherever the stress is nonzero, its normalized direction is found by dividing the stress by its length, so it is the unit vector that points in the same direction as the stress. The theorem below states that the constructed stress is nonzero on the marked open annulus $X_a < X < X_b$, so its normalized direction is defined there. It also extends that direction to the closed collars with the positive margin stated in Theorem 4.6(iii). The construction results used in that proof come after the theorem.

Theorem 4.6. *There are fixed numbers $h \in (0, 1/100)$, $\lambda > 0$, $C > 1$ and $0 < X_a < X_b$, together with profiles E, U, Π for $X \geq 0$ and $-1 \leq \eta \leq 1$, with the properties listed below. The quantities $F, H, V_0, Q_s, N_s, p_s, a, b_s$ and $\mathcal{T}_0$ are defined by Equations (4.7) to (4.11). Wherever $a > 0$, the quantities t_s, v_s are defined by (4.20). The symbols A, d come from (4.1). Here the closed annulus means the profile rectangle $[X_a, X_b] \times [-1, 1]$ in (X, η) . Every claim below holds for every $\eta \in [-1, 1]$.*

- (i) For every finite $R > 0$, the scalars $F = \phi/C, U, \Pi, V_0/X$ are smooth on $[0, R] \times [-1, 1]$. Every fixed mixed X, η derivative has a uniform bound there, with one-sided derivatives at the boundary. Also, $\phi > 0$, and $E = \sqrt{2X}F > 0$ whenever $X > 0$. By Equations (4.4)–(4.5), these factors give a smooth Cartesian field across the axis. There is one fixed $X_{\text{an}} \in (X_a, X_b)$ with the following property. On $[0, X_{\text{an}}] \times [-1, 1]$, these scalars and each fixed radial derivative are analytic in η on one common complex neighborhood of $[-1, 1]$. The inner collar $[X_a, X_{\text{an}}] \times [-1, 1]$, understood as in (4.24), has the positive directional margin stated in part (iii). The pressure is normalized to vanish at radial infinity, so

$$\Pi(X, \eta) = - \int_X^\infty \frac{E(x, \eta)^2}{2x} dx. \quad (4.25)$$

- (ii) The radial pressure balance $\Pi_X = E^2/(2X)$ in (4.7), with its smooth extension $\Pi_X = F^2$ at $X = 0$, holds exactly. So do the tangential residual identities in Proposition 4.2, with physical stress $q^{-A-1/2}\mathcal{T}_0$. The stress profile $\mathcal{T}_0$ is zero when $0 \leq X \leq X_a$ and $X \geq X_b$, and it is nonzero at every point with $X_a < X < X_b$. On $0 \leq X \leq X_a$, the profiles solve (4.13).
- (iii) Each of $F, a, v_s - 2$ has a positive lower bound on the closed annulus $[X_a, X_b] \times [-1, 1]$. The unit direction $n = \mathcal{T}_0/|\mathcal{T}_0|$ starts out defined on the radial interior $(X_a, X_b) \times [-1, 1]$, and it extends smoothly to the whole closed annulus, including the boundaries $X = X_a, X_b$. There is one fixed $\kappa \in (0, 2)$ such that everywhere on this rectangle,

$$n_\theta + t_s n_z \geq \kappa, \quad (v_s - 2)(n_z - t_s n_\theta)^2 \leq (2 - \kappa)(n_\theta + t_s n_z)^2. \quad (4.26)$$

For each η , the vector $n(X_a, \eta)$ is parallel to $(a(X_a, \eta), -b_s(X_a, \eta))$. At $X = X_b$, $n(X_b, \eta) = (1, 0)$ and $b_s(X_b, \eta) = 0$.

- (iv) There is a smooth weight $\zeta(X)$. It is positive when $X_a < X < X_b$ and zero for $X \leq X_a$ and $X \geq X_b$. Every radial derivative of ζ vanishes at X_a, X_b . In the radial interior, put $\delta = \min\{1, \log(X/X_a), \log(X_b/X)\}$. There is a constant $c > 0$ that does not depend on α . For each fixed multi-index $\alpha = (\alpha_1, \alpha_2)$, there are finite constants C_α, m_α such that, with $\partial^\alpha = \partial_X^{\alpha_1} \partial_\eta^{\alpha_2}$, the following bounds hold throughout $(X_a, X_b) \times [-1, 1]$.

$$|\mathcal{T}_0| \geq c\zeta, \quad |\partial^\alpha \mathcal{T}_0| \leq C_\alpha \zeta \delta^{-m_\alpha}. \quad (4.27)$$

- (v) The limits $M(\infty, \eta), J(\infty, \eta), S(\infty, \eta)$ of the cumulative integrals in (4.15) exist. The angular momentum is made convergent by subtracting the exterior power H_{pow} below from H before integrating. These four convergent quantities satisfy

$$M(\infty, \eta) = J(\infty, \eta) = S(\infty, \eta) = 0, \quad \int_0^\infty (H - H_{\text{pow}}) dX = 0, \quad H_{\text{pow}} = \sqrt{2X} c_\infty X^{-A}, \quad (4.28)$$

where $c_\infty > 0$ does not depend on η . For $X \geq X_b$,

$$U = V_0 = 0, \quad E = c_\infty X^{-A} \mathcal{H}(2d/X), \quad \mathcal{H}(Z) = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h (1+Zv)^{-h} dv. \quad (4.29)$$

In this region the physical swirl is the exact radial heat solution $c_\infty s^{-A} \mathcal{H}(2\tau/s)$. The profile $\mathcal{H}$ is positive and smooth for $Z \geq 0$, including the one-sided endpoint at zero. There is also a fixed $X_v \in (X_a, X_b)$ such that $U = V_0 = 0$ on all of $[X_v, \infty)$, including the outer collar $[X_v, X_b]$.

- (vi) Two fixed, disjoint intervals $\mathcal{I}_{\text{pos}}$ and $\mathcal{I}_{\text{mean}}$ are left available for later use. Lemma 5.2 uses the first for positive-order background corrections, and Lemma 8.7 uses the second for corrections to the averaged flow,

with $\sup \mathcal{I}_{\text{pos}} < \inf \mathcal{I}_{\text{mean}}$. Here positive order means a relative power q^{2nh} , $n \geq 1$ in the background expansion (5.1). Both intervals lie in (X_a, X_v) . For every X in either interval,

$$U = 0, \quad E = c_{\text{patch}}(1 + \eta^2)^{-1} X^{-1/2-\lambda}, \quad (4.30)$$

where the positive constant c_{patch} does not depend on X, η . Building the leading profile does not change either interval.

Every constant in this theorem depends only on the fixed choices used to build the profile and, for a derivative bound, on the order of that derivative. None depends on the physical variable q , on any later dyadic band, or on any later correction stage.

4.5 Construction results used in the proof

Building the profile takes three choices that have to work together. The outer flow fixes the pressure on the axis, the regular inner flow matches the outer flow's five cumulative integrals, and the shear on the interval between them satisfies the admissible cone condition. Lemma 4.7 first solves the finite-dimensional moment problem. Lemma 4.8 and Proposition 4.10 then build the outer and inner flows, and Lemma 4.11 builds the periodic shear. These results provide the pieces used in the proof of Theorem 4.6. The appendices prove each construction in full at the references given below.

In these statements, a *parameter derivative* means a derivative with respect to η , with every construction constant held fixed. For a function on $[-1, 1]$, write $\|f\|_{C^k} = \max_{0 \leq j \leq k} \sup_{\eta} |\partial_{\eta}^j f|$. For vectors, use the largest component norm. Only finitely many of these norms have to be small when the parameters are chosen. After the parameters have been fixed, every other fixed derivative order has a finite bound.

Lemma 4.7. *Let $\alpha_1, \dots, \alpha_m$ be distinct real numbers. Let $\beta_1, \dots, \beta_m$ be nonnegative, nonzero smooth functions supported in ordered, disjoint compact intervals of $(0, \infty)$. Then the matrix*

$$B_{ij} = \int_0^{\infty} x^{\alpha_i} \beta_j(x) dx$$

is invertible. Its inverse and every fixed parameter derivative stay bounded for any smooth compact family that keeps these assumptions true.

There is also a more general statement. Let $B(\eta)$ be a smooth invertible matrix on $[-1, 1]$, let $\mathcal{Q}_{\eta}$ be a smooth bilinear map, and let $d \in C^{\infty}([-1, 1]; \mathbb{R}^m)$. Fix an integer $k \geq 0$. Let v_j be the norm of multiplication by B^{-1} on C^j , and let q_j be the bilinear norm of $\mathcal{Q}$ on C^j , for $j = 0, k$. If

$$8v_j^2 q_j \|d\|_{C^j} \leq 1 \quad (j = 0, k),$$

then the equation

$$B(\eta)c(\eta) + \mathcal{Q}_{\eta}(c(\eta), c(\eta)) = d(\eta)$$

has a smooth solution. It is the only solution with $\|c\|_{C^0} \leq 2v_0 \|d\|_{C^0}$, and it satisfies $\|c\|_{C^k} \leq 2v_k \|d\|_{C^k}$. When $\mathcal{Q} = 0$, no smallness condition is needed and the solution is $c = B^{-1}d$.

Proof. These two claims are Lemmas A.1 and A.2. For the first, multilinearity gives

$$\det B = \int \det[x_j^{\alpha_i}] \prod_j \beta_j(x_j) dx_1 \cdots dx_m.$$

When $x_1 < \dots < x_m$, the determinant inside this integral has one fixed nonzero sign. To see why, any nonzero linear combination of m distinct powers has at most $m - 1$ positive zeros, by induction and Rolle's theorem. So the integral is nonzero.

For the second claim, on the ball of radius $r_j = 2v_j \|d\|_{C^j}$, the iteration $c \mapsto B^{-1}(d - \mathcal{Q}(c, c))$ satisfies

$$\|B^{-1}(d - \mathcal{Q}(c, c))\|_{C^j} \leq r_j/2 + v_j q_j r_j^2 \leq r_j, \quad 2v_j q_j r_j \leq \frac{1}{2}.$$

It is a contraction in C^0 and C^k . Starting the iteration from zero picks the same solution in both spaces. The pointwise implicit function theorem then shows that this solution is smooth. $\square$

The exterior is built before the part near the axis. Requiring its pressure to vanish at infinity fixes the pressure value that the axis problem must use. The family below contains the heat-flow exterior and keeps several intervals available, so later changes can restore the required moments.

Lemma 4.8. *There are finite choices of the parameters M_d, P_*, λ, h and a constant $R_* > 0$ with the following properties. First choose M_d . Then P_* may depend on M_d ; λ may depend on M_d, P_* ; and h may depend on M_d, P_*, λ . The choices satisfy*

$$T_d = e^{M_d} + 10, \quad P_* > e^{T_d}, \quad 0 < h < \min\{1/100, \lambda, e^{-T_d}\}.$$

For every $X_R \geq R_*$, there are smooth profiles (U_o, E_o) on $(0, \infty) \times [-1, 1]$, with $E_o > 0$, that satisfy the conditions below. Put $x = X/X_R$ and $f(\eta) = (1 + \eta^2)^{-1}$.

(i) When $0 < x \leq 1$,

$$U_o = 4\eta, \quad E_o = P_* f(\eta) x^{1/10}.$$

The pressure value

$$\Pi_0(\eta) = - \int_0^\infty \frac{E_o(X, \eta)^2}{2X} dX, \quad \Pi_o(X, \eta) = \Pi_0(\eta) + C_{p,o}(X, \eta) \quad (4.31)$$

does not depend on X_R . It is analytic on one complex neighborhood of $[-1, 1]$, it is even, and it satisfies

$$\Pi_0 \leq -\frac{5}{2} P_*^2 f^2, \quad \eta \Pi'_0(\eta) > 0 \quad (\eta \neq 0).$$

For this temporary profile, define the five cumulative integrals by (4.15), $V_{0,o}$ by (4.7), and $Q_{s,o}, N_{s,o}$ by the explicit formulas (4.16). Equations (4.11) and (4.20) then define its shear, stress, and cone coordinates. These are the same definitions used in Lemma 4.4. They make sense on $X > 0$, even though this temporary inner branch does not have to be regular at the physical axis.

(ii) There are four nonempty, ordered, disjoint intervals $\mathcal{I}_j = X_R(\alpha_j, \beta_j)$ and radii satisfying

$$X_R < X_{\text{good}} < \inf \mathcal{I}_1, \quad \sup \mathcal{I}_4 < X_v < X_{\text{tail}} < X_b = e^3 X_{\text{tail}},$$

and the endpoints of those intervals, after division by X_R , do not depend on X_R, η . On all four intervals, $U_o = 0$, and

$$E_o = c_{\text{patch}} f(\eta) X^{-1/2-\lambda} \quad \text{on } \mathcal{I}_1, \mathcal{I}_3, \mathcal{I}_4, \quad c_{\text{patch}} > 0.$$

Only $\mathcal{I}_2$ is used for the heat compensation. The other three intervals remain available for corrections. Also,

$$U_o = M_o = J_o = 0 \quad (X \geq X_v),$$

$$M_o(\infty) = J_o(\infty) = S_o(\infty) = 0, \quad \int_0^\infty (H_o - H_{\text{pow}}) dX = 0,$$

where $H_o = \sqrt{2X} E_o$ and $H_{\text{pow}} = \sqrt{2X} c_\infty X^{-A}$. Both $c_{\text{patch}}, c_\infty > 0$ are independent of X, η .

(iii) The relaxed cone condition holds on $[e^{-5} X_R, e^{1/2} X_{\text{tail}}] \times [-1, 1]$. The admissible condition holds on $[X_{\text{good}}, e^{1/2} X_{\text{tail}}] \times [-1, 1]$. On each closed rectangle, every gap that defines the relevant condition has a positive minimum. These minima may depend on X_R . For $X \geq X_b$,

$$U_o = V_{0,o} = 0, \quad E_o = c_\infty X^{-A} \mathcal{H}(2d/X),$$

where $\mathcal{H}$ is defined in (4.29). The physical swirl $K = c_\infty (r^2/2)^{-A} \mathcal{H}(4\tau/r^2)$ satisfies $\partial_t K = (\partial_{rr} + r^{-1} \partial_r - r^{-2})K$ and $K_r < 0$ for $r > 0$.

Proof. Proposition A.4 and Lemma A.5 provide the reference profiles and the common pressure value in part (i). The four intervals defined after (A.9) are the intervals in part (ii). The parameter M_d controls the decrease in the axial direction, T_d controls how long that decrease lasts in logarithmic radius, P_* is the initial azimuthal amplitude, and λ, h are the intermediate and outer exponents.

Lemma A.6 and Proposition A.7 replace the tail by the heat profile from part (iii). The correction on $\mathcal{I}_2$ restores the pressure integral from part (i), along with the total S -moment and the renormalized angular moment from part (ii). Once the other parameters have been fixed, those results keep the cone inequalities from part (iii) true for every sufficiently large X_R . $\square$

The exterior is now prepared with the right moments and heat flow. Its stress was defined by integrating outward from the axis. To get the conclusions at the outer edge as well, it still needs a regular inner extension with the same moments. Once that condition is met, the stress can instead be integrated backward from infinity, and its limiting direction can be found explicitly.

Lemma 4.9. *For every $X_R \geq R_*$, the family in Lemma 4.8 can be chosen to have the following extra property. Suppose a leading field is regular at the axis, agrees with that family when $X \geq X_{\text{tail}}$, satisfies the four moment identities in (4.28) as functions of η , and has the canonical pressure (4.25). Then its physical stress $T = q^{-A-1/2} \mathcal{T}_0$ satisfies*

$$T_\theta(r) = r^{-2} \int_r^\infty r'^2 R_\theta^{(0)}(r') dr', \quad T_z(r) = r^{-1} \int_r^\infty r' R_z^{(0)}(r') dr',$$

so $\mathcal{T}_0 = 0$ whenever $X \geq X_b$. On $e^{1/2} X_{\text{tail}} \leq X < X_b$, the admissible cone condition holds and

$$b_s = 0, \quad 2 + h < a \leq 2 + 2h, \quad T_\theta > 0, \quad 2 - (a - 2)(T_z/T_\theta)^2 \geq \kappa_0 > 0.$$

Put $y_b = \log(X_b/X)$. There are smooth functions b_θ, b_z on a closed outer collar, with $\inf b_\theta > 0$, such that

$$\mathcal{T}_{0,\theta} = e^{-4/y_b^2} y_b^{-3} b_\theta(y_b, \eta), \quad \mathcal{T}_{0,z} = e^{-4/y_b^2} y_b^3 b_z(y_b, \eta). \quad (4.32)$$

For this fixed profile, all these claims hold uniformly for $\eta \in [-1, 1]$. Their constants may depend on X_R , but they do not depend on the physical variable q .

Proof. These claims are Lemma A.8 and Proposition A.10. The four moment identities make the total residual integrals cancel. This gives the backward primitives shown above, which are the integrals from the current radius to infinity. The prepared terminal profile then gives the cone inequalities and (4.32). $\square$

The missing inner extension can now be built. It must be smooth at the axis and must match the five cumulative integrals of the prepared exterior at a chosen joining radius. It uses the function $\Pi_0(\eta)$ from Lemma 4.8 as its pressure value at $X = 0$.

Proposition 4.10. *Fix the outer-profile data from Lemma 4.8 and any prescribed finite set of η -derivative orders. It is possible to choose finite axis parameters $j_0, \delta_*, \sigma_*, \Lambda$, a logarithmic transition length T_{sh} , and an amplitude threshold C_0 so that every $C \geq C_0$ allows the following construction. Set*

$$X_a = 4/\Lambda, \quad X_R = 110(\text{CP}_*)^{10}, \quad x = X/X_R, \quad X_h = X_R e^{-5}, \quad f = (1 + \eta^2)^{-1}.$$

After choosing C , choose a positive activation width t_1 in the logarithmic coordinate $\log(X/X_a)$, a positive shear factor κ_0 , and the remaining transition widths. There are profiles $E = \sqrt{2X}\phi/C, U$ and $\Pi = \Pi_0 + C_p$ on $[0, X_h] \times [-1, 1]$ with the following properties.

- (i) The scalar ϕ is positive, and $\phi, U, \Pi, V_0/X$ are smooth on $[0, X_h] \times [-1, 1]$, including at $X = 0$. So Equations (4.4)–(4.5) give a Cartesian velocity and pressure that are smooth across the axis. For some $X_{\text{an}} \in (X_a, X_h)$, these scalars and every fixed radial derivative are analytic in η on one common complex neighborhood throughout $[0, X_{\text{an}}] \times [-1, 1]$. The profiles solve (4.13) on $0 \leq X \leq X_a$, and $\mathcal{T}_0 = 0$ there.

- (ii) The inequalities $a > 0$, $P_c > 2$, and $v_s < \mathcal{U}(P_c, J_c)$ hold on $(X_a, X_h]$. On the first collar $(X_a, X_{\text{an}}]$, one also has $v_s > 2$. At its inner endpoint, $a(X_a, \eta) > 0$ and $v_s(X_a, \eta) > 2 + c_{\text{ex}}$ for a constant $c_{\text{ex}} > 0$. Write $y_a = \log(X/X_a)$. On the first collar, the stress then factors as

$$\begin{aligned} \mathcal{T}_0 &= e^{-t_1^2/y_a^2} g_a(y_a) B_a(y_a, \eta), \quad g_a(0) > 0, \\ B_a(0, \eta) &= F(X_a, \eta) \mathfrak{s}(X_a, \eta) \neq 0. \end{aligned} \quad (4.33)$$

The functions g_a, B_a stay smooth through $y_a = 0$. For every fixed multi-index α , there are finite constants such that

$$|\mathcal{T}_0| \geq ce^{-t_1^2/y_a^2}, \quad |\partial_{X,\eta}^\alpha \mathcal{T}_0| \leq C_\alpha e^{-t_1^2/y_a^2} y_a^{-N_\alpha}.$$

Everything stated in items (i) and (ii) holds for every $\eta \in [-1, 1]$.

- (iii) Near X_h , the fields are exactly $U_0 = 4\eta$, $E_0 = P_* f x^{1/10}$, and $\mathfrak{m}(X_h, \eta) = \mathfrak{m}_0(X_h, \eta)$. Here $\mathfrak{m}_0 = (M_0, I_0, J_0, S_0, C_{p,0})$ is the list of five integrals of this reference pair, each taken from zero. These are equalities between functions of η , and both pressures take the same prescribed value $\Pi_0(\eta)$ at $X = 0$.

This leaves X_R free to exceed any further prescribed lower bound, without changing the earlier choices of Λ, T_{sh} .

Proof. First choose axis data for Proposition B.2. Then use Proposition B.3 and Corollary B.10 to continue the profile and match its moments. Set $U_* = 4\eta + j_0$, where $0 < j_0 \leq .05$ is the small offset in the axis velocity, and define

$$H_* = D\eta + dU_*, \quad Z_* = -A(1 - 2\eta U_*)U_* - H_* U'_* - d\Pi'_0 + 4A\eta\Pi_0.$$

The pressure bounds in Lemma 4.8 show that Z_* is positive at the one point where H_* is zero. Choose a threshold $\delta_* > 0$, and then choose a positive regularization parameter $\sigma_* > 0$ that keeps the denominator in the next fraction away from zero, so that

$$\chi = \frac{H_*^2}{H_*^2 + \sigma_*^2} > .99 \quad \text{where} \quad |Z_*| \leq \delta_*.$$

With these choices, the positive analytic value on the axis is

$$\phi(0, \eta) = \exp\left(-\Lambda \int_0^\eta \frac{L(w)H_*(w)}{H_*(w)^2 + \sigma_*^2} dw\right).$$

The parameter Λ sets the radial scale $Y = \Lambda X$. Propositions B.2 and B.3 give the stress-free solution through $Y = 4.1$ and the exit inequality $p_{s,1} + p_{s,2}^2/p_{s,1} > 2 + c_{\text{ex}}$ at $Y = 4$. Proposition B.5 and Corollary B.6 give the regularity at the axis and on the collar in part (i), along with the first-collar claims in part (ii). The factor κ_0 reduces the reference shear, and t_1 is the width of its first transition in $\log(X/X_a)$.

For part (iii), calculate the reference integrals $\mathfrak{m}_0$. Under the normalization

$$\hat{\mathfrak{m}} = (M/X_R, I/X_R^{3/2}, J/X_R^{3/2}, S/X_R, C_p),$$

direct integration of the reference pair gives

$$\begin{aligned} \hat{M}_0(x) &= 4\eta x, \quad \hat{I}_0(x) = \frac{5\sqrt{2}}{8} P_* f x^{8/5}, \quad \hat{J}_0(x) = 4\eta \hat{I}_0(x), \\ \hat{S}_0(x) &= 16\eta^2 x - \frac{5}{12} P_*^2 f^2 x^{6/5}, \quad C_{p,0}(x) = \frac{5}{2} P_*^2 f^2 x^{1/5}. \end{aligned} \quad (4.34)$$

The first continuation from Proposition B.5 ends at

$$X_i = 110, \quad G_i(\eta) = U(X_i, \eta), \quad a(X_i, \eta) = 4/5, \quad \mathcal{D}_X U(X_i, \eta) = 0.$$

The next transition, (B.34), keeps $U = G_i$ while matching E to the azimuthal reference profile. It keeps $U = G_i$ until the restoration interval described below.

To set a matching tolerance before choosing C , put $\mathcal{R} = [e^{-8}, e^{-5}]$, and take all the following minima and maxima over $\mathcal{R} \times [-1, 1]$. The reference fields have $Q_{\min} = \min Q_{s,0} > 0$ and $e_* = \min E_0 > 0$.

They also give the finite number $w_* = 1 + \max|N_{s,0}/(E_0 Q_{s,0})|$. Here restoration means changing $U = G_i(\eta)$ to 4η on $e^{-8} < x < e^{-7}$. After that come the five moment corrections on $e^{-6} < x < e^{-5}$. At the entrance $x = e^{-6}$ to the correction interval, write $\Delta\hat{m} = \hat{m} - \hat{m}_0$, using (4.19) with $\rho = X_R$. For the prescribed finite order k , choose $\varepsilon_m > 0$ small enough that the bound

$$\|G_i - 4\eta\|_{C^{k+1}} + \|\Delta\hat{m}(e^{-6}, \cdot)\|_{C^{k+1}} < \varepsilon_m$$

makes the restoration and correction satisfy

$$Q_s \geq Q_{\min}/2, \quad E \geq e_*/2, \quad .7 \leq a \leq .9, \quad \left| \frac{N_s}{EQ_s} \right| \leq w_*, \quad |b_s| \leq \frac{.1}{1+w_*}.$$

It follows that $v_s = a + b_s^2/a \leq .9 + .01/.7 < 1$, while

$$G = Q_s - \frac{b_s N_s}{aE} \geq \frac{6}{7} Q_s \geq \frac{3Q_{\min}}{7}, \quad P_c = \frac{X_R x}{L} G > 2$$

for every sufficiently large X_R . The same fixed tolerance therefore keeps the relaxed condition true through both restoration and moment matching, as part (ii) requires.

For each required order k , Lemma B.7 gives

$$\|G_i - 4\eta\|_{C^k} \leq C_k j_0 + C_k^{\text{nat}}/\Lambda + C_k^{\text{join}}(\Lambda)(t_1 + \kappa_0 + \omega_{\text{fin}}),$$

where ω_{fin} is the sum of the last two widths in $\log X$. One width is used to set $\mathcal{D}_X U = 0$; the other is used to move a to $4/5$ before $X = X_i$. The same lemma bounds $\log(CE(X_i, \cdot))$ independently of the later amplitude. That bound fixes the length T_{sh} of the transition to E_0 . The endpoint of this transition satisfies

$$x_{\text{sep}} = e^{T_{\text{sh}}/(CP_*)^{10}} \rightarrow 0.$$

Choose j_0, Λ first, to control the first two errors. Then choose C large, and finally choose the widths and κ_0 small. The bound (B.39) then makes the normalized moment error at $x = e^{-6}$ smaller than the chosen tolerance. The prescribed orders include the extra derivative required by (B.35). Proposition B.8 restores U_0 and solves all five moment equations before $x = e^{-5}$, while keeping the relaxed inequalities above. Direct integration of (U_0, E_0) gives (4.34), which finishes part (iii). $\square$

The inner and outer constructions leave an interval where only the relaxed cone condition is known. On that interval, replace the shear by a periodic family with the same mean, with every value in the family satisfying the admissible condition. The following map measures how far the shear is from each of the four constraint boundaries. For $a > 0, b \in \mathbb{R}$, and $p = (p_1, p_2) \in \mathbb{R}^2$, define

$$\begin{aligned} v &= a + \frac{b^2}{a}, & c &= p_1 - \frac{b}{a} p_2, & j &= p_2 + \frac{b}{a} p_1, \\ \Psi(a, b, p) &= (a, v - 2, c - v, 2(c - v)^2 - (v - 2)j^2). \end{aligned} \tag{4.35}$$

For the profile data (a, b_s, p_s) , all four entries are positive exactly when $a > 0, v_s > 2$, and (4.22) hold. The map Ψ is smooth on $\{a > 0\}$.

Lemma 4.11. *Let $\mathcal{I} = [X_-, X_+] \Subset (0, \infty)$, and let a, b_s, p_s be smooth on $\mathcal{I} \times [-1, 1]$, with $a > 0$. Define t_s, v_s, P_c, J_c by (4.20). Suppose $P_c > 2$ and $v_s < \mathcal{U}(P_c, J_c)$ everywhere on this rectangle, and suppose $v_s > 2$ near both radial endpoints. Then there is a smooth period-one family $(a_L, -b_L)(X, \eta, \varphi)$, $\varphi \in \mathbb{R}/\mathbb{Z}$, whose mean is the original shear:*

$$\int_0^1 (a_L, -b_L)(X, \eta, \varphi) d\varphi = (a, -b_s)(X, \eta).$$

Every shear on this loop satisfies the admissible cone condition while the given vector $p_s(X, \eta)$ stays fixed. More precisely, there is a number $\mu_L > 0$ such that

$$\Psi_i(a_L, b_L, p_s) \geq \mu_L \quad \text{on } \mathcal{I} \times [-1, 1] \times \mathbb{R}/\mathbb{Z}, \quad 1 \leq i \leq 4. \quad (4.36)$$

Near both radial endpoints, the loop does not depend on φ and is equal to $(a, -b_s)$. Smoothness here includes one-sided derivatives at $\eta = \pm 1$. The number μ_L may depend on all the fixed input data.

Proof. The variance construction in Lemma C.1 gives a smooth 2π -periodic ratio $t_\ell(X, \eta, \theta')$ and a smooth function $\rho_\ell(X, \eta) \geq 0$. Write $\langle f \rangle = (2\pi)^{-1} \int_0^{2\pi} f d\theta'$. These functions satisfy

$$\langle t_\ell \rangle = t_s, \quad \langle (t_\ell - t_s)^2 \rangle = \rho_\ell / a, \quad v_\ell = v_s + \rho_\ell > 2.$$

For every θ' , they also satisfy

$$c_\ell = p_{s,1} + t_\ell p_{s,2} > 2, \quad j_\ell = p_{s,2} - t_\ell p_{s,1}, \quad v_\ell < \mathcal{U}(c_\ell, j_\ell).$$

These are (C.10) and (C.8), together with the cutoff choice (C.9). The construction stays smooth at points where $\rho_\ell = 0$. Near the radial endpoints it has $\rho_\ell = 0$ and $t_\ell = t_s$. Now introduce the lifted parameter φ , starting from $\varphi(0) = 0$, by

$$\frac{d\varphi}{d\theta'} = \frac{a(1+t_\ell^2)}{2\pi v_\ell}, \quad (a_L, -b_L) = \frac{v_\ell(1, t_\ell)}{1+t_\ell^2}. \quad (4.37)$$

Direct integration shows both that the new parameter has period one and that the shear has the required mean when it is written in this parameter:

$$\begin{aligned} \int_0^{2\pi} \frac{d\varphi}{d\theta'} d\theta' &= \frac{a}{v_\ell} (1+t_s^2 + \rho_\ell/a) = 1, \\ \int_0^1 (a_L, -b_L) d\varphi &= \frac{a}{2\pi} \int_0^{2\pi} (1, t_\ell) d\theta' = (a, at_s) = (a, -b_s). \end{aligned}$$

The positive derivative in (4.37) has a positive minimum on the compact parameter set, so the inverse change of parameter is smooth. The shear obtained this way satisfies $-b_L/a_L = t_\ell$ and $a_L(1+t_\ell^2) = v_\ell$. Lemma 4.5 then shows that every entry of $\Psi(a_L, b_L, p_s)$ is positive. The smallest value among all those entries on $\mathcal{I} \times [-1, 1] \times \mathbb{R}/\mathbb{Z}$ is therefore a positive number μ_L , which proves (4.36). Wherever $\rho_\ell = 0$, formula (4.37) becomes $(a_L, -b_L) = (a, -b_s)$, which proves the claim about the endpoints. $\square$

4.6 Proof of the leading profile theorem.

Proof of Theorem 4.6. First attach the two profiles from Lemma 4.8 and Proposition 4.10, keeping their five cumulative integrals unchanged at the radius where they meet. Then use Lemma 4.11 to change the shear on the interval where only the relaxed cone condition holds. Estimate the errors caused by that change, and use Lemma 4.7 to solve the five equations that restore the integrals. Every estimate below is for $\eta \in [-1, 1]$.

Step 1: choose the parameters and join the profiles. Choose the outer parameters $M_d, T_d, P_*, \lambda, h$ and Π_0 from Lemma 4.8, using the family with the endpoint property in Lemma 4.9. Apply Proposition 4.10 with this same function Π_0 . In that proposition, Λ and the logarithmic transition length T_{sh} are fixed before the amplitude normalization C . Let R_* be the radius threshold from Lemma 4.8, and let C_0 be a lower bound from the axis construction after these choices. Then take

$$C \geq \max \left\{ C_0, \frac{(R_*/110)^{1/10}}{P_*} \right\}, \quad X_R = 110(CP_*)^{10} \geq R_*.$$

The proof of Proposition 4.10 defines the tolerance ε_m and the total logarithmic width ω_{fin} . Using the smallness convention from Definition 3.3, choose the profile parameters so that

$$0 < C^{-1} \ll T_{\text{sh}}^{-1} \ll \Lambda^{-1} \ll \sigma_* \ll \delta_* \ll j_0 \ll \varepsilon_m \ll h \ll \lambda \ll P_*^{-1} \ll M_d^{-1} \ll 1, \\ 0 < \omega_{\text{fin}} \ll t_1 \ll \kappa_0 \ll C^{-1}.$$

The second line continues the order of choices in the first. In particular, M_d, P_* are large; $\lambda, h, \varepsilon_m, j_0, \delta_*, \sigma_*$ are small; $\Lambda, T_{\text{sh}}, C$ are large; and the three last transition parameters are small. The exact scales derived from them, along with the extra requirement on the amplitude, are still

$$T_d = e^{M_d} + 10, \quad P_* > e^{T_d}, \quad X_R = 110(CP_*)^{10}.$$

After making these choices, fix the periodic shear and the correction bumps used in Steps 2–3. Then choose $N^{-1} \ll \omega_{\text{fin}}$. In the same convention, this means taking N large enough after ω_{fin} , those functions, and every constant in the estimates below have been fixed. This order of choices fixes every radial scale and width before the last integer N is chosen.

Put $X_h = X_R e^{-5}$. Let $(U_{\text{in}}, E_{\text{in}})$ be the axis connection and $(U_{\text{out}}, E_{\text{out}})$ the prepared outer pair, and define

$$(U_0, E_0)(X, \eta) = \begin{cases} (U_{\text{in}}, E_{\text{in}})(X, \eta), & X \leq X_h, \\ (U_{\text{out}}, E_{\text{out}})(X, \eta), & X \geq X_h. \end{cases}$$

Both definitions equal $(4\eta, P_*(1 + \eta^2)^{-1}(X/X_R)^{1/10})$ near X_h , so U_0, E_0 are smooth there. These are the profiles before Step 2 changes the shear. Set $H_0 = \sqrt{2X}E_0$, and use (4.15) to define $\mathbf{m}_0 = (M_0, I_0, J_0, S_0, C_{p,0})$ from U_0, E_0 . Keep the prescribed axis pressure $\Pi_0(\eta)$, and put $\Pi_0^{\text{prof}}(X, \eta) = \Pi_0(\eta) + C_{p,0}(X, \eta)$. Use (4.7), (4.16), and (4.11) to calculate $V_0, Q_{s,0}, N_{s,0}, p_{s,0}, a_0, b_{s,0}, \bar{T}_{0,0}$ from $U_0, E_0, \Pi_0^{\text{prof}}$. Write $p_0 = p_{s,0}, b_0 = b_{s,0}$, and $\mathbf{s}_0 = (a_0, -b_0)$. The five equalities in Proposition 4.10(iii) give

$$\mathbf{m}_0(X_h, \eta) = \mathbf{m}_{\text{out}}(X_h, \eta).$$

For $X \geq X_h$, the five integrands agree as well. It follows that

$$\mathbf{m}_0(X, \eta) - \mathbf{m}_{\text{out}}(X, \eta) = \mathbf{m}_0(X_h, \eta) - \mathbf{m}_{\text{out}}(X_h, \eta) \\ + \int_{X_h}^X \begin{pmatrix} U_0 - U_{\text{out}} \\ H_0 - H_{\text{out}} \\ U_0 H_0 - U_{\text{out}} H_{\text{out}} \\ U_0^2 - U_{\text{out}}^2 - (E_0^2 - E_{\text{out}}^2)/2 \\ (E_0^2 - E_{\text{out}}^2)/(2x) \end{pmatrix} dx = 0.$$

Lemma 4.4(i) now shows that the pressures and the coefficients $V_0, Q_{s,0}, N_{s,0}, p_0, \mathbf{s}_0, \bar{T}_{0,0}$ agree with their outer counterparts on this interval. In particular, the joined pair has the same pressure normalization and the same four total moment identities as the pair in Lemma 4.8. Its angular identity follows from

$$\int_0^\infty (H_0 - H_{\text{out}}) dX = I_0(X_h) - I_{\text{out}}(X_h) = 0.$$

The joined pair is regular at the axis, so the conditional conclusions about the stress in Lemma 4.9 now apply. Its stress is zero outside $[X_a, X_b]$. It has the stated factorizations at the inner and outer edges, and the admissible cone condition holds near both edges. The relaxed cone condition holds on the whole interval between them.

It remains to specify the interval where the shear still has to change. Let $\mathcal{I}_1$ be the first correction interval reserved in Lemma 4.8. Choose a closed interval $\mathcal{I} = [X_-, X_+] \subset (X_a, \inf \mathcal{I}_1)$ that contains

every point of (X_a, X_b) where the joined pair might fail the admissible condition. Choose its left endpoint strictly between X_a and the end of the analytic collar. Choose its right endpoint so that $X_{\text{good}} < X_+ < \inf \mathcal{I}_1$. Shorten the analytic collar to an endpoint $X_{\text{an}} \in (X_a, X_-)$. Take $Y_0 < Y_1$ inside $\mathcal{I}_1$, with $X_+ < Y_0$, and set $\mathcal{J} = [X_-, Y_1]$. The original pair is admissible near both ends of $\mathcal{I}$ and everywhere on $[X_+, Y_1]$. Apply Lemma 4.11 on $\mathcal{I} \times [-1, 1]$ to the shear $(a_0, -b_0)$ and the inviscid vector p_0 of this pair.

For each modified pair (U_j, E_j) , $j = N, f$, used below, define m_j by (4.15), set $\Pi_j = \Pi_0 + C_{p,j}$, and calculate the remaining coefficients from (4.7), (4.16), and (4.11). Write $p_j = p_{s,j}$ and $b_j = b_{s,j}$, and let V_j denote the radial-velocity profile of the pair.

Step 2: realize the periodic shear at a finite frequency. The periodic shear points in the required cone direction. To get that shear from actual profiles, integrate the difference between it and the original shear, then evaluate the resulting antiderivatives at a large radial frequency. Let $(a_L, -b_L)(X, \eta, \varphi)$ be the loop just constructed. Its mean identities give periodic antiderivatives $\mathcal{A}, \mathcal{B}$ with mean zero, satisfying

$$\partial_\varphi \mathcal{A} = -\frac{1}{2}(a_L - a_0), \quad \partial_\varphi \mathcal{B} = \frac{1}{2}E_0(b_L - b_0), \quad \int_0^1 \mathcal{A} d\varphi = \int_0^1 \mathcal{B} d\varphi = 0.$$

They are zero near both ends of $\mathcal{I}$, because the loop is constant there. Extend them by zero outside $\mathcal{I}$. For an integer $N \geq 1$, define

$$E_N = E_0 \exp\left(\frac{\mathcal{A}(X, \eta, N \log X)}{N}\right), \quad U_N = U_0 + \frac{\mathcal{B}(X, \eta, N \log X)}{N}. \quad (4.38)$$

In the shear identities below, $\mathcal{D}_X \mathcal{A}$ and $\mathcal{D}_X \mathcal{B}$ mean that X is differentiated while φ is held fixed. After taking those derivatives, evaluate every loop expression at $\varphi = N \log X$. The chain rule gives the exact identities

$$a_N = 1 - 2\mathcal{D}_X \log E_N = a_L - \frac{2\mathcal{D}_X \mathcal{A}}{N},$$

$$b_N = \frac{2\mathcal{D}_X U_N}{E_N} = e^{-\mathcal{A}/N} \left(b_L + \frac{2\mathcal{D}_X \mathcal{B}}{NE_0} \right).$$

Every radius is now fixed, and E_0 has a positive minimum on $\mathcal{J} \times [-1, 1]$. The phase does not depend on η . It follows that, for every fixed k , there are constants independent of N for which

$$\|U_N - U_0, E_N - E_0\|_k \leq A_k/N, \quad \|a_N - a_L, b_N - b_L\|_k \leq B_k/N. \quad (4.39)$$

Here, and through the end of Step 3, $\|\cdot\|_k$ means the componentwise maximum of the η -derivatives of orders $0 \leq j \leq k$ on $\mathcal{J} \times [-1, 1]$, or on the stated subinterval of $\mathcal{J}$. The second bound is taken on $\mathcal{I}$. The shear is unchanged on $\mathcal{J} \setminus \mathcal{I}$.

Write $u_N = U_N - U_0$, $e_N = E_N - E_0$, and $\Delta m_N = m_N - m_0$. Nothing changes below X_- . For $X \in \mathcal{J}$, the exact formulas for the five increments are

$$\Delta m_N(X, \eta) = \int_{X_-}^X \begin{pmatrix} u_N \\ \sqrt{2x} e_N \\ H_0 u_N + U_0 \sqrt{2x} e_N + \sqrt{2x} u_N e_N \\ 2U_0 u_N + u_N^2 - E_0 e_N - e_N^2/2 \\ E_0 e_N/x + e_N^2/(2x) \end{pmatrix} dx.$$

The interval is fixed and stays away from zero. The product rule and (4.39) therefore give

$$\|\Delta m_N\|_k \leq D_k/N, \quad \|\Pi_N - \Pi_0^{\text{prof}}\|_k = \|\Delta C_{p,N}\|_k \leq D_k/N.$$

Apply Lemma 4.4(ii) with the common value $\Pi_0(\eta)$ on the axis, and take N large enough that $E_N \geq \frac{1}{2} \min_{\mathcal{J} \times [-1,1]} E_0$. The lemma gives

$$\|p_N - p_0\|_k \leq C_k(A_{k+1} + D_{k+1})/N =: P_k/N. \quad (4.40)$$

So the large change in radial shear has only a small effect on the integrated vector p_s . This estimate does not require the radial derivatives of $E_N - E_0$ or $U_N - U_0$ to be small.

The change in p_s is now controlled, while the shear follows the chosen loop to order N^{-1} . The remaining job is to show that these errors fit inside the loop's strict margin from the cone boundary. Use the map Ψ from (4.35). With $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, let

$$\mu_L = \min_{\mathcal{I} \times [-1,1] \times \mathbb{T}} \min_i \Psi_i(a_L, b_L, p_0) > 0, \quad \mu_R = \min_{[X_+, Y_1] \times [-1,1]} \min_i \Psi_i(a_0, b_0, p_0) > 0.$$

Both minima are positive, by Lemma 4.11 and the way $\mathcal{I}$ was chosen. Take a compact neighborhood of the values of (a_L, b_L, p_0) and (a_0, b_0, p_0) on the two domains shown above, with $a > 0$ throughout that neighborhood. Let C_Ψ be an upper bound for the Lipschitz constant of Ψ there. Equations (4.39) and (4.40) show that, for every sufficiently large N , the perturbed triples lie in this neighborhood and obey the following componentwise bounds:

$$\begin{aligned} \Psi_i(a_N, b_N, p_N) &\geq \mu_L - C_\Psi(B_0 + P_0)/N \geq \mu_L/2 && \text{on } \mathcal{I}, \\ \Psi_i(a_N, b_N, p_N) &\geq \mu_R - C_\Psi P_0/N \geq \mu_R/2 && \text{on } [X_+, Y_1]. \end{aligned} \quad (4.41)$$

The positivity of these four quantities says exactly that $a_N > 0$, $v_N > 2$, and (4.22) hold. The modulated pair is therefore admissible everywhere on $\mathcal{J}$. Its five cumulative integrals still differ from the original ones by $O(N^{-1})$. The next step restores them exactly.

Step 3: solve the five moment equations. On $(Y_0, Y_1) \subset \mathcal{I}_1$, the modulation has not changed the profiles, and

$$U_N = U_0 = 0, \quad E_N = E_0 = K(\eta)X^{-1/2-\lambda}, \quad K(\eta) = c_{\text{patch}}(1 + \eta^2)^{-1} > 0.$$

Choose two smooth, nonnegative, nonzero functions β_1, β_2 whose compact supports in (Y_0, Y_1) are disjoint and ordered. Choose three functions $\gamma_1, \gamma_2, \gamma_3$ with the same properties. Set

$$u_c = \sum_{i=1}^2 \alpha_i(\eta) \beta_i(X), \quad e_c = \sum_{j=1}^3 \xi_j(\eta) \gamma_j(X), \quad c = (\alpha_1, \alpha_2, \xi_1, \xi_2, \xi_3).$$

For $U_f = U_N + u_c$ and $E_f = E_N + e_c$, the exact changes at Y_1 , listed in the order (M, J, I, S, C_p) , are

$$\begin{pmatrix} \int u_c dX \\ \int (H_0 u_c + \sqrt{2X} u_c e_c) dX \\ \int \sqrt{2X} e_c dX \\ \int (u_c^2 - E_0 e_c - e_c^2/2) dX \\ \int (E_0 e_c/X + e_c^2/(2X)) dX \end{pmatrix} = B(\eta)c + \mathcal{Q}_\eta(c, c) = d_N(\eta). \quad (4.42)$$

All five integrals are taken over (Y_0, Y_1) , and d_N is $-\Delta m_N(Y_1, \eta)$, with its entries put in the same row order. The linear map has two diagonal blocks:

$$B_U = \begin{pmatrix} \int \beta_1 & \int \beta_2 \\ \int H_0 \beta_1 & \int H_0 \beta_2 \end{pmatrix}, \quad B_E = \begin{pmatrix} \int \sqrt{2X} \gamma_1 & \int \sqrt{2X} \gamma_2 & \int \sqrt{2X} \gamma_3 \\ -\int E_0 \gamma_1 & -\int E_0 \gamma_2 & -\int E_0 \gamma_3 \\ \int E_0 \gamma_1/X & \int E_0 \gamma_2/X & \int E_0 \gamma_3/X \end{pmatrix}.$$

The weights in B_U have exponents $0, -\lambda$, while those in B_E have exponents $1/2, -1/2 - \lambda, -3/2 - \lambda$. Since $\lambda > 0$, all these exponents are distinct. Lemma 4.7 then shows that both blocks are invertible,

and their inverses depend smoothly on η . The inverse bounds are finite because λ , the radii, and all the bump functions were fixed before N .

Let ν_k be the norm of multiplication by B^{-1} on $C^k([-1, 1])$. As in Lemma 4.7, let q_k be the bilinear norm of $Q : C^k \times C^k \rightarrow C^k$. Since $\|d_N\|_{C^k} \leq D_k/N$, it is enough to choose

$$N \geq 8 \max_{k=0,2} \nu_k^2 q_k D_k$$

to get a smooth solution of (4.42) with $\|c\|_{C^2} \leq 2\nu_2 D_2/N$. Because the bump shapes are fixed, this solution also satisfies

$$\max_{r \leq 1} \|\partial_X^r(u_c, e_c)\|_2 \leq C/N.$$

When the five correction integrands in (4.42) are integrated only up to $X \in [Y_0, Y_1]$, the result is

$$\max_{0 \leq j \leq 2} \sup_{[Y_0, Y_1] \times [-1, 1]} \left| \partial_\eta^j(m_f - m_N) \right| \leq C/N.$$

The shear formulas and Lemma 4.4(ii) show that increasing N , if needed, gives

$$E_f \geq \frac{1}{2} \min_{[Y_0, Y_1] \times [-1, 1]} E_0, \quad \|a_f - a_N, b_f - b_N, p_f - p_N\|_0 \leq C_{\text{rep}}/N.$$

Here C_{rep} does not depend on N . Increase N until the final triples stay in the fixed compact neighborhood used to define C_Ψ . The inequality in (4.41) then stays strict on the repair interval. More exactly, if $N \geq 4C_\Psi C_{\text{rep}}/\mu_R$, then

$$\Psi_i(a_f, b_f, p_f) \geq \mu_R/2 - C_\Psi C_{\text{rep}}/N \geq \mu_R/4 > 0.$$

This correction changes no profile and no cumulative integral before Y_0 .

Equations (4.42) give $m_f(Y_1) = m_0(Y_1)$. All the changes to the profiles are supported in (X_-, Y_1) , so

$$\begin{aligned} (U_f, E_f) &= (U_0, E_0) && \text{for } X \leq X_- \text{ and } X \geq Y_1, \\ m_f(X) &= m_0(X) && \text{for } X \geq Y_1, \\ (\Pi_f, V_f, Q_{s,f}, N_{s,f}, p_f, \mathcal{T}_{0,f}) &= (\Pi_0^{\text{prof}}, V_0, Q_{s,0}, N_{s,0}, p_0, \mathcal{T}_{0,0}) && \text{for } X \geq Y_1. \end{aligned} \tag{4.43}$$

The last two equalities come from Lemma 4.4(i). Here V_f and V_0 are the radial-velocity profiles calculated from the final and joined pairs. Every lower bound on N used above is finite and depends only on data that has already been fixed. One finite integer can therefore meet all of them at once. From now on, write (U, E, Π) for the final profiles and $\mathcal{T}_0$ for their stress.

Step 4: verify the six conclusions. The final profiles satisfy the cone inequalities, and beyond the correction interval their five cumulative integrals agree exactly with those of the joined pair. These two facts now give the regularity, support, and endpoint estimates in the theorem. For the regularity in part (i), the changes in Steps 2–3 have compact support above X_{an} . They therefore leave the analytic axis rectangle from Proposition 4.10 unchanged. Every other profile piece and every join between pieces is smooth. Positivity of E follows from positivity of E_0 , the exponential in (4.38), and the bound for E_f on the repair interval. Near the axis, $E = \sqrt{2X}\phi/C$. On that neighborhood, $F = \phi/C$, $U, \Pi, V_0/X$ have the regularity stated in part (i). This gives finite bounds for every fixed mixed derivative on every finite rectangle.

The preserved pressure increment satisfies $C_p(\infty, \eta) = -\Pi_0(\eta)$. It follows that

$$\Pi(X, \eta) = \Pi_0(\eta) + C_p(X, \eta) = C_p(X, \eta) - C_p(\infty, \eta) = - \int_X^\infty \frac{E(x, \eta)^2}{2x} dx.$$

This proves both the pressure normalization in part (i) and the radial balance in part (ii). In particular, $\Pi_X = E^2/(2X) = F^2$ extends smoothly to the axis. Proposition 4.2 gives the exact

tangential residual identities. The first edge has not changed, so the stress is zero for $X \leq X_a$, and (4.13) holds there. By (4.43), the outer stress has not changed either, and it is zero for $X \geq X_b$.

The cone condition holds when $X_a < X < X_b$. It was kept near both edges and beyond Y_1 , and Steps 2–3 proved it on $[X_-, Y_1]$. In particular, (4.23) gives

$$\mathcal{T}_{0,\theta} + t_s \mathcal{T}_{0,z} = F(P_c - v_s) > 0.$$

The stress is therefore nonzero at every interior point, which finishes part (ii).

Next come the uniform claims in part (iii). Set $y_a = \log(X/X_a)$, $y_b = \log(X_b/X)$. The factorizations (4.33) and (4.32) extend $n = \mathcal{T}_0/|\mathcal{T}_0|$ to the edges as follows:

$$n = \frac{B_a}{|B_a|} \quad \text{near } X_a, \quad n = \frac{(b_\theta, y_b^6 b_z)}{\sqrt{b_\theta^2 + y_b^{12} b_z^2}} \quad \text{near } X_b.$$

Their denominators stay positive on smaller collars. Since $B_a(0, \eta) = F(X_a, \eta)\mathfrak{s}(X_a, \eta)$,

$$n(X_a, \eta) = \frac{(1, t_s)}{\sqrt{1 + t_s^2}}, \quad n(X_b, \eta) = (1, 0), \quad t_s(X_b, \eta) = 0.$$

The smooth functions $F, a, v_s - 2$ are positive in the interior. The endpoint bounds from Lemma 4.9 and Proposition 4.10 make them positive at X_a, X_b as well. Each one therefore has a positive minimum on the closed annulus.

On that rectangle, define

$$A_n = n_\theta + t_s n_z, \quad B_n = n_z - t_s n_\theta.$$

Inside the annulus, (4.23) gives

$$A_n = \frac{P_c - v_s}{|p_s - \mathfrak{s}|} > 0.$$

At both edges, $B_n = 0$. Also, $A_n = \sqrt{1 + t_s^2}$ at X_a , while $A_n = 1$ at X_b . It follows that $A_n > 0$ everywhere on the closed rectangle, so define

$$G_n = 2 - (v_s - 2)B_n^2/A_n^2.$$

Equation (4.22) gives

$$G_n = 2 - (v_s - 2)J_c^2/(P_c - v_s)^2 > 0$$

in the interior, while $G_n = 2$ at both edges. The functions A_n, G_n are therefore smooth and positive everywhere. Taking

$$\kappa = \min \left\{ 1, \min_{[X_a, X_b] \times [-1, 1]} A_n, \min_{[X_a, X_b] \times [-1, 1]} G_n \right\} > 0$$

gives (4.26), with $\kappa < 2$. This proves part (iii), including the directions at the endpoints.

For part (iv), let $c_a = t_1^2 > 0$ be the inner exponential constant from (4.33), and put

$$\zeta(X) = \exp(-c_a/y_a^2 - 4/y_b^2) \quad (X_a < X < X_b), \quad \zeta = 0 \quad \text{otherwise.}$$

This function is smooth, and it is flat at both edges. On a fixed inner collar, the ratio $\zeta/e^{-c_a/y_a^2}$ stays between two positive constants. On a fixed outer collar, the same is true of $\zeta/e^{-4/y_b^2}$. The smooth, nonzero factors in (4.33) and (4.32) then give

$$\begin{array}{lll} |\mathcal{T}_0| \geq c\zeta, & |\partial^\alpha \mathcal{T}_0| \leq C_\alpha \zeta y_a^{-m_\alpha} & \text{on the inner collar,} \\ |\mathcal{T}_0| \geq c\zeta, & |\partial^\alpha \mathcal{T}_0| \leq C_\alpha \zeta y_b^{-m_\alpha} & \text{on the outer collar.} \end{array}$$

Each derivative of an exponential factor contributes only finitely many inverse powers of y_a or y_b . On the compact interval between the collars, $\zeta > 0$, $|\mathcal{T}_0| > 0$, and every fixed derivative is bounded. With $\delta = \min\{1, y_a, y_b\}$, these bounds combine to give (4.27), which proves part (iv).

Finally, both exact connections leave the total moment identities in part (v) unchanged. For the last correction, this follows directly from

$$(M, J, S)_f(\infty) = (M, J, S)_0(\infty) = (0, 0, 0),$$

$$\int_0^\infty (H_f - H_0) dX = I_f(Y_1) - I_0(Y_1) = 0.$$

Together with Step 1, these equalities give (4.28). The exterior pair is still the heat profile from Lemma 4.8, with the same c_∞ . The identity

$$q^{-A} c_\infty X^{-A} \mathcal{H}(2d/X) = c_\infty s^{-A} \mathcal{H}(2\tau/s)$$

proves the formula for the physical exterior. Since $Y_1 < \sup \mathcal{I}_4 < X_v$, equation (4.43) keeps $U = M = J = 0$ for $X \geq X_v$, with the same X_v as in Lemma 4.8. Equation (4.7) then gives

$$V_0 = \frac{2\eta XU - 2D\eta M - dM_\eta}{L} = 0.$$

The radius X_v comes before the last collar and after all four reserved intervals. This proves part (v). For part (vi), the heat compensation used $\mathcal{I}_2$, while Step 3 used only $\mathcal{I}_1$. The intervals $\mathcal{I}_{\text{pos}} = \mathcal{I}_3$ and $\mathcal{I}_{\text{mean}} = \mathcal{I}_4$ therefore keep the exact profile (4.30), proving part (vi). Every choice was made in profile coordinates before the physical q or any later correction stage was introduced, as required. $\square$

The profiles now produce the leading field from Section 3, with its stress supported in the prescribed annulus. The next section corrects the higher-order residual terms found after (4.7).

5 Correcting the base flow to every order

Near the axis, the leading profiles (E, U, V_0, Π) from Theorem 4.6 satisfy the balances in Equations (4.7) and (4.13). Axial viscosity and the remaining radial terms still appear in the residual, however. Add axisymmetric corrections to obtain the background (u_B, p_B) described in Section 3 and Proposition 5.5. The velocity u_B is exactly divergence-free. Its momentum residual is the negative cylindrical divergence of the physical annular stress $\mathcal{T}_{\text{phys}}$, as in (5.41), plus an error $\mathcal{E}_B$ that is bounded by every power of q as $q \downarrow 0$. The same bounds hold for every fixed physical derivative on each compact range of profile coordinates. The annular stress becomes the input to the wave construction in Section 7.

These remaining terms appear at successive powers of q^{2h} in Equations (5.3)–(5.6). At each order, Lemma 5.1 solves for a correction near the axis. Lemma 5.2 then extends that correction in the radial direction and imposes the five integral conditions that keep its stress supported in the prescribed annulus. The induction in Proposition 5.3 finishes the current order before using its coefficients to build the next one.

The leading profiles and all constants used to construct them stay fixed throughout. The resulting sequence of coefficients is the formal expansion (5.1); this does not assert that the unmodified infinite series converges. After this sequence is built, Lemma 5.4 produces smooth fields for $q > 0$ with the same asymptotic coefficients. The proof of Proposition 5.5 checks that this summation keeps incompressibility, the exterior velocity, and the required bounds on the residual.

5.1 Higher coefficients near the axis.

First solve the coefficient equations (5.2)–(5.6) near the axis, where the stress must remain zero. For $n \geq 0$, set $\lambda_n = 2nh$, $E_0 = E$, $U_0 = U$, and $\Pi_0 = \Pi$. Every coefficient profile is a function of (X, η) . Here the subscript zero now marks the order in the expansion. In particular, $\Pi_0(X, \eta)$ is the whole leading pressure profile, and its trace

$\Pi_0(0, \eta)$ is the axis pressure function defined in (4.31). Use the similarity variables from (4.1), and write the physical coefficients as

$$\begin{aligned} u_{\theta,n} &= q^{-A+\lambda_n} E_n = r q^{-A-1/2+\lambda_n} \phi_n / C, & E_n &= \sqrt{2X} \phi_n / C, \\ u_{z,n} &= q^{-A+\lambda_n} U_n, & r u_{r,n} &= q^{\lambda_n} V_n, & p_n &= q^{-2A+\lambda_n} \Pi_n. \end{aligned} \quad (5.1)$$

Here ϕ_n , rather than E_n , is the azimuthal coefficient that is smooth at $X = 0$. The radial average is the operator from (4.6), applied with η held fixed: $\mathcal{A}_X(U_n)(X, \eta) = X^{-1} \int_0^X U_n(x, \eta) dx$.

The gap $2h$ between successive powers in (5.1) accounts for the two lower-order effects identified after (4.7). Recall from (4.1) that $A = \frac{1}{2} + h$ and $D = \frac{1}{2} - h$, so $1 - 2D = 2A - 1 = 2h$. For the first effect, the axial derivative identity in (4.2) contributes the factor $q^{-2D} = q^{-1} q^{2h}$ when two axial derivatives are taken. The radial relation $X = r^2/(2q)$ in (4.1) contributes q^{-1} when two radial derivatives are taken. This is why axial viscosity applied to order $n - 1$ appears in the angular and axial equations at order n . These are the order- $n - 1$ viscous terms in Equations (5.3) and (5.4) below.

For the second effect, recall that the leading radial balance $r \partial_r p^{(0)} = (u_\theta^{(0)})^2$ is the balance behind the pressure identity in (4.7). The physical powers in (4.3) give both sides the power q^{-2A} . After the other terms in the radial momentum equation are multiplied by r , they begin at $q^{-1} = q^{-2A} q^{2h}$, as noted after (4.7). This accounts for the shift by one order in the pressure equation (5.5). The remaining radial terms are collected in (5.6) and enter one order later.

To include derivatives of the powers of q , recall the operators T_a, Z_a from (4.2), and abbreviate

$$T_{a,n} = T_{a+\lambda_n}, \quad Z_{a,n} = Z_{a+\lambda_n}, \quad Z_{a,n}^{[2]} = Z_{a+\lambda_n-D} Z_{a+\lambda_n}.$$

The order of composition in the last expression keeps track of the change in the power after the first axial derivative. Any field with a negative index is taken to be zero. With the integration constant fixed by regularity at the axis, incompressibility is exactly the pair of identities

$$\partial_X V_n = -Z_{-A,n} U_n, \quad \frac{V_n}{X} = \frac{2\eta U_n - 2\eta(D + \lambda_n) \mathcal{A}_X(U_n) - d \partial_\eta \mathcal{A}_X(U_n)}{L}. \quad (5.2)$$

Fix $n \geq 1$. At this point every coefficient of order $j < n$ is known, including its radial cutoffs and the integral corrections from Lemma 5.2. On one common inner interval, solve for ϕ_n, U_n, Π_n . Equation (5.2) then determines V_n from U_n . The pressure coefficient Π_n has to be solved at the same time as ϕ_n, U_n .

Set $b = -A - \frac{1}{2}$ and $c = -A$. In the system below, primes mean ∂_X , with η held fixed. The following equations cancel the momentum residual of order n on the inner interval:

$$2(X\phi_n'' + 2\phi_n') = T_{b,n}\phi_n + \sum_{i+j=n} \left\{ V_i(\phi_j' + \phi_j/X) + U_i Z_{b,j} \phi_j \right\} - Z_{b,n-1}^{[2]} \phi_{n-1}, \quad (5.3)$$

$$2(XU_n'' + U_n') = T_{c,n}U_n + \sum_{i+j=n} \left\{ V_i U_j' + U_i Z_{c,j} U_j \right\} + Z_{-2A,n} \Pi_n - Z_{c,n-1}^{[2]} U_{n-1}, \quad (5.4)$$

$$\Pi_n' = C^{-2} \sum_{i+j=n} \phi_i \phi_j - \frac{\Omega_{n-1}}{2X}, \quad (5.5)$$

$$\Omega_k = T_{0,k} V_k + \sum_{i+j=k} \left\{ V_i(V_j' - V_j/(2X)) + U_i Z_{0,j} V_j \right\} - 2X V_k'' - Z_{0,k-1}^{[2]} V_{k-1}. \quad (5.6)$$

Every Ω_{n-1}, ϕ_{n-1} , or U_{n-1} that appears on the right-hand sides uses coefficients already constructed. For example, the pressure equation can be written as

$$\Pi'_n = 2C^{-2}\phi_0\phi_n + C^{-2} \sum_{i=1}^{n-1} \phi_i\phi_{n-i} - \frac{\Omega_{n-1}}{2X}.$$

When $n = 1$, the sum is empty. Every term in that sum and Ω_{n-1} is already known, while the first term is linear in the current unknown. The same point applies to the other equations after their transport sums are split into the cases $(i, j) = (0, n), (n, 0)$ and $1 \leq i, j < n$. The system is therefore linear at every positive order. The pressure equation will also be enforced when the current profiles are extended beyond the inner interval.

Here is how the powers of q and the terms from the cylindrical basis arise. For $rq^{b+\lambda_n}\phi_n/C$, angular radial transport contains $u_r(\partial_r + r^{-1})u_\theta$, which gives $V_i(\partial_X\phi_j + \phi_j/X)$. The angular vector Laplacian gives $2(X\partial_{XX}\phi_n + 2\partial_X\phi_n)$ after the factor $rq^{b-1+\lambda_n}/C$ is removed. The corresponding axial scalar Laplacian is $2(X\partial_{XX}U_n + \partial_XU_n)$. Every transport product has the same power because $A + D = 1$. Axial viscosity moves one step forward in the q^{2h} expansion because $1 - 2D = 2h$. In the radial equation, multiplying by r puts pressure and centrifugal acceleration at the power $q^{-2A+\lambda_n}$. The other radial terms begin at $q^{-1+\lambda_k}$, so $k = n - 1$. Finally, applying the radial operator $\partial_{rr} + r^{-1}\partial_r - r^{-2}$ to $V(X)/r$ gives $2X\partial_{XX}V/(qr)$, which gives the last two terms in (5.6).

Equation (5.2) gives $V_j = Xv_j$ with a smooth function v_j . It follows that every term in Ω_k is divisible by X . For example, $V_i(\partial_XV_j - V_j/(2X)) = Xv_i(v_j/2 + X\partial_Xv_j)$. Thus Ω_k/X is regular at the axis, including after any fixed number of parameter derivatives.

The linear equations must be solvable on the same radial interval at every order. For that to work, the recursion needs bounds on its η -derivatives even as the coefficient bounds grow. The analytic dependence of the leading inner profile gives those bounds.

Lemma 5.1. *There is an interval $0 \leq \xi \leq a$, where $\xi = \sqrt{X}$, that starts at the axis and extends into the inner collar from Theorem 4.6(i). The stress cone inequalities hold there with one uniform positive margin. On this interval, every positive order has exactly one solution of Equations (5.2)–(5.6) with $\phi_n(0, \eta) = U_n(0, \eta) = \Pi_n(0, \eta) = 0$. The profiles are smooth in X . For each fixed radial derivative, they are holomorphic on a neighborhood of $[-1, 1]$ in the parameter η , which means that they have a complex derivative at every point of that neighborhood. The neighborhood and its bounds may depend on the order and the radial derivative, but a does not. The uniqueness statement applies among solutions whose profiles extend holomorphically in η to a neighborhood of $[-1, 1]$, with bounds uniform for $0 \leq \xi \leq a$.*

Because every positive-order coefficient is zero at the axis, the leading traces of ϕ, U , and Π stay unchanged.

Proof. Step 1: write a linear system on the fixed inner interval. The analytic axis solution and its first collar, supplied by Proposition B.2 and Theorem 4.6, have the stated parameter regularity on one fixed radial rectangle. Choose a inside this rectangle and beyond the shorter collar that will remain uncut. At order n , use the lower-order coefficients already constructed, including their cutoffs and moment corrections. The corrections are supported to the right of this rectangle, and the radial cutoffs keep their analytic dependence on the parameter. A small enough complex neighborhood $S_\rho = \{\eta \in \mathbb{C} : \text{dist}(\eta, [-1, 1]) < \rho\}$ therefore contains no zero of L and gives bounded holomorphic coefficients and source terms everywhere on $0 \leq \xi \leq a$.

Introduce the extra variable K_n and the vector W_n so that the system can be written as

$$\begin{aligned} K_n &= \mathcal{A}_X(U_n) - U_n, & W_n &= (\phi_n, U_n, K_n, \Pi_n, \partial_\xi\phi_n, \partial_\xi U_n)^\top, \\ \partial_\xi W_n + \xi^{-1} \text{diag}(0, 0, 2, 0, 3, 1)W_n &= A_0W_n + A_1\partial_\eta W_n + f_n, & W_n(0) &= 0. \end{aligned} \quad (5.7)$$

The matrices A_0, A_1 are 6×6 matrices of coefficients in (ξ, η) , with n fixed. The known source f_n has six components and is built from the lower-order profiles. The extra unknown K_n replaces the radial average with a first-order equation, so no radial integral of a current unknown is left unevaluated in the system. In fact, $\partial_{\xi} K_n + 2K_n/\xi = -\partial_{\xi} U_n$, and the two second-order radial operators become $(\partial_{\xi\xi} + 3\xi^{-1}\partial_{\xi})/2$ and $(\partial_{\xi\xi} + \xi^{-1}\partial_{\xi})/2$. At every positive order, every term that contains an unknown of order n is linear. The pressure row is $\partial_{\xi}\Pi_n = 4\xi\phi_0\phi_n/C^2 + \text{known terms}$. Use the same row for $X\partial_X\Pi_n$ in the axial equation. With this form, every apparent division by X is regular.

Step 2: solve without shrinking the radial interval. Iterating the system takes derivatives in η . First control how many of these derivatives can appear, and only then estimate the repeated radial integrals. The coefficient of ∂_{η} has the following sparse block form. Set $H_c = D\eta + dU_0$. Its only entries that may be nonzero are

$$\begin{aligned} (A_1)_{51} &= 2H_c/L, & (A_1)_{52} &= (A_1)_{53} = -2d(\phi_0 + X\partial_X\phi_0)/L, \\ (A_1)_{62} &= 2(H_c - dX\partial_X U_0)/L, & (A_1)_{63} &= -2dX\partial_X U_0/L, & (A_1)_{64} &= 2d/L. \end{aligned}$$

Thus A_1 , at every radius and after any number of parameter derivatives, maps the first four coordinates into the last two and sends the last two coordinates to zero.

For $c_i = (0, 0, 2, 0, 3, 1)_i$, invert the diagonal part of (5.7), with zero data at the axis, by using the integral operator

$$(Gg)_i(\xi) = \int_0^{\xi} (s/\xi)^{c_i} g_i(s) ds, \quad \mathcal{K} = G(A_0 + A_1\partial_{\eta}).$$

The kernels of this operator do not depend on η , and they keep the two coordinate subspaces just described separate. If D_0 is any diagonal kernel that occurs between two factors, then

$$A_1(\xi)D_0A_1(s) = 0, \quad A_1(\xi)D_0\partial_{\eta}A_1(s) = 0.$$

So two neighboring derivative operators give zero even when the derivative on the left lands on a variable matrix coefficient. Any nonzero product of k factors can therefore contain at most $p_k = \lceil k/2 \rceil$ parameter derivatives.

Bound A_0, A_1, f_n on the larger neighborhood S_{ρ} , uniformly across the whole radial interval. On $S_{\rho'}$, where $0 < \rho' < \rho$, Cauchy's estimate bounds each parameter derivative by using the distance from the smaller complex neighborhood to the edge of the larger one. The available loss in radius is $\Delta = \rho - \rho'$, and it is divided among the at most p_k derivatives. The diagonal kernels have norm at most one. The nested radial integrals are ordered so that every new integration variable is no larger than the preceding one. Their region is an ordered simplex, whose volume is $a^{k+1}/(k+1)!$. Adding the at most 2^k compositions gives

$$\|\mathcal{K}^k Gf_n\|_{S_{\rho'}} \leq \frac{C_n^{k+1} a^{k+1}}{(k+1)!} (\max\{1, p_k/\Delta\})^{p_k}. \quad (5.8)$$

The k th root of the right-hand side of (5.8) tends to zero. The Picard series $W_n = \sum_{k \geq 0} \mathcal{K}^k Gf_n$ therefore converges on $S_{\rho'}$ for every finite a , no matter how large C_n is. Apply the same bound after iterating the homogeneous equation for the difference of two solutions; this proves uniqueness. In particular, the radial interval does not depend on the size of the coefficient bounds at later orders.

Step 3: recover smooth profiles at the axis. Radial smoothness follows without differentiating a singular expression:

$$(Gg)_i = \xi \int_0^1 t^{c_i} g_i(t\xi) dt, \quad \partial_{\xi}(Gg)_i = g_i(\xi) - c_i \int_0^1 t^{c_i} g_i(t\xi) dt.$$

The second identity is continuous at zero. Repeating this argument, and using a smaller neighborhood of the parameter when needed, gives every fixed radial derivative. Extend the integral

equation across $\zeta = 0$. The original equations keep the first four coordinates even and the last two odd, and uniqueness means that these are exactly the parities of the solution. For an even smooth function of ζ , Taylor's formula shows at every finite order that the function is smooth in $X = \zeta^2$ on the half-interval. This also gives the regularity needed to form the next source Ω_n / X . $\square$

5.2 Making the total radial residual integrals zero

The inner profiles $(\phi_n, U_n, V_n, \Pi_n)$ from Lemma 5.1 are not yet ready to be used as coefficients at the next order. First, extend E_n, U_n to every $X \geq 0$. Then recompute V_n, Π_n from Equations (5.2) and (5.5), and control the residual that the extension creates. Cutting off only E_n, U_n in the radial direction does not make the fields found by radial integration compactly supported. In particular, the pressure can be left with a nonzero constant outside the support, and the stress $\mathcal{T}_n$ defined below in (5.9) can be left with a nonzero tail there. To remove these terms, impose five moment conditions, which here means setting the five integrals in Equations (5.10) and (5.11) equal to zero. Lemma 5.2 proves that these conditions do remove the terms.

Here is how the residual coefficients are collected before the moment conditions are imposed. Use $R = \sqrt{2X}$. To find the part of order n , substitute (5.1) into the differential polynomial $\mathcal{R}$, take the physical derivatives, divide out the common tangential power q^{-A-1} , and keep the coefficient of q^{2nh} . This calculation is finite. Products use the pairs $i + j = n$, while axial viscosity uses order $n - 1$. More explicitly,

$$\begin{aligned} \mathfrak{r}_{\theta,n} &= \frac{R}{C} \left[T_{b,n} \phi_n + \sum_{i+j=n} \left\{ V_i \left(\phi'_j + \frac{\phi_j}{X} \right) + U_i Z_{b,j} \phi_j \right\} - 2 (X \phi_n'' + 2 \phi_n') - Z_{b,n-1}^{[2]} \phi_{n-1} \right], \\ \mathfrak{r}_{z,n} &= T_{c,n} U_n + \sum_{i+j=n} \left\{ V_i U'_j + U_i Z_{c,j} U_j \right\} + Z_{-2A,n} \Pi_n - 2 (X U_n'' + U_n') - Z_{c,n-1}^{[2]} U_{n-1}. \end{aligned}$$

These expressions are zero wherever the inner equations have been solved. Define the two needed stress components by integrating the negatives of the residuals with the indicated radial weights.

$$\mathcal{T}_{n,\theta}(R) = -R^{-2} \int_0^R q^2 \mathfrak{r}_{\theta,n}(q) dq, \quad \mathcal{T}_{n,z}(R) = -R^{-1} \int_0^R q \mathfrak{r}_{z,n}(q) dq. \quad (5.9)$$

Every profile inside these integrals uses the same value of η . The two physical stress components are $q^{-A-1/2+\lambda_n} \mathcal{T}_n$. For each forward integral to become zero after $R^2 \mathfrak{r}_{\theta,n}$ or $R \mathfrak{r}_{z,n}$ leaves the support of its source, the corresponding total integral must be zero. The five conditions below force both total integrals to be zero through the conservative momentum equations. They also remove the radial-velocity and pressure terms left by integration. Also write $F_n = \int_0^X U_n(x, \eta) dx = X \mathcal{A}_X(U_n)$ for the Stokes streamfunction coefficient. This indexed notation is different from the leading azimuthal profile $F = E / \sqrt{2X}$.

For an extension at order n that has not yet been corrected, call the five total moments

$$m_{n,1} = \int_0^\infty R U_n dR, \quad m_{n,2} = \int_0^\infty R^2 E_n dR, \quad m_{n,3} = \int_0^\infty \partial_R \Pi_n dR. \quad (5.10)$$

$$\begin{aligned} m_{n,4} &= \int_0^\infty R^2 \sum_{i+j=n} U_i E_j dR, \\ m_{n,5} &= \int_0^\infty \left(R \sum_{i+j=n} U_i U_j - \frac{1}{2} R^2 \partial_R \Pi_n \right) dR. \end{aligned} \quad (5.11)$$

These five moments depend on η , with every lower-order factor kept fixed. Write

$$\mathbf{m}_n = (m_{n,1}, \dots, m_{n,5}), \quad \mathbf{P}_n = \left(\phi_n, U_n, \frac{V_n}{X}, \Pi_n, \frac{F_n}{X} \right).$$

Every radial support used below is uniform for $|\eta| \leq 1$. The stress estimates use the flat weight ζ from Item (iv) of Theorem 4.6, together with

$$\delta(X) = \min\{1, \log(X/X_a), \log(X_b/X)\} \quad (X_a < X < X_b).$$

Lemma 5.2. *There are radii $X_a < X_- < X_+ < X_b$, chosen independently of n , for the following induction. Fix $n \geq 1$. For every $1 \leq j < n$, suppose that $\mathbf{P}_j$ is smooth, satisfies Equations (5.2) and (5.5) everywhere, and obeys (5.12) with $n = j$. Suppose as well that these profiles agree with their inner solutions on $[0, X_-]$ and keep the parameter regularity from Lemma 5.1 throughout $[0, a^2]$. When $n = 1$, there are no lower positive-order hypotheses.*

Then the inner order- n solution from Lemma 5.1 extends smoothly to every $X \geq 0$ and $|\eta| \leq 1$. The extension agrees with the inner solution on $[0, X_-]$, keeps its parameter regularity throughout $[0, a^2]$, satisfies Equations (5.2) and (5.5) everywhere, and obeys

$$\mathbf{m}_n(\eta) = 0 \quad (|\eta| \leq 1), \quad \text{supp}_X \mathbf{P}_n \subset [0, X_+]. \quad (5.12)$$

Both V_n/X and F_n/X extend smoothly to $X = 0$. The stress defined in (5.9) satisfies

$$\text{supp}_X \mathcal{T}_1 \subset [X_-, X_b], \quad \text{supp}_X \mathcal{T}_n \subset [X_-, X_+] \quad (n \geq 2),$$

and every fixed profile derivative ∂^I satisfies

$$|\partial^I \mathcal{T}_n(X, \eta)| \leq C_{n,I} \zeta(X) \delta(X)^{-N_{n,I}} \quad (X_a < X < X_b, |\eta| \leq 1). \quad (5.13)$$

The constants $C_{n,I}, N_{n,I}$ may depend on the order and the derivative. When $n \geq 2$, it is possible to take $N_{n,I} = 0$.

Proof. Step 1. Extend the current profiles without changing the inner solution. Keep the inner solution unchanged on $[0, X_-]$, then choose the rest of the extension so that $\mathbf{m}_n(\eta) = 0$ for every $|\eta| \leq 1$. Setting $m_{n,1} = m_{n,3} = 0$ removes the streamfunction and pressure terms outside the support. Setting $m_{n,1} = m_{n,2} = 0$ cancels the integrated time and viscosity terms. Setting $m_{n,4} = m_{n,5} = 0$ cancels the integrated momentum fluxes, including the pressure term. Steps 3 and 4 show why each of these statements follows.

The inner profiles $(\phi_n^{\text{in}}, U_n^{\text{in}}, V_n^{\text{in}}, \Pi_n^{\text{in}})$ are defined for $0 \leq X \leq a^2$ by Lemma 5.1. Let $\mathcal{I}_{\text{pos}}$ be the reserved positive-order patch from Item (vi) of Theorem 4.6. Choose X_- inside the theorem's inner collar, where the stress cone inequalities have a positive margin that is uniform over the collar. Then fix, independently of n ,

$$X_- < X_{\text{keep}} < X_{\text{cut}} < a^2 < \inf \mathcal{I}_{\text{pos}}, \quad \kappa \in C^\infty([0, \infty)), \quad 0 \leq \kappa \leq 1,$$

$$\kappa(X) = 1 \quad (0 \leq X \leq X_{\text{keep}}), \quad \kappa(X) = 0 \quad (X \geq X_{\text{cut}}).$$

Start with the extensions

$$\tilde{\phi}_n = \kappa \phi_n^{\text{in}}, \quad \tilde{U}_n = \kappa U_n^{\text{in}}, \quad \tilde{E}_n = \frac{R}{C} \tilde{\phi}_n, \quad R = \sqrt{2X},$$

and extend the products by zero past $X = a^2$.

In the R coordinate, the reserved patch $\mathcal{I}_{\text{pos}}$ becomes $\mathcal{J}_{\text{pos}} = \{R > 0 : R^2/2 \in \mathcal{I}_{\text{pos}}\}$. Choose the following nonnegative smooth bump functions. Their supports are mutually disjoint, appear in a fixed order, and stay the same at every order.

$$b_1^U, b_2^U, b_1^E, b_2^E, b_3^E \in C_c^\infty(\mathcal{J}_{\text{pos}}), \quad \int_0^\infty b_j^U dR = 1 \quad (j = 1, 2), \quad \int_0^\infty b_j^E dR = 1 \quad (j = 1, 2, 3),$$

Take five unknown scalar functions $\alpha_{n,j}(\eta), \beta_{n,j}(\eta)$ and set

$$\begin{aligned} U_n(X, \eta) &= \tilde{U}_n(X, \eta) + \sum_{j=1}^2 \alpha_{n,j}(\eta) b_j^U(R), \\ E_n(X, \eta) &= \tilde{E}_n(X, \eta) + \sum_{j=1}^3 \beta_{n,j}(\eta) b_j^E(R), \quad \phi_n = \frac{C}{R} E_n. \end{aligned} \quad (5.14)$$

At $R = 0$, use the smooth extension of the quotient that defines ϕ_n . The bump functions vanish near the axis, where $\phi_n = \tilde{\phi}_n$.

For any choice of these five coefficient functions, define the remaining profiles by

$$\begin{aligned} F_n(X, \eta) &= \int_0^X U_n(x, \eta) dx, \\ V_n(X, \eta) &= - \int_0^X (Z_{-A,n} U_n)(x, \eta) dx, \\ \Pi_n(X, \eta) &= \int_0^X \left[C^{-2} \sum_{i+j=n} \phi_i \phi_j - \frac{\Omega_{n-1}}{2x} \right] (x, \eta) dx. \end{aligned} \quad (5.15)$$

Every lower-order profile and Ω_{n-1} stays fixed in these formulas. The quotient Ω_{n-1}/x is smooth at the axis, as shown after (5.6). Equations (5.2) and (5.5) hold everywhere by construction, because the three profiles in (5.15) were defined by integrating those equations from the axis.

On $[0, X_-]$, the cutoff is one and every bump is zero. Integrating forward from the zero values at the axis therefore gives

$$(\phi_n, U_n, V_n, \Pi_n) = (\phi_n^{\text{in}}, U_n^{\text{in}}, V_n^{\text{in}}, \Pi_n^{\text{in}}).$$

The radial cutoff does not depend on η , and the bump functions vanish throughout $[0, a^2]$. The reconstructed profiles therefore keep the parameter regularity from Lemma 5.1 on that whole interval, which is what the next order needs.

Step 2. Solve the five moment equations. On $\mathcal{I}_{\text{pos}}$,

$$U_0 = 0, \quad E_0 = e_* f(\eta) R^{-1-2\lambda}, \quad e_* > 0, \quad \lambda > 0,$$

where f is smooth and stays a positive distance away from zero. Once the lower-order profiles have been fixed, each of the five moments is an affine function of the five unknown coefficients. For the fifth moment, the pressure equation gives

$$m_{n,5} = \int_0^\infty \left(\sum_{i+j=n} U_i U_j - \frac{1}{2} \sum_{i+j=n} E_i E_j + \frac{1}{2} \Omega_{n-1} \right) dX.$$

The term Ω_{n-1} depends only on lower orders, so none of the five coefficients changes it. Using $R^2 = 2X$, the pressure equation also gives

$$\partial_R \Pi_n = \frac{2E_0 E_n}{R} + \frac{1}{R} \left(\sum_{i=1}^{n-1} E_i E_{n-i} - \Omega_{n-1} \right).$$

Define the constant moment matrices

$$\begin{aligned} (B_U)_{ij} &= \int_0^\infty R^{p_i} b_j^U(R) dR, \quad (p_1, p_2) = (1, 1 - 2\lambda), \\ (B_E)_{ij} &= \int_0^\infty R^{s_i} b_j^E(R) dR, \quad (s_1, s_2, s_3) = (2, -2 - 2\lambda, -2\lambda). \end{aligned}$$

The matrix B_U is 2×2 , and B_E is 3×3 . Let $m_{n,j}^0$ be the moments obtained by setting all five coefficients in (5.14) equal to zero and then rebuilding the profiles with (5.15). Put

$$\mathbf{d}_{U,n} = \begin{pmatrix} m_{n,1}^0 \\ m_{n,4}^0 / (e_* f) \end{pmatrix}, \quad \mathbf{d}_{E,n} = \begin{pmatrix} m_{n,2}^0 \\ m_{n,3}^0 / (2e_* f) \\ -m_{n,5}^0 / (e_* f) \end{pmatrix}.$$

With $\alpha_n = (\alpha_{n,1}, \alpha_{n,2})^\top$ and $\beta_n = (\beta_{n,1}, \beta_{n,2}, \beta_{n,3})^\top$, the five moment conditions say exactly that

$$\mathbf{m}_n = 0 \iff B_U \alpha_n = -\mathbf{d}_{U,n}, \quad B_E \beta_n = -\mathbf{d}_{E,n}.$$

The exponents in each block are different, and the bump supports are ordered, so Lemma A.1 applies. Both matrices are therefore invertible, and the needed coefficients are

$$\alpha_n = -B_U^{-1} \mathbf{d}_{U,n}, \quad \beta_n = -B_E^{-1} \mathbf{d}_{E,n}. \quad (5.16)$$

The discrepancies depend smoothly on η , and every fixed derivative of $1/f$ is bounded on $[-1, 1]$. The coefficients are therefore smooth whenever the discrepancies have finite size, however large that size is. This proves $\mathbf{m}_n = 0$. The support statement in (5.12) still has to be proved.

Step 3. Make the velocity and pressure zero outside the correction region. Choose X_+ after the leading axial-velocity perturbation U_0 and every positive-order patch, but before the terminal outer collar. The first moment then gives

$$F_n(X, \eta) = \int_0^\infty U_n(x, \eta) dx = m_{n,1}(\eta) = 0 \quad (X \geq X_+).$$

So $\mathcal{A}_X(U_n) = V_n = 0$ there at every positive order. At order zero, the leading axial-flux moment correction also gives $U_0 = V_0 = 0$ there. Every term in every Ω_k contains either a V factor or a derivative of a V field, so each such term is zero on the same outside region.

At positive order, every product $\phi_i \phi_j$ with $i + j = n$ contains a positive-order factor. It follows that $\partial_X \Pi_n = 0$ after X_+ . The third moment now gives

$$\Pi_n(X, \eta) = \Pi_n(0, \eta) + \int_0^\infty \partial_R \Pi_n(R, \eta) dR = m_{n,3}(\eta) = 0 \quad (X \geq X_+).$$

Thus $\phi_n, U_n, V_n, \Pi_n, F_n$ are all zero for $X \geq X_+$. They are smooth because the cutoff construction is smooth, the solve in (5.16) is finite, and the remaining profiles come from the integrals in (5.15). At the axis,

$$\frac{F_n(X, \eta)}{X} = \int_0^1 U_n(tX, \eta) dt$$

is smooth, and (5.2) gives the same result for V_n/X . Since $[0, X_+] \times [-1, 1]$ is compact, the coefficient bounds follow.

$$\max_{k+\ell \leq m} \sup_{X \geq 0, |\eta| \leq 1} |\partial_X^k \partial_\eta^\ell \mathbf{P}_n(X, \eta)| \leq C_{n,m} < \infty \quad (n \geq 1, m \geq 0). \quad (5.17)$$

The support radius is the same at every order, although these constants can grow with n and m .

On the reserved interval $\mathcal{I}_{\text{mean}}$, which will later hold the angular-mean corrections, the higher-order velocity coefficients are exactly zero. The inner cutoff and the patch $\mathcal{I}_{\text{pos}}$ both have their support to the left of $\mathcal{I}_{\text{mean}}$. Thus $E_n = U_n = 0$ there. The first moment makes $F_n = 0$ as soon as the positive-order patch has been passed, and (5.2) then gives $V_n = 0$. In particular,

$$E_n = U_n = F_n = V_n = 0 \quad \text{on the reserved mean patch, for every } n \geq 1. \quad (5.18)$$

It is essential to put X_+ after the leading axial-velocity perturbation if the pressure is to have compact support, because Ω_0 can still be nonzero after the positive-order patches.

Step 4. Make the total tangential residual integrals zero. The inner equations make both positive-order stress components zero on $[0, X_-]$. To show that they are also zero after the outer radius, use the following conservative forms of the tangential residual, valid for any smooth axisymmetric divergence-free field.

$$\begin{aligned} r^2 \mathcal{R}_\theta &= \partial_t(r^2 u_\theta) + \partial_r(r^2 u_r u_\theta) + \partial_z(r^2 u_z u_\theta) - \partial_r(r^2 \partial_r u_\theta - r u_\theta) - \partial_{zz}(r^2 u_\theta), \\ r \mathcal{R}_z &= \partial_t(r u_z) + \partial_r(r u_r u_z) + \partial_z(r(u_z^2 + p)) - \partial_r(r \partial_r u_z) - \partial_{zz}(r u_z). \end{aligned}$$

The parity at the axis and the compact supports at positive order make the radial boundary terms zero. At order n , the angular moments that remain before differentiation are

$$\begin{aligned} \int_0^\infty r^2 u_{\theta,n} dr &= q^{3/2-A+\lambda_n} \int_0^\infty R^2 E_n dR, \\ \int_0^\infty r^2 \sum_{i+j=n} u_{z,i} u_{\theta,j} dr &= q^{3/2-2A+\lambda_n} \int_0^\infty R^2 \sum_{i+j=n} U_i E_j dR. \end{aligned}$$

Both are zero because $m_{n,2} = m_{n,4} = 0$. The power in the product is the same for every split $i + j = n$, so the cancellation happens before any derivative in the physical variables z, t is taken. The lower angular moment used by axial viscosity is zero because $m_{n-1,2} = 0$ when $n \geq 2$.

When $n = 1$, the compensated leading profile supplies the following convergent identity instead.

$$\int_0^\infty r^2 (u_{\theta,0}(r, z, t) - P(r)) dr = 0, \quad (5.19)$$

Here P is the fixed pure-power exterior from (A.45), so $\partial_z^2 P(r) = 0$. Differentiation under the integral is valid. Beyond the moving terminal collar, $u_{\theta,0} = K(r, t)$ does not depend on z , and the difference moment itself converges. The order-one angular viscosity term therefore has zero total moment as well.

The axial moments are

$$\begin{aligned} \int_0^\infty r u_{z,n} dr &= q^{1-A+\lambda_n} \int_0^\infty R U_n dR, \\ \int_0^\infty r \left(\sum_{i+j=n} u_{z,i} u_{z,j} + p_n \right) dr &= q^{1-2A+\lambda_n} \int_0^\infty R \left(\sum_{i+j=n} U_i U_j + \Pi_n \right) dR. \end{aligned} \quad (5.20)$$

Integration by parts turns the pressure part into

$$\int_0^\infty R \Pi_n dR = -\frac{1}{2} \int_0^\infty R^2 \partial_R \Pi_n dR.$$

The identities $m_{n,1} = m_{n,5} = 0$ now make both integrals in (5.20) zero. The axial-viscosity term uses the order- $n - 1$ axial-flux integral, which is also zero, including when that order is zero.

Integrating the conservative equations now gives, one order at a time,

$$\begin{aligned} \int_0^\infty r^2 \mathcal{R}_{\theta,n} dr &= \partial_t \int_0^\infty r^2 u_{\theta,n} dr + \partial_z \int_0^\infty r^2 \sum_{i+j=n} u_{z,i} u_{\theta,j} dr - \partial_{zz} \int_0^\infty r^2 u_{\theta,n-1} dr = 0, \\ \int_0^\infty r \mathcal{R}_{z,n} dr &= \partial_t \int_0^\infty r u_{z,n} dr + \partial_z \int_0^\infty r \left(\sum_{i+j=n} u_{z,i} u_{z,j} + p_n \right) dr - \partial_{zz} \int_0^\infty r u_{z,n-1} dr = 0. \end{aligned} \quad (5.21)$$

Here $\mathcal{R}_{j,n} = q^{-A-1+\lambda_n} \mathbf{r}_{j,n}$, with $j \in \{\theta, z\}$, is the order- n tangential residual in physical variables. When $n = 1$, the last integral in (5.21) means the convergent integral of $u_{\theta,0} - P$ from

(5.19), and the differentiation is applied to that convergent, subtracted integral. Dividing out the common physical powers gives

$$\int_0^\infty R^2 \mathfrak{r}_{\theta,n} dR = 0, \quad \int_0^\infty R \mathfrak{r}_{z,n} dR = 0.$$

Equation (5.9) can therefore also be written as

$$\begin{aligned} \mathcal{T}_{n,\theta}(R) &= -R^{-2} \int_0^R \varrho^2 \mathfrak{r}_{\theta,n}(\varrho) d\varrho = R^{-2} \int_R^\infty \varrho^2 \mathfrak{r}_{\theta,n}(\varrho) d\varrho, \\ \mathcal{T}_{n,z}(R) &= -R^{-1} \int_0^R \varrho \mathfrak{r}_{z,n}(\varrho) d\varrho = R^{-1} \int_R^\infty \varrho \mathfrak{r}_{z,n}(\varrho) d\varrho. \end{aligned}$$

This is the representation for the higher-order stress itself, with its sign retained.

Step 5. Find the stress support and bound the order-one term at the outer edge. When $n \geq 2$, the residual itself is zero after X_+ . Each same-order term contains a positive-order factor, and axial viscosity acts on order $n - 1 > 0$. The formula that integrates backward from infinity then proves $\text{supp } \mathcal{T}_n \subset [X_-, X_+]$.

When $n = 1$, the only source left outside is $-\partial_{zz}u_{\theta,0}$ in the angular equation. Beyond X_b , the leading profile is the z -independent heat field, so $\mathcal{T}_1$ is still supported inside the closed active annulus.

To get the bound near the edge, restrict further to the terminal collar where the exterior heat-flow cutoff is one. There $u_{\theta,0} = K(r, t)f_o$, where $K(r, t) = c_\infty s^{-A} \mathcal{H}(2(1-t)/s)$ is the physical heat field from (4.29). The smooth terminal multiplier f_o is constant on its first plateau and equals one whenever $X \geq X_b$. Its exact construction is (A.12), shifted in the logarithmic radial coordinate. In the local coordinate $y = \log X$, the difference from one has the form $1 - f_o = e^{-4/\delta_b^2}$ times a smooth factor, where $\delta_b = \log(X_b/X)$. Direct differentiation gives

$$a_z(\eta) = -\frac{2\eta}{L}, \quad b_z(\eta) = \frac{-2D\eta a_z + d\partial_\eta a_z}{L}, \quad \partial_{zz}(Kf_o) = Kq^{-2D} (a_z^2 \partial_{yy} f_o + b_z \partial_y f_o).$$

Every fixed η -derivative of a_z, b_z is bounded on $[-1, 1]$. The derivatives $\partial_y f_o$ and $\partial_{yy} f_o$ are bounded, respectively, by $Ce^{-4/\delta_b^2} \delta_b^{-3}$ and $Ce^{-4/\delta_b^2} \delta_b^{-6}$. Backward integration improves the bound by three powers of δ_b by Lemma A.9, used with $c = 4$ and $j = 3, 6$. The radial weights and the change between R and δ_b stay bounded and smooth on this fixed collar. After the stress power $q^{-A-1/2+\lambda_1}$ has been factored out, Lemma A.9 gives

$$|\mathcal{T}_1| \leq Ce^{-4/\delta_b^2} \delta_b^{-3}, \quad |\partial^I \mathcal{T}_1| \leq C_I e^{-4/\delta_b^2} \delta_b^{-N_I}. \quad (5.22)$$

The second estimate in (5.22) comes from differentiating the backward integral. Each derivative adds only a finite inverse power of δ_b . Near this edge, the inner factor in ζ stays between two positive constants. This proves (5.13) when $n = 1$. At the inner edge, the same estimate follows from $\mathcal{T}_1 = 0$ on $[0, X_-]$. When $n \geq 2$, every fixed derivative of $\mathcal{T}_n/\zeta$ is bounded because all of these stresses have the same compact support inside the annulus. $\square$

5.3 The coefficient induction and the residual left by a finite sum

The last two lemmas build and extend one coefficient after all lower orders have been finished. Lemma 5.1 solves the inner equations (5.2) through (5.6). Lemma 5.2 extends that solution and gives its support properties while keeping those equations unchanged on $0 \leq X \leq X_-$. This subsection applies the two lemmas repeatedly and estimates the residual left by every finite truncation in (5.25), including its physical derivatives. The gain in decay must rise with the truncation order, while the loss in the exponent caused by physical differentiation must not depend on that order. Equation (5.26) gives the exact bound.

For a tuple V of physical fields, write

$$|V|_m = \max_{|\alpha|+b \leq m} |\partial_x^\alpha \partial_t^b V|. \quad (5.23)$$

At the point in question, this is the largest absolute value among all Cartesian space-time derivatives of the tuple through order m . Write $u_n = u_{r,n}e_r + u_{\theta,n}e_\theta + u_{z,n}e_z$ for the physical velocity coefficient in (5.1), and write $\mathcal{T}_0$ for the leading stress profile from (4.11).

Before putting any cutoff in q , define the finite tuple

$$\mathcal{U}^{[N]} = (u_{\text{slow}}^{[N]}, p_{\text{slow}}^{[N]}, T_{\text{slow}}^{[N]}) = \sum_{n=0}^N (u_n, p_n, q^{-A-1/2+\lambda_n} \mathcal{T}_n).$$

The profiles in this sum already contain their radial cutoffs and moment corrections. For a physical stress pair $T = (T_\theta, T_z)$, define

$$\mathcal{F}_{\text{slow}}(u, p, T) = \mathcal{R}(u, p) + (\partial_r + 2/r)T_\theta e_\theta + (\partial_r + 1/r)T_z e_z. \quad (5.24)$$

The last two terms are the tangential divergence of the stress that is being kept for the later wave construction.

Proposition 5.3. *Fix the leading profiles from Theorem 4.6. There is a sequence of positive-order profiles, with each order built first by Lemma 5.1 and then by Lemma 5.2 after all lower orders have been finalized. Every finite velocity sum $u_{\text{slow}}^{[N]}$ is divergence-free. Take any fixed compact profile range $0 \leq X \leq X_{\max}$, $-1 \leq \eta \leq 1$, with $0 < X_{\max} < \infty$. For every $N, m \geq 0$ and $0 < q \leq 1$, the augmented momentum residual obeys*

$$|\mathcal{F}_{\text{slow}}(\mathcal{U}^{[N]})|_m \leq C_{N,m} q^{2h(N+1)-K_m}, \quad K_m \text{ independent of } N. \quad (5.25)$$

The constants can depend on the fixed profile range. The constant $C_{N,m}$ can depend on N , while the loss K_m in the exponent does not.

Proof. **Step 1. Build the sequence and cancel every coefficient that has been kept.** At order n , apply Lemma 5.1 with the finalized orders $0, \dots, n-1$. Next apply Lemma 5.2. This extends and corrects the current profiles, rebuilds V_n, Π_n from Equations (5.2) and (5.5), and defines $\mathcal{T}_n$ by (5.9). These operations keep the equations unchanged on the inner interval that was retained, and they keep the regularity needed to form Ω_n/X at the next order. Both lemmas use the same fixed radial intervals, so this process continues for every n .

Equation (5.2) gives exact incompressibility at each order, and hence for every finite sum. The pressure equation holds everywhere. Differentiating the stress primitives gives

$$(\partial_R + 2/R)\mathcal{T}_{n,\theta} = -\mathfrak{r}_{\theta,n}, \quad (\partial_R + 1/R)\mathcal{T}_{n,z} = -\mathfrak{r}_{z,n}.$$

Thus the tangential residual at each retained order is exactly the negative of the stress divergence. Proposition 4.2 gives the same leading identities at order zero. Once the sum has been cut off at N , every product or shifted viscosity term that has not been canceled has an exponent increment of at least $2h(N+1)$. For example, when $N = 1$, both the product of two order-one velocity coefficients and axial viscosity applied to order one first appear at order two.

Step 2. Estimate the physical derivatives of the finite residual. The coordinate formulas show exactly how physical differentiation changes the powers of q . Suppose that $g(x_\perp/\sqrt{q}, \eta)$ is smooth in its Cartesian arguments on a compact profile range. If there are $a = |\beta|$ transverse derivatives, k axial derivatives, and l time derivatives, then

$$\left| \partial_{x_\perp}^\beta \partial_z^k \partial_t^l [q^b g(x_\perp/\sqrt{q}, \eta)] \right| \leq C_m (1 + |b|)^m \|g\|_{C^m} q^{b-a/2-Dk-l}, \quad m = a + k + l. \quad (5.26)$$

This follows from the chain rule and the identities $\partial_t = q^{-1}L^{-1}(-q\partial_q + X\partial_X + D\eta\partial_\eta)$ and $\partial_z = q^{-D}L^{-1}(2\eta q\partial_q - 2\eta X\partial_X + d\partial_\eta)$. Each transverse derivative of these smooth Cartesian profiles contributes a factor $q^{-1/2}$.

The powers of n produced by differentiating q^{λ_n} are absorbed into the coefficient constants. The reduction in the exponent depends only on how many physical derivatives of each kind are taken. On the common stress support, $X \geq X_a > 0$, so $r^{-1} = q^{-1/2}(2X)^{-1/2}$ also has derivative losses that do not depend on N . Near the axis, use the smooth Cartesian representative of the velocity instead. A fixed truncated residual contains only finitely many terms. Applying (5.26) to those terms proves (5.25), with K_m independent of N . $\square$

5.4 Adding the coefficients while keeping zero divergence

The finite tuples $\mathcal{U}^{[N]}$ have residuals with better and better decay in (5.25), but the constants $C_{n,m}$ in the coefficient bounds (5.17) can grow too quickly for the entire formal series (5.1) to converge. To obtain smooth fields, multiply the n th positive-order term by $\chi(c_n q)$. Here $\chi = 1$ near zero and $\chi = 0$ once its argument reaches one, and the constants are chosen with $c_n \rightarrow \infty$, as in (5.34). Each later term is then zero outside a successively smaller neighborhood of $q = 0$. On any compact set where q stays a positive distance away from zero, only finitely many terms remain. The remaining point is to show that the finite residual bound in Proposition 5.3 survives these cutoffs. Lemma 5.4 proves this in (5.36) for the augmented residual (5.24).

To keep the velocity incompressible, apply the cutoffs to vector potentials before differentiating them. An axisymmetric Stokes streamfunction S gives such a potential. If S/r^2 is smooth as a function of (r^2, z, t) , then

$$\mathcal{A} = \frac{S}{r}e_\theta = \frac{S}{r^2}(-x_2, x_1, 0), \quad \text{curl } \mathcal{A} = -\frac{\partial_z S}{r}e_r + \frac{\partial_r S}{r}e_z.$$

The smoothness of S/r^2 in (r^2, z, t) makes this potential smooth in Cartesian coordinates.

For the positive-order profiles already constructed, the physical streamfunctions and vector potentials are

$$F_n = X\mathcal{A}_X(U_n), \quad \mathcal{S}_n = q^{1-A+\lambda_n}F_n, \quad \mathcal{A}_n = (\mathcal{S}_n/r)e_\theta, \quad u_{z,n} = \partial_s \mathcal{S}_n, \quad ru_{r,n} = -\partial_z \mathcal{S}_n. \quad (5.27)$$

Here $s = r^2/2$, as in (4.1). In fact, $\text{curl } \mathcal{A}_n = u_{r,n}e_r + u_{z,n}e_z$, so the curl gives exactly the radial and axial velocity components. In Cartesian coordinates, $r^2 = 2qX$ gives

$$\mathcal{A}_n = \frac{\mathcal{S}_n}{r^2}(-x_2, x_1, 0) = \frac{1}{2}q^{-A+\lambda_n}\frac{F_n}{X}(-x_2, x_1, 0).$$

Since F_n/X is smooth at $X = 0$, this formula extends smoothly through the axis whenever $q > 0$. Lemma 5.2 gives common supports and bounds for every fixed derivative of each coefficient. No bound uniform in n is needed.

Cutting off $\mathcal{A}_n$ and then taking its curl includes the extra velocity term that comes from differentiating the cutoff, so the result remains divergence-free. The summation argument will be stated for tuples of physical fields, which also allows it to be used later for the wave and mean correction sequence. In the present case, the index $j = n$, the decay order is $g_n = 2nh$, and the entries are

$$A_n = \mathcal{A}_n, \quad B_n = u_{\theta,n}e_\theta, \quad p_n = q^{-2A+\lambda_n}\Pi_n, \quad T_n = q^{-A-1/2+\lambda_n}\mathcal{T}_n.$$

The differential polynomial is $\mathcal{F}_{\text{slow}}$ from (5.24). Inside the lemma, U_j names a tuple of physical fields, not the axial profile $U_n(X, \eta)$. Use the derivative notation in (5.23), and set $\Lambda_{\log}(q) = 1 + |\log q|$. Every estimate in the lemma is uniform on the domain stated there.

The general argument needs one common domain, decay orders that rise to infinity, and derivative losses that do not depend on the summation index. Let Ω be a domain on which $0 < q < q_0 \leq 1$. Suppose $q = q(z, t)$ is smooth, stays a positive distance away from zero on every compact subset of Ω , and satisfies

$$|\partial_z^a \partial_t^b q| \leq C_{a,b} q^{1-aD-b}, \quad a, b \geq 0, \quad 0 < D < 1. \quad (5.28)$$

Let $U_0 = (u_0, p_0, T_0)$ be smooth, with $\operatorname{div} u_0 = 0$. Suppose that for every m , there are finite constants C_m, K_m, P_m such that $|U_0|_m \leq C_m q^{-K_m} \Lambda_{\log}(q)^{P_m}$. The stress entries T_0 and T_j below can be left out. For $j \geq 1$, suppose the smooth physical tuples

$$Z_j = (A_j, B_j, p_j, T_j), \quad B_j = b_j(r, z, t)e_\theta, \quad U_j = (\operatorname{curl} A_j + B_j, p_j, T_j) \quad (5.29)$$

are all defined on this same domain. Their Cartesian representatives must be smooth at the axis, and extending them by zero across any side boundary of their spatial support must also be smooth.

Suppose that for every j, m , there are finite constants $C_{j,m}, P_{j,m}$ such that

$$|Z_j|_m \leq C_{j,m} q^{g_j - \ell_m} \Lambda_{\log}(q)^{P_{j,m}}, \quad 0 < g_1 \leq g_2 \leq \dots \longrightarrow \infty, \quad (5.30)$$

where the losses ℓ_m are nonnegative, do not decrease as the derivative order rises, and do not depend on j .

For each finite J , set $U^{[J]} = U_0 + \sum_{j=1}^J U_j$. Write the scalar components of a tuple as $U = (U^1, \dots, U^{N_U})$. A differential polynomial of fixed finite order and degree is a finite sum of products of field components and their derivatives. Component by component, it has the form

$$(\mathcal{F}(U))^a = c_{a0}(x, t) + \sum_{\nu=1}^{M_a} c_{a\nu}(x, t) \prod_{r=1}^{d_{a\nu}} \partial_{x,t}^{\beta_{avr}} U^{k_{avr}}. \quad (5.31)$$

The output index ranges over the fixed finite set $1 \leq a \leq N_F$, and every M_a is finite. Fixed integers $s \geq 0, d \geq 1$ bound the derivative order and the degree, with $1 \leq d_{a\nu} \leq d$, $|\beta_{avr}| \leq s$ and $1 \leq k_{avr} \leq N_U$. For $\beta = (\alpha, b) \in \mathbb{N}_0^4$, the notation means $\partial_{x,t}^\beta = \partial_x^\alpha \partial_t^b$. The coefficient functions, the indices, and the bounds s, d are all fixed independently of the summation and truncation indices.

For every nonconstant product in (5.31), choose one region $\Omega_{a\nu}$ that contains the supports of all its field factors, both in the base and in every increment, before and after the cutoffs are applied. For the term that does not depend on U , set $\Omega_{a0} = \Omega$. Every coefficient $c_{a\nu}$, with $0 \leq \nu \leq M_a$, must satisfy the following bound for every Cartesian space-time multi-index $I \in \mathbb{N}_0^4$.

$$|\partial_{x,t}^I c_{a\nu}(x, t)| \leq C_I q^{-K_I} \Lambda_{\log}(q)^{P_I} \quad ((x, t) \in \Omega_{a\nu}). \quad (5.32)$$

The regions do not depend on the truncation. Since there are only finitely many coefficients, the finite constants C_I, K_I, P_I can be chosen, for each fixed I , to work for the whole list. They do not depend on the truncation index. The bounds therefore hold throughout the comparison between the partial sums and the cutoff sums. The terms c_{a0} , which do not depend on U , cancel in that comparison.

The next assumption concerns the full $\mathcal{F}$, including the terms that do not depend on the fields. Suppose every finite partial sum satisfies

$$|\mathcal{F}(U^{[J]})|_m \leq C_{J,m} q^{\rho_J - K_m^{\mathcal{F}}} \Lambda_{\log}(q)^{P_{J,m}} + E_{J,m}, \quad \rho_J \longrightarrow \infty, \quad (5.33)$$

where $K_m^{\mathcal{F}}$ does not depend on J , and, for each fixed J, m, N , $0 \leq E_{J,m} \leq C_{J,m,N} q^N$. At that fixed stage, all of these bounds, including the flat remainder, are uniform on the entire domain.

Lemma 5.4. *There are numbers $a_{j+1} \geq 2a_j$, with $a_1^{-1} < q_0$, and one fixed smooth cutoff χ that equals one on $[0, 1/2]$ and zero on $[1, \infty)$, such that*

$$U = U_0 + \sum_{j \geq 1} (\operatorname{curl}(\chi(a_j q) A_j) + \chi(a_j q) B_j, \chi(a_j q) p_j, \chi(a_j q) T_j) \quad (5.34)$$

is locally finite and smooth, and its velocity component u satisfies $\operatorname{div} u = 0$. Every added term extends smoothly by zero across the outer boundary $q = q_0$. For every m and every $J \geq \max(1, m)$,

$$|U - U^{[J]}|_m \leq 2^{-J} q^{g_{J+1}/2 - \ell'_m} \quad \left(0 < q < \frac{1}{2a_J}\right), \quad (5.35)$$

where ℓ'_m does not depend on J .

Under the residual assumption (5.33), every Cartesian space-time derivative of $\mathcal{F}(U)$ vanishes to infinite order as $q \downarrow 0$. In other words, this says

$$\forall m, N \quad \exists \delta, C > 0 \quad |\mathcal{F}(U)|_m \leq C q^N \quad (0 < q < \delta). \quad (5.36)$$

Proof. First bound the derivatives of every cutoff. Then choose the cutoffs so that the tail remains summable after any fixed number of derivatives, and compare the residual with one finite partial sum before the cutoffs are applied. In these bounds, each prescribed Cartesian derivative reduces the power of q by an amount that does not depend on the truncation order.

Step 1. Bound the derivatives of the cutoffs and their curls. The chain rule writes any positive-order derivative of $\chi(aq)$ as a sum of terms of the form

$$a^k \chi^{(k)}(aq) \prod_{v=1}^k \partial_z^{a_v} \partial_t^{b_v} q, \quad \sum_v a_v = a', \quad \sum_v b_v = b'.$$

On the support of each such term, $1/2 \leq aq \leq 1$. The bounds in (5.28) supply factors q^k , which compensate for a^k because aq stays bounded on the cutoff support. It follows that

$$|\partial_z^{a'} \partial_t^{b'} \chi(aq)| \leq C_{a', b'} q^{-a'D - b'},$$

with constants that do not depend on a .

Because $q = q(z, t)$, both $\partial_{x_1} q = \partial_{x_2} q = 0$. In particular, the product rule gives

$$\operatorname{curl}(\chi(aq) A_j) = \chi(aq) \operatorname{curl} A_j + a \chi'(aq) \nabla q \times A_j$$

The second term is the commutator created by differentiating the cutoff, and the assumed derivative bounds for the potentials control it. The product rule now bounds all derivatives through order m of the j th summand in (5.34) by $\widehat{C}_{j,m} q^{g_j - \ell'_m} \Lambda_{\log}(q)^{\widehat{P}_{j,m}}$. The loss ℓ'_m depends only on m and ℓ_{m+1} .

Step 2. Choose the cutoffs and estimate the tail. Choose the scales a_j one at a time so that $a_{j+1} \geq 2a_j$, every cutoff $\chi(a_j q)$ is supported inside $q < q_0$, and

$$\widehat{C}_{j,m} \Lambda_{\log}(q)^{\widehat{P}_{j,m}} q^{g_j/2} \leq 2^{-j} \quad (0 < q \leq a_j^{-1}, 0 \leq m \leq j). \quad (5.37)$$

At step j , there are only finitely many requirements, and each one holds for all sufficiently small q because $g_j > 0$. All derivatives through order m of the resulting cutoff increment are therefore at most $2^{-j} q^{g_j/2 - \ell'_m}$ whenever $m \leq j$.

On a compact subset where $q \geq \delta > 0$, every term with $a_j > \delta^{-1}$ is zero, so the sum is locally finite. The curl of every vector field is divergence-free. The field $\chi(a_j q) B_j$ is also divergence-free because its angular amplitude does not depend on θ . Smoothness and the stated extensions across the support boundaries now follow one term at a time.

Fix m and $J \geq \max(1, m)$. When $q < (2a_J)^{-1}$, the first J cutoffs all equal one. Since $q < 1$ and g_j does not decrease as its index rises, the remaining terms satisfy

$$\sum_{j>J} 2^{-j} q^{g_j/2 - \ell'_m} \leq 2^{-J} q^{g_{J+1}/2 - \ell'_m},$$

which proves (5.35).

Step 3. Compare the residual with one finite partial sum. The cutoff sum U is now smooth, and its velocity is divergence-free. To show that $\mathcal{F}(U)$ is flat, compare it with $\mathcal{F}(U^{[J]})$ for a finite partial sum whose residual has sufficiently high decay. Then control the change caused by $U - U^{[J]}$. The assumed bound on $|U_0|_m$ and (5.30) give

$$|U^{[J]}|_m \leq C_{J,m} q^{-K_m} \Lambda_{\log}(q)^{P_{J,m}}, \quad (5.38)$$

where $K_m \geq 0$ does not depend on J . This is because every increment has $g_j > 0$, and taking the curl uses one additional derivative of the potential.

The bounds s, d on order and degree are the ones from (5.31). Subtract the two products one factor at a time, then use the product rule and (5.32). For $e = U - U^{[J]}$, this gives

$$|\mathcal{F}(U) - \mathcal{F}(U^{[J]})|_m \leq C_m q^{-H_m} |e|_{m+s} (1 + |U^{[J]}|_{m+s} + |e|_{m+s})^{d-1}, \quad (5.39)$$

where H_m does not depend on J . The finitely many logarithmic factors from the coefficients are absorbed into this fixed power of q^{-1} . A polynomial that does not depend on U has zero difference, so it needs no estimate here.

Fix a derivative order m and a decay order N . Choose $J \geq m + s$ large enough that

$$\frac{g_{J+1}}{2} - \ell'_{m+s} \geq N + H_m + (d-1)(K_{m+s} + 1), \quad \rho_J - K_m^{\mathcal{F}} \geq N + 1.$$

Also choose J large enough that $g_{J+1}/2 - \ell'_{m+s}$ is nonnegative, and then keep J fixed. Shrink the q -neighborhood until $\chi(a_j q) = 1$ for $j \leq J$. In this neighborhood, one power of q^{-1} absorbs the constants and logarithms in (5.38), and (5.35) gives $|e|_{m+s} \leq 1$. Formula (5.39) is then $O(q^N)$.

The same conclusion holds for (5.33), after one inverse power of the small parameter absorbs its logarithm and the flatness of the fixed $E_{J,m}$ is used. This proves (5.36). The choices are made in this order. First choose (m, N) , then J , and then the neighborhood and its constants. The comparison uses only the remainder $E_{J,m}$ at that fixed truncation. $\square$

A finite initial block whose decay orders are not positive can be given one common cutoff, applied as in (5.34), and included in the base U_0 . That cutoff is one near $q = 0$, so every comparison with a finite partial sum is unchanged there.

For smooth vector fields v, e and smooth scalar fields p, π , the physical residual changes by exactly

$$\mathcal{R}(v + e, p + \pi) - \mathcal{R}(v, p) = \partial_t e - \Delta e + \nabla \pi + (v \cdot \nabla) e + (e \cdot \nabla) v + (e \cdot \nabla) e.$$

For $\mathcal{F}_{\text{slow}}$, also add the change in the linear tangential stress-divergence term from (5.24). On the common annular stress support, the cylindrical coefficients and every prescribed Cartesian derivative of them are bounded by a fixed negative power of q . In the final correction sequence, $\mathcal{F} = \mathcal{R}$, and there is no stress variable.

The same cutoff choices can satisfy further normalized estimates once those estimates have been proved for the coefficients. More exactly, at step j , any finite list of conditions

$$B_{j,k} \Lambda_{\log}(q)^{P_{j,k}} q^{\gamma_{j,k}} \leq 2^{-j}, \quad \gamma_{j,k} > 0, \quad (5.40)$$

can be added to (5.37). For derivatives of the normalized expansion coefficients and boundary weights, impose these extra coefficient estimates through the corresponding product rules for the cutoffs. To recover a full expansion after a fixed truncation N and at a fixed derivative order m , choose $M \geq \max(N, m)$ with $g_{M+1}/2 \geq g_{N+1}$. The finite block from $N + 1$ through M keeps its original orders near $q = 0$. The bound (5.35) controls the rest of the tail at order g_{N+1} , with the fixed exponent reduction ℓ'_m .

5.5 The base field after summation

The finite tuples $\mathcal{U}^{[N]}$ from Proposition 5.3 now have the residual estimates needed by Lemma 5.4. Apply that lemma to the coefficient fields and their potentials $\mathcal{A}_n$ from (5.27) to build the smooth background (u_B, p_B) . The construction must also keep the leading profile as in (5.42), the exact heat field outside the active region from (4.29), and the weighted stress bounds in (5.43). Those bounds are used to choose the wave amplitudes and their corrections in Propositions 7.5 and 7.6.

Proposition 5.5. *There are smooth axisymmetric fields (u_B, p_B) for $q > 0$, together with two physical tangential stress components $\mathcal{T}_{\text{phys}}$ supported where $X_a \leq X(r, z, t) \leq X_b$, such that $\text{div } u_B = 0$ and*

$$\mathcal{R}(u_B, p_B) = -(\partial_r + 2/r)\mathcal{T}_{\text{phys},\theta}e_\theta - (\partial_r + 1/r)\mathcal{T}_{\text{phys},z}e_z + \mathcal{E}_B. \quad (5.41)$$

Fix $0 < X_{\max} < \infty$. Uniformly on the profile rectangle $[0, X_{\max}] \times [-1, 1]$ in (X, η) , every Cartesian space-time derivative ∂^α and every $M > 0$ satisfy $|\partial^\alpha \mathcal{E}_B| \leq C_{\alpha, M} q^M$ as $q \downarrow 0$. The constants can depend on $X_{\max}$. The formal expansion of $(u_B, p_B, \mathcal{T}_{\text{phys}})$ has the finite truncations $\mathcal{U}^{[N]}$ from Proposition 5.3, with the physical Stokes streamfunctions S_n from (5.27) summed before differentiation.

Fix radial endpoints $0 < X_{\text{lo}} < X_a < X_b < X_{\text{hi}} < \infty$. The rectangle $[X_{\text{lo}}, X_{\text{hi}}] \times [-1, 1]$ is a fixed radial enlargement of the closed active annulus $[X_a, X_b] \times [-1, 1]$ from Item (ii) of Theorem 4.6, and η runs through its full closed parameter interval. On this enlarged rectangle, take any fixed composition $\mathcal{D}^I$ of $q\partial_q, \partial_X, \partial_\eta$. Then, as $q \downarrow 0$,

$$\begin{aligned} \mathcal{D}^I(q^A u_{\theta, B} - E_0) &= O_I(q^{2h}), & \mathcal{D}^I(q^A u_{z, B} - U_0) &= O_I(q^{2h}), \\ \mathcal{D}^I(q^{1/2} u_{r, B} - V_0/\sqrt{2X}) &= O_I(q^{2h}), & \mathcal{D}^I(q^A u_{r, B}) &= O_I(q^h). \end{aligned} \quad (5.42)$$

The constants in these estimates can depend on the fixed radial endpoints. With $\widehat{\mathcal{T}} = q^{A+1/2}\mathcal{T}_{\text{phys}}$, the weighted comparison uses ζ, δ from Item (iv) of Theorem 4.6 and holds whenever $X_a < X < X_b$ and $-1 \leq \eta \leq 1$.

$$|\mathcal{D}^I(\widehat{\mathcal{T}} - \mathcal{T}_0)| \leq C_I q^{2h} \zeta \delta^{-N_I} \leq C_I q^h \zeta \delta^{-N_I} \quad (0 < q \leq 1). \quad (5.43)$$

Every fixed physical space-time derivative of $u_B, p_B, \mathcal{T}_{\text{phys}}, \mathcal{E}_B$ extends continuously to $\eta = \pm 1$ on compact sets where $q > 0$. The positive-order corrections $u_B - u^{(0)}$ and $p_B - p^{(0)}$ are zero whenever $X \geq X_+$, with X_+ from Lemma 5.2. In particular, the exact exterior of the leading profile from (4.29) stays unchanged. On the reserved interval $X \in \mathcal{I}_{\text{mean}} \subset (X_a, X_v)$ from Item (vi) of Theorem 4.6, for every $-1 \leq \eta \leq 1$, the base velocity agrees exactly with the leading profile.

$$u_B = u^{(0)}, \quad u_{z, B} = 0, \quad q^A u_{\theta, B} = E_0 = c_{\text{patch}}(1 + \eta^2)^{-1} X^{-1/2-\lambda}. \quad (5.44)$$

Here $u^{(0)}$ is the velocity determined by the leading profile in (4.3).

Proof. Step 1. Check the assumptions of the summation lemma and choose the cutoffs. Use the potentials in (5.27) and the physical tuples listed before Lemma 5.4, while leaving order zero unchanged. The Cartesian formula for $\mathcal{A}_n$ proves regularity at the axis, and Lemma 5.2 gives the common supports and all fixed derivative bounds for the coefficients. By (5.26), because $D < 1/2$, the reduction in the power of q for the coefficient tuple in (5.27) and (5.1), including the stress components, is bounded by $\ell_m = 2A + m$. The coefficient constants absorb the powers of n produced by differentiating q^{λ_n} . For the cutoffs, set $\sigma = c_n q$. Then every integer $j \geq 0$ satisfies

$$\begin{aligned} (q\partial_q)^j \chi(c_n q) &= (\sigma\partial_\sigma)^j \chi(\sigma) \Big|_{\sigma=c_n q}, \\ \sup_{n \geq 1, q > 0} |(q\partial_q)^j \chi(c_n q)| &\leq \sup_{\sigma > 0} |(\sigma\partial_\sigma)^j \chi(\sigma)| =: C_j < \infty. \end{aligned}$$

Because of this, the same derivative estimates still hold uniformly after the cutoffs are applied. Proposition 5.3 already checked the bounds on the stress-divergence coefficients. Its estimate (5.25) is exactly the finite-residual hypothesis in the lemma, with $\rho_N = 2h(N+1)$ and no extra remainder. This gives every hypothesis of Lemma 5.4 on each fixed compact range of the profile variables. Applying the lemma gives $u_B, p_B, \mathcal{T}_{\text{phys}}$. Its residual $\mathcal{E}_B = \mathcal{F}_{\text{slow}}(u_B, p_B, \mathcal{T}_{\text{phys}})$ is bounded by every power of q , even after taking any of the prescribed physical derivatives.

Step 2: keep the field divergence-free, keep its support where it belongs, and keep the velocity on the mean patch. Differentiating the cutoff produces an extra radial term, which appears explicitly in

$$ru_{r,n}^{\text{cut}} = q^{2nh} \left[\chi(c_n q) V_n - \frac{2\eta}{L} (c_n q) \chi'(c_n q) F_n \right], \quad u_{z,n}^{\text{cut}} = q^{-A+2nh} \chi(c_n q) U_n. \quad (5.45)$$

The term containing $\chi'(c_n q) F_n$ carries the same q^{2nh} factor, and Lemma 5.2 gives it the same radial support as the other terms. Physical derivatives commute, so the divergence remains identically zero across every cutoff transition. Equation (5.18) makes both terms inside the radial brackets zero on the mean patch. The positive-order axial and azimuthal terms vanish there as well. This proves (5.44), including the terms created when the cutoff is applied to the Stokes streamfunction.

Step 3: prove the estimates for the normalized velocity and stress. The extra diagonal bounds in (5.40) control all normalized coefficient derivatives and Stokes streamfunction derivatives through order n . They also control the extra streamfunction derivative and, when $n \geq 2$, every derivative of $\mathcal{T}_n/\zeta$. The normalized velocity estimates are taken on the fixed enlarged rectangle from the statement. The weighted stress estimates are taken where $X_a < X < X_b$ and $-1 \leq \eta \leq 1$. When an estimate reaches the axis, the profiles are written in smooth Cartesian coordinates. The finite constants include the powers of n and the fixed derivatives of the cutoff. In addition to the physical requirements in the lemma, the chosen cutoff gives the following bound.

$$\left| \mathcal{D}^I \left[\chi(c_n q) q^{2nh} a_n \right] \right| \leq 2^{-n} q^{nh} \quad (n \geq \max\{2, |I|\}), \quad (5.46)$$

Here a_n is any one of $E_n, U_n, \Pi_n, V_n/\sqrt{2X}, F_n/\sqrt{2X}$ on the enlarged rectangle. It may instead be a component of $\mathcal{T}_n/\zeta$ on the active radial interior when $n \geq 2$. The same estimate holds for the expression inside the brackets in (5.45) after multiplication by q^{2nh} . At each order, the diagonal choice in the summation lemma also includes these finitely many extra seminorms. The analytic neighborhoods may depend on the order.

Fix a derivative order. Keep the finite initial block, whose positive-order terms are all $O(q^{2h})$, and sum (5.46) starting at an index that is at least two and larger than that derivative order. This proves (5.42). Under the common normalization q^A , the positive-order radial correction is actually $O(q^{3h})$. For the stress components with $n \geq 2$, the same argument carries the factor ζ . The one term with $n = 1$ is controlled by (5.22); each derivative costs only a finite inverse power of δ . This proves (5.43). The wave construction can now use the fixed positive covariance representation of $\mathcal{T}_0$, while treating the higher-order stress through signed corrections.

Step 4: recover the full expansion and the endpoint extensions. Fix a formal order N and a derivative order m . Split the summed remainder at $J = \max\{2N+2, m+1\}$. The finitely many terms with $N < n < J$ keep their original powers, all of which are at least $q^{2h(N+1)}$. The tail obeys the geometric estimate

$$\sum_{n \geq J} 2^{-n} q^{nh} \leq 2^{1-J} q^{Jh} \leq 2^{1-J} q^{2h(N+1)} \quad (0 < q \leq 1).$$

The coefficient constants for the finite block in the middle may depend on N, m . For small enough q , every cutoff in the finite initial block is identically one. This recovers every fixed order in the original formal expansion, even after physical differentiation causes its fixed loss in the exponent. The geometric bounds are used only to sum the series, so they leave the asymptotic coefficients unchanged. Step 1 already proved the residual claim by comparing the sum with a finite partial sum taken far enough out.

Finally, the cutoff sequence grows at least geometrically. On any compact set where q has a positive lower bound, only finitely many positive-order terms remain. Fix a physical space-time derivative. Intersect the finitely many parameter neighborhoods for the positive-order coefficients that contribute to it. Their profiles and Stokes streamfunctions extend to the resulting neighborhood. The order-zero term has the smooth one-sided extension in η supplied by the leading profile and the heat construction. Since $L > 0$, the coordinate map carries these extensions into the physical variables, together with all the specified derivatives. This proves the claimed extensions at $\eta = \pm 1$. $\square$

6 The auxiliary torus and the separation of oscillatory supports

Fix the background velocity u_B , the pressure p_B , and the annular stress $\mathcal{T}_{\text{phys}}$ from Proposition 5.5. The momentum residual is the divergence of that stress, with the sign used in (5.41), plus a flat remainder. The waves in Proposition 7.5 will supply the leading stress. Their amplitudes change under the local shear of the background according to (7.5). Before those waves can be built, they must be confined to regions where the local estimates control the variation of the shear. Their supports must also be arranged so that different waves do not create extra quadratic interactions. At the same time, this localization must allow rapid changes in time without creating radial derivatives large enough to ruin the residual estimates.

Introduce the periodic variable $Y \in \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, independently of the physical coordinates (r, θ, z, t) . Physical fields are then obtained by evaluating it at the fixed map depending on (r, t) in (6.3). This map lets time vary rapidly while adding much smaller radial derivatives. The time variation drives the pulse equation in Section 7, and the bounds on the radial derivatives control the errors caused by localization. Oscillations whose supports overlap in the remaining variables are given disjoint auxiliary supports, so their cross products vanish after evaluation. Different harmonics of the same localized oscillation still interact.

First, derive the evaluated derivative operators in (6.6). Lemma 6.1 separates the supports, and Lemma 6.2 puts overlapping bands into one common representation. Section 6.4 then defines the coefficient classes, and Proposition 6.6 gives their rules for products and derivatives.

Every preliminary geometric choice is made on the one domain $0 < q < q_{\text{big}}$. The number q_{big} may be made smaller when the fixed base and pulse coefficients are chosen, before the correction iteration starts.

6.1 Dyadic charts and derivatives after phase evaluation

Recall from (4.1) that $q = q(z, t)$, $A = 1/2 + h$, and $D = 1/2 - h$. Work on dyadic regions where q is comparable to a fixed scale Q . Rescale the radial, axial, and time coordinates by $Q^{1/2}, Q^D, Q$, using those three factors in that order. This keeps the concentrating annulus at a bounded size in every region. For a dyadic index ℓ , set

$$Q = 2^{-\ell}, \quad \varepsilon = Q^h, \quad S_* = \ell^2, \quad R = \frac{r}{\sqrt{Q}}, \quad Z = \frac{z}{Q^D}, \quad T = \frac{\tau}{Q}, \quad \tau = 1 - t. \quad (6.1)$$

In these charts, the radius R uses the fixed band scale Q . In the earlier profile calculations, the same letter meant $\sqrt{2X} = r/\sqrt{q}$. The two versions are related by $R_{\text{chart}} = \sqrt{q/Q} R_{\text{profile}}$. The quantities Q, ε, S_* stay constant throughout one chart. To get chart representatives, multiply a physical velocity, pressure, and residual by $Q^A, Q^{2A}, Q^{2A+1/2}$, using those three factors in that order. An integer matrix acts on the torus without breaking periodicity. Its two distinct eigenvalues will make the auxiliary phase vary at different rates in the radial and time directions. Fix

$$\begin{aligned} J_g &= \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, & \Lambda_g &= 4 - \sqrt{2}, & T_g &= 4 + \sqrt{2}, & b_g &= \sqrt{2} - 1, \\ v_r &= (1, -b_g), & v_t &= (b_g, 1), & \rho_g &= \frac{\log \Lambda_g}{\log T_g}, \\ \kappa_s &= 10^{-5}, & d_r &= 2((1+h)\rho_g - h\kappa_s) > 0. \end{aligned} \quad (6.2)$$

The vectors satisfy $J_g v_r = \Lambda_g v_r$ and $J_g v_t = T_g v_t$, with $1 < \Lambda_g < T_g$. Define the physical evaluation map by

$$Y = v_r r^{d_r} + v_t t \pmod{\mathbb{Z}^2}. \quad (6.3)$$

Start with a field on the extended domain, where Y is an independent variable, and then restrict it to this map. The chain rule says that its physical derivatives are taken by the operators

$$\begin{aligned} N_{\text{abs}} &= v_t \cdot \partial_Y, & \mathcal{L}_{\text{abs}} &= v_r \cdot \partial_Y, \\ \mathfrak{t} &= \partial_t + N_{\text{abs}}, & \mathfrak{r} &= \partial_r + d_r r^{d_r-1} \mathcal{L}_{\text{abs}}. \end{aligned} \quad (6.4)$$

Here ∂_t and ∂_r on the extended domain keep Y fixed. Before restriction, $\mathfrak{t}$ represents physical differentiation in time, while $\mathfrak{r}$ represents physical differentiation in radius.

Call Y the *absolute* auxiliary variable. For band ℓ , its *band representative* is the image Y_i under the following integer covering.

$$Y_i = J_g^i Y \pmod{\mathbb{Z}^2}, \quad i = i(\ell) = \left\lfloor \log_{T_g} \frac{Q^{-1-h}}{S_*} \right\rfloor. \quad (6.5)$$

If f is a function on the band torus, its pullback to the absolute torus is $f(J_g^i Y)$. A function descends to that band torus, meaning that it gives one well-defined function there, when every translation $Y \mapsto Y + a$ with $J_g^i a \in \mathbb{Z}^2$ leaves its value unchanged. These translations are called deck translations. Only large ℓ , for which $i(\ell) \geq 0$, are used. Write $N_i = v_t \cdot \partial_{Y_i}$ and $\mathcal{L}_i = v_r \cdot \partial_{Y_i}$. The eigenvector identities then give $N_{\text{abs}} = T_g^i N_i$ and $\mathcal{L}_{\text{abs}} = \Lambda_g^i \mathcal{L}_i$ on band pullbacks. So the physical derivatives have the chart forms below.

$$\begin{aligned} \mathfrak{t}_* &= Q^{1+h} \mathfrak{t} = -\varepsilon \partial_T + c_i N_i, & c_i &= T_g^i Q^{1+h} \asymp S_*^{-1}, \\ D_r &= \sqrt{Q} \mathfrak{r} = \partial_R + M_i d_r R^{d_r-1} \mathcal{L}_i, & M_i &= \Lambda_g^i Q^{d_r/2} \asymp \varepsilon^{-\kappa_s} S_*^{-\rho_g}, \\ D_z &= \sqrt{Q} \partial_z = \varepsilon \partial_Z, & D_\theta &= R^{-1} \partial_\theta. \end{aligned} \quad (6.6)$$

The velocity and residual normalizations introduced after (6.1) are used from this point on. In (6.6), the *slow* derivative terms act on R, Z, T while keeping the auxiliary variable Y_i fixed. The *fast* terms act on Y_i ; the chain rule along the physical phase map creates them. For example, $-\varepsilon \partial_T$ is the slow time term, while $c_i N_i$ is the fast time term. The velocity and residual normalizations produce the time factor $Q^{2A+1/2} Q^{-A} = Q^{1+h}$ and the viscous factor $Q^{2A+1/2} Q^{-A} Q^{-1} = \varepsilon$.

The integer $i(\ell)$ in (6.5) satisfies

$$\frac{1}{T_g} \frac{Q^{-1-h}}{S_*} < T_g^{i(\ell)} \leq \frac{Q^{-1-h}}{S_*}, \quad \Lambda_g^{i(\ell)} = (T_g^{i(\ell)})^{\rho_g}.$$

These inequalities give both comparisons in (6.6). The chosen covering index makes c_i comparable to S_*^{-1} . The choice of d_r leaves only the factor $\varepsilon^{-\kappa_s}$ in M_i , apart from its power of S_* . These are the two derivative scales used below.

The absolute coordinate derivatives $\mathfrak{r}, \partial_z, \partial_\theta, \mathfrak{t}$ commute. Their only variable coefficient is a function of r multiplying one constant direction on the torus. Cylindrical frame coefficients such as $1/R$ still have to be kept when vector components are differentiated. Every correction that depends on the auxiliary variable is supported away from the axis, so r^{d_r} does not need to be smooth at $r = 0$. Periodicity in the auxiliary variable does not create spatial periodicity after restriction to (6.3).

The construction also integrates along each of the two torus directions. On a Fourier mode with frequency n , this means dividing by the scalar product of that direction with n . Both directions satisfy

$$|v_r \cdot n| \geq \frac{c}{1 + |n|}, \quad |v_t \cdot n| \geq \frac{c}{1 + |n|} \quad (n \in \mathbb{Z}^2 \setminus \{0\}). \quad (6.7)$$

For the first direction, $v_r \cdot n = (n_1 + n_2) - \sqrt{2}n_2$. Changing the sign in front of the square root of two gives its algebraic conjugate. Multiplying the two gives the nonzero integer $(n_1 + n_2)^2 - 2n_2^2$, while the conjugate has absolute value at most $C|n|$. Every nonzero integer has absolute value at least one. Since the product is such an integer and the conjugate has the stated upper bound, the original factor has the lower bound in (6.7). The same argument handles v_t , using $(n_2 - n_1) + \sqrt{2}n_1$. Dividing by either dot product costs at most one power of the frequency size, which is why the inverse directional operators on zero-mean torus functions lose only finitely many torus derivatives. These operators are used in Lemma 8.2 and in the equations that correct the mean.

6.2 Slow cutoffs and disjoint auxiliary rectangles

Next, assign supports to the localized oscillations. The slow cutoffs in (6.9) form a squared partition of unity. When their covariances are summed in (7.30), that partition recovers the target stress. Different local oscillations are put in disjoint auxiliary rectangles whenever their slow supports meet. Lemma 6.1 proves this, and the separation removes the cross terms between those oscillations.

Choose a smooth squared partition $\sum_\ell \chi_\ell(q)^2 = 1$, with χ_ℓ supported where $q/Q \in [1/2, 2]$. In each band, choose a product squared partition $\sum_a \chi_{\ell,a}(R, Z, T)^2 = 1$. Its mesh is S_*^{-3} in every coordinate, and each support reaches at most one mesh length past its grid point on either side. Both partitions can be made by translating one smooth bump and dividing each translate by the square root of the sum of their squares. Write

$$\mathcal{C}_\ell(r, z, t) = \left(\frac{r}{\sqrt{Q}}, \frac{z}{Q^D}, \frac{1-t}{Q} \right), \quad B_{\ell,a} = \text{supp } \chi_{\ell,a}.$$

Fix a large enough starting band ℓ_0 , and make q_{big} smaller if needed so that $q_{\text{big}} \leq 2^{-\ell_0}$. In this chart, the closed active part of band ℓ where $\tau \geq 0$ is

$$\mathcal{A}_\ell = \left\{ \mathcal{C}_\ell(r, z, t) : \begin{array}{l} 0 < q < q_{\text{big}}, \quad \frac{1}{2} \leq q/Q \leq 2, \quad \tau \geq 0, \\ X_a \leq r^2/(2q) \leq X_b \end{array} \right\}.$$

Select the indices, representatives, and labels by

$$\begin{aligned} \mathcal{I}_\ell &= \{a : B_{\ell,a} \cap \mathcal{A}_\ell \neq \emptyset\}, \quad x_{\ell,a}^0 \in B_{\ell,a} \cap \mathcal{A}_\ell, \\ \Gamma &= \{(\ell, a, \sigma) : \ell \geq \ell_0, a \in \mathcal{I}_\ell, \sigma \in \{+, -\}\}. \end{aligned} \quad (6.8)$$

For $\gamma = (\ell, a, \sigma) \in \Gamma$, define its slow cutoff and its closed support in the physical slow coordinates by

$$\begin{aligned}\eta_\gamma(R, Z, T) &= \chi_\ell(q) \chi_{\ell, a}(R, Z, T), \\ K_\gamma &= \text{supp}_{(r, z, t)}(\eta_\gamma \circ \mathcal{C}_\ell).\end{aligned}\tag{6.9}$$

The supports are taken in the coordinate domain, including its smooth extension near $\tau = 0$. The two signs copy the same cutoff. So the partition identity on the active shell is

$$\eta_{(\ell, a, +)} = \eta_{(\ell, a, -)}, \quad \sum_{\ell} \sum_{a \in \mathcal{I}_\ell} (\eta_{(\ell, a, +)} \circ \mathcal{C}_\ell)^2 = 1.$$

The representatives and every other discrete choice are fixed before any derivatives are taken.

On these supports, R stays in one fixed compact subinterval of $(0, \infty)$, while Z, T stay in fixed bounded intervals. These bounds may depend on the fixed profile scale C . Derivatives of these cutoffs, of any fixed order in (R, Z, T) , are bounded by powers of S_* . The same remains true on fixed small enlargements inside the coordinate domain, using the smooth extension of the base from Proposition 5.5.

Give each label a small rectangle in its band torus. Choose its coordinates along v_r, v_t , so the pulse coordinate moves at unit speed under t_* . A center $c_\gamma \in \mathbb{T}^2$ will be chosen for every label. Write $\pi_j(Y) = J_g^j Y \pmod{\mathbb{Z}^2}$ for the covering in (6.5). Given $r_0 > 0$, define the band rectangle, its enlargement, and their preimages on the absolute torus by

$$\begin{aligned}\mathcal{R}_\gamma &= \{c_\gamma + \xi v_r + \eta v_t : |\xi|, |\eta| < r_0\} \pmod{\mathbb{Z}^2}, \\ \mathcal{R}_\gamma^+ &= \{c_\gamma + \xi v_r + \eta v_t : |\xi|, |\eta| < 2r_0\} \pmod{\mathbb{Z}^2}, \\ \mathcal{R}_\gamma^{\text{abs}} &= \pi_{i(\ell)}^{-1}(\mathcal{R}_\gamma), & \mathcal{R}_\gamma^{\text{abs},+} &= \pi_{i(\ell)}^{-1}(\mathcal{R}_\gamma^+).\end{aligned}\tag{6.10}$$

Take r_0 small enough that the enlarged parametrizations are injective, so they never give the same point from two different coordinate pairs. Choose a representative of c_γ in $\mathbb{R}^2$. On a local lift of $\mathcal{R}_\gamma$, the coordinates are

$$\begin{aligned}Y_i - c_\gamma - k_{\text{copy}} &= \xi_g v_r + \eta_g v_t, & |\xi_g|, |\eta_g| &< r_0, \\ v &= \frac{\eta_g + r_0}{c_i}, & L_s &= \frac{2r_0}{c_i} \asymp S_*.\end{aligned}\tag{6.11}$$

Here $k_{\text{copy}} \in \mathbb{Z}^2$ chooses the local lift, and the same coordinates extend to $\mathcal{R}_\gamma^+$. The dual coordinates satisfy $\mathcal{L}_i \eta_g = N_i \xi_g = 0$ and $N_i \eta_g = \mathcal{L}_i \xi_g = 1$. Since c_i is constant within the chart, (6.6) gives

$$D_r v = D_z v = 0, \quad t_* v = 1, \quad N_i \xi_g = 0.\tag{6.12}$$

So v measures normalized time along a rectangle and stays constant under the normalized spatial derivatives. The centers will now be chosen so that these local coordinates can be used without creating cross products between different local oscillations.

Lemma 6.1. *For the labels Γ and slow supports K_γ in Equations (6.8) and (6.9), there are centers c_γ and one radius $r_0 > 0$, shared by every label and independent of the band, with two properties. First, the enlarged rectangles in (6.10) are parametrized injectively. Second,*

$$\gamma \neq \gamma', \quad K_\gamma \cap K_{\gamma'} \neq \emptyset \quad \implies \quad \overline{\mathcal{R}_\gamma^{\text{abs},+}} \cap \overline{\mathcal{R}_{\gamma'}^{\text{abs},+}} = \emptyset.\tag{6.13}$$

Proof. Step 1: bound how many interactions are possible. If two dyadic supports meet, then $q/Q, q/Q' \in [1/2, 2]$, so $|\ell - \ell'| \leq 2$. Enlarge every mesh box by one fixed factor. Join two different labels when their enlarged slow boxes meet in physical coordinates and their band indices differ by at most four. Also join the two opposite signs of the same box. The resulting graph has

bounded degree. In fact, when $|\ell - \ell'| \leq 4$, the change of physical scale in each coordinate is bounded, and so is the ratio $\ell^2/(\ell')^2$. A box can then meet only a bounded number of enlarged boxes in any of those bands. Neither the degrees nor the number of colors needed depends on the band.

The differences between the covering indices $i(\ell)$ also have one fixed bound. If every band has $\ell \geq \ell_0$, then $|\ell - \ell'| \leq 4$ gives

$$|i(\ell) - i(\ell')| \leq 1 + \frac{4(1+h)\log 2 + 8/\ell_0}{\log T_g} \leq \Delta_{\max}, \quad (6.14)$$

after choosing an integer upper bound. To get this estimate, subtract $((1+h)\ell \log 2 - 2 \log \ell) / \log T_g$ at the two band indices and use the mean value theorem. Taking the floor adds at most one.

Step 2: separate the centers under the coverings that matter. Color the countable graph greedily, using only finitely many colors. Choose one rational center $c_\nu \in \mathbb{T}^2$ for each color ν , and set $c_\gamma = c_\nu$ whenever γ has that color. Choose the color centers so that all the following ordered constraints hold.

$$c_\mu \neq J_g^\Delta c_\nu \pmod{\mathbb{Z}^2} \quad (0 \leq \Delta \leq \Delta_{\max}), \quad \text{unless } (\Delta, \nu, \mu) = (0, \nu, \nu). \quad (6.15)$$

Every excluded equality is a proper closed condition on the finite tuple of centers. When the two colors are the same and $\Delta > 0$, this follows because $J_g^\Delta - I$ is invertible. The complement of all these conditions is open and dense, and it contains a rational tuple. This leaves every one of the finitely many forbidden differences a positive distance from the integer lattice.

Step 3: choose one size for every rectangle. Take rectangles in the eigenvector coordinates around these centers. Make them small enough both to be injective on the band torus and to keep the separation after the fixed enlargement. If a point belonged to the lifted rectangles of two adjacent labels at levels i and $i + \Delta$, it would give $c_\mu - J_g^\Delta c_\nu = J_g^\Delta e_\nu - e_\mu$ modulo the lattice, with $|e_\nu| + |e_\mu| \leq Cr_0$. Since Δ has a fixed finite bound, one fixed r_0 small enough makes this contradict (6.15). Step 1 joined every pair whose slow supports overlap. This proves (6.13), including the separation needed for the supports of derivatives. $\square$

For the rest of the construction, fix the centers c_γ and the radius r_0 from Lemma 6.1, along with one integer $\Delta_{\max}$ that satisfies (6.14). In particular, take a family $(F_\gamma)_{\gamma \in \Gamma}$ whose supports in (r, z, t, Y) are uniform in θ . Then

$$[\text{supp } F_\gamma \subset K_\gamma \times \mathcal{R}_\gamma^{\text{abs},+} \text{ for all } \gamma \in \Gamma] \implies F_\gamma F_{\gamma'} = 0 \quad (\gamma \neq \gamma').$$

The same identity holds for derivatives of smoothly extended fields, and it still holds after evaluation on (6.3).

Choose a nonzero smooth transverse cutoff $\chi_g \in C_c^\infty((-r_0, r_0))$, with $0 \leq \chi_g \leq 1$. Rescale one fixed function to get the time cutoff $0 \leq \psi \leq 1$, with

$$\psi(v) = 1 \quad \text{if } |v - L_s/2| \leq L_s/5, \quad \text{supp } \psi \subset \{|v - L_s/2| < L_s/3\}. \quad (6.16)$$

These cutoffs stay strictly inside the enlarged rectangles. Once the pulse equation has been solved there, multiplication by ψ keeps its time support inside the rectangle.

6.3 One common torus for overlapping bands

The rectangle $\mathcal{R}_\gamma$ from (6.10), for a label in band ℓ , is written in the band coordinate $Y_i = \pi_{i(\ell)}(Y)$ from (6.5). Before fields from overlapping bands are added, put all these rectangles and fields on the same torus H without changing their physical values. Lemma 6.2 gives uniformly bounded derivative factors

for this change of representation and keeps normalized Haar averages unchanged. Fix (z, t) with $0 < q(z, t) < q_{\text{big}}$. In a small enough neighborhood U of this point, set

$$\begin{aligned} \mathcal{D}_\ell &= \text{supp}_{(z,t)} \chi_\ell(q(z, t)), \quad \mathcal{L}(U) = \{\ell : \mathcal{D}_\ell \cap U \neq \emptyset\}, \\ i_0 &= \min_{\ell \in \mathcal{L}(U)} i(\ell), \quad H = \pi_{i_0}(Y). \end{aligned} \quad (6.17)$$

Choose U small enough that any two of its band indices differ by at most four. Equation (6.14) then gives the factorization

$$\begin{aligned} \Delta_\ell &= i(\ell) - i_0 \in \{0, \dots, \Delta_{\max}\}, \quad \pi_{i(\ell)} = \pi_{\Delta_\ell} \circ \pi_{i_0}, \\ Y_i &= J_g^{\Delta_\ell} H, \quad \mathcal{R}_\gamma^H = \pi_{\Delta_\ell}^{-1}(\mathcal{R}_\gamma), \quad \mathcal{R}_\gamma^{\text{abs}} = \pi_{i_0}^{-1}(\mathcal{R}_\gamma^H). \end{aligned} \quad (6.18)$$

Define $\mathcal{R}_\gamma^{H,+}$ by taking the same preimage with $\mathcal{R}_\gamma^+$. The map π_{i_0} is onto, so the separation in (6.13) also holds for these rectangles on the common torus. A sum of fields from different bands can now be written as

$$\sum_{\ell \in \mathcal{L}(U)} f_\ell(Y_{i(\ell)}) = \sum_{\ell \in \mathcal{L}(U)} f_\ell(J_g^{\Delta_\ell} H).$$

Lemma 6.2. *Take U, i_0, H as in (6.17). Every change $H \mapsto Y_i$ has derivative factors bounded by one constant that works for every band. The set $\mathcal{R}_\gamma^H$ is made of $14^{\Delta_\ell} \leq 14^{\Delta_{\max}}$ disjoint lifts of $\mathcal{R}_\gamma$. Whenever a function is pulled back from the torus in question, its Haar average is the same on the absolute, common, and band tori. Radial integration at fixed (z, t) , translation on the torus, averaging, and directional Fourier inversion on zero-mean functions all preserve the property of descending to the common torus and introduce no new dyadic band.*

Proof. Step 1: compare the covering maps and their averages. Write $\Delta = \Delta_\ell$. The matrix norms of J_g^Δ are bounded because $0 \leq \Delta \leq \Delta_{\max}$. The preimage rectangles are indexed by the finite quotient below.

$$\begin{aligned} \mathcal{R}_\gamma^H &= \bigsqcup_{[k] \in \mathbb{Z}^2 / J_g^\Delta \mathbb{Z}^2} \{J_g^{-\Delta}(c_\gamma + k + \xi v_r + \eta v_t) \pmod{\mathbb{Z}^2} : |\xi|, |\eta| < r_0\}, \\ \#(\mathbb{Z}^2 / J_g^\Delta \mathbb{Z}^2) &= |\det J_g|^\Delta = 14^\Delta. \end{aligned}$$

Here c_γ is the representative already chosen in $\mathbb{R}^2$. Compatibility of the Haar averages can be checked on Fourier characters.

$$\int_{\mathbb{T}^2} f(J_g^\Delta H) dH = \int_{\mathbb{T}^2} f(Y_i) dY_i. \quad (6.19)$$

A character with frequency n pulls back to one with frequency $(J_g^\Delta)^T n$. This new frequency is zero exactly when $n = 0$, which proves the equality.

Step 2: keep the common representation under all the stated operations. Radial integration fixes (z, t) , so it also fixes $q = q(z, t)$. It leaves every band cutoff unchanged, even if the integral passes through many radial grid boxes. Translations on the torus commute with deck translations. Averaging and directional Fourier multipliers keep the lattice of pullback frequencies unchanged. This proves every preservation claim in the lemma. $\square$

Fields are defined on the extended domain, where Y is independent, and use the original physical normalization. The common torus only gives a local way to represent them. Take two overlapping neighborhoods U, V , fix the same normalized band chart, and let f_U, f_V be the two common-torus representations. They agree by the identity

$$f_U(R, Z, T, J_g^{i_0(U)} Y) = f_V(R, Z, T, J_g^{i_0(V)} Y) \quad (6.20)$$

at every point over $U \cap V$. Both sides are the original field in that chart, so the identity still holds on the support closures where the set of overlapping dyadic supports changes.

Fix a band $\ell \in \mathcal{L}(U)$ and write $\Delta = \Delta_\ell$. On its chart, a field that descends to $H = J_g^{i_0} \gamma$ uses the operators in (6.6) with i replaced by i_0 ; the value of Q does not change. This gives $c_{i_0} = T_g^{-\Delta} c_i$ and $M_{i_0} = \Lambda_g^{-\Delta} M_i$. Equation (6.18) gives a fixed bound on Δ , so changing tori does not change the powers of ε in the derivative bounds.

Radial integration may collect more terms than pointwise multiplication. The next lemma counts the sums over slow labels that can appear. When several bands are compared, write $S_\ell = \ell^2$.

Lemma 6.3 (Number of relevant labels). *There is a constant C , independent of the band and the point, such that*

$$\sup_{(r,z,t)} \#\{\gamma \in \Gamma : (r,z,t) \in K_\gamma\} \leq C.$$

Fix a band ℓ , a compact chart set $B \subset \mathbb{R}^3$, and a bounded normalized radial interval I_R . Then

$$\begin{aligned} \#\{(\ell, a, \sigma) \in \Gamma : B_{\ell,a} \cap B \neq \emptyset\} &\leq C_B S_\ell^9, \\ \sup_{Z,T} \#\{(\ell, a, \sigma) \in \Gamma : B_{\ell,a} \cap (I_R \times \{Z\} \times \{T\}) \neq \emptyset\} &\leq C_{I_R} S_\ell^3. \end{aligned}$$

The same bounds hold after enlarging the mesh boxes by any fixed factor.

Proof. At one point, the dyadic partition and every one-dimensional grid partition have bounded overlap. The two signs contribute a factor of two. On a bounded interval, a mesh of size S_ℓ^{-3} has $O(S_\ell^3)$ positions. A compact chart lets all three coordinates vary, so it contains $O(S_\ell^9)$ boxes. A radial line lets only one coordinate vary, so it meets $O(S_\ell^3)$ boxes. Enlarging the boxes by a fixed factor changes only the constants. $\square$

The possible bands stay fixed during radial integration.

$$\mathcal{L}(z, t) = \{\ell : (z, t) \in \mathcal{D}_\ell\}, \quad q = q(z, t) \text{ is independent of } r.$$

The chosen slow supports lie in one uniformly bounded normalized radial interval. Lemma 6.3 limits the number of labels met by the integral to $O(S_{\text{ref}}^3)$, where $S_{\text{ref}} = \ell_{\text{ref}}^2$ for any $\ell_{\text{ref}} \in \mathcal{L}(U)$. All these S_ℓ are comparable. Counting the lifts on the common torus adds at most the fixed factor $14^{\Delta_{\text{max}}}$. The sum over the labels can grow by a polynomial in S_* , but it does not change the bands or the differences between their covering indices.

To solve an amplitude equation from the beginning of a pulse up to its current coordinate v , use the earlier points on that same lifted rectangle. The slow and transverse coordinates stay fixed along this path. A source built from several bands may be a function of H without being a function of one particular Y_i , because it may take different values at the different preimages of that point on the band torus. For this reason, describe the path separately on every lifted rectangle. The linear functional $\lambda_t(w) = v_t \cdot w / (1 + b_g^2)$ satisfies $\lambda_t(v_r) = 0$ and $\lambda_t(v_t) = 1$. Inside one lifted band rectangle, write $\eta_g(H) = \lambda_t(J_g^\Delta H - c_\gamma - k_{\text{copy}})$. For $w \in [0, L_s]$, define

$$\begin{aligned} H_w &= H + T_g^{-\Delta} c_i (w - v(H)) v_t \\ &= H_0 + T_g^{-\Delta} c_i w v_t, \quad H_0 = H - T_g^{-\Delta} (\eta_g(H) + r_0) v_t. \end{aligned} \tag{6.21}$$

This point has the same slow coordinates and the same band transverse coordinate ξ_g , while its pulse coordinate is w . Along this path, a source $f(R, Z, T, H)$ is evaluated as $f(R, Z, T, H_w)$. The other preimages are not averaged together. Since $\lambda_t J_g^\Delta = T_g^\Delta \lambda_t$, fixing w gives

$$D_H H_w = I - v_t \lambda_t, \quad D_H v = \frac{T_g^\Delta}{c_i} \lambda_t = O(S_*). \tag{6.22}$$

Every higher derivative of these affine maps is zero. In the original auxiliary coordinate Y , the same path is $Y' = Y + T_g^{-i}(\eta'_g - \eta_g)v_t$. A deck translation that keeps H unchanged translates this path by the same amount and relabels the copy of the rectangle. It follows that the unique solution with zero initial data descends to the common torus. Proposition 7.2 proves the exact differentiated inverse estimate. Equation (6.22) supplies the needed coordinate change without adding another power of ε .

6.4 Coefficient bounds and their algebra

The residual estimates have to track both the size of a correction and the way it vanishes at the edge of its support. Let f, g stand for angular mean coefficients, and let a, b stand for labelled wave amplitudes. Their products and normalized derivatives occur in the correction equations, for example

$$fg, \quad fa, \quad ab, \quad D_r f, \quad D_r a.$$

Definitions 6.4 and 6.5 introduce three classes. They record the size of a mean coefficient, the size of a wave amplitude, and the size of a radial moment, in that order.

$$f \in M_\alpha, \quad a_{\gamma, m} \in W_\alpha, \quad F(Z, T) \in S_\alpha.$$

The exponent α records a power of ε . The weight ζ in (6.23) measures how the function vanishes in the radial direction. An envelope P_v measures its decay in time and satisfies $0 < P_v \leq 1$ when $0 \leq v \leq L_s$. Equation (7.12) will give the formula for this envelope. The estimates are required at every fixed derivative order. This makes the extensions across the shell smooth and keeps the bounds valid under the products and derivatives in Proposition 6.6.

Recall the flat edge weight from Theorem 4.6. On the fixed active interval $X_a < X < X_b$, it can be written as

$$\zeta(X) = \exp\left(-\frac{a_a}{\log^2(X/X_a)} - \frac{a_b}{\log^2(X_b/X)}\right), \quad \delta(X) = \min\{1, \log(X/X_a), \log(X_b/X)\}, \quad (6.23)$$

where $a_a, a_b > 0$ are fixed. Extend ζ by zero outside the interval. For every $b > 0$ and every finite N , the quantity $\zeta^b \delta^{-N}$ goes to zero faster than every power at either endpoint. Both ζ and δ depend only on X , so they remain constant along the path in (6.21).

The phase varies rapidly, but its amplitude has a different derivative scale. A *coefficient derivative* differentiates only the amplitude after the oscillatory exponential has been factored out. Fix a finite stage j and use normalized chart representatives. For a coefficient a , write

$$\partial^I a = \partial_R^{I_1} \partial_Z^{I_2} \partial_T^{I_3} \partial_{H_1}^{I_4} \partial_{H_2}^{I_5} a, \quad I \in \mathbb{N}_0^5.$$

These derivatives keep the band, label, representative, and harmonic integer fixed. The derivatives in the auxiliary coordinates measure each representative separately, and the chain rule transforms them when i_0 changes. In the definitions below, $C_{j,I} > 0$ and $b_{j,I}, d_{j,I} \geq 0$ may depend on j, I and the fixed profile data. They are uniform in the band, label, rectangle copy, and point.

Definition 6.4 (Mean and moment coefficients). A smooth coefficient $f = f(R, Z, T, H)$ belongs to M_α when it satisfies all of the following conditions. It descends to every common torus that applies, and its representatives satisfy (6.20). It extends smoothly by zero outside the active radial shell. On $X_a < X < X_b$, it satisfies

$$\begin{aligned} \partial_\theta f &= 0, \quad \forall I \in \mathbb{N}_0^5 \exists C_{j,I}, b_{j,I}, d_{j,I} : \\ |\partial^I f| &\leq C_{j,I} \varepsilon^\alpha S_*^{b_{j,I}} \zeta \delta^{-d_{j,I}}. \end{aligned} \quad (6.24)$$

The coefficient may depend on H . It does not have to stay inside an auxiliary rectangle or satisfy a pulse-envelope bound. For a function $F = F(Z, T)$, define

$$F \in S_\alpha \iff \forall I \in \mathbb{N}_0^2 \exists C_{j,I}, b_{j,I} : \quad (6.25)$$

$$|\partial_{Z,T}^I F| \leq C_{j,I} \varepsilon^\alpha S_*^{b_{j,I}}.$$

Normalization of radial moments. The moments measured in S_α include the radial measure and the radial weight in their normalization. If $f_* = Q^{a_f} f_{\text{phys}}$, then

$$M_*(Z, T) := \int R^e \langle f_* \rangle_Y dR = Q^{a_f - (e+1)/2} \int r^e \langle f_{\text{phys}} \rangle_Y dr. \quad (6.26)$$

The auxiliary Haar average removes the dependence on Y . By (6.19), the average can be taken on any common torus that applies. The same scaling identity holds before averaging.

The phase, envelope, and support of a wave. For every label γ , prescribe the local data

$$\begin{aligned} \Phi_\gamma &= p_\gamma \theta + \varphi_\gamma(R, Z, T, H), & k &= \lceil \varepsilon^{-1/2} \rceil, & kp_\gamma &\in \mathbb{Z} \setminus \{0\}, \\ 0 < P_v &\leq 1 \quad (0 \leq v \leq L_s), & \mathcal{H}_j &\subset \mathbb{Z} \setminus \{0\}, & |\mathcal{H}_j| &< \infty. \end{aligned} \quad (6.27)$$

Here p_γ is fixed for the label, and φ_γ is defined on its enlarged lifted rectangles. The harmonic set $\mathcal{H}_j$ does not depend on the band. Equations (7.3) and (7.12) give the phases and envelopes used in the pulse construction.

Write $x_* = (R, Z, T)$ and $K_\gamma^* = \mathcal{C}_\ell(K_\gamma)$. On each component of $\mathcal{R}_\gamma^H$, use the coordinates ξ_g, v from (6.11). The domains and the supports allowed inside them are

$$\begin{aligned} \mathcal{E}_\gamma &= \{(x_*, H) : H \in \mathcal{R}_\gamma^H\}, \\ \Omega_\gamma &= \{(x_*, H) \in \mathcal{E}_\gamma : x_* \in K_\gamma^*, \xi_g(H) \in \text{supp } \chi_g, X_a \leq X \leq X_b\}, \\ \Omega_\gamma^{\text{cut}} &= \Omega_\gamma \cap \{v(H) \in \text{supp } \psi\}. \end{aligned} \quad (6.28)$$

The slow coordinates range over the chart domain, and $X = QR^2/(2q)$. Replace the closed rectangle in $\mathcal{E}_\gamma$ by $\mathcal{R}_\gamma^{H,+}$ to get the open enlarged domain $\mathcal{E}_\gamma^+$.

Definition 6.5 (Labelled wave coefficients). Take $\gamma \in \Gamma$ and $m \in \mathbb{Z} \setminus \{0\}$. A smooth amplitude $a_{\gamma,m}$ on $\mathcal{E}_\gamma^+$ belongs to W_α when its local formulas agree whenever two lifts represent the same point on the common torus, satisfy (6.20) on overlapping neighborhoods, and obey

$$\partial_\theta a_{\gamma,m} = 0, \quad \text{supp}(a_{\gamma,m}|_{\mathcal{E}_\gamma}) \subset \Omega_\gamma.$$

It extends smoothly by zero across the edges of the slow and transverse supports and across both edges of the shell. On $\mathcal{E}_\gamma \cap \{X_a < X < X_b\}$, its bounds are

$$\begin{aligned} \forall I \in \mathbb{N}_0^5 \exists C_{j,I}, b_{j,I}, d_{j,I} : \\ |\partial^I a_{\gamma,m}| \leq C_{j,I} \varepsilon^\alpha S_*^{b_{j,I}} \sqrt{\zeta} \delta^{-d_{j,I}} P_v, \quad 0 \leq v \leq L_s. \end{aligned} \quad (6.29)$$

These derivatives apply to the amplitude, not to the oscillatory phase. The factor P_v is only a pointwise weight. Beyond $v = 0, L_s$, the continuation has neither an envelope bound nor a requirement to extend by zero in time.

Cutoff fields and sums. An actual cutoff coefficient also satisfies

$$\text{supp } a_{\gamma,m}^{\text{cut}} \subset \Omega_\gamma^{\text{cut}} \quad \text{and smooth zero extension in } v.$$

No divisibility by $\eta_\gamma, \chi_g, \psi$ is assumed.

Let a_ν be a finite or locally finite family of amplitudes, with labels $\gamma_\nu \in \Gamma$ and harmonics $m_\nu \in \mathcal{H}_j$, representing the real wave

$$w = \sum_\nu a_\nu e^{ik_{\gamma_\nu} m_\nu \Phi_{\gamma_\nu}}.$$

Here k_γ is the integer k from (6.27) for the band containing γ . Before checking whether the wave belongs to W_α , combine all terms with the same label and harmonic.

$$w = \sum_\gamma \sum_{m \in \mathcal{H}_j} A_{\gamma,m} e^{ik_\gamma m \Phi_\gamma}, \quad A_{\gamma,m} = \sum_{\nu: (\gamma_\nu, m_\nu) = (\gamma, m)} a_\nu.$$

Then $w \in W_\alpha$ means that every $A_{\gamma,m}$ belongs to W_α , with bounds that are uniform as required in (6.29).

The bounds at every fixed derivative order make smooth zero extension across the shell edges possible. The weights ζ and $\sqrt{\zeta}$ absorb every allowed inverse power of δ , including the powers created by differentiation. The size and support of the products in the momentum residual can now be read from the following proposition.

Proposition 6.6. *Each class in Section 6.4 is closed under finite sums at a fixed exponent. If $\alpha, \beta \in \mathbb{R}$, then products satisfy*

$$M_\alpha M_\beta \subset M_{\alpha+\beta}, \quad M_\alpha W_\beta \subset W_{\alpha+\beta}. \quad (6.30)$$

If $a_{\gamma,m} \in W_\alpha$ and $b_{\gamma,m'} \in W_\beta$ have the same label, then

$$a_{\gamma,m} b_{\gamma,m'} \in \begin{cases} W_{\alpha+\beta}, & m + m' \neq 0, \text{ with harmonic } m + m', \\ M_{\alpha+\beta}, & m + m' = 0, \text{ after temporal cutoff.} \end{cases} \quad (6.31)$$

Before the cutoff, a zero harmonic satisfies the mean bounds on its labelled rectangle. Actual waves with different labels satisfy, for every pair of derivative multi-indices I, J ,

$$(\partial^I w_\gamma)(\partial^J w_{\gamma'}) = 0 \quad (\gamma \neq \gamma').$$

Let $\mathcal{C} \in \{M, W\}$. Every coefficient derivative in one fixed common-torus representation keeps the same ε exponent in its defining bounds. The operators below also preserve the compatibility condition (6.20), so they act on the classes as shown below.

operator	mapping	
$\partial_{R,Z,T}$	$\mathcal{C}_\alpha \longrightarrow \mathcal{C}_\alpha$	
D_r	$\mathcal{C}_\alpha \longrightarrow \mathcal{C}_{\alpha-\kappa_s}$	
$D_z = \varepsilon \partial_Z$	$\mathcal{C}_\alpha \longrightarrow \mathcal{C}_{\alpha+1}$	
$Q^{1+h} N_{\text{abs}} = c_{i_0} N_{i_0}$	$\mathcal{C}_\alpha \longrightarrow \mathcal{C}_\alpha$	
$-\varepsilon \partial_T$	$\mathcal{C}_\alpha \longrightarrow \mathcal{C}_{\alpha+1}$	(6.32)

These mappings act on amplitudes after the phase has been factored out. Bounds for the local coordinate derivatives, including those for ∂_H , keep the stated supports unchanged. Before cutoff, the wave rules are local to the labelled rectangles. After cutoff, they hold for fields extended smoothly on the common torus. At every fixed stage, products, derivatives, and angular averages keep the harmonic set finite and independent of the band.

Proof. The triangle inequality proves closure under finite sums at a fixed exponent.

Step 1: multiply the coefficient bounds. Leibniz' formula bounds every fixed derivative of a product by finitely many derivative bounds for its two factors. The powers of S_* and δ^{-1} add

and may increase. The weight bounds in Equations (6.30) and (6.31) follow from $0 \leq \zeta \leq 1$, $0 < P_v \leq 1$, and

$$\zeta^2 \leq \zeta, \quad \zeta^{3/2} P_v \leq \sqrt{\zeta} P_v, \quad \zeta P_v^2 \leq \begin{cases} \zeta, & m + m' = 0, \\ \sqrt{\zeta} P_v, & m + m' \neq 0. \end{cases}$$

A mean coefficient may have wider radial or auxiliary support than a wave coefficient. Their product stays inside the wave's support and keeps its envelope bound.

Step 2: separate the labels and select the angular harmonics. The support of any derivative of a smoothly extended function stays inside the function's original closed support. Once the time cutoff has supplied that smooth extension, Lemma 6.1 removes every cross-label product of the actual fields. Before the cutoff, the coefficient algebra is used only on the labelled band rectangle.

Recall that $\langle \cdot \rangle_\theta$ is the physical angular average, while $\langle \cdot \rangle_Y$ is the normalized Haar average in the auxiliary variable. Lemma 6.2 allows the second average to be computed in any torus coordinates that apply. Within one label, the product has harmonic $m + m'$. Since kp_γ is a nonzero integer, angular averaging gives

$$\left\langle a_{\gamma,m} a_{\gamma,m'} e^{ik(m+m')\Phi_\gamma} \right\rangle_\theta = \begin{cases} a_{\gamma,m} a_{\gamma,m'}, & m + m' = 0, \\ 0, & m + m' \neq 0. \end{cases}$$

In particular,

$$\langle a_{\gamma,m} e^{ikm\Phi_\gamma} \rangle_\theta = 0 \quad (m \neq 0), \quad f^\circ = f - \langle f \rangle_Y \quad \text{for an angular mean } f.$$

The second formula is the auxiliary zero-mean projection from (3.9). It is different from both the physical angular mean and the radial prefix average $\mathcal{A}_X(f)$ in (4.6). A mean coefficient can still have a nonzero auxiliary projection f° . Differentiation leaves every harmonic integer unchanged, while products add the integers in pairs. It follows that finitely many operations keep the harmonic set finite at any fixed stage.

Step 3: apply the normalized derivative operators. On the active annulus, the coordinate coefficients of D_r and all their coefficient derivatives are bounded by $C_I \varepsilon^{-\kappa_s} S_*^{C_I}$. Together with (6.6), these bounds give the exponent estimates in (6.32). Passing from a band torus to a common torus adds only the bounded matrices from Lemma 6.2. The local ∂_H bounds follow directly from (6.24) and (6.29).

In a fixed band chart, $\partial_{R,Z,T}$ preserves (6.20) because the covering maps do not depend on the slow coordinates. On every common torus, the auxiliary terms $M_{i_0} \mathcal{L}_{i_0}$ and $c_{i_0} N_{i_0}$ represent the same absolute operators. The first is $Q^{d_r/2} \mathcal{L}_{\text{abs}}$, and the second is $Q^{1+h} N_{\text{abs}}$. The other operators in (6.32) also preserve compatibility. These derivative rules apply to the amplitude coefficients. Differentiating $e^{ikm\Phi_\gamma}$ also differentiates the phase and brings down the factor km , which remains explicitly present.

Differentiation, integration, translation, averaging, and Fourier multipliers in (R, Z, T, H) all preserve independence from θ . □

The convention for wave supports also works with the initial-value equations used to build the pulses. The path (6.21) keeps the same slow and transverse variables. If a source vanishes along that whole rectangle, then its linear solution with zero initial data is identically zero there. Multiplying by ψ then keeps it inside the original rectangle. A smooth extension of the source, together with smooth dependence of the linear equation on its data, gives a smooth extension at those boundaries. Proposition 7.2 proves the mapping estimates at every fixed derivative order. The

The correction cycle in Proposition 9.6 keeps every correction inside these prescribed supports, and none of its estimates divides by the original cutoff factors.

7 Building the oscillations and correcting the residual stress

Fix the background (u_B, p_B) from Proposition 5.5, along with the charts, slow cutoffs, and auxiliary rectangles from Equations (6.1), (6.9), and (6.10). The next step is to build waves whose averaged quadratic velocity products give the prescribed radial fluxes of azimuthal and axial momentum in the leading stress $\mathcal{T}_0$, with the physical normalization in (7.30). Their principal linearized evolution also has to match the background shear and viscous diffusion in (7.5). The supports make it possible to solve these problems locally and add the waves without cross terms between different labels, by Lemma 6.1. Later corrections use the same amplitude equation with a prescribed harmonic source, solved with the inverse from Proposition 7.2. A second linear operator, $\mathcal{L}$ from Proposition 7.6, prescribes a signed change in the leading covariance.

The same linear amplitude equation, set by the background shear and viscosity, lies behind each part of the construction. Choose phases for which the homogeneous solution t_σ^h in Lemma 7.4 first grows and then decays. Its Gaussian bound (7.16) makes it exponentially small near both ends of its interval. This solution is called a *pulse*. Compute the covariance, meaning the averaged quadratic velocity product, of two pulses with different phase gradients, and choose their amplitudes so that this covariance matches the stress. The strict cone condition (7.1) makes the squared amplitudes positive in Proposition 7.5. Once those positive amplitudes are fixed, linearizing there allows later stress increments with either sign. Finally, take curls of supported vector potentials so that the velocity is exactly incompressible, as Lemma 7.7 does. The extra velocity terms and the cutoff errors remain in the covariance estimate in Corollary 7.8 and in the residual identity (7.40).

Every calculation uses normalized representatives in the charts until the physical factors are shown. Before taking any derivative, fix the band, its representative, its discrete labels, and its integer frequencies. Constants for derivatives of any fixed order may depend on the fixed profiles and on that order. In applications, they may also depend on the finite correction stage. They stay uniform across the bands, labels, and copies of the rectangles.

7.1 Building the phase and a frame perpendicular to its gradient

Start with the phase Φ in (7.3). It follows transport by the background up to the defect controlled in (7.9). For a harmonic $e^{ikm\Phi}$, the leading incompressibility condition on its amplitude t_m is $\nabla_* \Phi \cdot t_m = 0$. The frame B in (7.8) spans this two-dimensional plane and will compare the undamped principal equation with a diagonal system that has one positive and one negative eigenvalue, as in (7.10).

Fix a label $\gamma = (\ell, a, \sigma)$ from (6.8). On each lifted rectangle with that label, use the slow point $x_* = (R, Z, T)$, the transverse coordinate ξ_g , and the pulse coordinate $v \in [0, L_s]$, where $L_s = 2r_0/c_i \asymp S_*$, as in (6.11). Recall from (6.1) that $Q = 2^{-\ell}$, $\varepsilon = Q^h$, and $S_* = \ell^2$. The normalized spatial derivatives are the ones in (6.6), and $\nabla_* f = (D_r f, R^{-1} \partial_\theta f, D_z f)$. A prime means ∂_v , with x_* and ξ_g held fixed. In coefficient estimates, D^I means ordinary derivatives in the listed slow and auxiliary coordinates. The separate operators D_r, D_z keep their normalized meanings.

Write the velocity at the base of the chart as $be_r + Ve_\theta + Ge_z$, and set $F = V/R$ and $g = (RF_R, G_R)$. Here F is the angular velocity, while g is the radial shear, with its tangential components ordered as (θ, z) . A tangential two-vector is identified with the vector in $\text{span}(e_\theta, e_z)$ that has those two components. Equation (5.42) gives $b = O(\varepsilon)$, and the tangential velocity differs from its order-zero value by $O(\varepsilon^2)$ in

every fixed slow derivative. A subscript 0 means that an order-zero quantity is evaluated at the chosen representative $x_{\ell,a}^0$.

The positive lower bounds in Theorem 4.6(iii) make the following tangential frame and growth parameters well defined.

$$N = \frac{g_0}{|g_0|}, \quad K = N^\perp, \quad \lambda_0^2 = -2F_0N_\theta(2F_0N_\theta + |g_0|) > 0, \quad c_0 = \frac{\lambda_0}{2F_0N_\theta} < 0.$$

Use the quarter turn $K = (-N_z, N_\theta)$, so N, K form an orthonormal pair in the tangential plane, and take $\lambda_0 > 0$. The definitions of a, b_s, v_s in Equations (4.11) and (4.20) give $g_0 = F_0(-a, b_s)$ and $\lambda_0^2 = 2aF_0^2(1 - 2/v_s)$. So the profile bounds already imply $\lambda_0^2 > 0$ and $c_0 < 0$; these are not extra choices.

The phase also has to keep the stress directions allowed by Lemma 4.5. Before N, K, c_0 are frozen, that pointwise condition says exactly the same thing as

$$\mathcal{T}_{0,*} \cdot N < 0, \quad \left| c_0 \frac{\mathcal{T}_{0,*} \cdot K}{\mathcal{T}_{0,*} \cdot N} \right| < 1, \quad \mathcal{T}_{0,*} = (Q/q)^{A+1/2} \mathcal{T}_0. \quad (7.1)$$

For this equivalence, evaluate the frame, c_0 , and the target at the same point. The reason is that $c_0^2 = (v_s - 2)/2$, while the absolute value of the quotient without c_0 is $|J_c|/(P_c - v_s)$. The stress vanishes at the boundary of an annulus. At that boundary, the quotient means the one-sided limit coming from the smooth unit stress direction in Theorem 4.6(iii).

Because the inequality has a strict margin on the closed annulus, choose one $u_* > 0$ for which $u_*/\sqrt{1+u_*^2}$ is larger than the supremum of $|c_0(\mathcal{T}_{0,*} \cdot K)/(\mathcal{T}_{0,*} \cdot N)|$, using the interpretation above at the edges. Now freeze N, K, c_0 at each representative. Uniform continuity of the leading stress direction keeps the inequalities strict, with a uniform margin, throughout slow boxes that are small enough. The frequency and shear bounds, the positive lower bound for λ_0 , and the frame bounds also hold on fixed small enlargements of the active slow supports. Take these slow neighborhoods to be enlargements of the mesh boxes by one fixed factor relative to $\tau \geq 0$. Then every point stays within CS_*^{-3} of its representative. Derivatives at $\tau = 0$ are understood from one side. On the part of each neighborhood inside the closed active annulus, the uniform cone margin remains valid, with the limiting stress direction used at the radial edges. All of these bounds and margins are uniform across bands and labels.

The stress has two components, so each slow box uses two phases with different covariance directions. As a pulse moves along its rectangle, the phase gradient changes until viscous damping becomes larger than shear amplification. On the rectangle with sign $\sigma \in \{+1, -1\}$, define

$$k = \left\lceil \varepsilon^{-1/2} \right\rceil, \quad B_s^2 = \frac{2\lambda_0}{\varepsilon k^2 (1 + u_*^2)^{3/2}}, \quad s(v) = \sigma \left(\frac{u_*}{2} + \frac{u_* v}{L_s} \right). \quad (7.2)$$

Take $B_s > 0$. The carrier frequency k keeps εk^2 of order one, so viscosity stays in the leading amplitude equation. To choose the angular and axial wave numbers p, p_z , first set

$$(\tilde{p}/R_0, p_z) = B_s \left(K - \frac{\sigma u_* g_0}{L_s |g_0|^2} \right).$$

Angular periodicity requires kp to be an integer. Choose kp as a nearest nonzero integer to $k\tilde{p}$, using a fixed rule when there is a tie, and leave p_z unchanged. These wave numbers stay constant for the label. With $x_0 = \sigma B_s u_*/2$, define the phase on a lifted rectangle by

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v(pF + p_z G), \quad (7.3)$$

$$n_\Phi := \nabla_* \Phi = (x_0 - v(pF_R + p_z G_R), p/R, p_z - \varepsilon v(pF_Z + p_z G_Z)). \quad (7.4)$$

Given a source coefficient $f_m \in \mathbb{C}^3$ and a nonzero integer m , the principal amplitude problem asks for $t_m \in \mathbb{C}^3$ and a pressure coefficient $\pi_m \in \mathbb{C}$ such that $n_\Phi \cdot t_m = 0$ and

$$t'_m + \mathcal{K}t_m + m^2 dt_m + ikmn_\Phi \pi_m = -f_m, \quad d = \varepsilon k^2 |n_\Phi|^2. \quad (7.5)$$

The operator $\mathcal{K}$ is defined below. This equation keeps the transport in the pulse coordinate, the tangential velocity and radial shear of the base, and the viscous term in which both derivatives land on the exponential. The two factors $2F$ in $\mathcal{K}$ include the cylindrical connection terms, since radial motion changes both the tangential frame and the base swirl. Eliminating the normal pressure term while differentiating $n_\Phi \cdot t_m = 0$ gives the projected evolution operator $\mathcal{A}_\Phi$.

$$\mathcal{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad \mathcal{A}_\Phi = -\mathcal{K} + \frac{n_\Phi(n_\Phi^T \mathcal{K} - (n'_\Phi)^T)}{|n_\Phi|^2}. \quad (7.6)$$

Slow transport, the error in phase transport, and the remaining viscous terms belong to the residual beyond (7.5); Proposition 9.1 estimates them. The geometric coordinates and frame will also be needed later, so fix them now. Write $n_{\Phi, \tan} = (n_{\Phi, \theta}, n_{\Phi, z})$. Since $kp \in \mathbb{Z} \setminus \{0\}$ and $R > 0$, the component $n_{\Phi, \theta} = p/R$ is nonzero. This makes the following definitions valid.

$$K_a = \frac{n_{\Phi, \tan}}{|n_{\Phi, \tan}|}, \quad N_a = ((K_a)_z, -(K_a)_\theta), \quad s_a = \frac{n_{\Phi, r}}{|n_{\Phi, \tan}|}. \quad (7.7)$$

The corresponding basis of $n_\Phi^\perp$ and the frame built from it are

$$U = [e_r - s_a K_a, N_a], \quad B = U \begin{pmatrix} 1 & 1 \\ c_0 \sqrt{1+s^2} & -c_0 \sqrt{1+s^2} \end{pmatrix}. \quad (7.8)$$

The next lemma compares the actual phase normal and projected operator with their reference values. In these coordinates, the amplitude equation has two scalar reference growth rates plus a small matrix error.

Lemma 7.1. *For the labels and phases in Equations (6.8), (7.2), and (7.3), there is a sufficiently small $q_* > 0$, common to all labels and fixed derivative orders, with the following properties on every rectangle whenever $0 < q < q_*$. Every $e^{ikm\Phi}$, $m \in \mathbb{Z} \setminus \{0\}$ is single-valued in θ and has nonzero angular frequency. On the rectangle $0 \leq v \leq L_s$,*

$$|n_\Phi - B_s(s(v), K)| + |n'_\Phi| \leq C/S_*, \quad E_{ik} := (t_* + bD_r + F\partial_\theta + GD_z)\Phi = O(\varepsilon S_*^C). \quad (7.9)$$

The estimate for E_{ik} in (7.9) also holds after any fixed coefficient derivative, with a corresponding polynomial degree. The coefficients extend smoothly to the fixed enlargement. Every fixed mixed derivative in the slow, transverse, and pulse coordinates of n_Φ , and of the frames used here, is bounded by a polynomial in S_* .

The real 3×2 matrix B in (7.8) has columns that form a basis of $n_\Phi^\perp$. It is also viewed as a map $\mathbb{C}^2 \rightarrow \{t \in \mathbb{C}^3 : n_\Phi \cdot t = 0\}$. It has a left inverse B^ℓ that is uniformly bounded, and

$$B^\ell(\mathcal{A}_\Phi B - B') = \text{diag}(\lambda, -\lambda) + E, \quad \lambda(v) = \frac{\lambda_0}{\sqrt{1+s(v)^2}}, \quad |E| \leq C/S_*. \quad (7.10)$$

For the damping coefficient d in (7.5), set $d_{\text{ref}} = \varepsilon k^2 B_s^2(1+s^2)$. Then

$$|d - d_{\text{ref}}| \leq C/S_*. \quad (7.11)$$

Proof. Step 1: periodicity and estimates for the phase. Because $kp \in \mathbb{Z} \setminus \{0\}$, the angular frequency kmp is a nonzero integer for every $m \neq 0$. This proves the periodicity claim.

Write $H_\Phi = pF + p_z G$. Since $D_r v = D_z v = 0$ and the base does not depend on the auxiliary variable,

$$D_r \Phi = x_0 - v(H_\Phi)_R, \quad R^{-1} \partial_\theta \Phi = p/R, \quad D_z \Phi = p_z - \varepsilon v(H_\Phi)_Z.$$

These are exactly the three entries of n_Φ . Before rounding, $(p/R_0, p_z) \cdot g_0 = -\sigma B_s u_*/L_s$. The slow box has diameter $O(S_*^{-3})$, comparing with the base contributes $O(\varepsilon^2)$, and rounding contributes $O(k^{-1})$. The error in $pF_R + p_z G_R$ is therefore $O(S_*^{-3} + \varepsilon^2 + k^{-1})$. Multiply this by $v = O(S_*)$ and include the tangential error $O(S_*^{-1} + \varepsilon S_*)$; after one choice of q_* , this proves the first estimate in (7.9). Differentiating in the pulse coordinate gives $|n'_\Phi| \leq C/S_*$. Any fixed number of derivatives introduces only powers of S_* ; the discrete rounding is not differentiated. For the phase transport, compute each term.

$$t_* \Phi = -H_\Phi + \varepsilon v(H_\Phi)_T, \quad F \partial_\theta \Phi + G D_z \Phi = H_\Phi - \varepsilon v G(H_\Phi)_Z.$$

It follows that

$$E_{ik} = \varepsilon v((H_\Phi)_T - G(H_\Phi)_Z) + b n_{\Phi,r}.$$

Every term contains either a factor ε or a factor $b = O(\varepsilon)$, and all remaining derivatives of fixed order are bounded by polynomials in S_* . This proves both the value estimate and the estimates for the differentiated phase defect. Since $\varepsilon k^2 \asymp 1$ and B_s is bounded above and below, (7.11) follows.

Step 2: coordinates on the moving plane. Next, bound the frame in (7.8) and write the projected evolution in its coordinates. The phase comparison and the lower bound on B_s keep $|n_{\Phi,\tan}|$ bounded away from zero. The vector N_a in (7.7) is the perpendicular defined there, and it is close to N . Let M be the 2×2 matrix in (7.8). Then

$$B^\ell = M^{-1} \begin{pmatrix} e_r^T \\ N_a^T \end{pmatrix}, \quad \begin{pmatrix} e_r^T \\ N_a^T \end{pmatrix} U = I_2.$$

The matrices U, M, B and these left inverses are all uniformly bounded.

Step 3: diagonalization at the reference normal. For $n_{\text{ref}} = B_s(s, K)$, write $t = x(e_r - sK) + yN \in n_{\text{ref}}^\perp$, and form $\mathcal{K}_0$ from the reference coefficients. Since $g_0 = |g_0|N$,

$$\begin{aligned} (\mathcal{K}_0 t)_r &= 2F_0 s K_\theta x - 2F_0 N_\theta y, \\ n_{\text{ref}} \cdot \mathcal{K}_0 t &= 2B_s F_0 \{K_\theta(1 + s^2)x - sN_\theta y\}. \end{aligned}$$

Let $e = e_r - sK$. The vectors e, N are perpendicular, with squared lengths $1 + s^2, 1$. A direct calculation gives

$$-\mathcal{K}_0 e = (-2F_0 s K_\theta, -2F_0 e_\theta - g_0), \quad -\mathcal{K}_0 N = (2F_0 N_\theta, 0),$$

where each ordered pair lists the radial part first and the tangential part second. It follows that

$$e \cdot (-\mathcal{K}_0 e) = sK \cdot g_0 = 0, \quad e \cdot (-\mathcal{K}_0 N) = 2F_0 N_\theta, \quad N \cdot (-\mathcal{K}_0 e) = -(2F_0 N_\theta + |g_0|).$$

Divide the first coordinate by $1 + s^2$. In the (x, y) coordinates, the reference undamped matrix is then

$$\begin{pmatrix} 0 & 2F_0 N_\theta / (1 + s^2) \\ -(2F_0 N_\theta + |g_0|) & 0 \end{pmatrix}.$$

Its eigenvectors are the columns of M , and its eigenvalues are $\lambda, -\lambda$. To compare this matrix with the actual evolution, first use the value estimates for the base coefficients and for $n_\Phi - n_{\text{ref}}$. They change the algebraic projection $-\mathcal{K} + n_\Phi n_\Phi^T \mathcal{K} / |n_\Phi|^2$ by $O(S_*^{-1})$. Also, $K_a - K$ and $s_a - s$ are $O(S_*^{-1})$, so the actual frame and its left inverse differ from their reference values by the same order. Writing the algebraic projection in these frames therefore changes the reference matrix by $O(S_*^{-1})$. The moving-normal term $-n_\Phi(n'_\Phi)^T / |n_\Phi|^2$ has the same bound because $|n'_\Phi| = O(S_*^{-1})$ and $|n_\Phi|$ is bounded below. Also $s' = O(S_*^{-1})$, so the definitions of U, M give

$U', M' = O(S_*^{-1})$. So $B' = U'M + UM' = O(S_*^{-1})$, and multiplying by the bounded B^ℓ gives $-B^\ell B' = O(S_*^{-1})$. Together, these contributions give the error E in (7.10).

The same choice of q_* also has to work for every fixed derivative order. The value comparisons in Step 1 need only finitely many smallness conditions. For example, require

$$S_*^2(\varepsilon + \varepsilon^2 + k^{-1}) \leq 1$$

so that errors multiplied by $v = O(S_*)$ are at most C/S_* . Since $\varepsilon = 2^{-h\ell}$, $k^{-1} \leq \varepsilon^{1/2}$, and $S_* = \ell^2$, this holds for every sufficiently large band index. The bound $1 \leq \varepsilon k^2 \leq 4$ when $\varepsilon \leq 1$ gives fixed positive upper and lower bounds for B_s . Make the error in the phase normal smaller than half of this lower bound. Then $|n_{\Phi, \tan}| \geq B_s - C/S_*$ keeps the tangential part of the phase normal uniformly nonzero, and the same lower bound holds for $|n_\Phi|$. In the same way, $\det M = -2c_0\sqrt{1+s^2}$ and the fixed lower bound for $|c_0|$ keep $|\det M|$ bounded away from zero. So one lower bound on the band index, or equivalently one upper bound q_* on q , keeps every denominator in the frame and its left inverse uniformly nonzero.

Once that domain is fixed, the coefficient derivatives of the base coefficients and the phase normal n_Φ , up to any prescribed order, are bounded by $C_I S_*^{b_I}$. Apply the product and quotient rules to the displayed frame formulas. The lower bounds just proved then give bounds of the same form for U, M, B, B^ℓ and their pulse derivatives. Differentiating the phase defect keeps its factor ε , because the band is fixed and every fixed derivative of b is $O(\varepsilon)$. These arguments use bounds on higher derivatives, with constants and polynomial degrees allowed to depend on I , but they need no further smallness condition on those derivatives. The same q_* therefore works. $\square$

7.2 Solving the pulse equation and controlling its derivatives

The frame B from Lemma 7.1 turns (7.5) into an ordinary differential equation along each pulse. First solve it with zero initial amplitude and a prescribed source f_m . This gives the linear inverse used in Proposition 7.2 to remove later harmonics. Then solve the equation with zero source and a specified nonzero initial amplitude. That solution produces the pulses used for the leading covariance in Lemma 7.4. Both solutions must decay at both ends of the interval before they can be localized. Subtract the reference damping from the growing eigenvalue and define the shared envelope

$$P(v) = \exp \left(\int_{L_s/2}^v (\lambda(w) - d_{\text{ref}}(w)) dw \right). \quad (7.12)$$

Also write $P_v = P(v)$, as in the coefficient class (6.29). The estimates in (7.14) show that if a source is bounded by this envelope, then its response is bounded by the same envelope, apart from polynomial factors in S_* . Multiplying a solution by ψ changes its principal equation only where both the solution and the source are exponentially small, as shown after (7.40).

Proposition 7.2. *Fix $\alpha \in \mathbb{R}$, a finite set of harmonics $0 < |m| \leq M$, and a label at level i on a common torus $H = Y_{i_0}$, where $i - i_0$ is bounded as in Lemma 6.2. For each harmonic, let $f_m(x_*, H) \in \mathbb{C}^3$ be a coefficient that descends to H and is smooth on the fixed enlargement of the rectangles with that label. Assume that it has a fixed closed slow support inside the label's slow support, a compact transverse support inside $\text{supp } \chi_g$, and support in the closed active shell $X_a \leq X \leq X_b$. Across every one of these support boundaries, its extension by zero is smooth. Assume as well that, on $X_a < X < X_b$ and $0 \leq v \leq L_s$, every fixed multi-index I of derivatives in (R, Z, T, H) satisfies*

$$|D^I f_m| \leq C_I \varepsilon^\alpha S_*^{b_I} \sqrt{\zeta} \delta^{-a_I} P(v).$$

On the domain $0 < q < q_*$ fixed in Lemma 7.1, there is exactly one amplitude t_m with zero initial data that satisfies $n_\Phi \cdot t_m = 0$, and there is a pressure coefficient π_m for which (7.5) holds. They are given by

$$\begin{aligned} t'_m &= \mathcal{A}_\Phi t_m - m^2 dt_m - \text{proj}_{n_\Phi^\perp} f_m, & t_m(0) &= 0, \\ \pi_m &= \frac{i}{km} \frac{n_\Phi \cdot \mathcal{K} t_m - n'_\Phi \cdot t_m + n_\Phi \cdot f_m}{|n_\Phi|^2}. \end{aligned} \quad (7.13)$$

Here $\text{proj}_{n_\Phi^\perp} f = f - n_\Phi(n_\Phi \cdot f)/|n_\Phi|^2$ is the orthogonal projection. Evaluate the source along the path H_w from Equation (6.21), while holding $x_* = (R, Z, T)$ and the transverse coordinate fixed. The amplitude and pressure coefficients descend to the same common torus and keep the stated slow, transverse, and shell supports, including smooth extension by zero across their boundaries. The solution is defined on the whole closed pulse interval $0 \leq v \leq L_s$. On $X_a < X < X_b$, every fixed I satisfies

$$|D^I t_m| \leq C'_I \epsilon^\alpha S_*^{b'_I} \sqrt{\zeta} \delta^{-a'_I} P(v), \quad (7.14)$$

$$|D^I \pi_m| \leq C'_I \epsilon^{\alpha+1/2} S_*^{b'_I} \sqrt{\zeta} \delta^{-a'_I} P(v). \quad (7.15)$$

The constants and degrees may depend on M, I and on the prescribed source bounds. The threshold q_* does not depend on M, α or on the correction stage. If the source has uniform one-sided derivatives in the slow variables at $\tau = 0$ on compact sets where $q > 0$, then the solution and pressure have the same one-sided derivatives.

Proposition 7.2 requires only descent to the common torus H . The source may have different values at different preimages of one point on the band torus. It does not have to be perpendicular to n_Φ , and it does not have to be divisible by either rectangle cutoff. The support conclusion covers the slow, transverse, and shell variables. It does not claim extension by zero in time before multiplication by ψ .

Proof. The moving frame gives a two-dimensional equation. First bound its forward propagator by $P(v)/P(w)$. This propagator is the matrix that carries data at an earlier pulse coordinate to the homogeneous solution at a later one. Since the initial data are zero, Duhamel's formula says that the solution at the current coordinate is the integral of the earlier source values after the propagator carries each one forward. Use the same formula for derivatives. Finally, check the support, descent, and normal pressure equation.

Step 1: the envelope and forward propagator. The scalar reference growth rate is

$$a_{\text{net}}(y) = \frac{\lambda_0}{\sqrt{1+y^2}} - \frac{\lambda_0(1+y^2)}{(1+u_*^2)^{3/2}}, \quad y = |s(v)|.$$

It is zero at $y = u_*$, and its derivative in the pulse coordinate is

$$\frac{d}{dv} a_{\text{net}}(|s(v)|) = -\frac{u_* \lambda_0 |s(v)|}{L_s} \left((1+s(v)^2)^{-3/2} + 2(1+u_*^2)^{-3/2} \right).$$

Because $u_*/2 \leq |s(v)| \leq 3u_*/2$, this derivative lies between $-C/L_s$ and $-c/L_s$. By (7.12), $(\log P)' = a_{\text{net}}(|s(v)|)$, and both $\log P$ and its first derivative vanish at $v = L_s/2$. Integrating these derivative bounds twice from the midpoint gives

$$e^{-C(v-L_s/2)^2/L_s} \leq P(v) \leq e^{-c(v-L_s/2)^2/L_s} \leq 1. \quad (7.16)$$

The envelope therefore decays on both sides of the midpoint. To prove that the sourced amplitude t_m has the same decay, write $t_m = Bz_m$ in the actual moving frame. The coordinate vector z_m satisfies

$$z'_m = A_m z_m + g_m, \quad A_m = \text{diag}(\lambda, -\lambda) + E - m^2 dI_2, \quad g_m = -B^\ell \text{proj}_{n_\Phi^\perp} f_m. \quad (7.17)$$

Let $V_m(v, w)$ be the forward fundamental matrix for the homogeneous equation $z' = A_m z$, so it carries data from the coordinate in its second argument forward to the coordinate in its first. Normalize it by $V_m(w, w) = I_2$; then $\partial_v V_m(v, w) = A_m(v) V_m(v, w)$. Because the damping is scalar, there is an exact identity

$$V_m(v, w) = \exp\left(-(m^2 - 1) \int_w^v d(a) da\right) V_1(v, w). \quad (7.18)$$

For the $m = 1$ equation, the Euclidean norm satisfies $\partial_v |z| \leq (\lambda - d_{\text{ref}} + C/S_*)|z|$. Since L_s/S_* is bounded, this gives

$$\|V_m(v, w)\| \leq CP(v)/P(w), \quad 0 \leq w \leq v \leq L_s, \quad m \neq 0. \quad (7.19)$$

This value bound is uniform for every nonzero integer m , because the exponential in (7.18) is at most one.

Step 2: values and coefficient derivatives. The zero initial condition gives Duhamel's formula. The bound on $g_m(w)$ contains $P(w)$, which cancels the denominator in (7.19). Integrating then contributes $L_s = O(S_*)$.

$$z_m(v) = \int_0^v V_m(v, w) g_m(w) dw, \quad |z_m(v)| \leq C \varepsilon^\alpha S_*^{b_0+1} \sqrt{\zeta} \delta^{-a_0} P(v),$$

after increasing b_0 to cover the frame factors. To differentiate this formula, begin with the slow variables, the transverse coordinate at $v = 0$, and the pulse coordinate. In the next equation, D^I differentiates only the slow variables and the initial transverse coordinate, with v held fixed. These parameter derivatives commute with ∂_v , so differentiating the two-dimensional equation gives

$$(D^I z_m)' = A_m D^I z_m + D^I g_m + \sum_{0 < J \leq I} \binom{I}{J} (D^J A_m) D^{I-J} z_m. \quad (7.20)$$

All of these parameter derivatives of the zero initial data are zero. Suppose the coefficient derivatives satisfy $|D^J A_m| \leq C_J S_*^{c_J}$. An induction using (7.19) raises the polynomial degree by at most

$$1 + \max\{b_I, \max_{0 < J \leq I} (c_J + b'_{I-J})\}$$

at the I th step. The inverse-distance exponent can be increased to the largest of the source and lower-order exponents. The slow point stays fixed in the integral, so $\sqrt{\zeta}$ and δ stay fixed as well. Derivatives in the pulse coordinate then follow from (7.17). This proves every fixed-order derivative estimate without reducing α . Differentiating in the moving frame keeps the constraint $n_\Phi \cdot t_m = 0$ and includes the derivatives of the plane $n_\Phi^\perp$.

Step 3: common-torus coordinates and support. The estimates above used $\tilde{\zeta}_g, v$ on one lifted rectangle. To write the derivative estimates in H , use the directions v_r, v_t from (6.2) and recall the linear functional $\lambda_t(w) = v_t \cdot w / (1 + b_g^2)$. It has $\lambda_t(v_r) = 0$ and $\lambda_t(v_t) = 1$. If the current point is H and its pulse coordinate is $v(H)$, then the point on the same path with pulse coordinate w is

$$H_w = H + T_g^{-(i-i_0)} c_i (w - v(H)) v_t.$$

The vector v_t is an eigenvector of J_g with eigenvalue T_g , so

$$D_H H_w = I_2 - v_t \lambda_t, \quad D_H^a H_w = 0 \quad (a \geq 2), \quad D_H v = T_g^{i-i_0} \lambda_t / c_i = O(S_*).$$

Thus the pullback $H \mapsto H_w$ has bounded derivatives when w is held fixed. Returning to the current common coordinate adds only powers of S_* multiplying fixed derivatives in the pulse coordinate. Together with Step 2, this proves (7.14) in the stated coefficient variables. Deck translations reindex the copies and commute with this path. Uniqueness for the initial-value

equation with zero data proves descent to H ; it does not identify sources at different preimages of the finer band torus. The path keeps the slow and transverse points fixed. If the source vanishes along their whole rectangle, the solution vanishes there too. Apply the differentiated equation at the support boundaries to get smooth extension by zero and containment of the support.

Step 4: the constraint and pressure. The definition of $\mathcal{A}_\Phi$ gives

$$n_\Phi \cdot t'_m = -n'_\Phi \cdot t_m - m^2 d(n_\Phi \cdot t_m), \quad (n_\Phi \cdot t_m)' = -m^2 d(n_\Phi \cdot t_m).$$

Zero initial data therefore keep $n_\Phi \cdot t_m = 0$. Substituting this into the projected equation gives

$$t'_m + \mathcal{K}t_m + m^2 dt_m + f_m = \frac{n_\Phi}{|n_\Phi|^2} (n_\Phi \cdot \mathcal{K}t_m - n'_\Phi \cdot t_m + n_\Phi \cdot f_m).$$

Multiply (7.13) by $ikmn_\Phi$; the result is the negative of this normal vector. Thus (t_m, π_m) satisfies the full principal cancellation (7.5). The factor $(km)^{-1} = O(\varepsilon^{1/2})$, together with the fixed bounds on coefficient derivatives, proves (7.15). This cancels the normal components of the source as well. On a compact set where $q > 0$, the base has the one-sided endpoint derivative data from Proposition 5.5. The same parameter-dependent equations and differentiated Duhamel formulas carry those endpoint derivatives from the source to t_m , and then to π_m . $\square$

The harmonic factor in (7.18) only adds damping. Later higher harmonics therefore do not require a smaller space-time domain for these inverses.

Corollary 7.3. *The solution operators $f_m \mapsto (t_m, \pi_m)$ for the projected amplitude equation with $t_m(0) = 0$ are defined at every finite correction stage on the same domain $0 < q < q_*$. Enlarging the finite set of harmonics or raising the derivative order changes the constants and polynomial degrees, but it does not require a smaller domain.*

Proof. The frame and damping comparisons need only the one finite choice made in Lemma 7.1. The exact factor in (7.18) makes every higher harmonic at least as damped in value as the first harmonic. In (7.20), the extra factors of m and the coefficient derivatives raise only the constants and polynomial degrees in the source bounds. The already chosen threshold q_* therefore remains valid. $\square$

The zero-data inverse can now handle prescribed harmonic sources. The leading stress needs something different, a nonzero real solution whose radial component stays comparable to $P(v)$. Choose its initial value along the growing coordinate of the frame B . Its amplitude and direction must stay controlled while it later decays.

Lemma 7.4. *For each sign σ , let $t_\sigma^h = B(z_+, z_-)^T$ solve $(t_\sigma^h)' = \mathcal{A}_\Phi t_\sigma^h - dt_\sigma^h$, with $z_+(0) = P(0)$ and $z_-(0) = 0$. This fixes the real amplitude for the homogeneous $m = 1$ problem. Its pressure is given by (7.13) with $f_m = 0$. Using K_a, N_a, s_a from (7.7), write $t_\sigma^h = x(e_r - s_a K_a) + y N_a$. Then*

$$cP(v) \leq x(v) \leq CP(v), \quad \frac{y(v)}{x(v)} = c_0 \sqrt{1 + s(v)^2} + O(S_*^{-1}). \quad (7.21)$$

Every fixed derivative of this homogeneous solution in either the coefficient variables or the pulse coordinate is bounded by $C_1 S_^{b_1} P(v)$. Its homogeneous pressure, given by (7.13) with $f = 0$, has the extra factor $\varepsilon^{1/2}$.*

Proof. First control the ratio between the decaying and growing coordinates of the frame, then convert those coordinates to the radial and tangential coordinates x, y . Write $E = (E_{ab})$. As long as $z_+ > 0$, the ratio $r = z_-/z_+$ satisfies

$$r' = E_{21} + (-2\lambda + E_{22} - E_{11})r - E_{12}r^2, \quad r(0) = 0.$$

Let $c_\lambda > 0$ be a uniform lower bound for λ . Choose a sufficiently large fixed K_b . Then, whenever S_* is sufficiently large,

$$\pm r' \big|_{r=\pm K_b/S_*} \leq \frac{-2c_\lambda K_b + C(1 + K_b/S_* + K_b^2/S_*^2)}{S_*} < 0.$$

It follows that $|r| \leq K_b/S_*$. Since $z_+(0) = P(0) > 0$, the definition of the envelope gives

$$\left(\log \frac{z_+}{P}\right)' = -(d - d_{\text{ref}}) + E_{11} + E_{12}r = O(S_*^{-1}), \quad \left|\log \frac{z_+(v)}{P(v)}\right| \leq \frac{Cv}{S_*} \leq \frac{CL_s}{S_*} \leq C.$$

This keeps z_+ from reaching zero and proves $cP \leq z_+ \leq CP$ on the whole interval. Multiplying by the matrix in (7.8) gives $x = z_+ + z_- = z_+(1 + r)$ and $y = c_0\sqrt{1+s^2}(z_+ - z_-) = c_0\sqrt{1+s^2}z_+(1 - r)$. These identities give (7.21). Differentiate the coordinate equation and use (7.19); this proves the derivative estimates in the same way as (7.20). The initial value $P(0)$ is fixed with the label. The pressure formula supplies the extra factor $\varepsilon^{1/2}$. $\square$

How the shear transfers energy. The pulse equation also shows where the growth of the homogeneous solution comes from. For the real amplitude $t = t_\sigma^h$, the constraint $n_\Phi \cdot t = 0$ makes the normal term in $\mathcal{A}_\Phi t$ perpendicular to t . Also,

$$t \cdot \mathcal{K}t = -2Ft_r t_\theta + (2F + RF_R)t_r t_\theta + G_R t_r t_z = g \cdot (t_r(t_\theta, t_z)).$$

Take the scalar product of $t' = \mathcal{A}_\Phi t - dt$ with t . This gives

$$\frac{1}{2} \frac{d}{dv} |t|^2 = -g \cdot (t_r(t_\theta, t_z)) - \varepsilon k^2 |n_\Phi|^2 |t|^2. \quad (7.22)$$

The first term is the exchange of energy with the base shear, while the second is viscous dissipation. The vector $t_r(t_\theta, t_z)$ records the transport of azimuthal and axial momentum across a cylinder. At the representative point, write such a flux vector as $T = T_N N + T_K K$. The energy transfer is $-g_0 \cdot T = -|g_0| T_N$. The covariance construction also fixes the transverse component T_K . Both components of the covariance must therefore match the prescribed momentum flux.

Viscosity remains a leading-order term in the pulse equation. With viscosity equal to one, the base velocity and radial length give a Reynolds number of order q^{-h} . The carrier wavelength is $\sqrt{Q}/k \asymp Q^{1/2+h/2}$, while the physical amplitude of the leading oscillations built below is of order $Q^{-A}\sqrt{\varepsilon} = Q^{-1/2-h/2}$, apart from powers of S_* . The powers of Q cancel in their product. Thus $\varepsilon k^2 \asymp 1$ throughout the correction process in the pulse equation.

7.3 The covariance of the leading oscillations and its linearization

The two real homogeneous pulses $t_\pm^h$ from Lemma 7.4 give two covariance directions in each slow box. Proposition 7.5 chooses their scalar amplitudes $a_\pm$ so that a positive combination of those directions equals the leading stress. This subsection computes the two covariances, solves for the squared amplitudes, and then differentiates that relation. The differentiated relation gives signed stress corrections through $\mathcal{L}$ from Proposition 7.6.

Recall the angular and auxiliary averages in (3.7), with normalized Haar measure on the auxiliary torus. Take these averages at fixed slow variables, before evaluating at the physical phase map. For real velocity fields w , v on the extended domain, write $w_{\text{tan}} = (w_\theta, w_z)$.

The product $w_r w_{\tan}$ records radial transport of azimuthal and axial momentum. At each fixed slow point, define the covariance $\mathcal{C}(w)$ and the symmetric cross term $\mathcal{B}(w, v)$ by

$$\mathcal{C}(w) = \langle \langle w_r w_{\tan} \rangle_\theta \rangle_\gamma, \quad \mathcal{B}(w, v) = \langle \langle w_r v_{\tan} + v_r w_{\tan} \rangle_\theta \rangle_\gamma. \quad (7.23)$$

Lemma 6.2 allows these averages to be computed on any common torus that applies. Expanding the product gives $D\mathcal{C}(w)[v] = \mathcal{B}(w, v)$.

Write $\beta = (\ell, a)$ for a slow box and $\gamma = (\beta, \sigma)$ for one of its two rectangle labels. The two signs use the same slow cutoff, $\eta_\beta = \eta_{(\beta, +)} = \eta_{(\beta, -)}$. The squared partition therefore sums over β , counting each box once. On the slow neighborhood of β , define

$$b_\sigma = \chi_g(\xi_g) \psi(v) t_\sigma^h \cos(k\Phi_\sigma), \quad \sigma \in \{+, -\},$$

and extend it by zero outside the auxiliary rectangle with that sign. The factors χ_g, ψ make this extension smooth. The field b_σ contains neither the slow cutoff η_β nor a scalar amplitude. The auxiliary supports for the two signs are disjoint.

Let $H = [\mathcal{C}(b_+) \mid \mathcal{C}(b_-)]$. This is a real 2×2 matrix that depends on the slow point, and its columns use the component order $(r\theta, rz)$. For any real numbers a_+, a_- , the disjoint supports give $\mathcal{C}(a_+ b_+ + a_- b_-) = H(a_+^2, a_-^2)^T$. The stresses produced by these two waves are therefore exactly the nonnegative combinations of its two columns. The strict cone condition will make the two coefficients for the leading stress positive; their square roots define the real amplitudes of the waves.

Proposition 7.5. *For each slow box β , use the homogeneous pulses from Lemma 7.4 to form $b_\pm$ and H as above. Use the order-zero target $\mathcal{T}_{0,*} = (Q/q)^{A+1/2} \mathcal{T}_0$. After decreasing q_* once more, the matrix H is invertible throughout every enlarged slow neighborhood. Both components of $H^{-1} \mathcal{T}_{0,*}$ are positive where that neighborhood meets $X_a < X < X_b$. Componentwise square roots can therefore be used to define*

$$y = H^{-1} \mathcal{T}_{0,*}, \quad a_\sigma = \sqrt{y_\sigma}, \quad W_0 = \sqrt{\varepsilon} \sum_{\sigma=\pm} a_\sigma b_\sigma. \quad (7.24)$$

For every fixed slow multi-index I , the squared amplitudes satisfy the following bounds on $X_a < X < X_b$.

$$y_\sigma \geq c \sqrt{S_*} \zeta, \quad |D^I y_\sigma| \leq C_I S_*^{b_I} \zeta \delta^{-a_I}. \quad (7.25)$$

The coefficients a_σ extend smoothly by zero at the edges of the shell. On the slow neighborhood of the label, the harmonic coefficients of W_0 satisfy the derivative and envelope estimates of $W_{1/2}$. After multiplication by the slow cutoff, $\eta_\beta W_0 \in W_{1/2}$. Before that multiplication, the covariance is

$$\mathcal{C}(W_0) = \varepsilon \mathcal{T}_{0,*}. \quad (7.26)$$

This decomposition with positive coefficients uses only the order-zero target $\mathcal{T}_{0,*}$.

Proof. The proof of Lemma 7.1 works throughout these neighborhoods. Their diameter is still $O(S_*^{-3})$, and the comparison with the base holds on the enlarged annulus. The phase and homogeneous-pulse estimates, and therefore the estimates for the columns below, hold there with uniform constants.

Step 1: compute and estimate the covariance columns. The angular frequency kp is a nonzero integer, so the angular average of $\cos^2(k\Phi_\sigma)$ is exactly $1/2$. The auxiliary average can be computed on the band torus. On that torus, the rectangle parametrization has area element $|\det(v_r, v_t)| d\xi_g d\eta_g$, where $d\eta_g = c_i dv$. Recall that x, y are the geometric coordinates of t_σ^h from Lemma 7.4. This amplitude does not depend on ξ_g , and its radial component is x . The column H_σ is therefore

$$H_\sigma = \frac{|\det(v_r, v_t)|}{2} \left(\int \chi_g^2 d\xi_g \right) c_i \int_0^{L_s} \psi^2 x t_{\sigma, \tan}^h dv. \quad (7.27)$$

Define the corresponding positive scalar by

$$h_\sigma = \frac{|\det(v_r, v_t)|}{2} \left(\int \chi_g^2 d\zeta_g \right) c_i \int_0^{L_s} \psi^2 x^2 dv.$$

The Gaussian envelope and the region where $\psi = 1$ give

$$c\sqrt{L_s} \leq \int_0^{L_s} \psi^2 x^2 dv \leq C\sqrt{L_s}, \quad \int_0^{L_s} \frac{|v - L_s/2|}{L_s} \psi^2 x^2 dv \leq C.$$

To see this, make the substitution $w = (v - L_s/2)/\sqrt{L_s}$. The two expressions reduce to integrals of e^{-cw^2} and $|w|e^{-cw^2}$. It follows that $h_\sigma \asymp c_i \sqrt{L_s} \asymp S_*^{-1/2}$, and the normalized $x^2 \psi^2$ average of $|v - L_s/2|/L_s$ is $O(S_*^{-1/2})$. The definitions of K_a, N_a, s_a in (7.7) give the tangential part of the pulse, $t_{\sigma, \tan}^h = -s_a x K_a + y N_a$. Lemma 7.4 says that its ratio to x differs by $O(S_*^{-1})$ from $-s(v)K + c_0 \sqrt{1 + s(v)^2 N}$. The weighted average above is concentrated near $v = L_s/2$, where $s = \sigma u_*$. Set $A_c = -c_0 \sqrt{1 + u_*^2} > 0$. Then

$$H_\sigma = h_\sigma (-A_c N - \sigma u_* K + e_\sigma), \quad |e_\sigma| \leq C S_*^{-1/2}. \quad (7.28)$$

Lemma 7.4 also gives polynomial bounds for the derivatives of the covariance columns and the homogeneous solutions.

Step 2: solve for positive squared amplitudes. First leave out the errors e_σ in (7.28). For a target $T = T_N N + T_K K$, the two scalar equations then give

$$h_+ y_+ = \frac{1}{2} (-T_N / A_c - T_K / u_*), \quad h_- y_- = \frac{1}{2} (-T_N / A_c + T_K / u_*).$$

The reference cone is therefore given by $T_N < 0$ and $|T_K| < (u_*/A_c)(-T_N)$. The two formulas above then give a decomposition with positive coefficients. The choice of u_* , together with the uniform margin in the leading cone, gives $|T_K|/u_* \leq (1 - \eta_c)(-T_N/A_c)$ for some fixed $\eta_c > 0$ when $T = \mathcal{T}_{0,*}$, by (7.1). Freezing N, K, c_0 at the representative changes this normalized cone inequality by an error that tends uniformly to zero. At the radial edges, the smooth limiting stress direction gives the same comparison. The errors in (7.28) keep this strict inequality valid and give

$$|\det H| \geq c/S_*, \quad \|H^{-1}\| \leq C\sqrt{S_*}, \quad c\sqrt{S_*}|T| \leq y_\sigma \leq C\sqrt{S_*}|T|. \quad (7.29)$$

Step 3: derivatives, shell edges, and covariance. Apply the bounds for the leading stress in (4.27) inside the chart. Differentiating $HH^{-1} = I_2$ gives polynomial bounds for every fixed derivative of H^{-1} . Differentiating $y = H^{-1}\mathcal{T}_{0,*}$ then proves (7.25). When $|I| \geq 1$, the chain rule gives

$$D^I a_\sigma = \sum_{n=1}^{|I|} \sum_{\substack{I_1 + \dots + I_n = I \\ |I_\nu| \geq 1}} c_{I_1, \dots, I_n} y_\sigma^{1/2-n} \prod_{\nu=1}^n D^{I_\nu} y_\sigma.$$

The bounds in (7.25) therefore imply

$$|D^I a_\sigma| \leq C_I S_*^{b'_I} \sqrt{\zeta} \delta^{-a'_I}.$$

The same bound holds when $I = 0$. For every fixed derivative order, its right-hand side tends to zero at each edge of the shell, so extending a_σ by zero is smooth. The bounds for a_σ , Lemma 7.4, and the cutoffs χ_g, ψ give the local derivative and envelope bounds for W_0 . Multiplying by η_β then gives $\eta_\beta W_0 \in W_{1/2}$. The two rectangles are disjoint, and (7.24) gives $\mathcal{C}(W_0) = \varepsilon H y$. This is exactly (7.26). $\square$

Each slow box now has the required local covariance. Multiplying by the shared slow cutoff η_β multiplies (7.26) by η_β^2 . Return to physical velocities, then add the slow boxes and bands. This gives the leading physical stress pair

$$\sum_{\beta=(\ell,a)} Q^{-2A} \eta_\beta^2 \varepsilon \mathcal{T}_{0,*} = q^{-A-1/2} \mathcal{T}_0. \quad (7.30)$$

The scale can be checked directly. Since $A = 1/2 + h$,

$$Q^{-2A} \varepsilon \mathcal{T}_{0,*} = Q^{-2A+h} (Q/q)^{A+1/2} \mathcal{T}_0 = Q^{-A+h+1/2} q^{-A-1/2} \mathcal{T}_0 = q^{-A-1/2} \mathcal{T}_0.$$

Now use $\sum_\beta \eta_\beta^2 = 1$. Cross products between different labels vanish because their auxiliary supports are disjoint.

This sum gives the leading physical stress. Later residuals need stress increments with either sign. Obtain them by varying the amplitudes to first order around this fixed positive representation. Keep the positive coefficients a_σ fixed at the values from Proposition 7.5. Let $\alpha \in \mathbb{R}$, and let $\Sigma(R, Z, T) \in M_\alpha$ be a real two-vector that does not depend on either the angular or the auxiliary variables. This independence from the auxiliary variables is an extra requirement here; membership in M_α by itself allows such dependence. On the slow neighborhood of the label, define

$$d_\Sigma = H^{-1}(\Sigma/\varepsilon), \quad \delta a_\sigma = \frac{(d_\Sigma)_\sigma}{2a_\sigma}, \quad \mathcal{L}\Sigma = \sqrt{\varepsilon} \sum_{\sigma=\pm} \delta a_\sigma b_\sigma. \quad (7.31)$$

Each division is taken component by component. It is defined first on $X_a < X < X_b$, where $a_\sigma > 0$. The estimates below show that the extension by zero at the edges of the shell is smooth.

Proposition 7.6. *For every $\alpha \in \mathbb{R}$ and every real two-vector $\Sigma(R, Z, T) \in M_\alpha$ that is independent of the angular and auxiliary variables, (7.31) defines a linear operation $\mathcal{L}\Sigma$. On the slow neighborhood, its harmonic coefficients satisfy the local derivative and envelope bounds of $W_{\alpha-1/2}$, with smooth extension by zero at the edges of the shell. When the slow cutoffs are written out, its bounds and covariance are*

$$\eta_\beta \mathcal{L}\Sigma \in W_{\alpha-1/2}, \quad \mathcal{B}(W_0, \mathcal{L}\Sigma) = \Sigma, \quad \mathcal{C}(\eta_\beta \mathcal{L}\Sigma) \in M_{2\alpha-1}. \quad (7.32)$$

At every correction stage, the denominator a_σ remains this same fixed coefficient.

Proposition 7.6 places no restriction on the sign or relative size of Σ .

Proof. Apply Leibniz' rule to $d_\Sigma = H^{-1}(\Sigma/\varepsilon)$, using (7.29) and the bounds for the differentiated inverse proved in Proposition 7.5. The weighted derivative bounds on Σ give $|D^I(d_\Sigma)_\sigma| \leq C_I \varepsilon^{\alpha-1} S_*^{b'_I} \zeta \delta^{-a'_I}$. Every derivative of $y_\sigma^{-1/2}$ is a sum of terms of the form $C y_\sigma^{-k-1/2} \prod_{\nu=1}^k D^{I_\nu} y_\sigma$. By (7.25), the k weights in the numerator cancel all but $\zeta^{-1/2}$ in the denominator. Leibniz' rule therefore gives

$$|D^I \delta a_\sigma| \leq C_I \varepsilon^{\alpha-1} S_*^{b'_I} \sqrt{\zeta} \delta^{-a'_I}. \quad (7.33)$$

The remaining square-root weight at the edge proves smooth extension by zero. Lemma 7.4 gives the local estimates for the wave. Multiplying by η_β puts it in the supported wave class. Write $a = (a_+, a_-)^T$ and $\delta a = (\delta a_+, \delta a_-)^T$, and let $\odot$ mean multiplication component by component. These scalar coefficients do not depend on the averaging variables, and the rectangles are disjoint, so the following bilinear identity is exact.

$$\mathcal{B}(W_0, \mathcal{L}\Sigma) = \varepsilon H(2a \odot \delta a) = \varepsilon H d_\Sigma = \Sigma. \quad (7.34)$$

The factor 1/2 for the real cosine already appears in (7.27). The factor 2 in this cross identity is why (7.31) has its denominator. In $\mathcal{C}(\eta_\beta \mathcal{L}\Sigma)$, both wave factors have exponent $\alpha - 1/2$. The product rule for the classes then gives the last claim in (7.32). $\square$

In particular, before the curl correction the covariance identity is

$$\mathcal{C}(W_0 + \mathcal{L}\Sigma) = \varepsilon \mathcal{T}_{0,*} + \Sigma + \mathcal{C}(\mathcal{L}\Sigma).$$

This makes $\mathcal{L}$ a right inverse of $D\mathcal{C}(W_0)$, meaning that this operation first turns the chosen stress change into a wave correction, and the covariance derivative then turns that correction back into the same stress change. The quadratic remainder $\mathcal{C}(\mathcal{L}\Sigma)$ stays in the stress. The same squared partition that gives the leading covariance also puts this linear inverse together across the slow boxes. Write $W_{0,\beta}$ and $\mathcal{L}_\beta$ for the local constructions before the slow cutoff is applied. Take a real two-vector stress $\sigma(r, z, t)$ that does not depend on the angular or auxiliary variables, and whose normalized representatives $Q^{2A}\sigma$ belong to M_α . Set

$$\begin{aligned} W_{0,\text{phys}}^{\text{as}} &= \sum_{\beta=(\ell,a)} Q^{-A} \eta_\beta W_{0,\beta}, \\ \mathcal{L}_{\text{phys}}^{\text{as}} \sigma &= \sum_{\beta=(\ell,a)} Q^{-A} \eta_\beta \mathcal{L}_\beta (Q^{2A} \sigma). \end{aligned} \tag{7.35}$$

Each summand uses the scale $Q = 2^{-\ell}$ from its label $\beta = (\ell, a)$. The sums are locally finite when $q > 0$. Where slow supports overlap, the assigned rectangles are disjoint, the partitions are squared, and the physical factor is Q^{-2A} . Together, these facts give $\mathcal{B}(W_{0,\text{phys}}^{\text{as}}, \mathcal{L}_{\text{phys}}^{\text{as}} \sigma) = \sigma$. In a chart with scale Q , set $\Sigma = Q^{2A} \sigma$ and define the normalized representatives by

$$W_0^{\text{as}} = Q^A W_{0,\text{phys}}^{\text{as}}, \quad \mathcal{L}^{\text{as}} \Sigma = Q^A \mathcal{L}_{\text{phys}}^{\text{as}} (Q^{-2A} \Sigma).$$

To check whether these fields belong to the wave classes, collect all coefficients with the same label and harmonic, as in Definition 6.5. The slow cutoffs have bounded overlap, and the common-torus bounds give $W_0^{\text{as}} \in W_{1/2}$ and $\mathcal{L}^{\text{as}} \Sigma \in W_{\alpha-1/2}$, together with

$$\mathcal{B}(W_0^{\text{as}}, \mathcal{L}^{\text{as}} \Sigma) = \Sigma. \tag{7.36}$$

These definitions agree wherever the charts overlap. Because the input does not depend on the auxiliary variables, the scalar coefficients can be moved outside the covariance integral, exactly as the identity in Proposition 7.6 requires.

7.4 Exact curls, the tails left over, and the covariance remainder.

For the harmonic velocity $t_m e^{ikm\Phi}$, the condition $n_\Phi \cdot t_m = 0$ says that the amplitude is orthogonal to the phase gradient, and this cancels the leading term in its divergence. Derivatives of the amplitude t_m still contribute. Taking the full curl of the vector potential $A_m = C_m e^{ikm\Phi}$ in (7.38), rather than keeping only that leading term, gives the corrected amplitude $t_m + r_m$ needed for exact incompressibility. Differentiating the exponential gives the amplitude t_m , while differentiating the potential coefficient C_m and the cylindrical frame gives r_m . This subsection first estimates r_m in Lemma 7.7, then estimates the temporal cutoff residual in (7.40). This is the unused part of the source, multiplied by one minus the time cutoff, plus the time derivative of that cutoff multiplied by the pulse amplitude. The subsection finally finds the covariance of a stress increment built from the actual velocities in Corollary 7.8.

For a normalized potential $A = (A_r, A_\theta, A_z)$, define

$$\text{curl}_* A = (R^{-1} \partial_\theta A_z - D_z A_\theta, D_z A_r - D_r A_z, (D_r + R^{-1}) A_\theta - R^{-1} \partial_\theta A_r). \tag{7.37}$$

The normalized derivatives already include the chain-rule terms that appear when the potential is evaluated at (6.3). This means that $\text{curl}_* A$ is the normalized physical curl of the evaluated potential.

Fix a label, a nonzero integer m , and a harmonic coefficient $t_m \in W_\alpha$ that does not depend on θ and satisfies $n_\Phi \cdot t_m = 0$. The coefficient t_m must be supported in the prescribed slow and transverse supports and compactly inside the pulse interval, and its extension by zero must be smooth across all those boundaries. Choose the potential so that taking its curl and differentiating the exponential gives the amplitude

t_m . The potential coefficient C_m and the oscillating potential A_m are

$$C_m = \frac{i n_\Phi \times t_m}{km |n_\Phi|^2}, \quad A_m = C_m e^{ikm\Phi}, \quad \text{curl}_* A_m = (t_m + r_m) e^{ikm\Phi}. \quad (7.38)$$

The last identity defines r_m as the difference between the full curl amplitude and the prescribed transverse amplitude. Because C_m does not depend on θ , the remainder r_m is

$$r_m = (-D_z(C_m)_\theta, D_z(C_m)_r - D_r(C_m)_z, (D_r + R^{-1})(C_m)_\theta).$$

Lemma 7.7. *For every $\alpha \in \mathbb{R}$ and every nonzero integer m , let $t_m \in W_\alpha$ satisfy $n_\Phi \cdot t_m = 0$. Assume that it is supported in the prescribed slow and transverse supports and compactly inside the pulse interval, and that its extension by zero is smooth across all those boundaries. The construction in (7.38) gives $C_m \in W_{\alpha+1/2}$ and $r_m \in W_{\alpha+1/2-\kappa_s}$. The velocity $(t_m + r_m) e^{ikm\Phi} = \text{curl}_* A_m$ is exactly divergence-free, both for the normalized operators and after evaluation at the phase map. The potential and velocity extend smoothly by zero across the boundaries of their supports. For the full amplitude $a_m = t_m + r_m$, its scalar product with the phase normal satisfies*

$$ikm n_\Phi \cdot a_m = -((D_r + R^{-1})(a_m)_r + D_z(a_m)_z), \quad n_\Phi \cdot a_m \in W_{\alpha+1/2-\kappa_s}. \quad (7.39)$$

The physical potentials and pressures come from the chart versions by multiplying them by $Q^{1/2-A}$ and Q^{-2A} , respectively.

Proof. When the exponential in $\text{curl}_* A_m$ is differentiated, the amplitude it produces is

$$ikm n_\Phi \times C_m = -\frac{n_\Phi \times (n_\Phi \times t_m)}{|n_\Phi|^2} = t_m - \frac{n_\Phi (n_\Phi \cdot t_m)}{|n_\Phi|^2} = t_m.$$

Every other term in $\text{curl}_* A_m$ comes from differentiating the potential coefficient C_m or the cylindrical frame. The factor $(km)^{-1}$ raises the ε exponent by $1/2$. Applying D_r lowers it by no more than κ_s , by (6.32). Lemma 7.1 gives polynomial bounds for the coefficients of the multiplier $n_\Phi / |n_\Phi|^2$. Together, these estimates prove the two class bounds, including all derivatives of any fixed order. The normalized divergence is $\text{div}_* a = (D_r + R^{-1})a_r + R^{-1}\partial_\theta a_\theta + D_z a_z$. Since $D_r(R^{-1}) = -R^{-2}$, it follows that $(D_r + R^{-1})(R^{-1}f) = R^{-1}D_r f$. Substituting (7.37) gives

$$\begin{aligned} \text{div}_* \text{curl}_* A &= R^{-1}D_r \partial_\theta A_z - (D_r + R^{-1})D_z A_\theta \\ &\quad + R^{-1}D_z \partial_\theta A_r - R^{-1}\partial_\theta D_r A_z \\ &\quad + D_z(D_r + R^{-1})A_\theta - R^{-1}D_z \partial_\theta A_r = 0, \end{aligned}$$

where each positive term cancels its matching negative term because the operators commute. The chain rule keeps this identity true after evaluation at the phase map. Expanding the divergence of $a_m e^{ikm\Phi}$, whose amplitude coefficient does not depend on θ , gives (7.39). Dividing by km and applying D_r once gives the improvement $1/2 - \kappa_s$ in the exponent. The coefficient bounds and cutoffs give the claimed support and smooth extensions. $\square$

Build real velocities by pairing conjugate terms. For example, $t \cos(k\Phi)$, with real t , has coefficients $t/2$ at the two harmonics $m = 1, -1$. The matching potential and pressure coefficients obey the same conjugacy condition. In particular, the pressure that goes with a real cosine velocity mode keeps the factor $i/(km)$ from (7.13).

The temporal cutoff and its residual. For the solution in Proposition 7.2, set $\hat{t}_m = \psi t_m$ and $\hat{\pi}_m = \psi \pi_m$, then apply Lemma 7.7 to $\hat{t}_m$. Write $\hat{r}_m$ for the curl remainder of this cutoff amplitude. Before the solution is multiplied by ψ , it may reach both ends of the pulse interval. The cutoff makes the potential and pressure zero on an open neighborhood of each endpoint. On the labelled rectangle

$0 \leq v \leq L_s$, the product rule in (7.5) gives

$$\hat{t}_m' + \mathcal{K}\hat{t}_m + m^2 d\hat{t}_m + ikmn_\Phi \hat{\pi}_m + f_m = (1 - \psi)f_m + \psi' t_m. \quad (7.40)$$

Equation (6.16) says that $\psi = 1$ on $|v - L_s/2| \leq L_s/5$, and both terms on the right are zero there. Wherever either term is left, $|v - L_s/2| \geq L_s/5$. Equation (7.16) then adds a factor $S_*^C e^{-cS_*}$ to every estimate for a fixed coefficient derivative. During the iteration, $f_m e^{ikm\Phi}$ is an actual supported harmonic source whose extension by zero is smooth, so this local identity also describes the part left uncanceled in the global residual.

At any fixed correction stage, changing a fixed derivative estimate to physical variables introduces a fixed power of Q^{-1} and a polynomial in S_* . Since $S_* = \ell^2$, $Q = 2^{-\ell}$, and $q \asymp Q$, every fixed choice of M, C, N satisfies

$$q^{-N} Q^{-M} S_*^C e^{-cS_*} \leq C_N \exp(-c\ell^2 + (M + N)\ell \log 2 + 2C \log \ell) \longrightarrow 0.$$

So every resulting physical derivative is $O(q^N)$ for every fixed N . Later corrections keep these terms as separate additive flat residuals. The same argument handles the homogeneous cutoff tails. The pressure coefficient still satisfies $\hat{\pi}_m \in W_{\alpha+1/2}$, while the exact velocity amplitude is $\hat{t}_m + \hat{r}_m$, with $\hat{r}_m \in W_{\alpha+1/2-\kappa_s}$.

Covariance of the actual velocities. The linear covariance inverse was defined at the fixed primary wave. During the iteration, its increment is added to a divergence-free wave field that already includes earlier corrections, and the new increment has a curl remainder of its own. The exact expansion below picks out the prescribed stress increment. It also keeps the interaction with earlier corrections, the interaction with the new curl remainder, and the square of the new velocity.

Corollary 7.8. *Fix $\alpha, \rho \in \mathbb{R}$, and use the normalized representatives from (7.35). Suppose U is a real divergence-free wave field, including its curl remainders, with*

$$U = W_0^{\text{as}} + E, \quad U \in W_{1/2}, \quad E \in W_\rho.$$

Let $\Sigma(R, Z, T) \in M_\alpha$ be a real two-vector that does not depend on the angular or auxiliary variables. Apply Lemma 7.7 to the signed amplitudes $\mathcal{L}^{\text{as}}\Sigma$. This gives the divergence-free increment $V = \mathcal{L}^{\text{as}}\Sigma + R$, where $R \in W_{\alpha-\kappa_s}$ is its curl remainder. Then

$$\mathcal{C}(U + V) - \mathcal{C}(U) = \Sigma + \mathcal{B}(E, \mathcal{L}^{\text{as}}\Sigma) + \mathcal{B}(U, R) + \mathcal{C}(V). \quad (7.41)$$

The three remainder terms satisfy

$$\begin{aligned} \mathcal{B}(E, \mathcal{L}^{\text{as}}\Sigma) &\in M_{\rho+\alpha-1/2}, \\ \mathcal{B}(U, R) &\in M_{\alpha+1/2-\kappa_s}, \\ \mathcal{C}(V) &\in M_{2\alpha-1}. \end{aligned} \quad (7.42)$$

Taking the radial divergence of any one of these remainder terms lowers the ε exponent by no more than another κ_s .

Proof. Expand the quadratic map and use $\mathcal{B}(W_0^{\text{as}}, \mathcal{L}^{\text{as}}\Sigma) = \Sigma$. This gives

$$\begin{aligned} \mathcal{C}(U + V) - \mathcal{C}(U) &= \mathcal{B}(U, V) + \mathcal{C}(V) \\ &= \mathcal{B}(W_0^{\text{as}}, \mathcal{L}^{\text{as}}\Sigma) + \mathcal{B}(E, \mathcal{L}^{\text{as}}\Sigma) + \mathcal{B}(U, R) + \mathcal{C}(V) \\ &= \Sigma + \mathcal{B}(E, \mathcal{L}^{\text{as}}\Sigma) + \mathcal{B}(U, R) + \mathcal{C}(V). \end{aligned}$$

The bounds for $\mathcal{B}(E, \mathcal{L}^{\text{as}}\Sigma)$ and $\mathcal{B}(U, R)$ come from multiplying the stated classes. Since $\kappa_s < 1/2$, it follows that $V \in W_{\alpha-1/2}$, and so $\mathcal{C}(V) \in M_{2\alpha-1}$. This proves (7.41) and (7.42) without placing any positivity condition

on an accumulated stress. The claim about radial divergence follows from (6.32); the cylindrical terms of order zero cause no further loss in the exponent. $\square$

8 Compactly supported corrections to the mean

The wave construction in Section 7 gives pulses whose averaged covariance produces the leading stress target in (7.26). It also gives an inverse for the principal equation of every nonzero angular harmonic in Proposition 7.2. The residual's angular mean, meaning its average over the physical angle, still needs to be corrected. This mean contains quadratic products of the waves, and it can depend on the auxiliary torus even though it does not depend on the physical angle, as Proposition 8.1 shows. This section builds the pressure, tangential stresses, and divergence-free velocity increments needed to correct it. All of them are supported inside the active annulus; see Proposition 8.3 and Corollary 8.5.

There are two different inversion problems. For an auxiliary-averaged source, a compactly supported solution φ of $(D_r + e/R)\varphi = f$, with $e \in \{0, 1, 2\}$, can exist only when $\int R^e f dR = 0$; see Lemma 8.2. For a source that depends on the auxiliary torus, the fast time derivative can be inverted only when the source has zero auxiliary average at each slow point, as Lemma 8.6 explains. The section first derives the mean momentum equations in Proposition 8.1 and builds the radial inverse in Lemma 8.2, including the remainder that appears when its source depends on the torus. It then uses that inverse for pressure and divergence-free axial increments in Proposition 8.3, and for tangential stresses in Corollary 8.5. The integrated equations in (8.16) pick out three scalar defects, $\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z$, defined in Equations (8.12) and (8.15). A final linear map chooses five velocity coefficients so that their linear changes cancel these three defects while two velocity moments stay unchanged; this is Lemma 8.7. Lemma 8.8 calculates the nonlinear changes explicitly. The correction cycle in Proposition 9.6 uses all these constructions after every wave update.

Any radial integral without stated limits runs from zero to infinity. The averaging and projection conventions are the ones in Equations (3.7) and (3.9). Recall that $\langle \cdot \rangle_\theta$ takes the angular average of each cylindrical component, while $\langle \cdot \rangle_Y$ takes the Haar average on the auxiliary torus from Lemma 6.2, and $f^\circ = f - \langle f \rangle_Y$. So a field can be angularly invariant and still depend on that torus. The coefficient classes M_α and S_α are the classes from Definition 6.4; their estimates hold at every fixed derivative order. Unless physical variables are written out, the variables and operators are the normalized chart versions from (6.6).

8.1 Momentum equations in divergence form and integral constraints.

Here, divergence form means writing the transport terms as radial and axial derivatives of momentum fluxes, with the extra terms required by cylindrical coordinates kept beside them. This subsection derives the mean momentum equations in divergence form in Proposition 8.1, and states the two velocity moments in (8.2) that the corrections must leave unchanged. Radial integration of these equations shows which compactly supported corrections are possible, as Proposition 8.4 and Corollary 8.5 explain. Writing the equations in divergence form also shows the boundary contributions explicitly. The calculation keeps the quadratic products of the actual divergence-free field w , including its curl remainders, as shown in (8.1). In a common chart, using the normalized operators from (6.6), the actual velocity splits as

$$u = (b + \beta, V + v, G + \gamma) + w, \quad \langle w \rangle_\theta = 0, \quad W_{ab} = \langle w_a w_b \rangle_\theta. \quad (8.1)$$

Here (b, V, G) is the slow base, (β, v, γ) is its correction that does not depend on the physical angle, and w is the sum of the divergence-free fields made by taking curls of the wave potentials. The indices in W_{ab} run over $a, b \in \{r, \theta, z\}$, so W includes every curl remainder. The correction components and W extend smoothly by zero outside the active shell. The mean correction is divergence-free, which means $(D_r + R^{-1})\beta + D_z\gamma = 0$, and it must satisfy the two integral constraints

$$M_\theta := \int_0^\infty R^2 \langle v \rangle_Y dR = 0, \quad M_z := \int_0^\infty R \langle \gamma \rangle_Y dR = 0. \quad (8.2)$$

These two identities hold at every (Z, T) . Up to normalization, they say that the auxiliary-averaged correction has zero angular-momentum moment and zero axial-flux integral. The base is handled separately. Compactly supported azimuthal potentials leave the axial-flux constraint unchanged, and the finite-dimensional moment correction leaves both constraints unchanged.

Let p_m be the perturbation's mean pressure. Put $\Delta_0 = D_r^2 + R^{-1}D_r + D_z^2$ and $\Sigma_a = Q^{2A}\mathcal{T}_{\text{phys},a}$ for $a = \theta, z$, where the full residual stress tensor is the one in (5.41).

Proposition 8.1. *Let u have the divergence-free decomposition in (8.1), with slow base (b, V, G) , correction (β, v, γ) independent of the physical angle, and $\langle w \rangle_\theta = 0$. Let p_m be its pressure correction independent of the physical angle, and let $\Sigma_a = Q^{2A}\mathcal{T}_{\text{phys},a}$ be the normalized components of the base stress from (5.41). Keep the base's flat residual as a separate additive error. All its derivatives of every fixed order vanish faster than every power of q . Then the angular mean of the rest of the normalized residual is $(D_r p_m - g_r, E_\theta, E_z)$, where*

$$\begin{aligned} E_\theta &= \mathfrak{t}_* v + (D_r + 2/R)(bv + \beta V + \beta v + W_{r\theta}) \\ &\quad + D_z(Gv + V\gamma + \gamma v + W_{z\theta}) - \varepsilon(\Delta_0 - R^{-2})v - (D_r + 2/R)\Sigma_\theta, \\ E_z &= \mathfrak{t}_* \gamma + (D_r + 1/R)(b\gamma + \beta G + \beta \gamma + W_{rz}) \\ &\quad + D_z(2G\gamma + \gamma^2 + W_{zz} + p_m) - \varepsilon\Delta_0\gamma - (D_r + 1/R)\Sigma_z, \\ g_r &= -\mathfrak{t}_* \beta - (D_r + 1/R)(2b\beta + \beta^2 + W_{rr}) \\ &\quad - D_z(b\gamma + G\beta + \beta\gamma + W_{zr}) + (2Vv + v^2 + W_{\theta\theta})/R + \varepsilon(\Delta_0 - R^{-2})\beta. \end{aligned} \tag{8.3}$$

Proof. Incompressibility lets the transport terms be written in divergence form. For the cylindrical components of any divergence-free field, the three transport expressions are

$$\begin{aligned} (u \cdot \nabla u)_r &= (D_r + 1/R)u_r^2 + R^{-1}\partial_\theta(u_\theta u_r) + D_z(u_z u_r) - u_\theta^2/R, \\ (u \cdot \nabla u)_\theta &= (D_r + 2/R)(u_r u_\theta) + R^{-1}\partial_\theta u_\theta^2 + D_z(u_z u_\theta), \\ (u \cdot \nabla u)_z &= (D_r + 1/R)(u_r u_z) + R^{-1}\partial_\theta(u_\theta u_z) + D_z u_z^2. \end{aligned}$$

The turning of the cylindrical frame produces the centrifugal term in the first row and the extra $u_r u_\theta / R$ in the second. Any angular derivative integrates to zero, and so does any product with exactly one factor of w . Once the pure base terms are subtracted, the fluxes left are the ones in (8.3). The angular-derivative terms in the vector Laplacian also have zero angular average. This leaves $\Delta_0 - R^{-2}$ in the radial and angular components and Δ_0 in the axial component. By (5.41), the residual stress tensor contributes the two radial divergences displayed in (8.3). The pulse tails left out have nonzero angular harmonics, so their angular mean is zero. The calculation keeps the full covariance of the divergence-free wave field, including all its curl remainders. $\square$

8.2 A compactly supported primitive for the cylindrical weights.

This subsection builds a radial primitive by integrating in the radial direction along the matching torus path, and keeps its support inside the active shell; see Lemma 8.2. A direct radial integral may still be nonzero after the source support has ended. To stop that from happening, subtract its full radial integral multiplied by a fixed cutoff, and keep the error created by this subtraction. After evaluation at the phase, the radial derivative also differentiates the auxiliary variable in (6.4), so both integrals must follow the matching path on the torus. That path is included in the physical definition below.

Start with the radial operations in physical variables, keeping (z, t) and the independent angular variable fixed. Choose a slow cutoff $\chi_m(X)$ that is zero on an inner collar, equals one on an outer collar, and changes only in a fixed interior part of the active shell. Let $g(r, z, t, Y)$ be a smooth source that is periodic in Y and extends smoothly by zero outside that shell. Set

$$\begin{aligned}
\mathcal{I}g(r, Y) &= \int_0^r g(r', Y + ((r')^{d_r} - r^{d_r})v_r) dr', \\
\mathcal{J}g(r, Y) &= \int_0^\infty g(r', Y + ((r')^{d_r} - r^{d_r})v_r) dr', \\
\mathcal{I}_c g &= \mathcal{I}g - \chi_m \mathcal{J}g.
\end{aligned} \tag{8.4}$$

The shift in Y follows the radial phase map while the physical variables (z, t) stay fixed. Below the shell, $\mathcal{I}g = 0$; above it, $\mathcal{I}g = \mathcal{J}g$. It follows that $\mathcal{I}_c g$ is zero when $X \leq X_a$ and when $X \geq X_b$, and it extends smoothly by zero across both radial edges. The subtraction creates an error supported where χ_m changes, and that error is kept in the radial equation.

For the normalized operators, use a common torus $y = Y_{i_0}$ and set $M = \Lambda_g^{i_0} Q^{d_r} / 2$. At fixed (Z, T) , the same symbols mean

$$\begin{aligned}
\mathcal{I}g(R, y) &= \int_0^R g(R', y + M((R')^{d_r} - R^{d_r})v_r) dR', \\
\mathcal{J}g(R, y) &= \int_0^\infty g(R', y + M((R')^{d_r} - R^{d_r})v_r) dR', \\
\mathcal{I}_c g &= \mathcal{I}g - \chi_m \mathcal{J}g.
\end{aligned} \tag{8.5}$$

Here $\chi_m = \chi_m(QR^2/(2q))$ is how the fixed physical cutoff appears in the chart. For $e \in \{0, 1, 2\}$, define

$$\mathcal{D}_e = D_r + e/R, \quad \mathcal{T}_e f = R^{-e} \mathcal{I}_c(R^e f), \quad \mathcal{A}_e f = R^{-e} (\partial_R \chi_m) \mathcal{J}(R^e f). \tag{8.6}$$

At matching physical and chart points, where $r = \sqrt{Q} R$, the normalization is

$$f_* = Q^a f_{\text{phys}}, \quad Q^{a-1/2} r^{-e} \mathcal{I}_{c, \text{phys}}(r^e f_{\text{phys}}) = R^{-e} \mathcal{I}_c(R^e f_*) = \mathcal{T}_e f_*.$$

Here $\mathcal{I}_{c, \text{phys}}$ is the physical operator defined above. The term $\mathcal{A}_e f$ is the cutoff remainder created by making the primitive compactly supported.

The inverse must keep the source's decay order. The next lemma assumes that the source is multiplied by the stated power of the radial variable, averaged over the auxiliary torus, and then integrated over the full radial line. At every fixed value of the other two slow variables, the result must be zero. Under that exact condition, the cutoff remainder must also be flat. The lemma gives both estimates.

Lemma 8.2. *Fix a correction stage j , an exponent α , a band, its common torus $y = Y_{i_0}$, and $e \in \{0, 1, 2\}$. Let $f \in M_\alpha$ be a smooth scalar coefficient. In particular, for every fixed multi-index I of derivatives in (R, Z, T, y) , its smooth extension by zero satisfies*

$$|\partial^I f| \leq C_{j,I} \varepsilon^\alpha S_*^{B_{j,I}} \zeta \delta^{-B_{j,I}}.$$

Then $\mathcal{T}_e f$ is supported in the same active radial shell and in the projection of the input support to (Z, T) . Under the primitive normalization defined above, it belongs to M_α . The exact identity is

$$\mathcal{D}_e \mathcal{T}_e f = f - \mathcal{A}_e f. \tag{8.7}$$

If $\int R^e \langle f \rangle_Y dR = 0$ at every (Z, T) , then $\langle \mathcal{A}_e f \rangle_Y = 0$. Also, for every fixed I and every integer $p \geq 1$,

$$|\partial^I \mathcal{A}_e f| \leq C_{j,I,p} \varepsilon^{\alpha+p\kappa_s} S_*^{B_{j,I,p}} \zeta \delta^{-B_{j,I,p}}. \tag{8.8}$$

For each (I, p) , only finitely many derivatives of the input are used. The constants and powers may depend on those indices, the stage, and the fixed profile, but not on the band, label, or point. If f does not depend on Y and $\int R^e f dR = 0$ at every slow point, the cutoff remainder is identically zero.

Proof. First estimate the primitive without differentiating the argument inside it, which shifts rapidly. Then check the exact radial identity and use Fourier inversion on the torus to estimate the cutoff remainder.

Step 1. Support and coefficient estimates. Work in a neighborhood with one fixed common index, as in Lemma 6.2, and put $\mathcal{L} = v_r \cdot \partial_y$ and $g = R^e f$. Multiplication by R^e leaves the input estimate unchanged on the shell. To estimate $\mathcal{I}_c g$, set $U = R^{d_r}$ and extend $F(U) = g(R)/(d_r R^{d_r-1})$ smoothly by zero. The shell stays inside one fixed compact subset of $R > 0$, so this change of variables and its inverse have bounded derivatives of every fixed order. The exact representation on the half-lines is

$$\begin{aligned} A_-(U, y) &= \int_{-\infty}^0 F(U + u, y + Muv_r) du, \\ A_+(U, y) &= \int_0^{\infty} F(U + u, y + Muv_r) du, \\ \mathcal{J}g &= A_- + A_+, \quad \mathcal{I}_c g = (1 - \chi_m)A_- - \chi_m A_+. \end{aligned} \tag{8.9}$$

For fixed u , the shift depends on neither U nor (Z, T) . So a coefficient derivative acts only on F and χ_m , and brings in no factor of M . The input's extension by zero includes derivatives of the moving shell, so taking derivatives creates no boundary terms.

Near the inner edge, only the left integral appears. Write $d_a = \log(X/X_a)$ and $d_b = \log(X_b/X)$, so $\zeta = \exp(-a_a/d_a^2 - a_b/d_b^2)$ and $\delta = \min(1, d_a, d_b)$, where a_a, a_b are fixed and positive. For every fixed B , the function $s \mapsto e^{-a_a/s^2} s^{-B}$ is increasing when s is positive and small enough. Along the left integration segment, this bounds the input weight by the same kind of weight at the endpoint; the factors from the opposite edge are bounded and comparable. Near the outer edge, use the right integral and the matching monotonicity for d_b . In the interior, the output weight has a positive lower bound. Each integration runs over a bounded length. The same facts apply after any fixed coefficient derivative, and prove the M_α estimate. Formula (8.9) also shows directly that the output is zero below and above the shell. Multiplying by $R^{\pm e}$ costs only fixed constants there.

Step 2. The radial equation. Acting on the integrand in (8.9), the operator $\partial_U + M\mathcal{L}$ is the same as taking its u -derivative. It therefore sends A_- to F , sends A_+ to $-F$, and sends $\mathcal{J}g$ to zero. Multiply by $d_r R^{d_r-1}$, then use the product rule for R^{-e} . This gives (8.7). In physical variables, the same calculation is $\mathfrak{r}\mathcal{I}_c g = g - (\partial_r \chi_m)\mathcal{J}g$.

Step 3. The cutoff remainder under the moment condition. The radial identity and the support bounds are now proved. To estimate the cutoff term still left, use the cancellation in the full radial integral. Haar measure does not change under translation, so $\langle \mathcal{J}g \rangle_Y = \int \langle g \rangle_Y dR$. Under the weighted moment condition, this is zero for $g = R^e f$. If H is a zero-mean function on the torus, the Diophantine inequality (6.7) says that the absolute value of either dot product is at least one fixed positive constant divided by one plus the size of the nonzero integer frequency. It gives

$$\|\mathcal{L}^{-p} H\|_{C_y^m} \leq C_{m,p} \|H\|_{C_y^{m+p+3}}, \quad p \geq 1. \tag{8.10}$$

The Fourier multiplier here is $(2\pi i v_r \cdot k)^{-p}$. After $m + p + 3$ derivatives of H have been estimated, the series still to be summed is bounded by $\sum_{k \in \mathbb{Z}^2} (1 + |k|)^{-3}$. Slow and radial derivatives commute with this multiplier. The zero-average part of the full-line integral therefore obeys the exact identity

$$\mathcal{J}g = (-M^{-1})^p \int_{\mathbb{R}} \partial_U^p \mathcal{L}^{-p} F^\circ(U + u, y + Muv_r) du.$$

This follows by integrating the u -derivative of $\mathcal{L}^{-1} F^\circ$, then repeating the same step; compact support makes every endpoint term zero. The zero Fourier mode left out of this formula has a full-line

integral that is identically zero at every slow point. Averaging, integration, and slow differentiation of the coefficients commute on the smooth extensions by zero, so every derivative of that mode also has zero integral. The same fixed-shift argument handles any further coefficient derivatives. For the output derivative indexed by I , it is enough to use radial input derivatives through $I_R + p$, torus derivatives through $|I_y| + p + 3$, and the slow derivatives named by I . Since $M^{-1} \leq C\varepsilon^{\kappa_s} S_*^{\ell_g}$ and $\partial_R \chi_m$ is supported in the interior, this proves (8.8). Fix a physical normalization and a physical derivative order, and let $L \geq 0$ bound the resulting loss of powers of q , including the loss from differentiating after evaluation at the phase. The graph operators give a finite L of this kind that does not depend on p . Given any flatness order $N \geq 0$, choose p so that $h(\alpha + p\kappa_s) > N + L$. Since $\varepsilon = Q^h$ and $q \asymp Q$, this strict excess absorbs the fixed powers of S_* and gives an $O(q^N)$ bound for that physical derivative. No estimate uniform in p is being claimed. Finally, when f does not depend on Y , $\mathcal{J}(R^e f) = \int R^e f dR$, which is zero under the stated weighted integral condition. $\square$

Whenever the weighted mean integral is zero, the inverse is exact apart from a flat cutoff remainder. This flatness uses estimates at every fixed derivative order. Frequencies comparable to M can come close to resonance with v_r , meaning that the dot product with the frequency can be small and division by it can be large. The extra derivatives in the torus variables in (8.10) control their small Fourier coefficients. A fixed finite regularity bound would give only a fixed finite order of flatness.

8.3 Reconstructing the pressure and making divergence-free velocity corrections.

This subsection applies the radial inverse from Lemma 8.2 first to the pressure p_m , then to an azimuthal vector potential that produces a divergence-free velocity increment; see Proposition 8.3. The axial component of that increment equals the prescribed source γ_d , apart from the cutoff remainder in (8.14). The pressure source may have a nonzero radial integral after its auxiliary average is taken. Subtracting a fixed bump times that integral separates out the obstruction, and the source left over can then be integrated with compact support, as in (8.12).

Let $\mathcal{I}_{\text{mean}}$ be the interval in X reserved by Theorem 4.6(vi), and set

$$I_m = \{\sqrt{2X} : X \in \mathcal{I}_{\text{mean}}\}.$$

The *mean patch* is the region $r/\sqrt{q} \in I_m$. Choose a fixed bump $\hat{\rho} \in C_c^\infty(I_m)$ with $\int \hat{\rho}(x) dx = 1$, and define

$$\rho_{\text{phys}}(r, z, t) = q^{-1/2} \hat{\rho}(r/\sqrt{q}), \quad \rho = Q^{1/2} \rho_{\text{phys}}.$$

This bump does not depend on the angular or auxiliary variables, and $\int \rho_{\text{phys}} dr = \int \rho dR = 1$. It is fixed independently of the velocity corrections.

The operator T_0 reconstructs the pressure, while T_1 reconstructs an azimuthal vector potential. The operators T_1 and T_2 also recover the radial fluxes of axial and azimuthal momentum. Write Ψ for the azimuthal coefficient of the vector potential. With a star marking a normalized representative, the physical normalizations are

$$\begin{aligned} u_* &= Q^A u_{\text{phys}}, & g_* &= Q^{2A+1/2} g_{\text{phys}}, & p_* &= Q^{2A} p_{\text{phys}}, \\ \Psi_* &= Q^{A-1/2} \Psi_{\text{phys}}, & \rho_* &= Q^{1/2} \rho_{\text{phys}}. \end{aligned} \tag{8.11}$$

In particular, $Q^{2A} \mathcal{I}_{c, \text{phys}} g_{\text{phys}} = \mathcal{I}_c g_*$. When the ε exponent is said to be preserved, it refers to these normalized representatives and already includes the length of integration.

Proposition 8.3. *Fix the radial operators from Lemma 8.2 and the bump ρ defined above.*

- (i) **Pressure.** For a smooth shell-supported scalar source $g_r(R, Z, T, y)$, define the radial defect $\mathcal{P}$ and the pressure p_m by

$$\mathcal{P} = \int \langle g_r \rangle_Y dR, \quad p_m = \mathcal{T}_0(g_r - \rho \mathcal{P}), \quad \int \rho dR = 1. \quad (8.12)$$

Then

$$D_r p_m - g_r = -\rho \mathcal{P} - \mathcal{A}_0(g_r - \rho \mathcal{P}), \quad \partial_R \langle p_m \rangle_Y = \langle g_r \rangle_Y - \rho \mathcal{P}. \quad (8.13)$$

If $g_r \in \mathcal{M}_\alpha$, then $\mathcal{P} \in \mathcal{S}_\alpha$ and $p_m \in \mathcal{M}_\alpha$. Changing g_r by a term in $\mathcal{M}_\alpha$ changes the pressure by a term in the same class. Under this class assumption, the cutoff remainder in (8.13) has zero auxiliary mean and is flat as $q \downarrow 0$ at every fixed derivative order.

- (ii) **Axial velocity.** Let $\gamma_d(R, Z, T, y)$ be a smooth, shell-supported desired axial increment such that $\int R \langle \gamma_d \rangle_Y dR = 0$ at every (Z, T) . Using (8.4), define the physical azimuthal coefficient of the vector potential by $\Psi = -r^{-1} \mathcal{I}_c(r \gamma_{d, \text{phys}})$. Define its normalized velocity increments by

$$\Psi_* = \mathcal{T}_1 \gamma_d, \quad \Delta \beta = -\varepsilon \partial_Z \Psi_*, \quad \Delta \gamma = (D_r + 1/R) \Psi_* = \gamma_d - \mathcal{A}_1 \gamma_d. \quad (8.14)$$

The actual increment $(\Delta \beta, 0, \Delta \gamma)$ is divergence-free and satisfies $\langle \Delta \gamma \rangle_Y = \langle \gamma_d \rangle_Y$. If $\gamma_d \in \mathcal{M}_\alpha$, then $\Psi_*, \Delta \gamma \in \mathcal{M}_\alpha$ and $\Delta \beta \in \mathcal{M}_{\alpha+1}$. If γ_d does not depend on Y , its zero weighted integral makes $\Delta \gamma = \gamma_d$ at every point. The pressure and vector potential are compactly supported inside the active shell, and their support in (Z, T) stays inside the projection of the source support.

Proof. **Part (i). Pressure.** The source in (8.12) has zero radial integral after its auxiliary mean is taken, since

$$\int \langle g_r - \rho \mathcal{P} \rangle_Y dR = \mathcal{P} - \mathcal{P} \int \rho dR = 0.$$

Apply Lemma 8.2 with $e = 0$, then take the auxiliary average. The moment $\mathcal{P}$ belongs to $\mathcal{S}_\alpha$ when $g_r \in \mathcal{M}_\alpha$. To see this, integrate its derivatives of any fixed order over the bounded shell, using the smooth extension by zero and the fact that $\zeta \delta^{-B}$ is bounded for every finite B . The interior bump $\rho \mathcal{P}$ belongs to $\mathcal{M}_\alpha$. Since ρ is fixed independently of the current velocity, the same argument proves the statement about changes and completes part (i).

Part (ii). Axial velocity. The azimuthal potential produces the two physical velocity components $-\partial_Z \Psi$ and $(\tau + 1/r) \Psi$. The operators τ and ∂_Z commute when they differentiate after evaluation at the phase map, so the divergence of these two components is exactly zero:

$$(\tau + r^{-1})(-\partial_Z \Psi) + \partial_Z((\tau + r^{-1}) \Psi) = 0.$$

The weighted primitive identity gives the last equality in (8.14), including its sign. The zero weighted axial-flux integral makes the auxiliary average of its cutoff remainder zero. Finally, $Q^{1/2-D} = Q^h = \varepsilon$, which supplies the extra factor ε in $\Delta \beta$ in (8.14). The remaining claims follow from Lemma 8.2, and part (ii) is complete. $\square$

8.4 The three compatibility defects in the integrated equations.

This subsection finds the integral defects left by the pressure reconstruction in (8.13) and by the two tangential equations in (8.3). These defects are the quantities that must be removed before the required compactly supported corrections can exist. Tangential stresses are then built so that their radial divergences cancel the auxiliary-averaged residuals apart from two explicit bump terms; see Corollary 8.5. The pressure reconstruction leaves the integral $\mathcal{P}$ from (8.12) to be corrected. In the same way, the auxiliary averages of the tangential residuals must have zero weighted radial integrals before compactly supported stresses can represent them, by Lemma 8.2. The moment constraints in (8.2) remove the time terms and the axial-viscosity terms from these integrals. The axial fluxes left after that give two more defects, as Equations (8.15) and (8.16) show.

Along with $\mathcal{P}$ from (8.12), define the two flux defects and the second radial moment c_ρ of ρ by

$$\begin{aligned}\mathcal{J}_\theta &= \int R^2 \langle Gv + V\gamma + \gamma v + W_{z\theta} \rangle_Y dR, \\ \mathcal{J}_z &= \int R \langle 2G\gamma + \gamma^2 + W_{zz} \rangle_Y dR - \frac{1}{2} \int R^2 \langle g_r \rangle_Y dR, \quad c_\rho = \frac{1}{2} \int R^2 \rho dR.\end{aligned}\tag{8.15}$$

Proposition 8.4. *Let (β, v, γ) and w be the smooth corrections from (8.1), supported in the shell. Suppose that the two moments in (8.2) are zero at every (Z, T) , and reconstruct p_m from their radial source g_r by (8.12). For the residual components E_θ, E_z from (8.3) and the defects $\mathcal{J}_\theta, \mathcal{J}_z, c_\rho$ from (8.15),*

$$\int R^2 \langle E_\theta \rangle_Y dR = \varepsilon \partial_Z \mathcal{J}_\theta, \quad \int R \langle E_z \rangle_Y dR = \varepsilon \partial_Z (\mathcal{J}_z + c_\rho \mathcal{P}).\tag{8.16}$$

Proof. After auxiliary averaging, D_r becomes ∂_R , and $\mathfrak{t}_*$ becomes $-\varepsilon \partial_T$. Multiply each radial divergence in (8.3), including the one from the residual stress tensor, by its stated weight and integrate. Each one becomes a term at the endpoints, and compact support makes those terms zero. Two integrations by parts make the radial viscous integrals zero.

$$\int R^2 (v_{RR} + R^{-1}v_R - R^{-2}v) dR = (2 - 1 - 1) \int v dR = 0,$$

$$\int R (\gamma_{RR} + R^{-1}\gamma_R) dR = - \int \gamma_R dR + \int \gamma_R dR = 0.$$

These formulas are applied to the auxiliary averages. The time terms and axial-viscosity terms are derivatives of the two integrals that are already zero. The compactly supported pressure reconstruction fixes the pressure moment as follows.

$$\int R \langle p_m \rangle_Y dR = -\frac{1}{2} \int R^2 \partial_R \langle p_m \rangle_Y dR = -\frac{1}{2} \int R^2 \langle g_r \rangle_Y dR + c_\rho \mathcal{P}.$$

Putting this expression into the axial fluxes that remain gives (8.16). Since ρ is scaled using the moving q , c_ρ may depend on (Z, T) . For this reason, its full product with $\mathcal{P}$ stays inside ∂_Z . $\square$

The moments in physical variables and their normalized representatives are related by

$$\begin{aligned}(M_\theta)_* &= Q^{A-3/2} (M_\theta)_{\text{phys}}, & (M_z)_* &= Q^{A-1} (M_z)_{\text{phys}}, \\ \mathcal{P}_* &= Q^{2A} \mathcal{P}_{\text{phys}}, & (c_\rho)_* &= Q^{-1} (c_\rho)_{\text{phys}}, \\ (\mathcal{J}_\theta)_* &= Q^{2A-3/2} (\mathcal{J}_\theta)_{\text{phys}}, & (\mathcal{J}_z)_* &= Q^{2A-1} (\mathcal{J}_z)_{\text{phys}}.\end{aligned}\tag{8.17}$$

The constraints that set an integral to zero do not change under normalization. Every nonzero correction target uses the normalization for its own row.

The two tangential residual moments are therefore axial derivatives of $\mathcal{J}_\theta$ and $\mathcal{J}_z + c_\rho \mathcal{P}$. Subtract a fixed normalized bump times each weighted residual integral, then apply the radial inverse. These sources do not depend on the auxiliary variable, and their weighted integrals are zero, so the inverse is exact. The resulting pair will be the target for covariance added by the waves. The weighted radial divergence of that covariance enters (8.3) with a positive sign through $W_{r\theta}$ and W_{rz} . Fix the interior profiles $\hat{\sigma}_\theta, \hat{\sigma}_z$ so that $\int \xi^2 \hat{\sigma}_\theta(\xi) d\xi = \int \xi \hat{\sigma}_z(\xi) d\xi = 1$, and define

$$\sigma_{\theta, \text{phys}} = q^{-3/2} \hat{\sigma}_\theta(r/\sqrt{q}), \quad \sigma_{z, \text{phys}} = q^{-1} \hat{\sigma}_z(r/\sqrt{q}).$$

Their supports lie inside the mean patch. In each chart, use $\sigma_\theta = Q^{3/2} \sigma_{\theta, \text{phys}}$ and $\sigma_z = Q \sigma_{z, \text{phys}}$, so $\int R^2 \sigma_\theta dR = \int R \sigma_z dR = 1$. This gives the same physical bump functions wherever charts overlap.

Corollary 8.5. *Under the assumptions of Proposition 8.4, define the covariance targets*

$$\begin{aligned} H_\theta &= -\mathsf{T}_2(\langle E_\theta \rangle_Y - \sigma_\theta \varepsilon \partial_Z \mathcal{J}_\theta), \\ H_z &= -\mathsf{T}_1(\langle E_z \rangle_Y - \sigma_z \varepsilon \partial_Z (\mathcal{J}_z + c_\rho \mathcal{P})). \end{aligned} \quad (8.18)$$

They do not depend on the auxiliary variable, are compactly supported inside the active shell, and satisfy

$$\begin{aligned} (D_r + 2/R)H_\theta &= -\langle E_\theta \rangle_Y + \sigma_\theta \varepsilon \partial_Z \mathcal{J}_\theta, \\ (D_r + 1/R)H_z &= -\langle E_z \rangle_Y + \sigma_z \varepsilon \partial_Z (\mathcal{J}_z + c_\rho \mathcal{P}). \end{aligned}$$

The radial primitives act while (Z, T) is held fixed. It follows that H_θ and H_z are zero whenever their matching integrands in (8.18) are zero for every R . The construction does not enlarge the support in the axial variable Z .

Proof. The integrated identities and the normalization of the two bumps give

$$\begin{aligned} \int R^2 (\langle E_\theta \rangle_Y - \sigma_\theta \varepsilon \partial_Z \mathcal{J}_\theta) dR &= 0, \\ \int R (\langle E_z \rangle_Y - \sigma_z \varepsilon \partial_Z (\mathcal{J}_z + c_\rho \mathcal{P})) dR &= 0. \end{aligned}$$

Both sources are independent of the auxiliary variable. Lemma 8.2 makes their cutoff remainders zero, which proves the two identities and the statements about support. $\square$

These radial-divergence identities leave the factors $\varepsilon \partial_Z \mathcal{J}_\theta$ and $\varepsilon \partial_Z (\mathcal{J}_z + c_\rho \mathcal{P})$ in the bump terms that remain. The three defects can therefore be corrected at fixed (Z, T) without losing the displayed factors of $\varepsilon \partial_Z$.

8.5 Inverting the auxiliary-time derivative on zero-mean functions.

Corollary 8.5 reduces the auxiliary-averaged correction to the three integral defects and a compactly supported stress. The part of the mean residual left after its auxiliary average is removed, $(E_\theta^\circ, E_z^\circ)$, has zero auxiliary average at every slow point by (3.9). Its correction comes from the fast time derivative. The Fourier multiplier for that derivative is invertible at every nonzero torus frequency by (6.7). The axial velocity increment is then made from its azimuthal potential through (8.14), which keeps the velocity incompressible.

Recall from Equations (6.4) and (6.6) that $N_{\text{abs}} = v_t \cdot \partial_Y$ acts on the original auxiliary coordinate. On a common torus $y = Y_{i_0}$, the normalized fast time derivative is $c_{i_0} N_{i_0}$, where $N_{i_0} = v_t \cdot \partial_y$ and $c_{i_0} = T_g^{i_0} Q^{1+h}$. The normalized physical time derivative splits as $\mathfrak{t}_* = -\varepsilon \partial_T + c_{i_0} N_{i_0}$. Its slow part $-\varepsilon \partial_T$ differentiates the coefficient's explicit dependence on time while y stays fixed. Its fast part $c_{i_0} N_{i_0}$ differentiates the auxiliary oscillations along the physical phase map. The fast part is inverted here, while the slow part stays in the residual.

Lemma 8.6. *Let $N = v_t \cdot \partial_Y$ on $\mathbb{T}^2$, with v_t as in (6.2), and let F be a smooth function with zero Haar mean. The equation $N\varphi = F$ has exactly one smooth solution with zero mean, namely $\varphi = N^{-1}F$. For every integer $m \geq 0$, this solution is given by the Fourier series below and satisfies the stated norm bound.*

$$N^{-1}F = \sum_{k \in \mathbb{Z}^2 \setminus \{0\}} \frac{\widehat{F}(k)}{2\pi i v_t \cdot k} e^{2\pi i k \cdot y}, \quad \|N^{-1}F\|_{C_y^m} \leq C_m \|F\|_{C_y^{m+4}}. \quad (8.19)$$

For a pair $E_\theta, E_z \in \mathsf{M}_\alpha$ of normalized residual coefficients that do not depend on the physical angle, use this inverse in the original auxiliary coordinate Y . It defines the physical tangential update $-N_{\text{abs}}^{-1}(E_{\theta, \text{phys}}^\circ, E_{z, \text{phys}}^\circ)$.

The residuals and velocities here have the normalizations in (8.11). Defining the update in physical coordinates makes its values agree wherever charts overlap. Its chart representatives are

$$\Delta v = -c_{i_0}^{-1} N_{i_0}^{-1} E_{\theta}^{\circ}, \quad \gamma_d = -c_{i_0}^{-1} N_{i_0}^{-1} E_z^{\circ}, \quad c_{i_0}^{-1} \leq CS_*. \quad (8.20)$$

Both desired increments $\Delta v, \gamma_d$ belong to M_{α} , and each has zero auxiliary mean at every slow and radial point. Realizing γ_d through (8.14) gives the actual velocity increment

$$\begin{aligned} (\Delta\beta, \Delta v, \Delta\gamma) &= (-\varepsilon \partial_Z T_1 \gamma_d, \Delta v, \gamma_d - \mathcal{A}_1 \gamma_d), \\ (\Delta\beta, \Delta v, \Delta\gamma) &\in M_{\alpha+1} \times M_{\alpha} \times M_{\alpha}. \end{aligned} \quad (8.21)$$

This field is divergence-free, and adding it leaves both integral constraints in (8.2) unchanged. Let $a = -\mathcal{A}_1 \gamma_d$ be the cutoff remainder in the axial increment. Its cancellation equations for the fast-time derivative are

$$c_{i_0} N_{i_0} \Delta v = -E_{\theta}^{\circ}, \quad c_{i_0} N_{i_0} \Delta\gamma = -E_z^{\circ} + c_{i_0} N_{i_0} a. \quad (8.22)$$

The angular cancellation is exact. The term $c_{i_0} N_{i_0} a$ is flat at every fixed order of coefficient derivatives.

Proof. The divisor bound in (6.7) gives $|v_t \cdot k|^{-1} \leq C(1 + |k|)$. Taking four more derivatives of F than the number requested for the output leaves a summable $(1 + |k|)^{-3}$ bound for the two-dimensional Fourier series. This proves (8.19), smoothness, and uniqueness once the zero Fourier coefficient is fixed. The multiplier commutes with every slow coefficient derivative and preserves real-valued functions. It acts while (R, Z, T) is held fixed, so it preserves slow and radial support and their smooth extensions by zero, though it does not have to preserve support on the auxiliary torus.

Let $P_i F(Y) = F(J_g^i Y)$. A mode k on the i -torus has frequency $(J_g^i)^T k$ in the original auxiliary coordinate Y . Since $J_g v_t = T_g v_t$,

$$N_{\text{abs}} P_i = T_g^i P_i N_i, \quad N_{\text{abs}}^{-1} P_i = T_g^{-i} P_i N_i^{-1}. \quad (8.23)$$

The finite covering preserves the Haar average. The same fact also follows from its injective map between the frequency lattices. Applying (8.19) in the common frequency coordinate k gives the same fixed loss of derivatives in every chart.

In physical variables, the update is $-N_{\text{abs}}^{-1} E_{\text{phys}}^{\circ}$. Its value in chart coordinates has the factor $Q^{A-(2A+1/2)} T_g^{-i_0} = Q^{-1-h} T_g^{-i_0} = c_{i_0}^{-1}$, which proves (8.20). The choice of covering level and its bounded difference from a band level give the bound CS_* for this factor. For each output derivative, four additional derivatives in the torus variables are enough, so the class estimates at every fixed derivative order keep their ε exponent. Applying the slow-time term $-\varepsilon \partial_T$ to these increments supplies one more factor of ε .

The desired increments have zero auxiliary mean at every radial point. This gives both desired integral constraints, and making the increment from the azimuthal potential keeps them exactly. The bounds for $\Delta\beta$ and $\Delta\gamma$ in (8.21) follow from Proposition 8.3(ii). The fast derivative commutes with $\mathcal{I}$, $\mathcal{J}$, and $\mathcal{I}_c$, because the shifts are translations and χ_m does not depend on the torus variable. Apply that derivative to $\Delta\gamma = \gamma_d + a$ to get (8.22). Its last term is flat at every fixed order of coefficient derivatives by (8.8). Full slow differentiation does not have to commute with $\mathcal{I}_c$; Lemma 8.2 estimates those derivatives instead. $\square$

8.6 A five-dimensional correction that preserves the moment constraints.

At this point, there are inverses for the pressure in Proposition 8.3, the auxiliary-averaged tangential residual in Corollary 8.5, and the part with zero auxiliary average in Lemma 8.6. What remains are the three integral defects $\mathcal{P}, \mathcal{J}_{\theta}, \mathcal{J}_z$ from Equations (8.12) and (8.15). This subsection corrects their linear changes by adding a velocity increment that also leaves the two constraints in (8.2) unchanged. Three azimuthal coefficients and two axial coefficients solve the five equations in (8.25).

On the reserved mean patch, Proposition 5.5 says that the background still follows the power law (4.30). This form turns the equations into separate power moments of bumps with compact support. Lemma 8.7 first builds one fixed linear map that works for any target values. Lemma 8.8 then finds the terms that remain when those targets are the defects in the current fields.

For $x = r/\sqrt{q} \in I_m$, normalizing the velocity at q gives the base

$$G_q = 0, \quad V_q = a(\eta)x^{-1-2\lambda}, \quad |a(\eta)| \geq a_0 > 0. \quad (8.24)$$

Here $a(\eta) = 2^{1/2+\lambda}c_{\text{patch}}(1+\eta^2)^{-1}$. Equation (5.44) gives the exact statement for the whole summed base. The other tangential coefficients, beyond the leading profile, have their support elsewhere by (5.18). Multiplying their azimuthal potentials by a cutoff in q cannot create an axial component on this patch, because q does not depend on the radius. This means that (8.24) holds for the summed base used in the balances. The fixed parameter λ is positive.

Lemma 8.7. *Suppose the fixed slow base satisfies (8.24) on the reserved interior interval I_m , where $\lambda > 0$ and $|a(\eta)| \geq a_0 > 0$. It is possible to choose three fixed azimuthal bump profiles and two fixed axial bump profiles, all supported in this interval, with the following property. At each (Z, T) , take any three scalar targets $(\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z)$. There is then exactly one linear combination $(\Delta v, \gamma_d)$ of the physically rescaled profiles that satisfies*

$$\begin{aligned} \int R^2 \Delta v \, dR &= 0, & \int R \gamma_d \, dR &= 0, \\ \int (2V/R) \Delta v \, dR &= -\mathcal{P}, \\ \int R^2 (G \Delta v + V \gamma_d) \, dR &= -\mathcal{J}_\theta, \\ \int (2RG \gamma_d - RV \Delta v) \, dR &= -\mathcal{J}_z. \end{aligned} \quad (8.25)$$

Normalize the three targets one row at a time as in (8.17). The resulting increments do not depend on either the angular variables or the auxiliary variables, and they depend linearly on the targets. For every α , targets in S_α give increments in M_α for every fixed coefficient derivative. The constants in these bounds may depend on the fixed λ . The azimuthal-potential construction in (8.14) gives $\Delta \gamma = \gamma_d$ exactly and $\Delta \beta \in M_{\alpha+1}$. It also preserves both integral constraints exactly. When the targets are the current defects from Equations (8.12) and (8.15), the last three equations cancel both the current defects and the displayed terms that involve the fixed base.

Proof. Step 1, choose the five profiles. Normalize each physical row and its target using q . On this patch, (8.24) separates the rows into an angular block with the powers x given by $2, -2-2\lambda, -2\lambda$, and an axial block with the powers $1, 1-2\lambda$. Since the fixed λ is positive, the powers in each list are all different.

Here is a direct way to see that the map is invertible. Choose a nonnegative bump η_0 , not identically zero, whose support lies near some $x_0 > 0$ inside the patch. Next choose a small fixed $d > 0$ and make the support narrow enough that the scaled copies $\eta_j(x) = a_j^{-1}\eta_0(x/a_j)$, $a_j = e^{jd}$, $j = 0, 1, 2$ lie in disjoint subintervals of the patch. Their power moments are

$$\int x^p \eta_j(x) \, dx = \mu_p e^{jdp}, \quad \mu_p = \int x^p \eta_0(x) \, dx > 0.$$

For the three angular powers, divide each row of the matrix by μ_p . What remains is the usual Vandermonde matrix in the different numbers e^{dp} , so its determinant is not zero. Using two copies gives the same result for the axial block. Every factor left over is a nonzero multiple of $a(\eta)$.

Step 2, find the coefficients by inverting the matrices. Use η_0, η_1, η_2 for the azimuthal profiles and η_0, η_1 for the axial profiles. Write $\mathcal{P}_q, \mathcal{J}_{\theta,q}, \mathcal{J}_{z,q}$ for the physical targets after multiplying them by the powers in (8.17), with q in place of Q . Define the two matrices

$$A_\theta = \begin{pmatrix} \mu_2 & \mu_2 e^{2d} & \mu_2 e^{4d} \\ 2a\mu_{-2-2\lambda} & 2a\mu_{-2-2\lambda} e^{d(-2-2\lambda)} & 2a\mu_{-2-2\lambda} e^{2d(-2-2\lambda)} \\ -a\mu_{-2\lambda} & -a\mu_{-2\lambda} e^{-2d\lambda} & -a\mu_{-2\lambda} e^{-4d\lambda} \end{pmatrix},$$

$$A_z = \begin{pmatrix} \mu_1 & \mu_1 e^d \\ a\mu_{1-2\lambda} & a\mu_{1-2\lambda} e^{d(1-2\lambda)} \end{pmatrix}, \quad a = a(\eta).$$

Step 1 shows that both matrices are invertible. Use their inverses to set

$$\begin{aligned} (u_0, u_1, u_2)^T &= A_\theta^{-1}(0, -\mathcal{P}_q, -\mathcal{J}_{z,q})^T, \\ (s_0, s_1)^T &= A_z^{-1}(0, -\mathcal{J}_{\theta,q})^T, \\ \Delta v_{\text{phys}}(r, z, t) &= q^{-A} \sum_{j=0}^2 u_j \eta_j(r/\sqrt{q}), \\ \gamma_{d,\text{phys}}(r, z, t) &= q^{-A} \sum_{j=0}^1 s_j \eta_j(r/\sqrt{q}). \end{aligned}$$

The matrix A_θ makes the angular-momentum integral zero and gives the targets $-\mathcal{P}_q, -\mathcal{J}_{z,q}$. The matrix A_z makes the axial-flux integral zero and gives the target $-\mathcal{J}_{\theta,q}$. Because $G_q = 0$, these two blocks cover all five equations. Their inverses give one physical correction, depending linearly on the three targets. The correction is smooth in the slow variables whenever the targets are smooth. The inverse can get worse as $\lambda \downarrow 0$, but no bound uniform in that limit is needed here.

Step 3, check the normalization and bounds, then build the axial field. Moving from rows normalized at q to a chart based at a fixed Q multiplies each row and its target by the same powers from (8.17), along with bounded smooth powers of q/Q . Every fixed-order derivative of those ratios is bounded. Inverting and differentiating the matrices therefore costs only fixed constants. The radial weight is also bounded above and below on the fixed interior support. This proves the map from S_α to M_α . The required axial field is slow, and its weighted axial-flux integral is zero. Lemma 8.2 therefore gives a cutoff remainder that vanishes identically. Its azimuthal potential and radial velocity component remain inside the mean patch, including the intervals between the finitely many bump supports. $\square$

The linear map is now fixed without using the target values. To use it in the nonlinear equations, the defects must be recomputed from the updated field. The next lemma gives the exact change and the extra decay supplied by the bounds used in the iteration. The base estimates needed for that gain hold on the whole support of the azimuthal potential, including the intervals between the bump supports.

Lemma 8.8. *Suppose the slow base satisfies (8.24), with the fixed $\lambda > 0$ and the lower bound $|a(\eta)| \geq a_0 > 0$. Let (β, v, γ) be a smooth divergence-free mean correction supported in the active shell, and keep the base and the wave field w fixed. Use Equations (8.3), (8.12), and (8.15) to form g_r and the three defects $(\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z)$. Feed those three defects into Lemma 8.7, and construct the resulting γ_d through (8.14). The updated mean velocity is*

$$(\beta_{\text{new}}, v_{\text{new}}, \gamma_{\text{new}}) = (\beta + \Delta\beta, v + \Delta v, \gamma + \Delta\gamma),$$

Recompute $g_{r,\text{new}}$ from (8.3), and define $\mathcal{R}_g = g_{r,\text{new}} - g_r - (2V/R)\Delta v$. Then

$$\begin{aligned}\mathcal{R}_g = & -\mathfrak{t}_*\Delta\beta - (D_r + 1/R)(2b\Delta\beta + 2\beta\Delta\beta + (\Delta\beta)^2) \\ & - D_z(b\Delta\gamma + G\Delta\beta + \beta\Delta\gamma + \gamma\Delta\beta + \Delta\beta\Delta\gamma) \\ & + (2v\Delta v + (\Delta v)^2)/R + \varepsilon(\Delta_0 - R^{-2})\Delta\beta.\end{aligned}\tag{8.26}$$

The defects recomputed from the updated field are

$$\begin{aligned}\mathcal{P}_{\text{new}} &= \int \langle \mathcal{R}_g \rangle_Y dR, \\ (\mathcal{J}_\theta)_{\text{new}} &= \int R^2 \langle \gamma\Delta v + v\Delta\gamma + \Delta\gamma\Delta v \rangle_Y dR, \\ (\mathcal{J}_z)_{\text{new}} &= \int R \langle 2\gamma\Delta\gamma + (\Delta\gamma)^2 \rangle_Y dR - \frac{1}{2} \int R^2 \langle \mathcal{R}_g \rangle_Y dR.\end{aligned}\tag{8.27}$$

Assume in addition that, everywhere on the support of the azimuthal potential for this correction, every fixed coefficient derivative of b is bounded by $C_I \varepsilon S_*^{B_I}$, while the corresponding derivatives of the slow fields V, G are bounded by $C_I S_*^{B_I}$. Suppose the current corrections satisfy $v, \gamma \in M_{0.9}$ and $\beta \in M_{1.9}$, and suppose the three targets lie in S_α for $\alpha \geq 0.9$. Then

$$\mathcal{R}_g \in M_{\alpha+0.9-2\kappa_s}, \quad (\mathcal{P}_{\text{new}}, (\mathcal{J}_\theta)_{\text{new}}, (\mathcal{J}_z)_{\text{new}}) \in S_{\alpha+0.9-2\kappa_s}.$$

Proof. Subtract the two radial sources in (8.3). The covariance W does not change. Expanding every quadratic product gives (8.26). In the azimuthal flux defect, the terms linear in the base are $\int R^2 (G\Delta v + V\Delta\gamma) dR$. In the axial flux defect, they are $\int (2RG\Delta\gamma - RV\Delta v) dR$. The second term here comes from the radial-source term $-\frac{1}{2} \int R^2 (2V/R)\Delta v dR$. Since $\Delta\gamma = \gamma_d$, the last three rows of (8.25) cancel the old defects together with these linear terms. The products left after that cancellation give (8.27).

For the estimates, Lemma 8.7 gives $\Delta v, \Delta\gamma \in M_\alpha$ and $\Delta\beta \in M_{\alpha+1}$. None of these three increments depends on the auxiliary variable. In particular, $\mathfrak{t}_*\Delta\beta = -\varepsilon\partial_T\Delta\beta$. The terms in $\mathcal{R}_g$ lie in the following classes.

term	class
$-\mathfrak{t}_*\Delta\beta$	$M_{\alpha+2}$
$(D_r + 1/R)(2b\Delta\beta)$	$M_{\alpha+2-\kappa_s}$
$D_z(b\Delta\gamma + G\Delta\beta)$	$M_{\alpha+2}$
$2v\Delta v/R$	$M_{\alpha+0.9}$
$(\Delta v)^2/R$	$M_{2\alpha}$
$\varepsilon(\Delta_0 - R^{-2})\Delta\beta$	$M_{\alpha+2-2\kappa_s}$

The other transport products lie in $M_{\alpha+2-\kappa_s}$. Because $\alpha \geq 0.9$, every exponent in the table is at least $\alpha + 0.9 - 2\kappa_s$. Radial integration keeps these estimates after each moment has been normalized by the factors given above. The radial weight has a positive lower bound on the interior support of the azimuthal potential, so the local base bounds in the statement are enough. $\square$

8.7 Recomputing the fields and matching them between charts.

This subsection gives the order for recomputing the residuals and defects in Equations (8.3), (8.12), and (8.15) after a mean correction. It also checks that the constructions match between charts, using Equations (8.4) and (8.23) and Lemma 6.2. Each correction changes the quadratic fluxes, so the next source has to be computed from the updated divergence-free velocity. A cutoff remainder whose auxiliary average is zero can have a nonzero average after multiplication by another term. Its contributions to the pressure and to the compatibility integrals in Equations (8.3) and (8.15) therefore have to be kept in the next step.

Fix the base coefficients and the stress. An actual velocity tuple (β, v, γ, w) then determines the pressure and the mean residuals in this order.

- (1) Using the full wave velocities, form $W_{ab} = \langle w_a w_b \rangle_\theta$. Then define g_r by the third row of (8.3).
- (2) Define $\mathcal{P} = \int \langle g_r \rangle_Y dR$ and $p_m = T_0(g_r - \rho \mathcal{P})$.
- (3) With this pressure, define E_θ, E_z by the first two rows of (8.3).
- (4) Using the same velocity and radial source, define $\mathcal{J}_\theta, \mathcal{J}_z$ by (8.15).

Repeat these definitions after every change to the velocity. For either kind of mean update, the new tuple is

$$(\beta, v, \gamma, w)_{\text{new}} = (\beta + \Delta\beta, v + \Delta v, \gamma + \Delta\gamma, w).$$

For the temporal update, use the current $E_\theta^\circ, E_z^\circ$ in (8.20). For the moment update, use the current $(\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z)$ in (8.25). In both updates, $\Delta\gamma$ means the actual axial increment from (8.14). Its cutoff remainder is therefore included before any of the products in the four definitions above are evaluated. After the moment update, Equations (8.26)–(8.27) give the recomputed defects exactly.

Proposition 9.6 and its proof give the correction cycle and the estimates for its exponents.

Because the definitions use physical variables, they match across charts. The radial integrals keep (z, t) fixed, and $q = q(z, t)$ keeps them from creating new dyadic bands. It is therefore possible to fix one common index on a neighborhood that contains the closures of all the supports in question. The Fourier covariance in (8.23) makes the formulas agree where the charts overlap, and the radial formulas agree because they have the same physical definition. Changing representations shifts the covering indices by only bounded amounts. The integer-valued covering indices are held fixed while derivatives are taken, and the coordinate changes preserve the derivative bounds already proved. All the estimates above hold on the same domain $0 < q < q_*$, which was fixed before the iteration. On any closed region $q \geq c > 0$, including $\eta = \pm 1$, only finitely many bands occur. On this region, the fixed-shift integrals, the inverse Fourier transform, and the finite-dimensional moment correction preserve every one-sided derivative of the source, including the derivatives at the endpoints.

9 Improving the residual and building the local field

The wave and mean constructions from Sections 7 and 8 correct different parts of the momentum residual. This section puts them together to prove Theorem 3.1. Start with the fixed background (u_B, p_B) from Proposition 5.5 and the primary fields assembled in (7.35), which Lemma 7.7 makes divergence-free. The construction must produce a local velocity u_{loc} whose residual and every derivative of that residual vanish to every order at the singularity, while the inner velocity growth and the exterior heat flow remain unchanged. Every correction creates new nonlinear errors. A full cycle therefore has to improve the decay before the next cycle begins.

One cycle works in the following order. Proposition 7.2 first removes the nonzero angular harmonics. Equation (7.31) then changes the primary amplitudes to adjust the averaged stress. The mean equations use (8.20) to remove the part whose auxiliary average is zero. Finally, the five-equation system (8.25) corrects the three compactly supported defects while preserving the two moment constraints. Proposition 9.6 proves that a cycle gains a fixed positive power of ε . Every inverse uses the same fixed background and the same primary amplitudes. This lets every finite stage live on one common domain, as Lemma 9.7 shows. Summing the stages there gives the local field in Proposition 9.9.

Table 1 collects the scales, operators, and field classes that occur repeatedly. In this section, $\kappa_s = 10^{-5}$, and $W_\alpha, M_\alpha, S_\alpha$ have the meaning given in Section 6.4 at every derivative order. Each exponent is a power of $\varepsilon = Q^h$, with the normalization

used for those classes. At a fixed correction stage and a fixed order of amplitude differentiation, the constants, the powers of S_* , and the inverse powers of distance are allowed to depend on that stage and order. They are uniform in the band, label, copy, and point. The angular mean $\langle \cdot \rangle_\theta$ and the auxiliary mean $\langle \cdot \rangle_Y$ are different operations, both defined in (3.7). Every class estimate is an estimate for normalized chart coefficients. The factors that recover the physical velocity, pressure, and residual are Q^{-A} , Q^{-2A} , and $Q^{-2A-1/2}$, respectively. In particular, the physical fields have residual

$$\mathcal{R}(u, p) = \partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p, \quad \mathcal{R}_* = Q^{2A+1/2} \mathcal{R}(u, p).$$

The residual identities below use physical fields, while their estimates use normalized coefficients. A Cartesian space-time jet collects a field and all its Cartesian space-time derivatives up to the stated order. Equation (5.23) writes $|\cdot|_m$ for the largest absolute value among them through that order.

9.1 Residual estimates for velocities built from curls

The amplitude equations (7.5) cancel the main part of a prescribed source. The physical velocity also contains the terms created by taking a curl, as Lemma 7.7 describes. Its residual contains derivatives of the amplitudes and the background as well. Equation (9.2) expands those terms. Proposition 9.1 and Lemma 9.2 first estimate the linear terms left after the main cancellation and the nonlinear interactions. Those estimates give the gains needed when Proposition 9.6 adds one correction after another.

For a fixed label, write an angular harmonic as $a \exp(ikm\Phi)$, where $m \in \mathbb{Z} \setminus \{0\}$ and a is the amplitude. Equation (7.4) defines the phase normal n_Φ , and (6.6) gives the chain-rule operators for evaluation at $Y = Y(r, t)$. The background coefficients (b, V, G) are the ones used in Sections 6 and 7.

For a transverse amplitude t_m , the potential in (7.38) gives

$$C_m = \frac{in_\Phi \times t_m}{km|n_\Phi|^2}, \quad \text{curl}_*(C_m e^{ikm\Phi}) = (t_m + r_m) e^{ikm\Phi}, \quad a_m = t_m + r_m.$$

Here r_m is the correction created by taking the curl. If $t_m \in W_\alpha$, then Lemma 7.7 gives $r_m \in W_{\alpha+1/2-\kappa_s}$. Here longitudinal means measured in the direction of the phase normal. Equation (7.39) estimates the scalar product with that normal for the full amplitude a_m . This difference matters in the quadratic transport term, because the factor doing the advecting is the full divergence-free velocity.

Separate the principal linear operator, which the amplitude equation handles, from all the other transport, pressure, and viscous terms. For comparison with the residual later, write the principal operator acting on an amplitude and its pressure as

$$\mathcal{L}_m(a, \pi) = Q^{1+h} N_{\text{abs}} a + \mathcal{K}a + \varepsilon k^2 m^2 |n_\Phi|^2 a + ikmn_\Phi \pi. \quad (9.1)$$

The inverse from Proposition 7.2 constructs (t_m, π_m) so that $\mathcal{L}_m(t_m, \pi_m) = -f_m$ and $n_\Phi \cdot t_m = 0$, before the pulse cutoff is applied. The amplitude t_m does not have to extend by zero in time before that cutoff is applied.

Proposition 9.1. *Fix a label, $m \in \mathbb{Z} \setminus \{0\}$, and $\alpha \in \mathbb{R}$. Let $f_m \in W_\alpha$ satisfy the support and smooth-extension assumptions in Proposition 7.2. Also assume that the source harmonic $f_m e^{ikm\Phi}$ extends smoothly by zero outside the fixed local rectangle, including across the ends of its pulse. Let (t_m, π_m) solve $\mathcal{L}_m(t_m, \pi_m) = -f_m$. The same conclusion holds for a homogeneous pulse $t_m \in W_\alpha$, with $f_m = 0$. Multiply the potential and the pressure by the pulse cutoff, then take the curl as in (7.38). After the prescribed source $f_m e^{ikm\Phi}$ has been added, the normalized linear residual lies in $W_{\alpha+1/2-3\kappa_s}$. The only terms left outside that class are additive tails from the pulse cutoff, and every Cartesian space-time derivative of each tail vanishes to infinite order as $q \downarrow 0$. The pressure amplitude π_m lies in $W_{\alpha+1/2}$.*

Proof. Expand the linearized residual into its principal operator (9.1) and the terms left over. For this calculation, a is any harmonic amplitude and π is its pressure amplitude. Let

$E_{ik} = (t_* + bD_r + (V/R)\partial_\theta + GD_z)\Phi$ be the phase transport defect from (7.9). The terms left after linearizing around (b, V, G) are

$$\begin{aligned} \mathcal{L}_m^{\text{rem}}(a, \pi) = & (-\varepsilon\partial_T + bD_r + GD_z)a + ikmE_{ik}a \\ & + a_r(\partial_R b)e_r + a_z D_z(b, V, G) + (b/R)a_\theta e_\theta + (D_r\pi, 0, D_z\pi) \\ & - \varepsilon \left\{ \begin{aligned} & (D_r^2 + R^{-1}D_r + D_z^2)a - R^{-2}(a_r, a_\theta, 0) \\ & + 2ikm(n_{\Phi,r}D_r + n_{\Phi,z}D_z)a \\ & + ikm(D_r n_{\Phi,r} + n_{\Phi,r}/R + D_z n_{\Phi,z})a \\ & + 2ikm(n_{\Phi,\theta}/R)(-a_\theta, a_r, 0) \end{aligned} \right\}. \end{aligned} \quad (9.2)$$

For a cylindrical vector v transported by another cylindrical vector u , differentiation of the moving cylindrical basis contributes $(-u_\theta v_\theta, u_\theta v_r, 0)/R$. Linearizing that term gives the connections in $\mathcal{K}$ and the displayed b/R term. Applying the scalar Laplacian to the harmonic gives the quadratic phase-gradient term in (9.1) and the two mixed phase–amplitude derivative terms in (9.2). Applying the vector Laplacian also gives the last angular connection and the R^{-2} term. The amplitude coefficients do not depend on θ .

The transverse-amplitude equation in Proposition 7.2 gives $\mathcal{L}_m(t_m, \pi_m) = -f_m$, with $\pi_m \in W_{\alpha+1/2}$. For each kind of term in (9.2), the gain over α is listed below.

term	gain
slow transport	$1 - \kappa_s$
phase transport defect	$1/2$
base derivatives and connections	1
pressure-amplitude gradient	$1/2 - \kappa_s$
viscous amplitude derivatives	$1 - 2\kappa_s$
mixed derivatives and phase divergence	$1/2 - \kappa_s$
viscous angular connection	$1/2$

The bound for the phase defect allows polynomial factors of S_* . Applying the principal velocity operator to the curl correction does not change its exponent. The fast-time derivative differentiates an amplitude coefficient, and $\varepsilon k^2 = O(1)$. At every derivative order, the potential formula and these chain-rule operators give the common lower bound $\alpha + 1/2 - 3\kappa_s$ for the exponents of these terms.

Apply the pulse cutoff to the potential and the pressure before taking the curl. The source left uncanceled, $(1 - \psi)f$, and the main derivative of the pulse cutoff, $\psi't$, together with its fixed multipliers, are supported where $P_v \leq e^{-cS_*}$. For every fixed physical derivative, their bounds have the form $CQ^{-M}S_*^P e^{-cS_*}$. They are therefore flat, because $S_* = \ell^2$ and $Q = 2^{-\ell}$. Keep these terms in the additive residual throughout the iteration. Later forward solves are applied only to the supported source terms.

□

The nonlinear terms need separate estimates for interactions between a wave and a mean field and for interactions between two waves. In the two-wave case, incompressibility controls the contraction of the advecting amplitude with the phase gradient.

In the next lemma, $(w \cdot \nabla_*)v$ means cylindrical transport with the normalized derivatives $(D_r, R^{-1}\partial_\theta, D_z)$, including the terms created by differentiating the cylindrical frame. The lemma therefore estimates the normalized nonlinear residual.

Lemma 9.2. *Let w be a wave in W_α . Let v be a mean field whose tangential components lie in M_μ and whose radial component lies in $M_{\mu+1}$. Then the nonzero harmonics of $(w \cdot \nabla_*)v + (v \cdot \nabla_*)w$ lie in $W_{\alpha+\mu-1/2}$.*

Take two waves w, w' built from curls, with the same label and with exponents α, α' . The nonzero harmonics of $(w \cdot \nabla_*)w' + (w' \cdot \nabla_*)w$ lie in $\mathbb{W}_{\alpha+\alpha'-\kappa_s}$, and the zero harmonic lies in the mean class $\mathbb{M}_{\alpha+\alpha'-\kappa_s}$. Waves with different labels have zero product on their closed supports.

Proof. When a mean field advects the phase of a wave, the phase derivative costs $k = O(\varepsilon^{-1/2})$. Radial transport of the amplitude contains a radial mean factor of order $\mu + 1$ and one D_r . Axial transport gains one power through D_z . When a wave transports a mean field, the radial cost is at most κ_s , and the cylindrical connection terms contain no derivatives. Each of these terms has at least the claimed mean-wave exponent.

Now take two harmonics with the same label. When the transport derivative falls on $a' \exp(ikm'\Phi)$ and the advecting harmonic is $a \exp(ikm\Phi)$, phase differentiation gives $ikm'(a \cdot n_\Phi)a'$. Equation (7.39) cancels what looks like a loss of one half-power. Terms where the derivative falls on the amplitude cost at most κ_s , and the connection terms cost nothing. The pulse weights satisfy $P_v^2 \leq P_v$ and their product has radial weight ζ . For a nonzero output, this is stronger than the required $\sqrt{\zeta}$ weight. Lemma 6.1 gives the support separation for different labels. Applying Leibniz' rule gives every statement at each fixed order of amplitude differentiation. $\square$

9.2 Sources and errors kept during the iteration

To apply Proposition 9.1 and Lemma 9.2 at one stage after another, every new source has to satisfy the support and derivative assumptions of the pulse inverse in Proposition 7.2. Mean operations can make the auxiliary support larger, so those assumptions have to be checked after a whole cycle. Equation (9.3) keeps the errors that are already flat at the singularity separate from the sources still waiting to be corrected. Even so, every nonlinear product uses the full velocity fields. The supported sources also have to stay inside the fixed enlarged rectangle. Proposition 9.3 proves all of these facts.

Use the source supports Ω_γ and the enlarged domains $\mathcal{E}_\gamma^+$ from (6.28). The flat term below contains the base error $\mathcal{E}_B$ from (5.41), the tails caused by pulse cutoffs, and the cutoff remainders that have not been canceled. The exact mean balances can include these flat terms, because their definition always uses the full current velocity. Angular means can fill the whole auxiliary torus. The rectangle and envelope conditions below apply only to the nonzero harmonics.

Proposition 9.3. Fix $j \in \mathbb{N}_0$. Suppose $(u^{[j]}, p^{[j]})$ is built from the fixed slow base and the primary waves by a finite sequence of the following operations, the pulse inverse and curl construction, the signed amplitude map for stresses that do not depend on the auxiliary variables, and the mean maps from Section 8. Reconstruct the pressure through (8.12) after every update, and use the full velocity fields in every product. Then the physical residual splits as

$$\mathcal{R}(u^{[j]}, p^{[j]}) = \mathcal{G}^{[j]} + \mathcal{F}^{[j]}. \quad (9.3)$$

In one common chart, set $\mathcal{G}_*^{[j]} = Q^{2A+1/2} \mathcal{G}^{[j]}$ and $\mathcal{F}_*^{[j]} = Q^{2A+1/2} \mathcal{F}^{[j]}$. The following three statements then hold.

(i) Supported sources. On the coarsest common auxiliary torus, the nonzero angular harmonics have the form

$$\mathcal{G}_*^{[j]} - \langle \mathcal{G}_*^{[j]} \rangle_\theta = \sum_{\gamma \in \Gamma} \sum_{m \in \mathcal{H}_j} f_{\gamma,m}^{[j]} e^{ik_\gamma m \Phi_\gamma}, \quad (9.4)$$

$$\mathcal{H}_j \subset \mathbb{Z} \setminus \{0\}, \quad |\mathcal{H}_j| < \infty.$$

The sum is locally finite. The set $\mathcal{H}_j$ depends on the finite construction, but not on the band. The coefficients satisfy

$$\text{supp}(f_{\gamma,m}^{[j]}|_{\mathcal{E}_\gamma}) \subset \Omega_\gamma, \quad f_{\gamma,m}^{[j]} \in C^\infty(\mathcal{E}_\gamma^+; \mathbb{C}^3),$$

and they extend smoothly by zero across the fixed slow support, the fixed transverse support, and the enlarged local torus rectangle. Let α, μ be any lower bounds for the wave and mean exponents obtained by

applying Propositions 9.1, 7.2, 6.6, and 7.6, Section 8, and Lemma 9.2 along this finite sequence. Then

$$f_{\gamma,m}^{[j]} \in W_\alpha, \quad \langle \mathcal{G}_*^{[j]} \rangle_\theta \in M_\mu.$$

The mean coefficients extend smoothly by zero outside the active shell. The next pulse inverse is applied exactly to the coefficients $f_{\gamma,m}^{[j]}$ in (9.4).

(ii) Flat remainder. For every fixed j, m, N ,

$$|\mathcal{F}^{[j]}|_m \leq C_{j,m,N} q^N. \quad (9.5)$$

This physical bound is uniform in the band, label, and point.

(iii) Exact mean equations. Let $(\beta^{[j]}, v^{[j]}, \gamma^{[j]}, w^{[j]})$ be the full current velocity correction from (8.1), and set $\mathcal{E}_{B,*} = Q^{2A+1/2} \mathcal{E}_B$. Then

$$\begin{aligned} W_{ab}^{[j]} &= \langle w_a^{[j]} w_b^{[j]} \rangle_\theta, \quad \mathcal{P}^{[j]} = \int \langle g_r^{[j]} \rangle_Y dR, \quad p_m^{[j]} = \mathcal{T}_0(g_r^{[j]} - \rho \mathcal{P}^{[j]}), \\ \langle \mathcal{G}_*^{[j]} + \mathcal{F}_*^{[j]} - \mathcal{E}_{B,*} \rangle_\theta &= (D_r p_m^{[j]} - g_r^{[j]}, E_\theta^{[j]}, E_z^{[j]}), \end{aligned} \quad (9.6)$$

Here $g_r^{[j]}, E_\theta^{[j]}, E_z^{[j]}$ are the values given by (8.3) for this tuple.

Proof. First prove (i). Then keep track of the additive errors in (ii) while preserving the exact mean equations in (iii).

Part (i), support and coefficient bounds. The primary pulses have the stated supports and weighted bounds at every derivative order. Differentiation preserves both the closed support and the amplitude-class bounds. Each explicit D_r costs κ_s . Taking more amplitude derivatives of that estimate can make its constants and polynomial degrees larger, but it does not charge that exponent loss a second time.

In a product of a mean and a wave, the product keeps the wave's slow support, transverse support, local torus rectangle, and P_v factor. Multiplying two waves with one label adds their integer phase multipliers. Their zero harmonic is exactly their angular mean, because the angular frequency kp is a nonzero integer. Every other output keeps the wave weight, since $P_v^2 \leq P_v$. Different labels do not interact. Taking curls, forming pressure coefficients, and propagating a particular pulse all preserve the phase integer. Mean operations cannot create a nonzero angular harmonic. Any nonzero harmonic produced later still contains a wave factor, so it keeps that factor's support and envelope.

The radial integration, pressure reconstruction, mean-vector-potential construction, and auxiliary-time inversion from Section 8 all preserve the bounds for mean amplitudes. For the stress correction in Proposition 7.6, subtract a supported interior profile with the same weighted radial moment as the source. The corrected source then has total moment zero. Its weighted radial integral can therefore be taken forward from the left edge of its support or backward from the right edge. Near either edge, let s be the matching logarithmic distance from (6.23). There, the weight ζ is e^{-a/s^2} times a smooth positive factor with fixed positive upper and lower bounds, where $a > 0$ is fixed. For every fixed $M \geq 0$ and every integer $k \geq 0$, sufficiently small positive δ, s satisfy

$$\int_0^\delta s^{-M} e^{-a/s^2} ds \leq C_{M,a} \delta^{3-M} e^{-a/\delta^2}, \quad \left| \frac{d^k}{ds^k} e^{-a/s^2} \right| \leq C_{k,a} s^{-3k} e^{-a/s^2}.$$

The substitution $v = a/s^2$ gives the first bound, and direct differentiation gives the second. On the normalized shell, the radial weights and the radial Jacobian are bounded. Differentiating the radial integral therefore leaves a bound of the form $\zeta\delta^{-M'}$, where M' can depend on the derivative. In the interior, ζ has a positive lower bound. A source of order α therefore gives a stress correction $\Sigma \in M_\alpha$.

For the amplitude division, use the same normalization as in (7.31), namely $d_\Sigma = H^{-1}(\Sigma/\varepsilon)$ and $\delta a_\sigma = (d_\Sigma)_\sigma / (2a_\sigma)$. The inverse-matrix bounds and (7.25) give the following estimates for every fixed coefficient multi-index I .

$$\begin{aligned} |D^I(d_\Sigma)_\sigma| &\leq C_I \varepsilon^{\alpha-1} S_*^{b_I} \zeta \delta^{-M_I}, \\ |D^I(a_\sigma^{-1})| &\leq C_I S_*^{b_I} \zeta^{-1/2} \delta^{-M_I}, \\ |D^I \delta a_\sigma| &\leq C_I \varepsilon^{\alpha-1} S_*^{b_I'} \sqrt{\zeta} \delta^{-M_I'}, \\ |D^I(\sqrt{\varepsilon} \delta a_\sigma b_\sigma)| &\leq C_I \varepsilon^{\alpha-1/2} S_*^{b_I''} \sqrt{\zeta} \delta^{-M_I''} P_v. \end{aligned}$$

This gives (7.33) again. The last line also uses the fundamental pulse bounds. After the slow cutoff, it is the $W_{\alpha-1/2}$ estimate from Proposition 7.6. The remaining square-root weight at the edge makes the extension by zero smooth.

Each new source now satisfies the assumptions of Proposition 7.2. Along its prescribed path, the slow and transverse variables are fixed. If the data are zero all along one of these paths, the solution is zero there as well. The final cutoff puts the support back inside the fixed enlarged local torus rectangle. A source with a smooth extension gives a solution with a smooth extension. This proves the support condition needed by the inverse. There is no need to divide by the original cutoff.

The set of local bands stays fixed. Radial paths keep (z, t) fixed, and q depends only on (z, t) . Temporal inversion and pulse propagation keep all slow variables fixed. The common-torus construction from Lemma 6.2 therefore applies at every step of the finite construction. A radial mean can collect a number of labels that grows polynomially in S_* , while only a bounded number of wave labels overlap at any one point. This radial collection changes only the polynomial factors in S_* . It does not change the common chart or the ε exponent.

Parts (ii) and (iii), flat errors and exact means. When an actual increment is (w, π) , the residual changes by

$$\begin{aligned} \mathcal{R}(u + w, p + \pi) &= \mathcal{R}(u, p) + \partial_t w - \Delta w + \nabla \pi \\ &\quad + (u \cdot \nabla)w + (w \cdot \nabla)u + (w \cdot \nabla)w. \end{aligned}$$

The old $\mathcal{F}$ is added directly. Every product in the display uses the actual fields, including the curls with their cutoffs and the remainders from the modified radial integral with compact support. Put every newly created flat term into $\mathcal{F}$. At any fixed stage, there are only finitely many kinds of these terms. The Gaussian bound in Proposition 9.1 is uniform over the labels. Lemma 8.2 bounds each cutoff remainder by any chosen power of q , at the cost of using finitely many more source derivatives. Summing a number of labels that grows polynomially does not spoil those bounds. Keep the Gaussian cutoff tails with nonzero harmonics in the additive residual, and do not feed them into later forward pulse solves. The mean residuals E_θ, E_z are the exact conservative balances and may contain flat cutoff remainders, which a fixed mean inverse may cancel. If a contribution is canceled, remove it from $\mathcal{F}$. The cutoff-remainder estimates put the remainder in M_α for every α . This update keeps both the stated classes and the exact splitting of the residual. It proves (9.5) at every finite stage, which proves (ii). Apply Proposition 8.1 to the full current tuple and use (8.12). This gives (9.6) and proves (iii). $\square$

In the realization step, the finite corrections are summed to build the infinite local field. Its residual is controlled by comparing that field with one fixed finite stage. The flat residuals from the separate stages are not added together.

9.3 Starting the iteration and running a full correction cycle

The primary pulses, followed by the first mean corrections, give the first residual bounds in Proposition 9.5. Proposition 9.6 then shows that a full cycle improves every one of those bounds while keeping the support and moment conditions. The induction statement below combines the residual orders in (9.8) with the bounds on the total velocity correction in (9.9). Those total bounds control how the existing correction interacts with every later increment.

Throughout the construction, keep the base (b, V, G) , the primary amplitudes, and every inverse operator fixed. Let $w_0^{\text{tan}} = W_0^{\text{as}}$ be the normalized representative of the assembled primary transverse field from (7.35). For each finite construction, write its normalized velocity and pressure as

$$u_* = (b + \beta, V + v, G + \gamma) + w, \quad p_* = p_{B,*} + p_m + p_w, \quad \langle w \rangle_\theta = \langle p_w \rangle_\theta = 0. \quad (9.7)$$

Here $p_{B,*}$ is the pressure of the base, p_w is the sum of the pressure harmonics supplied by the amplitude equations, and p_m is reconstructed from the current g_r through (8.12). Equations (8.3), (8.12), and (8.15) then determine $E_\theta, E_z, \mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z$.

Definition 9.4 (Finite correction state). For $j \in \mathbb{N} \cup \{0\}$, a state at stage j consists of real fields of the form (9.7) with the properties listed next. The wave w is a sum of curls of supported wave potentials with the fixed phases for each label. The mean correction (β, v, γ) is the sum of the curl of an azimuthal potential and an azimuthal field added directly. Both velocity corrections are therefore exactly divergence-free. The source supports, finite sets of harmonics, smooth extensions by zero, and physical residual splitting are the ones in Proposition 9.3.

The normalized nonzero harmonics $\mathcal{G}_{\text{wave}}^{[j]}$ of $\mathcal{G}_*^{[j]}$, the exact tangential mean residuals, and the three integral defects obey

$$\begin{aligned} \mathcal{G}_{\text{wave}}^{[j]} &\in W_{B_j}, & E_\theta, E_z &\in M_{C_j^*}, & (\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z) &\in S_{C_j^*}, \\ B_j &= \frac{1}{2} + \sigma_j, & C_j^* &= 1 + \sigma_j, & \sigma_j &= \frac{1}{5} + \frac{j}{10}, \end{aligned} \quad (9.8)$$

and the total corrections to velocity and pressure satisfy

$$w \in W_{1/2}, \quad w - w_0^{\text{tan}} \in W_{0.68}, \quad v, \gamma, p_m \in M_{0.9}, \quad \beta \in M_{1.9}. \quad (9.9)$$

Each wave statement applies to every grouped nonzero harmonic for each label. The two normalized moments for angular momentum and axial flux stay exactly equal to zero,

$$\int_0^\infty R^2 \langle v \rangle_Y dR = 0, \quad \int_0^\infty R \langle \gamma \rangle_Y dR = 0. \quad (9.10)$$

Reconstructing the pressure is part of the state. It follows that the radial residual of every such state is $-\rho\mathcal{P}$ plus the flat cutoff remainder from Proposition 8.3.

Proposition 9.5. *Start with the fixed base and the curls of the primary potentials. Apply the temporal mean update (8.20), then apply the five-equation correction (8.25). Reconstruct the pressure through (8.12) both before and after each update. The resulting fields form a stage 0 state in the sense of Definition 9.4. In particular, their residual orders are $B_0 = 0.7$ and $C_0^* = 1.2$.*

Proof. First estimate the primary waves. Then remove the means that depend on the auxiliary torus and the three moment defects. This first stage uses the faster decay of the primary residual after taking its auxiliary average. The primary transverse amplitudes have exponent $1/2$. The supported parts of their linear residual have exponent $1 - 3\kappa_s$, and the nonlinear residuals of the waves have

exponent $1 - \kappa_s$. Before any mean update, the exact mean balances in (8.3) give $g_r, p_m, E_\theta, E_z \in M_{1-\kappa_s}$ and put the defects in $S_{1-\kappa_s}$.

Now take the auxiliary means. The primary covariance from Proposition 7.5, assembled by (7.35), cancels the order-zero auxiliary radial-tangential stress tensor exactly. This also includes derivatives of the slow partition, because the squares of the partition functions add up to one. To write out this cancellation, let $\Sigma_a^{(0)}$ be the leading tensor and let Σ_a be the full tensor in (8.3). Before adding the mean velocities, those equations become

$$\begin{aligned} \langle E_\theta \rangle_Y &= (\partial_R + 2/R)(\langle W_{r\theta} \rangle_Y - \Sigma_\theta^{(0)}) - (\partial_R + 2/R)(\Sigma_\theta - \Sigma_\theta^{(0)}) + \varepsilon \partial_Z \langle W_{z\theta} \rangle_Y, \\ \langle E_z \rangle_Y &= (\partial_R + 1/R)(\langle W_{rz} \rangle_Y - \Sigma_z^{(0)}) - (\partial_R + 1/R)(\Sigma_z - \Sigma_z^{(0)}) + \varepsilon \partial_Z \langle W_{zz} + p_m \rangle_Y. \end{aligned} \quad (9.11)$$

Each first difference contains at least one curl correction. Before taking the divergence, the tensor cross term between a curl correction and a primary transverse amplitude lies in $M_{3/2-\kappa_s}$. The axial fluxes, the axial pressure term, and the higher-order auxiliary stress tensor all have residual exponent at least $2 - \kappa_s$. It follows that $\langle E_\theta \rangle_Y, \langle E_z \rangle_Y \in M_{1.49}$.

Set $H_0 = 1 - \kappa_s$ and apply the temporal update (8.20). Its tangential increment lies in M_{H_0} , and the radial velocity it creates lies in M_{H_0+1} . The pressure change has exponent H_0 , because $2V\Delta v/R$ occurs in Δg_r . In either tangential residual, the contribution from this pressure change contains an axial derivative and gains the matching extra power. Once the fast-time term has been canceled, a slow-time derivative supplies one more power of ε . Each radial linear flux contains either $b = O(\varepsilon)$ in every fixed-order amplitude derivative on the shell, or the radial velocity increment that was just created. Axial fluxes gain one power, while viscosity gains $1 - 2\kappa_s$. Every radial quadratic mean flux also contains a radial mean. All these changes to the residual have exponent at least $H_0 + 1 - 2\kappa_s$. Interaction of the mean correction with $w \in W_{1/2}$ has wave exponent at least H_0 . After pressure is recomputed, the defects still lie in S_{H_0} .

Next apply the five-equation correction map (8.25) at order H_0 , and recompute the pressure. Its slow increments have no fast-time term, so the tangential and wave estimates from the preceding paragraph still hold. In the changes to the defects, $2V\Delta v/R$ and its two contributions to the weighted flux moments are exactly the linear terms in $\Delta \mathcal{P}, \Delta \mathcal{J}_\theta, \Delta \mathcal{J}_z$ that this system cancels. Every term left contains either a slow-time derivative of the radial velocity increment, a radial flux with b or another radial mean, an axial derivative, radial viscosity, or one more tangential mean in $v^2, \gamma v, \gamma^2$. Each has a gain greater than 0.8 over H_0 . The remaining defects therefore have exponent greater than $1.8 - \kappa_s$, and the two tangential means keep exponent 1.49. These estimates give (9.8) with $B_0 = 0.7$ and $C_0^* = 1.2$.

The primary curl correction has exponent $1 - \kappa_s > 0.68$. Every mean increment and pressure change has exponent $H_0 > 0.9$, and its radial exponent is $H_0 + 1 > 1.9$. This proves (9.9). The prescribed temporal increments have auxiliary mean zero. The first two homogeneous equations in the correction system enforce (9.10), and reconstructing the mean field preserves the prescribed auxiliary mean exactly. $\square$

The initial stage now has the required bounds. Repeat the same four corrections, and compare every new error with the bound needed at the next stage. When estimating how the new increments interact with the existing fields, the total bounds in (9.9) remain available.

Proposition 9.6. *Let $(u^{[j]}, p^{[j]})$ be a stage j state as in Definition 9.4. Using the same base, primary amplitudes, supports, and inverse operators, it is possible to construct a stage $(u^{[j+1]}, p^{[j+1]})$ state whose residual orders are*

$$B_{j+1} = B_j + \frac{1}{10}, \quad C_{j+1}^* = C_j^* + \frac{1}{10}.$$

Construct it by applying the following corrections in this order.

- (i) Solve the inhomogeneous amplitude equation for each supported nonzero harmonic.
- (ii) Change the wave amplitudes so that they correct the tangential residual after taking its auxiliary average.
- (iii) Apply the temporal mean inverse to the part whose auxiliary average is zero.
- (iv) Apply the five-equation correction to the three current defects.

Reconstruct the pressure after every correction. Build the wave increments and the meridional velocity increments, meaning the radial and axial velocity increments, from their potentials, and add the azimuthal mean increments directly. The cumulative bounds (9.9) and the exact moments (9.10) are preserved.

Proof. The four steps below carry out (i)–(iv) in that order. For this cycle, shorten the notation to $B = \frac{1}{2} + \sigma_j$ and $C_* = B + \frac{1}{2} = 1 + \sigma_j$. Whenever g_r or p_m appears below, an improved order refers to the change in that quantity. The total corrections still satisfy (9.9). Within each step, w, β, v, γ are the fields at the start of that step. The increment replaces them by

$$w + \Delta w, \quad (\beta + \Delta\beta, v + \Delta v, \gamma + \Delta\gamma).$$

Use these full updated fields in (8.3) to compute $g_{r,\text{new}}$, and set

$$\mathcal{P}_{\text{new}} = \int \langle g_{r,\text{new}} \rangle_Y dR, \quad p_{m,\text{new}} = \mathcal{T}_0(g_{r,\text{new}} - \rho \mathcal{P}_{\text{new}}).$$

Compute every residual and defect in the next step from this updated state.

Step 1, cancel the supported harmonics. Apply Proposition 7.2 to each grouped supported source. Use zero initial data where the prescribed path enters the common auxiliary torus. For a source coefficient f_m , the normalized harmonic increments are

$$\Delta w_m = \text{curl}_* \left(\frac{i n_\Phi \times (\psi t_m)}{k m |n_\Phi|^2} e^{ikm\Phi} \right), \quad \Delta p_{w,m} = \psi \pi_m e^{ikm\Phi}, \quad \mathcal{L}_m(t_m, \pi_m) = -f_m.$$

Put these fields together with their physical rescalings over every label and harmonic, keeping each conjugate pair. The velocity increment lies in W_B , and its pressure lies in $W_{B+1/2}$. The new errors with nonzero harmonics have these exponents.

linear error	$B + 1/2 - 3\kappa_s$
cross with old exact waves	$B + 1/2 - \kappa_s$
particular-correction self-interaction	$2B - \kappa_s$
cross with total mean correction	$B + 0.4$

Proposition 9.1 gives the first row, and Lemma 9.2 gives the other three. The change in mean covariance caused by these increments has exponent at least $B + 1/2 = C_*$. Taking a radial divergence costs κ_s . The exact balances and the linear pressure map now give

$$E_\theta, E_z, \Delta g_r, \Delta p_m \in M_{C_* - \kappa_s}, \quad (\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z) \in S_{C_* - \kappa_s}.$$

The angular-momentum and axial-flux constraints remain unchanged. Together with the exact integrated identities in (8.16), this improves the weighted moments by one power,

$$\int R^2 \langle E_\theta \rangle_Y dR, \quad \int R \langle E_z \rangle_Y dR \in S_{C_* + 1 - \kappa_s}. \quad (9.12)$$

The axial identity contains $c_\rho \mathcal{P}$ inside its axial derivative. This is the term that accounts for the normalized correction $-\rho \mathcal{P}$ left in the radial equation after the pressure has been reconstructed.

Step 2, correct the auxiliary-averaged residual by changing the stress. Once its total weighted radial moments have been removed, a stress with compact support can cancel the tangential residual after taking its auxiliary average. Make this correction in physical coordinates, so the stress agrees between charts. Use the residual components E_θ, E_z from (8.3), supported in the prescribed annulus, and keep the background residual separate with its flatness

bound. In physical coordinates, the auxiliary averages are $F_2 = Q^{-2A-1/2}\langle E_\theta \rangle_Y$ and $F_1 = Q^{-2A-1/2}\langle E_z \rangle_Y$, and these definitions agree from one chart to another. For $e \in \{1, 2\}$, choose one fixed interior profile $\widehat{b}_e$ with $\int x^e \widehat{b}_e(x) dx = 1$. Set $b_e(r, z, t) = q^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q})$, and then set

$$M_e = \int_0^\infty r^e F_e(r) dr, \quad \sigma_e(r) = -r^{-e} \int_0^r (r')^e (F_e(r') - b_e(r') M_e) dr'.$$

These radial integrals are taken while (z, t) stays fixed. The total weighted moment of the integrand is zero because $\int r^e (F_e - b_e M_e) dr = M_e - M_e = 0$. Starting the integral at zero fixes the integration constant. Since the weighted moment is zero, σ_e becomes zero beyond the annulus that supports the source. Thus σ_e has compact support and $(\partial_r + e/r)\sigma_e = -F_e + b_e M_e$. Define the normalized tensor pair by $\Sigma = Q^{2A}(\sigma_2, \sigma_1)$. The weighted radial-integral estimate in the proof of Proposition 9.3 gives $\Sigma \in M_{C_* - \kappa_s}$. This construction uses the auxiliary averages F_e , so the answer does not depend on any auxiliary torus coordinate, even when the original fields have nonzero Fourier modes on that torus.

Apply the amplitude-correction operator $\mathcal{L}_{\text{phys}}^{\text{as}}$ from (7.35) to the pair $\sigma = (\sigma_2, \sigma_1)$. This operator adds up the localized operators in physical coordinates. Its normalized form $\mathcal{L}^{\text{as}}\Sigma$ lies in the supported class $W_{B-\kappa_s}$. By (7.36), its prescribed symmetrized cross covariance is Σ with $w_0^{\text{tan}} = W_0^{\text{as}}$. Each local summand uses (7.31) before the slow cutoff and physical rescaling are applied. For each sign $\sigma \in \{+, -\}$, the formula divides its known numerator by the fixed, positive primary amplitude $2\sqrt{y_\sigma}$. It can therefore produce an increment of either sign without taking a square root of the current covariance or of $y + d\Sigma$. For the homogeneous amplitude equation, whose source is zero, use the pressure coefficient. Build the velocity by taking the curl, as in Lemma 7.7. The covariance identity pairs the two signs for one slow label. When their covariances are added, each slow label is used once, together with the square of the slow partition. The correction's quadratic self-interaction stays in the residual.

The nonzero harmonic errors now have these exponents.

linear error	$B + 1/2 - 4\kappa_s$
mean interaction	$B + 0.4 - \kappa_s$
cross with old exact waves	$B + 1/2 - 2\kappa_s$
signed-correction self-interaction	$2B - 3\kappa_s$

For the auxiliary mean, split the old exact wave into its fixed primary transverse amplitude and a remainder in $W_{0.68}$. More specifically, write the signed increment as $s = t_s + r_s$, where t_s is the transverse amplitude and r_s is the curl remainder. Set $\mathcal{B}(a, b) = \langle \langle a \otimes b + b \otimes a \rangle_\theta \rangle_Y$. Its exact change in covariance is

$$\langle \Delta W \rangle_Y = \mathcal{B}(w_0^{\text{tan}}, t_s) + \mathcal{B}(w - w_0^{\text{tan}}, t_s) + \mathcal{B}(w, r_s) + \langle \langle s \otimes s \rangle_\theta \rangle_Y. \quad (9.13)$$

The radial-tangential components of the first term equal Σ , so their divergence cancels $F_e - b_e M_e$. Before taking the divergence, the tensor terms that remain have these exponents.

transverse correction with old-wave remainder	$B + 0.68 - \kappa_s = C_* + 0.18 - \kappa_s$
signed curl correction times old exact wave	$C_* + 1/2 - 2\kappa_s$
signed-correction self-interaction	$2B - 2\kappa_s = C_* + \sigma_j - 2\kappa_s$

These are exactly the remainder terms in the covariance expansion from Corollary 7.8. Taking their radial divergence loses one power κ_s . Axial fluxes gain one when the tensor change has order at least $C_* - \kappa_s$. The changes in g_r, p_m have order at least $C_* - 2\kappa_s$, and the effect of the pressure on E_z gains one. The moment-correction term $b_e M_e$, which has not been canceled, has order $C_* + 1 - \kappa_s$ by (9.12).

Since $0.18 - 2\kappa_s > 0.17$ and $\sigma_j - 3\kappa_s > 0.17$, the state after the pressure is rebuilt satisfies

$$\langle E_\theta \rangle_Y, \langle E_z \rangle_Y \in M_{C_*+0.17}, \quad E_\theta, E_z \in M_H, \quad (\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z) \in S_H, \quad H = C_* - 2\kappa_s. \quad (9.14)$$

Step 3: Remove the nonconstant auxiliary means. The auxiliary averages now have the better order in (9.14). Their parts with zero auxiliary average still have order H , and the temporal mean inverse can cancel them. For the current exact mean residuals, set $E_a^\circ = E_a - \langle E_a \rangle_Y$, with $a = \theta, z$. The temporal inverse from (8.20) gives

$$\Delta v = -c_{i_0}^{-1} N_{i_0}^{-1} E_\theta^\circ, \quad \gamma_d = -c_{i_0}^{-1} N_{i_0}^{-1} E_z^\circ, \quad \Psi_* = T_1 \gamma_d, \quad \Delta \beta = -\varepsilon \partial_Z \Psi_*, \quad \Delta \gamma = (D_r + R^{-1}) \Psi_*.$$

Here $N_{i_0}^{-1}$ is the inverse with zero average from Lemma 8.6. The prescribed tangential increments lie in M_H , and the radial velocity increment they produce lies in M_{H+1} . Rebuild the pressure. The term $2V\Delta v/R$ still has order H in Δg_r , so $\Delta p_m \in M_H$. Every other part of Δg_r has one of the following orders. The fast-time derivative of $\Delta \beta$ has order $H+1$. Each radial flux contains b , an old radial mean, or a new radial mean. The axial fluxes gain one. The other quadratic terms in the azimuthal velocity have order at least $H+0.9$. The viscous term contains a factor ε and loses at most two radial derivatives.

Write $\Delta \gamma$ for the axial increment that is actually produced, and write γ_d for the increment prescribed in (8.20). Since $t_* = -\varepsilon \partial_T + c_{i_0} N_{i_0}$, the fast-time identities are

$$\begin{aligned} c_{i_0} N_{i_0} \Delta v &= -E_\theta^\circ, \\ c_{i_0} N_{i_0} \Delta \gamma &= -E_z^\circ + \mathcal{F}_{ax}, \\ \mathcal{F}_{ax} &= c_{i_0} N_{i_0} (\Delta \gamma - \gamma_d). \end{aligned}$$

The last term is the flat remainder from reconstructing the axial increment in Lemma 8.6. This mean update does not change the wave covariance. Subtracting the two versions of (8.3) gives

$$\begin{aligned} E_{\theta, \text{new}} - \langle E_\theta \rangle_Y &= -\varepsilon \partial_T \Delta v - \varepsilon (\Delta_0 - R^{-2}) \Delta v \\ &\quad + (D_r + 2/R) ((b + \beta) \Delta v + (V + v) \Delta \beta + \Delta \beta \Delta v) \\ &\quad + D_z ((G + \gamma) \Delta v + (V + v) \Delta \gamma + \Delta \gamma \Delta v), \\ E_{z, \text{new}} - \langle E_z \rangle_Y &= -\varepsilon \partial_T \Delta \gamma - \varepsilon \Delta_0 \Delta \gamma + \mathcal{F}_{ax} \\ &\quad + (D_r + 1/R) ((b + \beta) \Delta \gamma + (G + \gamma) \Delta \beta + \Delta \beta \Delta \gamma) \\ &\quad + D_z (2(G + \gamma) \Delta \gamma + (\Delta \gamma)^2 + \Delta p_m). \end{aligned}$$

Use (9.9), the bounds $|D^I b| \leq C_I \varepsilon S_*^{b_I}$ on the shell, and the orders of the increments above. The four kinds of terms then have these lower bounds for their class exponents.

term	slow time	radial flux	axial flux	viscosity
exponent	$H+1$	$H+1-\kappa_s$	$H+1$	$H+1-2\kappa_s$

After $\mathcal{F}_{ax}$ is taken out of the axial component, the displayed changes in the residual therefore lie in $M_{H+1-2\kappa_s}$. Keep $\mathcal{F}_{ax}$ in the additive flat term. The change in the wave lies in W_H because it interacts with the waves already added, whose total obeys the $W_{1/2}$ bound. The remainders from the modified, compactly supported radial integral go into the additive flat term, while their actual velocities continue to be used. The construction has now reached

$$E_\theta, E_z \in M_{\min(C_*+0.17, H+1-2\kappa_s)}, \quad (\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z) \in S_H.$$

Step 4: Enforce the three conditions needed for compact support. The tangential residuals have improved, but the compatibility defects still have order H . Apply the five-equation correction map (8.25) to these defects at order H . The azimuthal increment and the prescribed axial increment are slow. They are supported on the fixed support of the test-function profiles in the five-equation correction system, and they lie in M_H . The radial velocity increment they produce lies in M_{H+1} . The first two rows preserve (9.10). The last three rows remove the linear contributions to the changes in $(P, \mathcal{J}_\theta, \mathcal{J}_z)$. Rebuild the pressure from the new fields that are actually used.

The earlier bounds for changes in the nonzero harmonics and the tangential mean residuals still hold. Unlike a fast-time correction, a slow correction from this system has no fast-time term to cancel. To estimate the defects that remain, subtract $2V\Delta v/R$ from Δg_r . What remains contains the slow-time derivative of the radial velocity just produced, radial derivatives of $b\Delta\beta$ and of products involving old or new radial means, axial flux derivatives, products of Δv with old or new means, and the radial viscous term. Every one of these terms has exponent at least $H + 0.9 - 2\kappa_s$. In $\Delta\mathcal{J}_\theta$ and $\Delta\mathcal{J}_z$, the terms left after this system cancels the linear ones are products of tangential means or moments of this same radial remainder. The exact formulas are in (8.27). Hence

$$(\mathcal{P}, \mathcal{J}_\theta, \mathcal{J}_z) \in S_{H+0.9-2\kappa_s} = S_{C_*+0.9-4\kappa_s}.$$

These inequalities close the cycle.

$$\begin{aligned} \min\{\tfrac{1}{2} - 3\kappa_s, \tfrac{1}{2} - \kappa_s, B - \kappa_s, 0.4\} &\geq 0.4, \\ \min\{\tfrac{1}{2} - 4\kappa_s, 0.4 - \kappa_s, \tfrac{1}{2} - 2\kappa_s, B - 3\kappa_s\} &\geq 0.4 - \kappa_s, \\ H - B = \tfrac{1}{2} - 2\kappa_s &> 0.1, \quad \min\{0.17, 1 - 4\kappa_s\} = 0.17 > 0.1, \\ 0.9 - 4\kappa_s &> 0.1. \end{aligned}$$

Here $B \geq 0.7$ was used. The orders of the wave, the full tangential mean, and the defects all improve by 0.1. The smallest order of a signed increment is $B - \kappa_s \geq 0.69999 > 0.68$, while the particular increments start at $B \geq 0.7$. Every new increment in the mean velocity or mean pressure has decay exponent greater than 0.9, and every increment in the radial mean velocity has exponent greater than 1.9. A finite sum therefore still satisfies (9.9). Reconstructing the pressure gives the claimed radial residual. Proposition 9.3 gives the source hypotheses for the next cycle, which completes the induction. $\square$

9.4 One common domain and estimates for physical derivatives.

To use Lemma 5.4, every finite state $(u^{[j]}, p^{[j]})$ must exist on the same domain. Lemma 9.7 provides that domain. The gain in decay must also survive differentiation in the physical variables. More exactly, once the derivative order is fixed, the loss in powers of q must not depend on the correction stage. The constants in the bounds may still depend on that stage, as Lemma 9.8 allows.

Lemma 9.7. *There is a number $q_{\text{big}} > 0$ that does not depend on the correction stage or the derivative order, and every finite partial sum above, before its summation cutoffs are applied, is defined on $0 < q < q_{\text{big}}$. Once the stage and the order of an output amplitude derivative are fixed, constructing and estimating that derivative needs only finitely many derivatives of the input amplitudes. The constants may depend on the correction stage.*

Proof. Choose $0 < q_{\text{big}} \leq q_*$ once and for all so that the pulse estimates hold whenever $q < q_{\text{big}}$. Make this choice with the fixed background, the stated positive lower bound for $|n_\Phi|$, the primary covariance, and the fixed supports of the normalized mean-correction profiles. Every later inhomogeneous amplitude problem is a linear equation whose coefficients are these fixed ones and whose source is already known. The same threshold works for every harmonic because (7.18) writes its propagator as the fundamental propagator times a forward damping factor whose magnitude is at most one. At each

fixed derivative order, differentiation puts powers of the harmonic integer into the constants and allows polynomial growth in S_* . It does not call for a smaller q -threshold, as Corollary 7.3 shows.

The signed updates divide by the original $2\sqrt{y_\sigma}$ in (7.31), and the finite-dimensional correction system uses the fixed linear map from Lemma 8.7. Pressure, the temporal inverses, and the modified radial integrals that define the mean potentials are all fixed linear maps. The primary construction fixes every inverse coefficient and every denominator. These finite operations are therefore defined on the same domain for sources of any size, including sources near q_{big} . Each cycle improves the decay exponent on that whole domain.

The number of input derivatives needed at a given stage can be counted with a directed acyclic graph. Each vertex is an intermediate expression, and each arrow records one of the sums, products, derivatives, or proved inverses used to build a later expression from an earlier one. Directed means that these arrows point from input toward output. Acyclic means that no chain of arrows loops back to where it started. The source vertices, where no earlier expression feeds in, are the input fields. For a source vertex a , let $D_{v,a}(m)$ be any sufficient number of derivatives of the input a for bounding, through derivative order m , the expression at vertex v . Suppose the operation at v needs $d_{v,w}$ extra derivatives of its input at w . Define

$$D_{v,a}(m) = \max_{w \rightarrow v} D_{w,a}(m + d_{v,w}),$$

starting with $D_{b,a}(m) = m$ when the source vertices satisfy $b = a$, and with $D_{b,a}(m) = 0$ when $b \neq a$. At every fixed correction stage the graph has only finitely many vertices, so every derivative count produced this way is finite. Differentiating a pulse equation to order m gives an equation containing amplitude derivatives of order at most m ; the lower-order derivatives are already controlled. The torus inverses need finitely many more amplitude derivatives, while the modified radial integrals keep the stated derivative bounds. To make the cutoff remainder flat to a chosen power, integrate the same exact integral by parts only finitely many times. These counts tell exactly how many source derivatives need estimates. The explicit ε losses instead lower the decay exponent. For example, estimates for more amplitude derivatives of $N^{-1}D_r f$ still have the one explicit radial loss belonging to D_r . First construct and estimate every finite-stage coefficient as a function of independent slow and auxiliary variables. Then set the auxiliary variables equal to the phase map $Y = Y(r, t)$ and differentiate with respect to (x, t) . □

Put the base and the finite initialization together as $U_0 = (u^{[0]}, p^{[0]})$. For $j \geq 1$, let A_j^w be the sum of the physical wave-potential increments made during the cycle from stage $j-1$ to stage j . Let Ψ_j be the sum of that cycle's physical azimuthal-potential coefficients, and let b_j be the sum of its direct azimuthal velocity coefficients in physical variables. In the rest of the summation argument, B_j means the azimuthal vector from (5.29). Define

$$\begin{aligned} A_j &= A_j^w + \Psi_j e_\theta, & B_j &= b_j e_\theta, \\ p_j &= p^{[j]} - p^{[j-1]}, & Z_j &= (A_j, B_j, p_j), \\ \Delta u_j &= \text{curl } A_j + B_j = u^{[j]} - u^{[j-1]}, \\ (u^{[j]}, p^{[j]}) &= U_0 + \sum_{j=1}^J (\Delta u_j, p_j). \end{aligned} \tag{9.15}$$

Thus Z_j is the tuple from (5.29), without the optional stress entry. In a fixed Q chart, the linearity of T_0 and the fixed choice of ρ give this explicit formula for the pressure increment.

$$Q^{2A} p_j = p_w^{[j]} - p_w^{[j-1]} + T_0(g_r^{[j]} - g_r^{[j-1]} - \rho(\mathcal{P}^{[j]} - \mathcal{P}^{[j-1]})). \quad (9.16)$$

Here $p_w^{[j]}, g_r^{[j]}, \mathcal{P}^{[j]}$ are the normalized quantities from (9.7) and (9.6). The pressure differences from all the steps in between telescope, leaving (9.16).

Lemma 9.8. *For every order m of Cartesian space-time derivatives,*

$$|Z_j|_m + |\Delta u_j|_m \leq C_{j,m} q^{g_j - \ell_m} (1 + |\log q|)^{P_{j,m}}, \quad g_j = \frac{h}{10} j \rightarrow \infty, \quad (9.17)$$

where ℓ_m does not depend on j and includes the extra spatial derivative used to recover velocities from the wave and mean vector potentials. The finite partial sums satisfy

$$|(u^{[j]}, p^{[j]})|_m \leq C_{j,m} q^{-K_m} (1 + |\log q|)^{P_{j,m}},$$

with K_m independent of j . Their residuals satisfy

$$|\mathcal{R}(u^{[j]}, p^{[j]})|_m \leq C_{j,m} q^{h\sigma_j - K_m} (1 + |\log q|)^{P_{j,m}} + E_{j,m}, \quad E_{j,m} \leq C_{j,m,N} q^N \quad \text{for every } N, \quad (9.18)$$

and here too K_m is independent of j .

Proof. On the supports of the perturbations, $r \asymp \sqrt{Q}$ and $q \asymp Q$. While differentiating in one fixed chart, keep Q fixed. For an amplitude coefficient, the graph operators in (6.6) lose the following powers of Q .

radial physical derivative	$1/2 + h\kappa_s$	(9.19)
axial physical derivative	$D = 1/2 - h$	
time physical derivative	$1 + h$	
derivative of the cylindrical frame	$1/2$.	

For example, the coefficient of an auxiliary-coordinate derivative in the physical radial derivative is $r^{d_r-1} \Lambda_g^{i_0} = O(Q^{-1/2} \varepsilon^{-\kappa_s} S_*^P)$, while $T_g^{i_0} = O(Q^{-1-h} S_*^P)$. Each additional fixed derivative lowers the power of Q by a fixed amount. The bounds for these exponents stay uniform when the chart indices differ by only a bounded amount.

The label phase satisfies $|\nabla_x \Phi| \leq CQ^{-1/2} S_*^P$ and $|\partial_t \Phi| \leq CQ^{-1-h} S_*^P$. The axial term $p_z Z / \varepsilon$ costs $Q^{-D} \varepsilon^{-1} = Q^{-1/2}$. More generally, for every fixed pair of nonnegative integers a, b with $a + b \geq 1$, (7.3) and (6.12) give

$$|\nabla_x^a \partial_t^b \Phi_\gamma| \leq C_{a,b} Q^{-a/2 - b(1+h)} S_*^{P_{a,b}},$$

uniformly over all bands and labels. In each chart, the pulse coordinate has no physical spatial derivatives and has one fixed first physical time derivative. Every other factor has the slow derivative bounds already stated. Set

$$s_x = \frac{1}{2} + \frac{h}{2}, \quad s_t = 1 + \frac{3h}{2}.$$

Since $k \leq 2Q^{-h/2}$, the chain rule gives

$$|\nabla_x^a \partial_t^b e^{ikm\Phi}| \leq C_{a,b,m} Q^{-as_x - bs_t} S_*^{P_{a,b,m}}.$$

These costs are at least as large as the ones in (9.19), so they control them. At any fixed stage there are only finitely many harmonic integers, and they change only the constants. A quadratic residual can at most double the current harmonic range, while inverse and curl operations keep each integer unchanged. The harmonic range is therefore finite independently of the band.

For a class exponent α , Leibniz' rule now bounds the Cartesian space-time derivatives of the normalized coefficient by $CQ^{h\alpha - as_x - bs_t} S_*^P$. The factors $\sqrt{\zeta} \delta^{-M}$ and $\zeta \delta^{-M}$ are bounded on the closed shell for every fixed M , so they do not cause any further loss in powers. The rescaling factors for physical velocity, pressure, wave potential, and mean potential are, respectively, Q^{-A} , Q^{-2A} , $Q^{1/2-A}$, $Q^{1/2-A}$. Each of them is bounded by Q^{-2A} . The normalized chart wave potential already contains the factor $(km)^{-1}$, before physical rescaling. Recovering a velocity from a wave or mean vector potential costs one more spatial derivative. One common choice that is large enough is

$$\ell_m = 2A + (m+1)(1+3h/2),$$

Every increment kept at stage j has normalized exponent at least $j/10$, and its physical prefactor is bounded by Q^{-2A} . This proves (9.17).

Apply the same calculation to (9.9), to the fixed background, and to (9.8). Converting the residual from normalized coordinates to physical coordinates adds one fixed power of q . The rest of the radial residual consists of $-\rho\mathcal{P}$, written in physical coordinates, and its cutoff remainder. Equation (9.5) gives $E_{j,m}$ uniformly over all labels at each fixed stage. This proves (9.18) and the bound for the whole correction. Where two coordinate representations overlap, the fields agree after the prescribed changes of torus coordinates and relabeling. Their smooth zero extensions then carry the estimates across support and band boundaries. $\square$

9.5 Building the local field.

The finite partial sums $(u^{[j]}, p^{[j]})$ now have residuals whose decay can be made arbitrarily high on one common domain. Lemmas 9.7 and 9.8 also give the bounds for physical derivatives that Lemma 5.4 needs. This subsection sums the potentials with cutoffs that shrink from one stage to the next, producing the local velocity u_{loc} from Proposition 9.9. The remaining work is to check that this sum keeps the endpoint regularity, the exterior heat flow, and the inner growth in Theorem 3.1(ii)–(iv).

Proposition 9.9. *There are smooth fields $(u_{\text{loc}}, p_{\text{loc}})$ on Ω_* , obtained by summing the finite corrections above, such that $\text{div } u_{\text{loc}} = 0$ and every conclusion of Theorem 3.1 holds.*

Proof. Apply the summation lemma with $\mathcal{F}(u, p) = \mathcal{R}(u, p)$. Use $q_* = q_{\text{big}}$ from Lemma 9.7, shrinking the earlier local thresholds to this common value. The inputs are the velocity field and the pressure field. Step 1 checks the hypotheses of Lemma 5.4. Steps 2–5 prove Theorem 3.1(i)–(iv), in that order. They prove Cartesian regularity, endpoint regularity, the exterior formula together with the flat residual, and the inner swirl asymptotic.

Step 1: Choose representatives and check the summation hypotheses. Take U_0 to be the fixed slow base together with the finite initialization block. If needed, apply one cutoff to that block inside $q < q_{\text{big}}$. Apply it to the block's wave and mean potentials before differentiation, and also to its direct angular means and pressures. The cutoff equals one near $q = 0$, so every finite state still agrees there with its original expression.

For $j \geq 1$, use the representatives $Z_j = (A_j, B_j, p_j)$ from (9.15). Their mean-potential part gives

$$\text{curl}(\Psi_j e_\theta) = (-\partial_z \Psi_j, 0, (\partial_r + r^{-1})\Psi_j).$$

Since $B_j = b_j e_\theta$ and $\partial_\theta b_j = 0$, multiplying B_j by any function of $q(z, t)$ keeps its divergence equal to zero. Every one of these representatives extends smoothly by zero and vanishes near the axis for each fixed $t < 1$. Because it vanishes near the axis, its Cartesian extension is smooth there.

Lemma 9.7 gives one domain for every finite partial sum before the summation cutoffs are applied. The similarity identities give (5.28), and $q \geq 1 - t$ gives the local lower bound that the summation lemma needs. Lemma 9.8 gives (5.30) with $g_j = hj/10$, and the losses in the exponents do not depend on j .

For the fixed background and initialization, all derivatives through order m are bounded by $C_m q^{-K_m}(1 + |\log q|)^{P_m}$. Finally, (9.18) is exactly (5.33) for the fixed differential polynomial

$$\mathcal{F}(u, p) = \mathcal{R}(u, p) = \partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p,$$

with $\rho_j = h\sigma_j \rightarrow \infty$. The estimate (9.18) concerns the actual residual at each point after the auxiliary variables have been set equal to $Y(r, t)$. At any finite stage, the flat term is bounded uniformly over the infinitely many dyadic labels. Cutting off the finite initialization adds errors only away from $q = 0$, and those errors satisfy the same fixed-stage flatness condition. The residual estimates hold uniformly over the whole local domain. To see this, choose one fixed, bounded interval in X that contains every slow support and every annular correction support. The estimates above and (5.41) apply on that interval. Outside it, every finite state is exactly the exterior heat field from (4.29), together with the centrifugal pressure, so its residual is zero there.

Step 2: Sum the fields and check their Cartesian regularity. Lemma 5.4 now gives potentials whose locally finite sum defines a velocity with $\operatorname{div} u_{\text{loc}} = 0$, and

$$\forall m, N \quad |\mathcal{R}(u_{\text{loc}}, p_{\text{loc}})|_m = O(q^N) \quad (q \downarrow 0). \quad (9.20)$$

The tail estimate compares the field just built with one fixed finite state and that state's flat remainder. The flat residuals separated at the finite stages are controlled through this comparison, not added together. To identify the potentials and pressure of the summed fields, let $\mathcal{A}_0, \mathcal{B}_0 e_\theta, p_0$ be the representatives of U_0 . These representatives are the realized slow base from (5.27), together with the finite initialization and its pressure. Define

$$\begin{aligned} \mathcal{A} &= \mathcal{A}_0 + \sum_{j \geq 1} \chi(a_j q) A_j, \\ \mathcal{B} e_\theta &= \mathcal{B}_0 e_\theta + \sum_{j \geq 1} \chi(a_j q) B_j, \\ p_{\text{loc}} &= p_0 + \sum_{j \geq 1} \chi(a_j q) p_j, \quad u_{\text{loc}} = \operatorname{curl} \mathcal{A} + \mathcal{B} e_\theta. \end{aligned} \quad (9.21)$$

These formulas give the required vector potential, azimuthal field, and pressure. At the axis, the base Stokes streamfunctions are r^2 times a smooth function of (r^2, z, t) , so the base vector potential is smooth in Cartesian coordinates. The base angular field is r times such a scalar, times e_θ . For each $t < 1$, every annular representative vanishes on a neighborhood of the axis. Equation (9.21) therefore proves Theorem 3.1(i).

Step 3: Check one-sided regularity away from the singular point. The endpoint regularity still has to be checked away from $q = 0$. Fix a compact spatial set and numbers $0 < c < c' < q_*$. Only finitely many terms in the expansion in powers of q^{2h} have nonzero cutoff factors on a neighborhood of $c \leq q \leq c'$, and only finitely many correction stages do, because both cutoff sequences tend to infinity. On the chosen compact set there are also only finitely many dyadic bands and slow labels. By Theorem 4.6 and Proposition 5.5, the base profiles and their Stokes streamfunctions have bounds for derivatives of every fixed order on the closed parameter range $-1 \leq \eta \leq 1$. The geometry chooses its labels, representatives, rounded frequencies, and enlarged local torus rectangles from the closed range $\tau \geq 0$, and it keeps those discrete choices fixed as $\tau \rightarrow 0$. The pulse inverse follows a path on which the physical slow variables stay fixed and $\tau \geq 0$. Differentiate its fixed linear ODE to get the bounds at every derivative order. Combine these bounds with the quotient bound (7.33), which uses the fixed positive primary amplitudes, and with the radial and temporal mean maps. The result is a finite one-sided bound for every fixed derivative of each wave potential, mean

vector potential, azimuthal mean component, and pressure whose cutoff is nonzero on this neighborhood. Their smooth extensions by zero cover every boundary of their supports.

The denominator in the coordinate conversion is $L \geq 1 - 2h > 0$, so the conversion keeps all these fixed-order derivative bounds on $q \geq c$. On the annular supports, r stays away from zero. At the axis, use the Cartesian factors described above. The finite sum then has uniformly bounded Cartesian space–time derivatives of every order all the way to $\tau = 0$, including inside the active shell. Bound one more time derivative and apply the fundamental theorem of calculus. This gives a uniform one-sided limit for every derivative. Applying the same theorem on compact spatial sets shows that the limits agree with spatial differentiation. Integrating in time shows that the limits of consecutive time derivatives agree with one another. This proves Theorem 3.1(ii) for $\mathcal{A}, \mathcal{B}e_\theta, p_{\text{loc}}$ themselves.

Step 4: Check the exterior heat field and the flat residual. Every annular wave or mean correction potential, every azimuthal mean component, and every pressure correction vanishes beyond X_b . The leading profile and every coefficient in the expansion in powers of q^{2h} have zero total axial moment. Their Stokes streamfunctions therefore vanish in the same exterior region. It follows that, for one fixed X_{ext} beyond all the slow supports, $\mathcal{A} = 0$ and $\mathcal{B} = K$. The leading-profile pressure is normalized to vanish at radial infinity, and all positive-order pressure corrections have compact support. The pressure there is therefore exactly $-\int_r^\infty K(\rho, \tau)^2 d\rho/\rho$. Using (4.29), define

$$\mathcal{H}_{\text{ext}}(s) = 2^A c_\infty \mathcal{H}(4s), \quad K(r, \tau) = r^{-1-2h} \mathcal{H}_{\text{ext}}(\tau/r^2).$$

This is (3.5). For every integer $m \geq 0$, the integral formula (A.34) gives

$$\sup_{s \geq 0} |\mathcal{H}_{\text{ext}}^{(m)}(s)| \leq 2^A c_\infty 4^m (h)_m (1+h)_m < \infty,$$

where $(b)_m$ is the rising factorial, the product of m factors starting at b and increasing by one each time, and $(b)_0 = 1$. The bound follows because the factor $(1 + Zv)^{-h-m}$ in that integral is at most one whenever $Z, v \geq 0$. The required derivative bounds therefore hold on the whole nonnegative half-line. Lemma A.6 says that K solves the radial heat equation, and its centrifugal pressure makes the Navier–Stokes residual exactly zero in the exterior. Combine this with the estimates for the background residual on compact sets of profile variables in (5.41) and (9.20). This proves all of Theorem 3.1(iii).

Step 5: Check the growth of the inner velocity. Finally, choose any fixed $X_{\text{in}} \in (0, X_a)$ at which the positive leading azimuthal velocity is $e_0 = E_0(X_{\text{in}}, 0) > 0$. Every annular correction vanishes there. The realized expansion in powers of q^{2h} in (5.1) has its leading value plus a remainder $O(q^{2h})$ after the velocity factor q^{-A} is removed, at the fixed point $(X, \eta) = (X_{\text{in}}, 0)$ in profile coordinates. At $z = 0$, the coordinates satisfy $\eta = 0$ and $q = \tau$, so

$$u_{\theta, \text{loc}}(\sqrt{2X_{\text{in}}\tau}, 0, 0, 1 - \tau) = \tau^{-A} (e_0 + O(\tau^{2h})).$$

This is Theorem 3.1(iv), and it completes the proof. □

10 A compactly supported force and breakdown on all of space

This section turns the fields from Theorem 3.1 into the solution and force on all of space in Theorem 1.1. Write the local fields as

$$u^{\text{loc}} = \text{curl } \mathcal{A} + \mathcal{B}e_\theta, \quad p^{\text{loc}}.$$

Here $\mathcal{A}$ and $\mathcal{B}$ are the representatives built in (9.21), and $\mathcal{B}$ does not depend on θ . Start with viscosity one. The local velocity has the growth in (3.6) required by Theorem 1.1. What remains is to give it zero initial data and compact spatial support while making its momentum residual agree, before time one, with a force in

$C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$. This is done in Proposition 10.1 and Lemma 10.3.

Multiply the vector potential, the azimuthal coefficient, and the pressure by cutoffs that equal one near $(0, 1)$. Lemma 10.2 then proves that every derivative of the new residual has a uniform limit as $t \uparrow 1$, and Lemma 10.3 builds a force for $t > 1$ whose derivatives attain all those limits. Before time one, the equation gives a uniform energy bound in Lemma 10.4. Lemma 10.5 then compares this field with any smooth solution of bounded kinetic energy. That comparison transfers the growth of the local velocity to the smooth solution and gives the contradiction needed for Theorem 1.1. Rescaling gives every positive viscosity and also gives the periodic construction in Corollary 10.6.

10.1 Keeping the construction inside the local domain.

This subsection places the velocity and pressure inside the domain from Theorem 3.1. First recall why the chosen vector potential $\mathcal{A}$ is smooth at the axis, as stated in Theorem 3.1(i), and why it vanishes in the heat exterior from (3.5). Because of these two facts, its curl can be taken after the localization, and the force at time one can be estimated in Lemma 10.2. For the background coefficients U_n from (5.1), the Stokes streamfunctions and their sum are

$$\begin{aligned} \mathcal{S}_n &= q^{1-A+2nh} \int_0^X U_n(X', \eta) dX', \quad \mathcal{S} = \mathcal{S}_0 + \sum_{n \geq 1} \chi(c_n q) \mathcal{S}_n, \\ \mathcal{A}_{\text{base}} &= \frac{\mathcal{S}}{r} e_\theta. \end{aligned} \tag{10.1}$$

The cutoffs $\chi(c_n q)$ are the ones used in Proposition 5.5. Equations (4.28) and (5.10) give $\int_0^\infty U_n(X, \eta) dX = 0$ for every $n \geq 0$. Each radial integral in (10.1) is therefore zero beyond the common support of the axial coefficients. Near the axis, $\mathcal{S} = r^2 a(r^2, z, t)$ for a smooth scalar a , so

$$(\mathcal{S}/r) e_\theta = a(r^2, z, t) (-x_2, x_1, 0).$$

This is the smooth Cartesian form of the background potential.

Every wave velocity is the curl of its supported physical potential. For a prescribed axial mean increment $\gamma_{d, \text{phys}}$, the mean construction uses the azimuthal potential coefficient

$$\Psi = r^{-1} \mathcal{I}_c(r \gamma_{d, \text{phys}}), \quad \mathcal{I}_c g = \mathcal{I} g - \chi_m \mathcal{J} g,$$

where the radial operations are defined in (8.4). Beyond both the source and the region where χ_m changes, the identities $\chi_m = 1$ and $\mathcal{I} g = \mathcal{J} g$ hold, so $\Psi = 0$. Together with their summation cutoffs, these wave and mean potentials are the terms added to $\mathcal{A}_{\text{base}}$ in (9.21). The direct azimuthal mean increments are part of $\mathcal{B}$. Setting the auxiliary variables equal to $Y(r, t)$ keeps the fields independent of θ . Every annular correction vanishes near the axis for each fixed $t < 1$. In particular, for the fixed X_{ext} from Theorem 3.1, the chosen potential satisfies $\mathcal{A} = 0$ on $X \geq X_{\text{ext}}$.

Now choose cutoffs supported inside the local domain $\Omega_* = \{\tau > 0, q < q_*\}$, where $\tau = 1 - t$. Apply the spatial cutoff before taking the curl, which keeps the velocity incompressible. The time cutoff gives zero initial data.

Proposition 10.1. *There are a compact set $\mathcal{K} \subset \mathbb{R}^3$, a smooth vector field u , and a smooth scalar field p on $\mathbb{R}^3 \times [0, 1)$. Their supports satisfy $\text{supp } u(\cdot, t) \cup \text{supp } p(\cdot, t) \subset \mathcal{K}$ for every $0 \leq t < 1$, and they also satisfy*

$$\text{div } u = 0, \quad u(\cdot, t) = p(\cdot, t) = 0 \quad \text{for all sufficiently small } t \geq 0. \tag{10.2}$$

These fields agree with $u^{\text{loc}}, p^{\text{loc}}$ on a spatial neighborhood of the origin for every time sufficiently close to 1.

Proof. A fixed spatial cutoff will stay inside the local domain once q is bounded above in terms of z and τ . The coordinates in (3.2) give such a bound uniformly in r . If $1 - \eta^2 \geq 1/2$, then $q \leq 2\tau$. Otherwise $|\eta| > 2^{-1/2}$, and $q \leq (\sqrt{2}|z|)^{1/D}$. It follows that

$$q \leq C_0(\tau + |z|^{1/D}). \quad (10.3)$$

Choose $0 < \tau_0 < 1/2$ and $z_0 > 0$ so that $C_0(\tau_0 + z_0^{1/D}) < q_*/2$. For any fixed $r_0 > 0$, choose a smooth axisymmetric cutoff $0 \leq \chi_x \leq 1$ that is a smooth function of r^2, z and satisfies

$$\chi_x = 1 \quad (r \leq r_0/2, |z| \leq z_0/2), \quad \text{supp } \chi_x \subseteq \{r < r_0, |z| < z_0\}.$$

Also choose a smooth function $0 \leq \chi_t \leq 1$ of t with

$$\chi_t(t) = 0 \quad (1 - t \geq \tau_0), \quad \chi_t(t) = 1 \quad (0 \leq 1 - t \leq \tau_0/2), \quad c(x, t) = \chi_x(x)\chi_t(t).$$

By (10.3), the support of c for $t < 1$ lies in $q < q_*/2$. It therefore stays a fixed distance, measured in the q coordinate, from the outer boundary of Ω_* . On that domain, define

$$u = \text{curl}(c\mathcal{A}) + c\mathcal{B}e_\theta, \quad p = cp^{\text{loc}}, \quad (10.4)$$

and extend both fields by zero outside the support of the cutoff. The product rule gives

$$u = cu^{\text{loc}} + \nabla c \times \mathcal{A}, \quad \text{div } u = \text{div curl}(c\mathcal{A}) + r^{-1}\partial_\theta(c\mathcal{B}) = 0.$$

The Cartesian forms recalled above make the fields smooth at the axis. The cutoffs, together with the margin from $q = q_*$, let the fields extend smoothly by zero everywhere else. Their support lies in $\mathcal{K} = \text{supp } \chi_x$, and the time cutoff gives (10.2). Both cutoffs equal one near $(0, 1)$, so they leave the local fields unchanged there. $\square$

For $0 \leq t < 1$, define

$$f = \mathcal{R}(u, p) = \partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p. \quad (10.5)$$

This definition includes every derivative of the cutoffs. Taking its divergence gives

$$-\Delta p = \sum_{i,j=1}^3 \partial_i \partial_j (u_i u_j) - \text{div } f.$$

The localized pressure therefore satisfies the Poisson equation with the matching term from the divergence of the force. The force is allowed to have nonzero divergence.

10.2 Limits of force derivatives at the final time.

So far, the force in (10.5) is defined only for $t < 1$. To extend it smoothly past time one, all of its derivatives must have compatible limits as $t \uparrow 1$. At the origin, those limits are controlled by the flatness of the local residual in (3.4). Away from the origin, the terms introduced by localization are controlled by the endpoint bounds in Theorem 3.1(ii) and the exact exterior heat flow in (3.5). Lemma 10.2 first proves the derivative limits. Lemma 10.3 then builds an extension with compact support in time.

Lemma 10.2. *For every spatial multi-index α and every integer $j \geq 0$, the derivative $\partial_x^\alpha \partial_t^j f$ converges uniformly on $\mathbb{R}^3$ as $t \uparrow 1$. There are functions $F_j \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, all supported in $\mathcal{K}$, such that*

$$\lim_{t \uparrow 1} \partial_x^\alpha \partial_t^j f(x, t) = \partial_x^\alpha F_j(x), \quad \partial_x^\alpha F_j(0) = 0. \quad (10.6)$$

These are the derivative limits of f from $t < 1$, and they agree with both spatial and time differentiation.

Proof. Start with endpoint bounds away from the origin. Take a compact spatial set on which q has a positive lower bound and stays away from q_* . Theorem 3.1(ii) bounds every physical derivative of the actual potentials and scalar fields there. More exactly, let K_0 be such a compact set, let $G = \mathcal{A}, \mathcal{B}e_\theta, p^{\text{loc}}$, and take $t < t' < 1$. The fundamental theorem of calculus gives

$$\sup_{K_0} |\partial_x^\alpha \partial_t^j G(\cdot, t') - \partial_x^\alpha \partial_t^j G(\cdot, t)| \leq |t' - t| \sup_{K_0 \times [t, t']} |\partial_x^\alpha \partial_t^{j+1} G|.$$

The last supremum has a bound near time one that does not depend on t, t' . Every fixed derivative is therefore uniformly Cauchy as $t \uparrow 1$. These constants may depend on the chosen positive lower bound for q .

The argument in Proposition 9.9 that leaves only finitely many nonzero cutoff terms gives these bounds on the closed parameter range $-1 \leq \eta \leq 1$. It gives them for the actual representatives and for the smooth Cartesian factors in their near-axis forms, while the labels and integer frequencies stay fixed. Multiplying by the fixed cutoffs and applying the finite differential expression (10.5) keep the bounds.

In the region that remains, r has a positive lower bound and q is small enough that $X \geq X_{\text{ext}}$. The velocity there is the exterior azimuthal field $K(r, \tau)e_\theta$ from (3.5). In terms of the heat profile $\mathcal{H}$ defined in (A.32),

$$K(r, \tau) = c_\infty s^{-A} \mathcal{H}(2\tau/s), \quad s = r^2/2. \quad (10.7)$$

Thus the function in Theorem 3.1 is $\mathcal{H}_{\text{ext}}(Z) = 2^A c_\infty \mathcal{H}(4Z)$. For each j , the integral in (A.32) can be differentiated uniformly for $Z \geq 0$ because its differentiated integrand is bounded by a constant times $e^{-v} v^{h+j}$. It follows that

$$|\partial_\tau^j K(r, \tau)| \leq C_j r^{-1-2h-2j}, \quad \left| \partial_\tau^j \frac{K(\rho, \tau)^2}{\rho} \right| \leq C_j \rho^{-3-4h-2j}. \quad (10.8)$$

The pressure is normalized to vanish at radial infinity. On a compact interval of positive radii $r \in [r_-, r_+]$, where $r_- > 0$, the second upper bound is integrable over $\rho \geq r_-$. Dominated convergence therefore gives

$$\partial_\tau^j p^{\text{loc}}(r, \tau) = - \int_r^\infty \partial_\tau^j \left(\frac{K(\rho, \tau)^2}{\rho} \right) d\rho.$$

It also gives a uniform bound and a one-sided limit for every derivative on each compact interval of positive radii. Here $\partial_t = -\partial_\tau$. Differentiate the lower endpoint to get the higher radial derivatives of the pressure, and differentiate (10.7) directly to get those of K . This gives the required smooth one-sided extension to $\tau = 0$, using the integral on $\tau \geq 0$. In this region $\mathcal{A} = 0$, so after localization the fields contain only these smooth exterior quantities and the fixed cutoffs. The heat equation for K and the identity $\partial_r p^{\text{loc}} = K^2/r$ also show directly that the residual of the uncut exterior field is exactly zero.

These two regions cover the final time slice away from the origin. If $z \neq 0$, then $q \geq |z|^{1/D} > 0$. If $z = 0$ and $x \neq 0$, then $r > 0$, and the exterior argument above applies on a small neighborhood. Every Cartesian space-time derivative of the force therefore converges uniformly on compact spatial sets that do not contain the origin.

Near the origin, both cutoffs equal one. Use (3.4) on $X \leq X_{\text{ext}}$, and use the exactly zero residual when X is larger. For every α, j, N , this gives, on one fixed neighborhood of $(0, 1)$,

$$|\partial_x^\alpha \partial_t^j f(x, t)| \leq C_{\alpha, j, N} (\tau + |z|^{1/D})^N. \quad (10.9)$$

Given any tolerance, first choose a small ball and then choose a time close enough to one that this bound is below the tolerance throughout the ball. A finite cover of the part of $\mathcal{K}$ outside that ball gives uniform convergence there. Every one of these derivatives is zero outside $\mathcal{K}$. They are therefore uniformly Cauchy on $\mathbb{R}^3$, and their limits are zero at the origin.

Define F_j as the uniform limit for $\alpha = 0$. Pass to the limit in the fundamental theorem of calculus along segments parallel to the spatial coordinate axes. This identifies every other limit as $\partial_x^\alpha F_j$. It follows that F_j is smooth and has the stated support. Passing to the limit in the matching identity in time gives

$$\partial_x^\alpha \partial_t^j f(x, t) = \partial_x^\alpha F_j(x) - \int_t^1 \partial_x^\alpha \partial_t^{j+1} f(x, s) ds. \quad (10.10)$$

This proves that the derivative limits agree with both spatial and time differentiation, and it finishes the proof of (10.6). $\square$

The limits F_j are the Taylor coefficients needed to continue the force through time one. Give the coefficients time cutoffs whose supports shrink as the derivative order grows, so the extension also has compact support in time. A smooth force with compact support has the decay required in [13, (5)], and the zero initial data has the required initial-data properties as well.

Lemma 10.3. *The force in (10.5) extends to $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$, with support in $\mathcal{K} \times [0, 2]$.*

Proof. Choose $\chi_0 \in C_c^\infty(\mathbb{R})$ so that it equals one near zero and equals zero whenever its argument is at least one. For $\sigma = t - 1 \geq 0$, set

$$f(x, 1 + \sigma) = \sum_{j=0}^{\infty} \chi_0(b_j \sigma) \frac{\sigma^j}{j!} F_j(x), \quad (10.11)$$

where the increasing integers $b_j \geq 1$ are chosen below. On the support of each summand, $0 \leq \sigma \leq b_j^{-1}$. The product rule then gives

$$\left\| \partial_x^\alpha \partial_\sigma^m \left(\chi_0(b_j \sigma) \frac{\sigma^j}{j!} F_j \right) \right\|_\infty \leq C_{j,m} b_j^{m-j} \|F_j\|_{C^{|\alpha|}}. \quad (10.12)$$

To see this, suppose a derivatives land on the cutoff. Their factor b_j^a is multiplied by a monomial of degree $j - m + a$, which is bounded by $b_j^{-(j-m+a)}$. If that monomial has been differentiated to zero, the term is absent. Set $b_0 = 1$. For $j \geq 1$, define

$$b_j = \min \left\{ b \in \mathbb{N} : b > b_{j-1}, \max_{\substack{a, m \geq 0 \\ a+m \leq \lfloor j/2 \rfloor}} C_{j,m} b^{m-j} \|F_j\|_{C^a} \leq 2^{-j} \right\},$$

where the maximum is taken over nonnegative integers a, m . This gives only finitely many conditions, and each one has $m - j < 0$. So the defining set contains at least one integer. With this choice, (10.12) is at most 2^{-j} whenever $|\alpha| + m \leq \lfloor j/2 \rfloor$.

After any fixed number of derivatives, the tail now converges uniformly. At $\sigma = 0$, the cutoff in each fixed summand is constant on a neighborhood of zero. Its m -th derivative therefore contributes only when $j = m$, and that contribution is F_m . Uniform convergence after differentiation, together with (10.10), shows that (10.11) matches every mixed space–time derivative limit from $t < 1$. Every summand keeps its support in $\mathcal{K}$ and is zero when $\sigma \geq 1$. The same uniform convergence proves smoothness at those support boundaries. Finally, the original force is zero near $t = 0$, by (10.2) and the construction of p .

For the decay bounds, choose R with $\mathcal{K} \subset B(0, R)$, and write $M_{\alpha,m} = \|\partial_x^\alpha \partial_t^m f\|_\infty$. Since the force has compact support, every integer $k \geq 0$ satisfies

$$|\partial_x^\alpha \partial_t^m f(x, t)| \leq M_{\alpha,m} (3 + R)^k (1 + |x| + t)^{-k}.$$

$\square$

10.3 Energy and comparison.

This subsection starts by proving that the smooth compactly supported force f from Lemma 10.3 gives an energy bound for the constructed velocity u that stays uniform up to time one (Lemma 10.4). Next it proves that any smooth solution with the same force, zero initial data, and bounded kinetic energy must agree with that velocity on every shorter time interval (Lemma 10.5). This comparison lets the proof carry the local velocity growth in (3.6) over to any supposed global smooth solution. In these energy estimates on all of space, $\|\cdot\|_p$ is the spatial $L^p(\mathbb{R}^3)$ norm, and every spatial integral with no domain written beside it is over $\mathbb{R}^3$.

Lemma 10.4. *Set $\mathcal{F}(t) = \int_0^t \|f(s)\|_2 ds$ for $0 \leq t \leq 1$. Then $\mathcal{F}(1) < \infty$, and*

$$\|u(t)\|_2^2 + 2 \int_0^t \|\nabla u(s)\|_2^2 ds \leq \mathcal{F}(t)^2 \quad (0 \leq t < 1). \quad (10.13)$$

In particular, the kinetic energy stays uniformly bounded, and the total dissipation on $[0, 1)$ is finite.

Proof. On any interval $[0, T]$ with $T < 1$, the localized velocity and pressure are smooth and have support in one fixed compact spatial set. Taking the inner product of the equation with u and integrating in space gives

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_2^2 + \|\nabla u(t)\|_2^2 = \langle f(t), u(t) \rangle. \quad (10.14)$$

For $\delta > 0$, divide this identity by $(\|u(t)\|_2^2 + \delta^2)^{1/2}$, drop the nonnegative dissipation term, and apply Cauchy-Schwarz. Integrate from the zero initial data and then let $\delta \downarrow 0$. The result is $\|u(t)\|_2 \leq \mathcal{F}(t)$. Put this bound back into (10.14) and integrate once more to get

$$\|u(t)\|_2^2 + 2 \int_0^t \|\nabla u(s)\|_2^2 ds \leq 2 \int_0^t \mathcal{F}'(s) \mathcal{F}(s) ds = \mathcal{F}(t)^2.$$

Because the force is smooth and compactly supported, $\mathcal{F}(1) < \infty$. Monotone convergence then shows that the integrated dissipation on $[0, 1)$ is finite, using only the bounds already proved for $t < 1$. $\square$

The comparison argument only needs smoothness and a uniform spatial L^2 bound on $[0, T]$, where $T < 1$. These assumptions put no restriction on how the spatial derivatives grow at infinity. The equation will first be used to find the pressure gradient, and only then will the proof take the limit in the localized energy identity.

Lemma 10.5. *Fix $T < 1$. Suppose v, P is a smooth solution of (1.1) with viscosity one on $\mathbb{R}^3 \times [0, T]$. Suppose it has the force f from Lemma 10.3, zero initial velocity, and $v \in L^\infty([0, T]; L^2(\mathbb{R}^3))$. Then $v = u$ on that whole interval, where u is the localized velocity from Proposition 10.1.*

Proof. Take the difference of the two solutions and estimate it on balls whose radii grow without bound. The pressure term needs its own estimate because nothing has been assumed about the spatial growth of P . Each constant written as C_T below may depend on T and on the stated bounds for the fields, but it does not depend on the cutoff radius. Set

$$w = v - u, \quad \pi = P - p, \quad g_{ij} = v_i v_j - u_i u_j = w_i w_j + w_i u_j + u_i w_j.$$

The convention is $(\operatorname{div} g)_i = \sum_j \partial_j g_{ij}$. On $[0, T]$, $\|w(t)\|_2 \leq C_T$ and $\sum_{i,j} \|g_{ij}(t)\|_1 \leq C_T$. The same force appears in both equations, so it cancels and leaves

$$\partial_t w + (v \cdot \nabla) w + (w \cdot \nabla) u = \Delta w - \nabla \pi, \quad \operatorname{div} w = 0. \quad (10.15)$$

Pressure gradient. Let R_i be the Riesz transform whose Fourier multiplier is $i\zeta_i/|\zeta|$, and define the pressure as a distribution, a continuous linear rule that assigns a number to each smooth compactly supported test function, by

$$\pi_* = \sum_{i,j=1}^3 R_i R_j g_{ij}. \quad (10.16)$$

The multiplier for this operator is $-\xi_i \xi_j / |\xi|^2$; its value at zero does not matter. The Fourier transform of an L^1 function is bounded, so π_* is uniformly in $H^{-s}(\mathbb{R}^3)$ for every $s > 3/2$. More exactly, its squared norm is at most a constant times $\|g\|_1^2 \int_{\mathbb{R}^3} (1 + |\xi|^2)^{-s} d\xi$. The multiplier also gives

$$\Delta \pi_* = - \sum_{i,j} \partial_i \partial_j g_{ij}.$$

The next step is to prove that $\nabla \pi = \nabla \pi_*$ in the space-time interior. Take $a \in C_c^\infty(0, T)$. Writing (10.15) in conservative form gives

$$\int_0^T a \nabla \pi dt = \Delta \int_0^T a w dt + \int_0^T a' w dt - \operatorname{div} \int_0^T a g dt.$$

The three terms on the right belong, in that order, to H^{-2} , L^2 , and H^{-3} . The last one has this property because $g \in L_t^\infty L_x^1$, and the Fourier estimate above can be used with $s = 2$. Also, $\pi_* \in L_t^\infty H_x^{-2}$, so

$$H_a := \int_0^T a (\nabla \pi - \nabla \pi_*) dt \in H^{-3}(\mathbb{R}^3).$$

Taking the divergence of the difference equation gives $\Delta \pi = - \sum_{i,j} \partial_i \partial_j g_{ij} = \Delta \pi_*$. It follows that $\Delta H_a = 0$. The Fourier transform of H_a is a weighted L^2 function supported at the single point $\{0\}$, so it must be zero. Allowing a spatial test function as well proves the stated equality. The equation has now determined the pressure gradient under these assumptions without putting any restriction on the spatial growth of P .

Pressure flux. Next, bound the pressure contribution on the boundary of an expanding ball by the weighted dissipation inside that ball. Start with $\int \pi w \cdot \nabla \chi_R$. It will be estimated by powers of a weighted L^6 norm of w , with powers low enough for the dissipation to absorb them. Choose a smooth, compactly supported function $0 \leq \varphi \leq 1$ that equals one on the unit ball. For $R \geq 1$, write $\varphi_R(x) = \varphi(x/R)$, set $\chi_R = \varphi_R^8$, and define

$$E_R = \int \chi_R |w|^2, \quad A_R = \left(\int \chi_R |\nabla w|^2 \right)^{1/2}, \quad B_R = \|\varphi_R^4 w\|_6.$$

The Sobolev inequality and the product rule give

$$B_R \leq C \|\nabla(\varphi_R^4 w)\|_2 \leq C(A_R + R^{-1} \|w\|_2). \quad (10.17)$$

Use the standard fact that Riesz transforms are bounded on L^p when $1 < p < \infty$; see [20]. If a is a scalar function and $\mathcal{T}$ is a linear operator, the commutator $[a, \mathcal{T}]g = a(\mathcal{T}g) - \mathcal{T}(ag)$ measures the difference between multiplying before and after applying the operator. Multiplying (10.16) by φ_R^4 gives

$$\varphi_R^4 \pi_* = \sum_{i,j} R_i R_j (\varphi_R^4 g_{ij}) + \sum_{i,j} [\varphi_R^4, R_i R_j] g_{ij}. \quad (10.18)$$

The first sum has $L^{3/2}$ norm no larger than $C_T(B_R + 1)$, because

$$\|\varphi_R^4 w_i w_j\|_{3/2} \leq B_R \|w\|_2, \quad \|\varphi_R^4 w_i u_j\|_{3/2} \leq \|w\|_2 \|u\|_6.$$

The nonlocal kernel of $R_i R_j$ is bounded by $C|x - y|^{-3}$. Any multiple of the identity in that operator cancels from the commutator, so the absolute value of the commutator kernel is bounded by

$$K_R(x - y) = C|x - y|^{-3} \min\{|x - y|/R, 1\}.$$

Integrating directly in the radial variable gives

$$\|K_R\|_{4/3}^{4/3} \leq C \left[R^{-4/3} \int_0^R r^{-2/3} dr + \int_R^\infty r^{-2} dr \right] \leq CR^{-1}.$$

Young's convolution inequality now bounds the second sum in (10.18) in $L^{4/3}$ by $C_T R^{-3/4}$. These identities can first be proved for compact smooth cutoffs of g , then passed to g by convergence in L^1 , the uniform commutator bound, and convergence of the Riesz transforms in H^{-s} . In particular, π_* is locally integrable in both space and time.

Test the equality of the pressure gradients against the compact vector field $a(t)\chi_R w$. This gives

$$\int \pi w \cdot \nabla \chi_R dx = \int \pi_* w \cdot \nabla \chi_R dx \quad \text{for almost every } t \in (0, T).$$

Because $|\nabla \chi_R| \leq CR^{-1}\varphi_R^7$, the two sums in (10.18) are paired with $CR^{-1}\varphi_R^3|w|$. Interpolation gives

$$\|\varphi_R^3 w\|_3 \leq \|\varphi_R^2 w\|_3 \leq B_R^{1/2} \|w\|_2^{1/2}, \quad \|\varphi_R^3 w\|_4 \leq B_R^{3/4} \|w\|_2^{1/4}.$$

It follows that

$$\left| \int \pi w \cdot \nabla \chi_R \right| \leq C_T R^{-1} \left[(B_R + 1) B_R^{1/2} + R^{-3/4} B_R^{3/4} \right]. \quad (10.19)$$

Difference energy. The pressure-flux bound can now be used in the energy estimate for the difference. Every integral has compact support, so smoothness is enough to justify the calculation. Pairing (10.15) with $\chi_R w$ gives

$$\begin{aligned} \frac{1}{2} E'_R + A_R^2 &= - \int \chi_R (w \cdot \nabla) u \cdot w + \frac{1}{2} \int |w|^2 \Delta \chi_R \\ &\quad + \frac{1}{2} \int |w|^2 v \cdot \nabla \chi_R + \int \pi w \cdot \nabla \chi_R. \end{aligned}$$

Choose R large enough that $\chi_R = 1$ on a neighborhood of $\text{supp } u$. Then $v = w$ on $\text{supp } \nabla \chi_R$, and the transport flux is no larger than

$$CR^{-1} \int \varphi_R^6 |w|^3 \leq C_T R^{-1} B_R^{3/2}.$$

The last bound follows from $\varphi_R^2 |w| = (\varphi_R^4 |w|)^{1/2} |w|^{1/2}$ and Hölder's inequality. The Laplacian term is at most $C_T R^{-2}$. Substitute (10.17) into (10.19). Every power of A_R that appears in these flux bounds is then no larger than $3/2 < 2$, so Young's inequality gives

$$\frac{1}{2} E'_R + \frac{1}{2} A_R^2 \leq \|\nabla u\|_\infty E_R + \frac{C_T}{R} \quad \text{for almost every } t \in (0, T).$$

For example, $R^{-1} A_R^{3/2} \leq \delta A_R^2 + C_\delta R^{-4}$; the terms with powers $1/2$ and $3/4$ obey the same required bound. The coefficient $\|\nabla u\|_\infty$ is bounded on $[0, T]$, and $E_R(0) = 0$. Gronwall's inequality now gives

$$E_R(t) \leq \frac{2C_T}{R} \int_0^t \exp \left(2 \int_s^t \|\nabla u(a)\|_\infty da \right) ds \leq \frac{C'_T}{R}.$$

For every fixed ball, $\chi_R = 1$ once R is large enough. Letting $R \rightarrow \infty$ therefore gives $w = 0$ throughout $[0, T]$. $\square$

10.4 Growth, lifespan, and viscosity.

The force extension and energy estimate for the localized fields (u, p) are now in place (Lemma 10.3, (10.13)). This subsection uses the inner growth asymptotic (3.6) from Theorem 3.1 to find the maximal classical lifespan. Together with the comparison lemma, Lemma 10.5, the same formula rules out a global smooth solution with bounded energy. A spatial rescaling then gives the result for every fixed positive viscosity without changing the time of the singularity.

Proof of Theorem 1.1. By (10.5), the field constructed above solves the equation exactly on $[0, 1)$. Its smoothness and fixed compact support give $u \in C([0, T]; H^3)$ for every $T < 1$. Lemma 10.3 gives the force the required smoothness and support, and Lemma 10.4 gives the bounded energy.

Fix the X_{in} from Theorem 3.1, and take the Cartesian points

$$x_\tau = (\sqrt{2}X_{\text{in}}\tau, 0, 0), \quad t = 1 - \tau. \quad (10.20)$$

Along this path, $z = 0$ and $q = \tau$. For every sufficiently small $\tau > 0$, the spatial and time cutoffs in (10.4) both equal one. Equation (3.6) then gives

$$u_\theta(x_\tau, 1 - \tau) = \tau^{-A}(e_0 + O(\tau^{2h})) \longrightarrow +\infty. \quad (10.21)$$

The annular corrections vanish in this fixed inner similarity region. In the expansion, the cutoff sum of the terms with $n \geq 1$, in powers of q^{2h} , has exactly the relative-error bound shown here, with the summation cutoffs still included. The points x_τ where the velocity diverges approach the origin. Since $H^3(\mathbb{R}^3) \hookrightarrow L^\infty(\mathbb{R}^3)$, there can be no classical H^3 continuation through time one. In the classical H^3 class, Sobolev approximation justifies the same difference energy calculation, and $H^3 \hookrightarrow W^{1,\infty}$ supplies the bound needed for Gronwall's inequality. This proves uniqueness on every shorter interval. The maximal classical existence interval is therefore $[0, 1)$.

Now suppose there were a global smooth solution v, P with the same data and uniformly bounded kinetic energy. For every $T < 1$, Lemma 10.5 applies on the closed interval $[0, T]$, so $v = u$ throughout $[0, 1)$. Equation (10.21) would then contradict the boundedness of the smooth field v on a compact neighborhood of $(0, 1)$. The force cannot be zero, because if it were, (10.13) would give $u = 0$.

Finally, fix $\nu > 0$ and define

$$\begin{aligned} u_\nu(x, t) &= \sqrt{\nu} u(x/\sqrt{\nu}, t), \\ p_\nu(x, t) &= \nu p(x/\sqrt{\nu}, t), \quad f_\nu(x, t) = \sqrt{\nu} f(x/\sqrt{\nu}, t). \end{aligned} \quad (10.22)$$

Put $y = x/\sqrt{\nu}$. The chain rule gives

$$\begin{aligned} \partial_t u_\nu + (u_\nu \cdot \nabla_x) u_\nu - \nu \Delta_x u_\nu + \nabla_x p_\nu &= \sqrt{\nu} [\partial_t u + (u \cdot \nabla_y) u - \Delta_y u + \nabla_y p](y, t) = f_\nu(x, t), \\ \nabla_x \cdot u_\nu(x, t) &= (\nabla_y \cdot u)(y, t) = 0. \end{aligned}$$

The initial data are still zero, and the common spatial support becomes $\mathcal{K}_\nu = \sqrt{\nu}\mathcal{K}$. Time has not been rescaled, and

$$\partial_x^\alpha \partial_t^m f_\nu(x, t) = \nu^{(1-|\alpha|)/2} (\partial_y^\alpha \partial_t^m f)(x/\sqrt{\nu}, t).$$

So the force is still smooth and compactly supported in $\mathcal{K}_\nu \times [0, 2]$, and it satisfies every required decay bound for this fixed $\nu > 0$. The energy and dissipation scale as

$$\|u_\nu(t)\|_2^2 = \nu^{5/2} \|u(t)\|_2^2, \quad \nu \int_0^1 \|\nabla u_\nu\|_2^2 dt = \nu^{5/2} \int_0^1 \|\nabla u\|_2^2 dt. \quad (10.23)$$

The growth occurs at $\sqrt{\nu}x_\tau$, still at time one. If a smooth bounded-energy solution v_ν, P_ν existed for the rescaled data, the inverse rescaling would give the viscosity-one solution

$$v(y, t) = \nu^{-1/2} v_\nu(\sqrt{\nu}y, t), \quad P(y, t) = \nu^{-1} P_\nu(\sqrt{\nu}y, t), \quad \|v(t)\|_2^2 = \nu^{-5/2} \|v_\nu(t)\|_2^2.$$

That viscosity-one solution has already been ruled out. This proves the theorem for every fixed positive viscosity. $\square$

10.5 The periodic equations.

The velocity, pressure, and force all have support in one fixed compact set (Proposition 10.1 and Lemma 10.3), so the construction can also be made periodic. First, rescale the supports until they lie inside a fundamental cube. Then add all their integer translates. The translated supports do not overlap, so the momentum equation, including its nonlinear term, stays intact. The pressure becomes periodic too, as the erratum to the problem statement requires [13].

The periodic domain and the interior of its fundamental cube are

$$\mathbb{T}^3 = \mathbb{R}^3 / \mathbb{Z}^3, \quad Q_0 = (-1/2, 1/2)^3.$$

The next corollary proves the periodic breakdown alternative (D) in [13].

Corollary 10.6. *For every $\nu > 0$, there is a smooth force on $\mathbb{T}^3 \times [0, \infty)$ with compact support in time such that the solution (U, P_{per}) of the periodic Navier–Stokes equations, with viscosity ν and zero initial velocity, is smooth on $[0, 1)$ and satisfies*

$$\limsup_{t \uparrow 1} \|U(t)\|_{L^\infty(\mathbb{T}^3)} = \infty.$$

Before time one, its velocity and pressure are supported in a fixed compact subset of the interior of Q_0 . No global smooth periodic solution exists for the same initial data and force.

Proof. Fix the viscosity ν , and call the whole-space fields and force already constructed u, p, f . Their common spatial support lies in $\mathcal{K}_\nu = \sqrt{\nu}\mathcal{K}$, just as in the viscosity rescaling (10.22). Choose $\lambda > 1$ large enough that $\lambda^{-1}\mathcal{K}_\nu \subseteq Q_0$, and set $t_0 = 1 - \lambda^{-2}$. For $t \geq t_0$, define

$$\begin{aligned} \tilde{u}(x, t) &= \lambda u(\lambda x, \lambda^2(t - t_0)), \\ \tilde{p}(x, t) &= \lambda^2 p(\lambda x, \lambda^2(t - t_0)), \\ \tilde{f}(x, t) &= \lambda^3 f(\lambda x, \lambda^2(t - t_0)). \end{aligned}$$

Use the first two formulas only for $t < 1$, and use the force formula for every $t \geq t_0$. Extend all three fields by zero when $t < t_0$. They are smooth across $t = t_0$ because the original fields and force vanish on an interval after their initial time. Every term in the momentum equation is λ^3 times the corresponding original term, so the viscosity is still ν . The original time one is reached exactly when $\lambda^2(t - t_0) = 1$, which is $t = 1$.

Define the periodic fields by

$$\begin{aligned} U(x, t) &= \sum_{k \in \mathbb{Z}^3} \tilde{u}(x + k, t), \quad 0 \leq t < 1, \\ P_{\text{per}}(x, t) &= \sum_{k \in \mathbb{Z}^3} \tilde{p}(x + k, t), \quad 0 \leq t < 1, \\ F_{\text{per}}(x, t) &= \sum_{k \in \mathbb{Z}^3} \tilde{f}(x + k, t), \quad t \geq 0. \end{aligned}$$

Different translates have disjoint supports, with a positive gap between them. These sums are locally finite, so derivatives pass through them, and at most one translate is nonzero at any point. In particular,

$$(U \cdot \nabla)U(x, t) = \sum_{k \in \mathbb{Z}^3} (\tilde{u}(x + k, t) \cdot \nabla) \tilde{u}(x + k, t).$$

The periodic velocity, pressure, and force therefore solve the equation exactly:

$$\partial_t U + (U \cdot \nabla)U - \nu \Delta U + \nabla P_{\text{per}} = F_{\text{per}}, \quad \operatorname{div} U = 0, \quad U(\cdot, 0) = 0.$$

The force is smooth, and its time support lies in $[t_0, 1 + \lambda^{-2}]$. On the torus, this gives every decay estimate $\|\partial_x^\alpha \partial_t^m F_{\text{per}}(t)\|_\infty \leq C_{\alpha,m,N}(1+t)^{-N}$. For the path x_τ from (10.20), the rescaled growth path is

$$\tilde{x}_\tau = \lambda^{-1} \sqrt{\nu} x_\tau, \quad \tilde{t}_\tau = 1 - \lambda^{-2} \tau.$$

This path stays inside Q_0 near $(0, 1)$, where only the zeroth translate is nonzero. Equations (10.21) and (10.22) give

$$U_\theta(\tilde{x}_\tau, \tilde{t}_\tau) = \lambda \sqrt{\nu} \tau^{-A} (e_0 + O(\tau^{2h})) \longrightarrow +\infty.$$

Finally, take any two smooth periodic solutions with the same force and initial data on an interval $[0, T]$ with $T < 1$. Their difference w satisfies

$$\frac{1}{2} \frac{d}{dt} \|w\|_{L^2(\mathbb{T}^3)}^2 + \nu \|\nabla w\|_{L^2(\mathbb{T}^3)}^2 \leq \|\nabla U\|_\infty \|w\|_{L^2(\mathbb{T}^3)}^2.$$

Periodic integration makes both the transport term and the pressure term disappear. Gronwall's inequality shows that the two solutions agree through T . Any global smooth solution would therefore equal U before time one, which is impossible because its velocity blows up at $t = 1$. $\square$

A Matching radial moments and constructing the heat exterior

This appendix constructs the outer part of the leading profile (U, E, Π) from Theorem 4.6. Its pressure integral gives the datum Π_0 for the regular axis problem (Lemma A.5). Its moment conditions, including (A.8), make the radially integrated leading residual stress $\mathcal{T}_0$ vanish in the exterior after the heat replacement (Lemma A.8). The construction has to meet both requirements and still keep the strict stress cone inequalities on the chosen radial intervals (Propositions A.4, A.7 and A.10).

It starts with finite-dimensional moment corrections (Lemmas A.1 and A.2). The same corrections will later attach the axis profile (Proposition B.8) and change the shear (Proposition C.2). Next, the outer profile is built from a given inner reference profile (Section A.2). At the end, its terminal power law E_{pow} is replaced by an exact heat flow. Compensating moment corrections keep both the pressure datum and the exterior stress conditions unchanged (Lemmas A.6 and A.8 and Proposition A.7).

A.1 Adjusting finitely many moments.

Changing a profile on a compact radial interval changes its cumulative moments in (4.15). Add fixed smooth bumps with adjustable coefficients to correct those changes. The next two results show why this works. Distinct power weights produce independent changes in the moments (Lemma A.1), and the resulting quadratic moment equations can be solved when the discrepancies are small (Lemma A.2). Together with Lemma 4.4, these corrections let each radial change preserve the data already prescribed farther out.

Lemma A.1. *Let $\alpha_1, \dots, \alpha_m$ be distinct real numbers. For every j , let β_j be a smooth function that is nonnegative, not identically zero, and supported in the interior of a compact interval $I_j \subset (0, \infty)$, where $I_1 < \dots < I_m$. Then the matrix*

$$B_{ij} = \int_0^\infty x^{\alpha_i} \beta_j(x) dx$$

is invertible. The same statement holds in a coordinate obtained by taking a logarithm, where the weights are exponential. If the data vary smoothly over a compact set of parameters and the stated separations remain in place, then the inverse and every fixed parameter derivative of the inverse stay bounded.

Proof. A nonzero linear combination of m distinct powers can have at most $m - 1$ distinct positive zeros. To see this, divide by the smallest power. If the quotient had m zeros, Rolle's theorem would give $m - 1$ zeros of its derivative. That would contradict the induction hypothesis for the other $m - 1$ powers. It follows that $\det[x_j^{\alpha_i}]$ is nonzero whenever $0 < x_1 < \dots < x_m$, and its sign stays the same throughout that connected set. Since the determinant is multilinear,

$$\det B = \int_{I_1 \times \dots \times I_m} \det[x_j^{\alpha_i}] \prod_j \beta_j(x_j) dx_1 \cdots dx_m \neq 0.$$

The integrand has one sign, and it is nonzero on a set of positive measure. The change of variables $x = e^y$ proves the logarithmic version. Compactness, together with differentiation of $B^{-1}B = \text{Id}$, proves the final statement. That statement does not give a uniform bound for the inverse when two exponents approach each other. $\square$

Some moment weights approach each other as λ tends to zero. The corrections have to stay small enough to preserve the cone, so the proof needs to know how much this costs in the inverse bound. Take identical translated bumps $\beta(y - c_1)$ and $\beta(y - c_2)$, and set $\Delta = c_2 - c_1 > 0$ and $b(s) = \int e^{st} \beta(t) dt$. Dividing row i by $e^{s_i c_1}$ gives

$$B = \begin{pmatrix} b(s_1) & e^{s_1 \Delta} b(s_1) \\ b(s_2) & e^{s_2 \Delta} b(s_2) \end{pmatrix}, \quad \det B = b(s_1)b(s_2)(e^{s_2 \Delta} - e^{s_1 \Delta}). \quad (\text{A.1})$$

Fix the bump and fix $\Delta > 0$. Then $\|B^{-1}\| = O(\lambda^{-1})$ when $s_1 - s_2 = \lambda$ and the slopes stay in a fixed compact interval. The inverse stays bounded if the limiting separation of the slopes is nonzero. The same $O(\lambda^{-1})$ bound holds for moment matrices with weights $1, x^{-\lambda}$ and bumps on ordered disjoint intervals. This follows by expanding $x^{-\lambda} = 1 - \lambda \log x + O(\lambda^2)$, because the two bump averages of $\log x$ stay strictly separated.

The rank calculation handles the linearized moment map. The actual moments are quadratic in the velocity profiles, so the correction coefficients have to solve a finite system with a quadratic remainder. The next estimate gives a small solution and also controls its parameter derivatives. It does not require all orders of derivatives to be small at the same time.

Lemma A.2. *Let $K = [-1, 1]$. Suppose a finite family of correction coefficients $c \in \mathbb{R}^m$ changes the required moments in the following way. First take the specified linear combinations of the moments, and divide each resulting component by its specified nonzero factor. The change is then*

$$F_\eta(c) = B(\eta)c + \mathcal{Q}_\eta(c, c), \quad \eta \in K. \quad (\text{A.2})$$

Here B and the bilinear map $\mathcal{Q}$ are smooth, B is invertible on K , and the given moment discrepancy is $d \in C^\infty(K; \mathbb{R}^m)$. Write $\beta_0 = \sup_K \|B^{-1}\|$, $\kappa_0 = \sup_K \|\mathcal{Q}\|$, and $d_0 = \|d\|_{C^0(K)}$. If

$$8\beta_0^2 \kappa_0 d_0 \leq 1,$$

there is exactly one solution of $F_\eta(c) = d(\eta)$ in the ball $\|c\|_{C^0} \leq 2\beta_0 d_0$. This solution is smooth in η , and iterating $c \mapsto B(\eta)^{-1}(d(\eta) - \mathcal{Q}_\eta(c, c))$ from zero selects it. If $\mathcal{Q} = 0$, no smallness condition is needed.

Now choose any finite k in advance. Let β_k be the operator norm for multiplication by B^{-1} on $C^k(K)$, and let κ_k be the corresponding bilinear norm of $\mathcal{Q}$. If $8\beta_k^2 \kappa_k \|d\|_{C^k} \leq 1$ as well, the same solution branch satisfies

$$\|c\|_{C^k} \leq 2\beta_k \|d\|_{C^k}. \quad (\text{A.3})$$

Proof. Set $r = 2\beta_0 d_0$. The map $c \mapsto B^{-1}(d - \mathcal{Q}(c, c))$ takes the closed C^0 ball of radius r into itself. Its Lipschitz constant on that ball is at most $2\beta_0 \kappa_0 r \leq 1/2$. The same inequality makes $B + D_c \mathcal{Q}(c, c)$ invertible. The pointwise implicit function theorem then gives smoothness, including one-sided derivatives at the endpoints. Apply the same contraction argument to

$C^k(K)$, the space of functions on the compact interval whose derivatives through the stated order are continuous. This space is complete in its usual norm, and pointwise multiplication makes it an algebra. That is the Banach algebra used here, and the argument gives (A.3). The iterations are the same ones as in C^0 , so they have the same limit. Every other fixed derivative is finite by implicit differentiation, with a constant that may depend on the corresponding data. Any extra parameter in a fixed compact set, such as the pulse amplitude below, is handled in exactly the same way. $\square$

The size of these correction estimates depends on the radial scale of the support. For example, if $\delta V = v_* \sum_j c_j(\eta) \beta_j(X/\rho)$, then

$$\|\partial_X^r \partial_\eta^k \delta V\|_\infty \leq C_r v_* \rho^{-r} \|c\|_{C^k}.$$

For bumps whose width in $\log X$ stays fixed, the matching estimate uses $(X\partial_X)^r$. Small coefficients therefore preserve any fixed finite collection of strict inequalities involving field derivatives on that patch. This estimate applies to the moment-correction bumps. The rapid modulation that comes before a correction has separate estimates for its radial derivatives.

For the five cumulative integrals in (4.15), the linearized system on a power-law interval splits into two axial corrections and three azimuthal corrections. The next corollary identifies their weights so that the rank calculation and the small-solution result above can be applied.

Corollary A.3. *Use the five radial moment functions (M, I, J, S, C_p) from (4.15), where C_p is the pressure increment. Under the rescaling $X = \rho x$, their normalizing factors are*

$$(\rho, \rho^{3/2}, \rho^{3/2}, \rho, 1), \quad H = \rho^{1/2} \sqrt{2x} E. \quad (\text{A.4})$$

On a fixed correction interval, suppose $U_0 = u_c(\eta)$ and $E_0 = e_* f_*(\eta) x^\alpha$, with $e_* > 0$ and smooth $f_* > 0$. Two additive U bumps and three additive E bumps give an invertible Jacobian for the normalized five-moment map as long as $\alpha \notin \{-1/2, 1/2, 3/2\}$. The moment changes have the exact quadratic form (A.2). When $U_0 = 0$, the three corrections to E can prescribe changes in (I, S, C_p) while leaving M, J unchanged.

Proof. In the normalized rows, replace J by $J - u_c I$ and S by $S - 2u_c M$, keeping u_c fixed at its base value. The differential of $J - u_c I$ is $\int \sqrt{2x} E_0 \delta U dx$, and the differential of $S - 2u_c M$ is $-\int E_0 \delta E dx$. Together with the other three rows, these give a matrix with two blocks. Apart from nonzero normalizing factors, their weights are

$$\begin{array}{l|l} U \text{ block} & 1, \quad x^{\alpha+1/2} \\ E \text{ block} & x^{1/2}, \quad x^\alpha, \quad x^{\alpha-1}. \end{array}$$

The factors e_*, f_* and $\sqrt{2}$ remain in the normalization of the moment components. Their inverses and every fixed parameter derivative stay bounded. The condition on α says exactly that the powers in each block are distinct, so Lemma A.1 applies. The original functionals have degree at most two in U, E , which makes every remaining term quadratic. In particular, the row operations do not assume that the corrected U stays constant. $\square$

For the axis moment correction $\alpha = 1/10$, the two blocks use the powers $(0, 3/5)$ and $(1/2, 1/10, -9/10)$. On the intermediate power-law interval, $\alpha = -1/2 - \lambda$ gives $(0, -\lambda)$ and $(1/2, -1/2 - \lambda, -3/2 - \lambda)$. The first of these last two blocks can bring in an inverse bound of λ^{-1} . The value of λ is fixed before the later radial frequency is chosen. These normalized inverse bounds, together with the exact restoration of matching data in Lemma 4.4, are what both corrections use.

A.2 Constructing and matching the outer radial profile.

This subsection gives a profile that connects an inner reference power law to a purely azimuthal exterior power law E_{pow} . Its inner data will be prescribed in (A.7). The intervals in between will set the required total moments in (A.8) while keeping the stress-cone inequalities. Four subintervals will be kept free for the later profile, heat, background, and mean corrections; they are listed after (A.9). First come the radial pieces. Their free amplitudes are chosen and their stated properties checked afterward, in Sections A.3 and A.5.

Fix, once and for all, the smooth step

$$\sigma(y) = \frac{e^{-1/y^2}}{e^{-1/y^2} + e^{-1/(1-y)^2}} \quad (0 < y < 1), \quad \sigma = 0 \ (y \leq 0), \quad \sigma = 1 \ (y \geq 1). \quad (\text{A.5})$$

The positive moment-correction bumps are rescaled copies of σ' placed on ordered, disjoint subintervals of the chosen patch. Each stage uses its own logarithmic radial coordinate y . Its origin is the start of that stage, and $d/dy = \mathcal{D}_X = X\partial_X$.

The parameters for the next profile are fixed in this order:

$$M_d, \quad T_d = e^{M_d} + 10, \quad P_* > e^{T_d}, \quad 0 < \lambda \ll 1, \quad 0 < h \ll \min\{\lambda, e^{-T_d}\} \quad (\text{A.6})$$

Here $\ll$ has the smallness meaning from Definition 3.3: choose λ small enough after fixing M_d, T_d, P_* , and after that choose h small enough compared with both λ and e^{-T_d} . The large radius X_R is chosen after all of them.

On the reference inner interval, write $x = X/X_R$ and $y = \log x$, and prescribe

$$U = 4\eta, \quad E = P_* f(\eta) e^{y/10}, \quad f = (1 + \eta^2)^{-1} = e^{-J_0}, \quad J_0 = \log(1 + \eta^2), \quad y \leq 0. \quad (\text{A.7})$$

This ideal region is only a temporary choice for the outer radial profile; Section B replaces its inner part with a regular axis. The goal outside is the power $E_{\text{pow}} = c_\infty X^{-A}$, where $c_\infty > 0$ does not depend on η , together with the moment identities

$$\int_0^\infty (U^2 - E^2/2) dX = 0, \quad \int_0^\infty (H - H_{\text{pow}}) dX = 0, \quad H_{\text{pow}} = \sqrt{2X} E_{\text{pow}}. \quad (\text{A.8})$$

These two identities remove the two integration constants that would otherwise remain in the exterior stress. The profile is now built one stage at a time.

Recall that $l = X\partial_X \log H$. Choosing l is therefore the same as choosing $(\log E)' = l - 1/2$ and integrating it.

Reducing the axial component to zero. Start at $x = 1$. Keep $U = 4\eta$, and for one unit of logarithmic radial length use $l = (3/5)(1 - \sigma(y))$. On the next interval, set

$$l = 0, \quad U = k(y)\eta, \quad k(y) = 4[1 - \sigma(\log(1 + y)/M_d)], \quad 0 \leq y \leq T_d.$$

Here $|k'| \leq e_d/(1 + y)$, where $e_d = 4\|\sigma'\|_\infty/M_d$. The function k is zero for the last eleven units of this stage. Let $P_1 \asymp P_*$ be the value of E/f at the start of the axial transition. Then $E = P_1 f e^{-y/2}$ throughout that interval.

The intermediate power-law interval. Set $U = 0$. Use $l = -\lambda\sigma(y)$ on a unit interval, then keep $l = -\lambda$ for

$$T_w = 60 \log(1/\lambda). \quad (\text{A.9})$$

In the local logarithmic radial coordinate on this interval, reserve $(T_w - 25, T_w - 20)$, $(T_w - 20, T_w - 15)$, $(T_w - 14, T_w - 9)$, and $(T_w - 8, T_w - 3)$. In that order, they will hold the correction after the cone is realized, the heat compensation, the moment conditions for higher-order background corrections, and the moment conditions for the phase-averaged corrections. Choose λ small enough that all four patches lie inside this interval.

The axial pulse. An axial pulse provides the adjustable contribution to the total S -moment. Two smaller corrections at the end of the pulse set $M = J = 0$. Let X_p be the radius at the end of the intermediate power-law interval, and write $E(X_p, \eta) = e_b f$. For $0 \leq y \leq 13/\lambda$, keep $l = -\lambda$ and set $U = ER_b$. With $\zeta_b = \lambda y$, define

$$\begin{aligned} \phi_b(\zeta_b) &= \int_0^{\zeta_b} \sigma(v/.02) dv & (\zeta_b \geq 0), \quad R_0(\zeta_b) &= \phi_b(\zeta_b)[1 - \sigma(\zeta_b - 10)], \\ R_b &= \text{Amp}(\eta)R_0(\zeta_b) + c_1\beta_1(y) + c_2\beta_2(y). \end{aligned}$$

Extend ϕ_b by zero when $\zeta_b < 0$. The bumps β_1, β_2 are identical translates of width .3, centered at $13/\lambda - 3$ and $13/\lambda - 1$. Their coefficients c_1, c_2 set $M = J = 0$ at the end of the pulse. These coefficients are affine functions of Amp . The first equality in (A.8) will determine that amplitude.

Profile interpolation and the angular momentum correction. Next, remove the parameter dependence from the azimuthal profile and adjust its angular momentum without changing the pressure integral. Set $U = 0$ from this point on. Let X_{end} be the radius at the end of the pulse, and write $E(X_{\text{end}}, \eta) = e_{\text{end}} f$. Set

$$\log E = \log e_{\text{end}} - (1/2 + \lambda)y - \vartheta_f(y)J_0 - (1 - \vartheta_f(y)) \log 2, \quad \vartheta_f(y) = 1 - \sigma(y/T_f). \quad (\text{A.10})$$

Choose a fixed, sufficiently large T_f . Then $-\lambda - .1 \leq l \leq -\lambda$. The factor that depends on η decreases to the constant $1/2$, so the amplitude cannot grow. Next, keep the resulting η -independent exponential for $30 \log(1/\lambda)$. Place two relative E bumps of width .3, centered three units and one unit before the end of this interval. They impose

$$I = \frac{XH}{1-\lambda} \quad \text{at the end,} \quad \int_{\text{two bumps}} (E^2 - E_{\text{unedited}}^2) dy = 0. \quad (\text{A.11})$$

The value of E at both endpoints stays unchanged. This part of the construction uses only E and does not depend on Amp .

Transition to the exterior power law. Let X_{rel} be the radius where the exterior transition starts, and let e_{rel} be the unchanged, uniform value of E there. Over a unit interval, vary l smoothly from $-\lambda$ to -1 . Keep $l = -1$ for $4 \log(1/h)$, then vary it from -1 to $-h$ over another unit interval. Each transition uses σ . On the last radial interval $0 \leq y \leq 3$, set

$$\psi_o(y) = 1 - \sigma((y-1)/2), \quad f_o(y) = 1 - \rho_o \psi_o(y), \quad \rho_o = c_o h, \quad l = -h + \frac{f'_o}{f_o}, \quad (\text{A.12})$$

where $c_o > 0$ is fixed small enough that $0 \leq f'_o/f_o < h/4$. Extend f_o as a constant to the left of this interval, and extend it by 1 to the right. Define

$$Q_p = \int_0^3 \exp\left(\int_0^v (1+l(s)) ds\right) \frac{f'_o(v)}{f_o(v)} dv. \quad (\text{A.13})$$

After the transition to $-h$, keep $l = -h$ until the solution of $Q' + (1+l)Q = -l - h$, which starts at $Q = (\lambda - h)/(1 - \lambda)$ when the exterior transition begins, has decreased to Q_p . Run the final transition over three units of logarithmic radial length, then keep $l = -h$ at every larger radius. On and after that interval, $E = E_{\text{pow}} f_o$.

Proposition A.4. *There are finite choices, made in the order (A.6), for which the profile from Section A.2 defines smooth U, E for $X > 0$ and $\eta \in [-1, 1]$, with $E > 0$, and satisfies (A.8). Every stage boundary divided by X_R is independent of both X_R and η . After the compact axial pulse, $U = M = J = 0$. After the final interval, $E = E_{\text{pow}}$. There are four disjoint, unchanged patches, listed after (A.9), where $U = 0$ and E is a positive multiple of $fX^{-1/2-\lambda}$. For every fixed*

$x_- \in (0, 1]$, a sufficiently large X_R , which may depend on x_- , makes the reference profile meet the strict relaxed cone condition from $x = x_-$ through the first half-unit of its final tail. It meets the admissible stress-cone condition throughout the intermediate power-law interval, the axial pulse, the profile interpolation, and the exterior transition. It also meets that condition as soon as $l < 0$ during the transition into the constant-slope interval. These statements cover a finite logarithmic interval ending at tail coordinate $1/2$. The heat construction below handles the rest of the endpoint collar.

A.3 Closing the moment integrals.

Section A.2 fixed the radial pieces but left the amplitudes in the axial pulse and the moment-correction bumps open. This subsection chooses those amplitudes so that the moment identities in Proposition A.4 hold. The same estimates control the size of the corrections and their parameter derivatives, which will be needed to check the stress-cone inequalities in Section A.5.

Write C_{pre} for constants that may depend on the fixed choices M_d, T_d, P_* and on the fixed steps. They are uniform for small λ , followed by sufficiently small h . Taking any prescribed finite number of parameter derivatives may change these constants. None of them depends on the later radius X_R or on the normalization C .

Directly integrating the axial transition and the intermediate power-law interval gives, at the start of the pulse,

$$e_b \leq C_{\text{pre}} \lambda^{30}, \quad \left\| \frac{A_X(U)}{E} \right\|_{C_\eta^1} \leq C_{\text{pre}} \lambda^{29}, \quad \left\| \frac{S(X_p)}{X_p e_b^2 f^2} \right\|_{C_\eta^1} \leq C_{\text{pre}} (1 + \log(1/\lambda)). \quad (\text{A.14})$$

Here is why these bounds hold. The moment M stays constant after the axial transition, while XE grows at rate $1/2 - \lambda$ on the intermediate power-law interval. The denominator XHE for the J radial moment grows at rate $1/2 - 2\lambda$, which gives the same kind of small bound for its moment discrepancy. Also, XE^2 is constant during the axial transition and changes by the factor $e^{-2\lambda y}$ on the intermediate power-law interval. That factor stays bounded above and below for $0 \leq y \leq T_w$. Integrating over the logarithmic length proves the last bound, including the fixed reference profile on the inner interval. The function f and its reciprocal have bounded fixed derivatives on $[-1, 1]$, so differentiating these integrals gives the stated parameter bounds.

For the two pulse corrections, divide by the matching normalizing factors $X_p e_b f$ and $X_p H(X_p) e_b f$ from M, J . In the dy coordinate, the weights are $e^{s_1 y}, e^{s_2 y}$, where $s_1 = 1/2 - \lambda$ and $s_2 = 1/2 - 2\lambda$. Normalize each weight at the first bump center. The main pulse ends at $11/\lambda$, which is at least $2/\lambda - 3$ before that center, and both slopes are larger than .4. The normalized moment discrepancy from the main pulse is therefore at most $C(1 + |\text{Amp}|)e^{-c/\lambda}$. The contribution from before the pulse picks up the factor $e^{-s_i(13/\lambda-3)}$ in addition to the bounds just proved. Formula (A.1) bounds the inverse matrix by $O(\lambda^{-1})$. The exponential absorbs that loss, giving

$$\|c_i\|_{C_\eta^1} \leq C_{\text{pre}} e^{-c/\lambda} (1 + |\text{Amp}|), \quad M = J = 0 \quad \text{after the pulse.} \quad (\text{A.15})$$

Both here and in the amplitude calculation, this estimate says that the affine coefficients of 1 and Amp have the stated parameter bounds. Every fixed y derivative has the same bound. Every fixed ζ_b derivative brings in only a power of λ^{-1} , so it is still exponentially small.

The axial corrections now fix M and J for each trial pulse amplitude. Next, set the angular moment needed at the start of the exterior transition. Put $r_I = I/(XH)$. It satisfies

$$r_I' + (1 + l)r_I = 1.$$

Before the exterior transition, $1 + l$ stays bounded below, and the explicit shapes give a C_η^1 bound C_{pre} at the end of the profile interpolation. On the interval with uniform profile, $r_I -$

$(1 - \lambda)^{-1}$ decays at rate $1 - \lambda$. The moment discrepancy at the first correction center is therefore $O_{C_\eta^1}(C_{\text{pre}}\lambda^{28})$. Make a relative change $E \mapsto E(1 + c_1\beta_1 + c_2\beta_2)$. The I row is linear with slope $1 - \lambda$, and the linearized pressure row has slope $-1 - 2\lambda$ in y . Divide these rows by XH and E^2 , respectively, at the first center. Equation (A.1) gives an inverse that stays bounded. The quadratic pressure increment is the only nonlinear term. Lemma A.2 gives coefficients $O_{C_\eta^1}(C_{\text{pre}}\lambda^{28})$ and the exact conditions in (A.11). The y derivatives of these corrections are much smaller than λ , so $E > 0$ and $l \leq -\lambda/2$ both remain true.

At the start of the exterior transition, $U = M = J = 0$ and $I = XH/(1 - \lambda)$ does not depend on η . The integral formula (4.16) for Q_s gives $Q_s = (\lambda - h)/(1 - \lambda)$ exactly. The source $-l - h$ is nonnegative from there until the final interval. During the first transition, that solution keeps the lower bound $c\lambda$. On the constant-slope interval with $l = -1$, $Q'_s = 1 - h$, so the value at the end is comparable to $\log(1/h)$. The next transition changes that size by only bounded factors. By contrast, $Q_p \asymp \rho_o \asymp h$, as (A.13) shows on a fixed interval. The constant-slope interval in between, where $l = -h$, therefore has a positive, finite logarithmic radial length. On the final interval, the exact solution is

$$Q_s(y) = \int_y^3 \exp\left(\int_y^v (1 + l(s)) ds\right) \frac{f'_o(v)}{f_o(v)} dv. \quad (\text{A.16})$$

It reaches zero at the endpoint and stays zero afterward. Throughout the exterior transition, I does not depend on η , and after that $Q_s = -1 + (1 - h)I/(XH)$. It follows that

$$I = \frac{XH}{1 - h} \quad \text{beyond the tail}, \quad \int_0^\infty (H - H_{\text{pow}}) dX = 0.$$

The subtracted moment is integrable at zero because $H_{\text{pow}} \propto X^{-h}$ and $h < 1$. Far enough out, its integrand is identically zero. The angular correction, its endpoint, and all the lengths in the exterior transition depend only on E , so they do not depend on the axial amplitude.

The angular moment has now been fixed independently of the pulse amplitude. The one scalar condition left is $S(\infty) = 0$. Before solving it, bound the part that comes from the transition to the exterior profile. Since $(XE^2)' = 2lXE^2$, the interval with constant slope $l = -1$ gives the exact factors

$$\frac{(XE^2)_{\text{hold end}}}{(XE^2)_{\text{hold start}}} = h^8, \quad \frac{E_{\text{hold end}}}{E_{\text{hold start}}} = h^6. \quad (\text{A.17})$$

Here “hold start” and “hold end” are the two endpoints of the constant-slope interval where $l = -1$. The label “release” in the next integral means the whole transition to the exterior power law. The first transition and the constant-slope interval contribute at most $CX_{\text{rel}}e_{\text{rel}}^2$ to $\int XE^2 dy$. From the end of the constant-slope interval onward, $l \leq -3h/4$, so the rest of the integral is at most $CX_{\text{rel}}e_{\text{rel}}^2 h^8/h$. Therefore

$$\int_{\text{release}} XE^2 dy \leq CX_{\text{rel}}e_{\text{rel}}^2 \quad (\text{A.18})$$

with a constant that is uniform in h . The h^8 estimate controls this integral. The separate h^6 estimate will later control the ratio $w = N_s/(EQ_s)$ near the outer endpoint. The profile interpolation and the constant-slope interval after it contribute at most $C(1 + \log(1/\lambda))X_{\text{end}}e_{\text{end}}^2$. The first η derivatives satisfy the same bounds because the exterior transition is uniform and the earlier logarithmic shapes have bounded derivatives.

On the pulse, $XE^2 = X_p e_b^2 f^2 e^{-2\lambda y}$. Change variables to $\zeta_b = \lambda y$. Equations (A.14), (A.15), and (A.18) then give the exact formula

$$\frac{\lambda S(\infty)}{X_p e_b^2 f^2} = \text{Amp}^2 K_b - \frac{1 - e^{-26}}{4} + \mathcal{E}(\text{Amp}, \eta), \quad K_b = \int_0^{13} e^{-2\zeta_b} R_0(\zeta_b)^2 d\zeta_b, \quad (\text{A.19})$$

where, on $[.9, 1.2] \times [-1, 1]$,

$$\|\mathcal{E}\|_{C_\eta^1} \leq C_{\text{pre}} \lambda (1 + \log(1/\lambda)).$$

Every term in this estimate that comes from the outer profile E was chosen without using Amp . Its only remaining dependence on the amplitude comes from the affine end bumps, and that dependence is exponentially small. Since $R_0 \leq \zeta_b$, $K_b \leq \int_0^\infty \zeta_b^2 e^{-2\zeta_b} d\zeta_b = 1/4$. On $.02 \leq \zeta_b \leq 9$, the bound goes the other way, $R_0 \geq \zeta_b - .02$, and integration gives $K_b > .20$. The main expression in (A.19) is less than $-.047$ at amplitude $.9$, and greater than $.038$ at amplitude 1.2 . Its derivative with respect to the amplitude is at least $.36$ throughout that bracket. It follows that, when λ is small, the full expression has exactly one root in the bracket. The linearization at that root is nonsingular, so it can be inverted locally. This makes the root smooth, and

$$|\partial_\eta \text{Amp}| \leq C_{\text{pre}} \lambda (1 + \log(1/\lambda)). \quad (\text{A.20})$$

Both equalities in (A.8) now hold. What remains is to check that these moment corrections keep the stress inside its cone. The bounds below prescribe smallness for only finitely many orders of η -derivatives. If smallness bounds are also set for any other finite list of derivative orders, differentiating the integrals and applying Lemma A.2 gives those bounds before the parameters are fixed. Once the parameters have been chosen, every other fixed derivative order still has a finite bound.

A.4 Fixing the pressure datum for the axis profile.

The azimuthal profile E now fixes the pressure Π_0 on the axis by requiring the pressure to vanish at radial infinity, as in (4.25). Proposition B.2 constructs the axis profile analytically. For that construction to work, this pressure datum must extend holomorphically to a neighborhood of $[-1, 1]$ and must obey the sign and size bounds in Lemma A.5. This subsection checks those facts before the axis is attached. It also shows that the datum does not depend on the later choice of radial scale X_R . The pressure integrals here use the global logarithmic coordinate $y = \log(X/X_R)$ on the full reference profile.

Lemma A.5. *Let $E_{\text{id,sched}}$ consist of the reference inner profile (A.7), followed by the outer profile E from Section A.2. Leave out the azimuthal corrections that preserve the total pressure increment, and leave every other transition interval unchanged. Then the function*

$$\Pi_0(\eta) = -\frac{1}{2} \int_{-\infty}^{\infty} E_{\text{id,sched}}(y, \eta)^2 dy \quad (\text{A.21})$$

does not depend on X_R and is analytic on a complex neighborhood of $[-1, 1]$. It is even. The derivative $\Pi_0'(\eta)$ has the same sign as η whenever $\eta \neq 0$. It also satisfies

$$\Pi_0(\eta) \leq -\frac{5}{2} P_*^2 f(\eta)^2. \quad (\text{A.22})$$

For the full reference radial profile, start with this datum and integrate $\Pi_X = E^2/(2X)$. The result is

$$\Pi(X, \eta) = -\frac{1}{2} \int_{\log(X/X_R)}^{\infty} E(v, \eta)^2 dv. \quad (\text{A.23})$$

Proof. Without the angular bumps, every part of E can be written as $c(y)f(\eta)^{\vartheta(y)}$. Here c and the transition lengths do not depend on η , and $0 \leq \vartheta \leq 1$. The reference inner profile, the axial transition, the intermediate power-law interval, and the pulse all have $\vartheta = 1$. The profile interpolation lowers this exponent to zero, and the exterior transition and tail have $\vartheta = 0$. The next step is to choose one simply connected neighborhood of the real interval in the complex plane, one connected region in which every closed loop can shrink to a point without leaving the region. Once the poles and zeros are excluded, this property lets one logarithm be chosen consistently throughout the region.

Choose it so that it avoids every pole and zero of f , and fix an analytic logarithm there, meaning a holomorphic function whose exponential is the original function. Then $f^\theta = \exp(\theta \log f)$ is holomorphic, with a bound that is uniform for $0 \leq \theta \leq 1$. The squared integrand is bounded by $Ce^{y/5}$ as $y \rightarrow -\infty$, and by $Ce^{-(1+2h)y}$ as $y \rightarrow +\infty$. The interval between those two tails is finite. Integrating the holomorphic functions with these uniform bounds proves that the pressure datum is analytic.

For real η , differentiating each factor gives $\partial_\eta f^{2\theta} = -2\theta J'_0 f^{2\theta}$. It follows that every term in Π'_0 has the same sign as η . The reference inner profile makes the sign strict. That profile contributes exactly $-(5/2)P_*^2 f^2$, which proves (A.22). Nothing in the construction in the logarithmic radial coordinate contains X_R , so the pressure datum does not depend on that scale. Finally, (A.11) says that putting the angular bumps back changes the total pressure increment by zero. So the full increment is $-\Pi_0$, which proves (A.23). $\square$

The axis construction can now use this pressure datum. Every later change to the profile must keep the total pressure increment unchanged so that the datum stays fixed. Matching all five radial integrals also keeps the radial velocity and the functions Q_s, N_s unchanged past the correction interval. Suppose an edit has zero total integral of $(E_{\text{new}}^2 - E_{\text{old}}^2)/(2X)$. Then the pressure found by integrating forward is unchanged before and after the edit's support, although it may change inside that support. A correction to the strict right of X can be left out of the backward integral (A.23) when it preserves the total pressure increment. Lemma 4.4 gives a stronger result when all five radial moment functions agree at a radius where the two fields meet again. From that radius onward, the radial velocity and the functions Q_s, N_s are restored on the whole later interval, along with their parameter derivatives. The axis construction uses the analytic function Π_0 , which is already fixed, and integrates pressure forward. Later edits of E may be smooth without being analytic, but they cannot change that datum. Their total pressure increment is either zero or is restored by the correction that matches all five radial moments. The heat replacement changes the exterior profile while preserving the required convergent differences of moments.

A.5 Uniform admissible stress-cone inequalities on the outer interval.

The moment identities and the pressure datum on the axis are now fixed. It remains to check the stress cone inequalities on the outer intervals named in Proposition A.4. Before choosing the large radial scale X_R , consider the ratios in the sufficient cone test from Lemma 4.5. The final choice of that scale will then make the inequalities strict, with the same margin throughout the required finite interval.

Write $w = N_s/(EQ_s)$ wherever $Q_s > 0$. The sufficient test from Lemma 4.5 is

$$a - b_s w > 0, \quad 2b_s w + b_s^2/a + (a - 2)w^2 < 2. \quad (\text{A.24})$$

The proof will give uniform positive lower bounds for $a - b_s w$ and $2 - 2b_s w - b_s^2/a - (a - 2)w^2$. Throughout this check, a, b_s, w stay in fixed compact ranges and $Q_s > 0$. Afterward, increasing X_R makes $p_{s,1} = XQ_s/L$ large enough throughout the chosen finite logarithmic interval. At points where $v_s \leq 2$, the remaining requirement is $P_c = p_{s,1}(1 - b_s w/a) > 2$.

The reference inner profile and axial transition. On (A.7), putting the profile directly into the equation for Q_s gives

$$Q_s = \frac{(3/5)(4L - 1) - h(1 - 8\eta^2) + (D + 4d)\eta J'_0}{8/5} \geq c > 0. \quad (\text{A.25})$$

There, $b_s = 0$ and $a = 4/5$. Split the source in the equation for N_s into its geometric part and its pressure part. At the beginning of the first transition, the geometric part is $O(|\eta|)$, and the pressure part is $O(P_*^2|\eta|)$. One parameter derivative removes the factor $|\eta|$. These bounds come from the constant ideal geometric source and the bounded pressure source.

Both sources are integrated against the kernel in the integral representation of N_s . The pressure and ideal shapes are even, so both sources vanish at $\eta = 0$.

During the first transition, $U = 4\eta$, $W = 1 - 4L < 0$, and $l \geq 0$, so the angular source is at least $-h$. Since the transition has finite length, Q_s keeps a positive lower bound. On this interval, $b_s = 0$ and $a \leq 2$. In the axial transition, the angular source S_q is

$$S_q = -h(1 - 2k\eta^2) + (D + dk)\eta J'_0, \quad Q_s \geq c(\eta^2 + e^{-y}) \quad (0 \leq y \leq T_d). \quad (\text{A.26})$$

Solving the linear equation by variation of constants—multiplying by its integrating factor and integrating—gives the second bound. The facts used here are $\eta J'_0 \asymp \eta^2$, $D + dk \geq D$, positivity of Q_s at the inner endpoint, and $h \ll e^{-T_d}$.

On this interval the geometric axial source is $O(|\eta|)$. Equation (A.23) makes its pressure source $O(E^2|\eta|)$. The reason is that the unmodified profile at larger radii decays at least as fast as a constant times $e^{-\Delta y/2}$ and satisfies $|\partial_\eta \log E| \leq |J'_0|$. The later correction to the azimuthal profile can be left out because it preserves the total pressure increment. Solve $N'_s + N_s = S_n$ and use $E^2 = P_1^2 f^2 e^{-y}$. This gives

$$\frac{|N_s|}{E^2} \leq C|\eta| \left(\frac{e^{T_d}}{P_*^2} + 1 + y \right) \leq C|\eta|(1 + y).$$

Now $b_s = 2k'\eta/E$, so (A.26) gives $|b_s w| \leq C e_d$. Also, $b_s^2 \ll 1$ because $P_* > e^{T_d}$. Since $a = 2$, first choose M_d so that e_d is small, then choose P_* . Both strict inequalities in (A.24) follow.

The intermediate power-law interval. During the axial transition, write $\mathcal{A}_X(U) = \bar{k}\eta$. Then $\bar{k}' = k - \bar{k}$. On the final eleven units, where $k = 0$, this gives $L\bar{k} < 1/2$. It follows that $W = 1 - L\bar{k} \geq 1/2$ on the next transition. Put $c_\eta = D\eta J'_0 \asymp \eta^2$. Since $U = 0$, the angular source on that transition is $(-l)W - h + c_\eta$. The function Q_s keeps a positive lower bound that does not depend on λ . The bound for w depends only on parameters chosen before λ . So $\sqrt{\lambda}|w|$ is small, while $b_s = 0$.

On the constant-slope interval, use its local logarithmic radial coordinate. There $\bar{k}(y) = \bar{k}(0)e^{-y}$, so the angular source is exactly $\lambda - h + c_\eta - \lambda L\bar{k}(0)e^{-y}$. Integrating the scalar equation for Q_s gives

$$\begin{aligned} Q_s &\geq c_{\text{pre}}(\eta^2 + \lambda + e^{-(1-\lambda)y}), \\ Q_s - \frac{\lambda - h + c_\eta}{1 - \lambda} &= O_{C_h^1} \left(C_{\text{pre}}(1 + y)e^{-(1-\lambda)y} \right), \\ |N_s/E| &\leq C_{\text{pre}}|\eta|(1 + y)e^{-(1/2-\lambda)y}. \end{aligned} \quad (\text{A.27})$$

For the last line, $U = 0$ makes the geometric source vanish, while the pressure source remains $O(E^2|\eta|)$. Convolving it with e^{-y} gives the stated bound. The same estimate holds after one η derivative, but without the factor $|\eta|$. Using $|\eta|/(\eta^2 + e^{-(1-\lambda)y}) \leq \frac{1}{2}e^{(1-\lambda)y/2}$ now gives

$$\sqrt{\lambda}|w| \leq C_{\text{pre}}\sqrt{\lambda}(1 + y)e^{\lambda y/2} = o(1) \quad (0 \leq y \leq 60 \log(1/\lambda)).$$

Since $a = 2 + 2\lambda$ and $b_s = 0$, the stress lies in the admissible cone on this interval. The same argument works on the preceding transition wherever $l < 0$; at its first endpoint it gives the relaxed condition.

The pulse. Both shear components change here, so the cone test needs a more exact relation between the averaged axial velocity and the pulse profile. Set $m = \mathcal{A}_X(U)/E$ and $\beta = 1/2 - \lambda$. The exact equation for m is $m' + \beta m = R_b$. Extend the main pulse by zero to the left and expand its exponential convolution twice. The result is

$$m = \frac{R_b}{\beta} - \frac{\lambda \partial_{\zeta_b} R_b}{\beta^2} + O(\lambda^2) + m(0)e^{-\beta y}. \quad (\text{A.28})$$

Taylor's formula gives this expansion because it bounds the convolution remainder by $C\lambda^2 \|\partial_{\zeta_b}^2 R_b\|_\infty \int_0^\infty e^{-\beta v} v^2 dv$. The pulse starts flat, and the end bumps have fixed ζ_b derivatives that are exponentially small. The same expansion remains true after one η derivative. Equation (A.14) bounds the initial value.

In the angular equation, $W = 1 - 2D\eta Em - d(Em)_\eta$. Together with (A.20), this gives $S_q = \lambda - h + c_\eta + O(C_{\text{pre}}E)$. Equations (A.14) and (A.27) then give

$$Q_s = \frac{\lambda - h + c_\eta}{1 - \lambda} + O(C_{\text{pre}}\lambda^{28}). \quad (\text{A.29})$$

The pressure is normalized to vanish at radial infinity. It and its first η derivative are both $O(E^2)$. Also, integrate $XE^2(R_b^2 - 1/2)$ together with the difference of moments already built up before the pulse. This shows that $S/(XE^2) = O_{C_\eta^1}(C_{\text{pre}}e^{26}/\lambda)$ throughout the pulse. Formula (4.16) writes Q_s, N_s in terms of the radial moment functions and their η -derivatives; it does not use radial derivatives of E, U . Substituting the estimates into that formula gives

$$\frac{N_s}{E} = -R_b + (D + c_\eta)m - D\eta m_\eta + O(C_{\text{pre}}\lambda^{27}). \quad (\text{A.30})$$

Every constant here is independent of C . Since $|\eta|/(\lambda + c_\eta) \leq C/\sqrt{\lambda}$, equation (A.20) makes the term from $\eta m_\eta/Q_s$ an $o(1)$ contribution. Combining (A.28)–(A.30) gives

$$w = 2R_b - C_d \partial_{\zeta_b} R_b + o(1), \quad C_d = \frac{\lambda(D + c_\eta)(1 - \lambda)}{\beta^2(\lambda - h + c_\eta)} \leq 2 + o(1).$$

This estimate for w turns the cone test into bounds on the pulse shape and its derivative. For the coefficient of R_b , use $D + c_\eta - \beta = \lambda - h + c_\eta$. To bound C_d , maximize the quotient at $c_\eta = 0$ and use $h/\lambda \rightarrow 0$. Also, $a = 2 + 2\lambda$ and $b_s = -R_b + O(\lambda)$. Apart from the exponentially small end bumps, $R_b \geq 0$ and $\partial_{\zeta_b} R_b \leq 1.2$, so

$$\begin{aligned} b_s w &\leq \sup_{R \geq 0} (-2R^2 + 2.41R) + o(1) < .74, \\ 2b_s w + b_s^2/a + (a - 2)w^2 &\leq \sup_{R \geq 0} (-3.5R^2 + 4.82R) + o(1) < 1.68. \end{aligned}$$

These quadratic cross terms use only the upper bound on the pulse derivative. Near the cutoff the derivative is negative, which makes the left sides smaller. Every derivative remains bounded by a fixed constant. The first margin gives $a - b_s w > 1$, while $v_s \geq a > 2$. This proves that the stress stays in the admissible cone throughout the pulse.

Profile interpolation and exterior transition. After the pulse, $U = M = J = 0$, so $W = 1$. During the profile interpolation, the angular source is $-l - h + \vartheta_f c_\eta$. It follows that $Q_s \geq c\lambda$, and the small C_η^1 angular correction keeps that bound. Since $S(\infty) = 0$,

$$\frac{S(X)}{X} = \frac{1}{2X} \int_X^\infty E(X')^2 dX'. \quad (\text{A.31})$$

Until the exterior transition begins, $l \leq -\lambda/2$; the remaining integral over the exterior transition is bounded by (A.18). This gives $(|S| + |S_\eta|)/X \leq CE^2/\lambda$, including the angular edit, while $|\Pi| + |\Pi_\eta| \leq CE^2$. Formula (4.16) now gives $|w| \leq CE/\lambda^2 \ll 1$, because the end of the pulse made E exponentially small in $1/\lambda$. Here $b_s = 0$ and $2 < a \leq 2 + 2\lambda + .2 + o(1)$, so both cone margins remain positive.

Once the exterior transition has begun, its fields and the remaining radial moment functions are uniform in η . The only exceptions are the explicit coefficients in the formula for Q_s, N_s . Equations (A.23) and (A.31) together give the following exact expression.

$$N_s = 2\eta \int_y^\infty (he^{v-y} - A)E(v)^2 dv = O(E(y)^2).$$

Throughout the exterior transition and at every larger radius, $l \leq -3h/4$. Since $(\log E)' = l - 1/2$, for $v \geq y$ this gives

$$E(v)^2 = E(y)^2 \exp\left(\int_y^v (2l(s) - 1) ds\right) \leq E(y)^2 e^{-(1+3h/2)(v-y)}.$$

It follows that

$$\begin{aligned} h \int_y^\infty e^{v-y} E(v)^2 dv &\leq hE(y)^2 \int_0^\infty e^{-3hs/2} ds = \frac{2}{3} E(y)^2, \\ \int_y^\infty E(v)^2 dv &\leq E(y)^2 \int_0^\infty e^{-(1+3h/2)s} ds = \frac{E(y)^2}{1+3h/2}. \end{aligned}$$

Together with $A = 1/2 + h$ and $|\eta| \leq 1$, these estimates prove the stated bound for N_s , uniformly in h . Before the steep-decay interval, $Q_s \geq c\lambda$. After that interval, and through the first half-unit of the terminal interval, $Q_s \geq ch$. On the last constant-slope interval it falls only to $Q_p \asymp h$, and (A.16) gives the same lower bound for $0 \leq y \leq 1/2$. Equation (A.17) gives $E \leq Ce_{\text{rel}}h^6$ after the steep-decay interval. So $E/Q_s \ll 1$, even when h is much smaller than e_{rel} . It follows that $|w| \ll 1$, $b_s = 0$, and $2 < a \leq 4$ on the part of the exterior transition being considered. This proves the strict true cone there.

Every lower bound for Q_s used above holds on a fixed compact logarithmic interval after all the choices in (A.6). On such an interval, the fields and the functions Q_s, N_s do not depend on X_R under the normalizations (A.4). One final increase of X_R makes the test that needs large $p_{s,1}$ hold uniformly. This completes the proof of Proposition A.4.

A.6 Replacing the exterior power law with a heat flow.

The reference profile (U, E) from Proposition A.4 has the required moments. Past the terminal interval it also has $U = 0$ and $E = E_{\text{pow}} = c_\infty X^{-A}$. This exterior power law E_{pow} still leaves a viscous residual. The next step is to replace it outside the annulus with the exact radial swirl heat flow K from Lemma A.6. After the replacement, the exterior momentum residual is zero, and away from the axis the field has smooth limits at the singular time. The replacement changes three moments. Proposition A.7 restores them with a correction in the second reserved interval $\mathcal{I}_2$; this also restores the pressure datum Π_0 already prescribed for the axis in (A.21).

For a profile that is regular at the axis, let $\mathcal{R}_\theta^{(0)}, \mathcal{R}_z^{(0)}$ be its leading azimuthal and axial momentum residuals. In the notation of (4.12), this means $\mathcal{R}_j^{(0)} = R_j^{(0)}$. These residuals include radial viscosity but leave out the higher-order axial viscosity. As in Proposition 4.2, the stress components are $T_\theta(r) = -r^{-2} \int_0^r s^2 \mathcal{R}_\theta^{(0)}(s) ds$ and $T_z(r) = -r^{-1} \int_0^r s \mathcal{R}_z^{(0)}(s) ds$. The next subsection uses the corrected moment conditions to turn these into integrals from r to infinity, in Lemma A.8. It then finds the sign and limiting direction of the stress as the outer endpoint is approached, in Proposition A.10.

Throughout this subsection, $0 < h < 1/2$, $A = 1/2 + h$, and $D = 1/2 - h$. Keep $d = 1 - \eta^2$, $\tau = 1 - t$, and $L = 1 - 2h\eta^2$ from (4.1). Every parameter in the outer radial profile is fixed in the order shown in (A.6), before X_R is increased. On the tail from (A.12), use the local logarithmic radial coordinate $y = \log(X/X_{\text{tail}})$. Write $X_b = e^3 X_{\text{tail}}$ and $E_{\text{pow}} = c_\infty X^{-A}$, where $c_\infty > 0$ does not depend on η . The reference tail is then $E_{\text{pow}} f_o(y)$.

Set $a_K = 1 + h$. Define the heat factor by

$$\mathcal{H}(Z) = \frac{1}{\Gamma(a_K)} \int_0^\infty e^{-v} v^{a_K-1} (1 + Zv)^{-h} dv, \quad Z \geq 0, \quad (\text{A.32})$$

Here $\Gamma(a_K) = \int_0^\infty e^{-v} v^{a_K-1} dv$ is the gamma integral, so $\mathcal{H}(0) = 1$. The heat profile is $E_{\text{heat}}(X, \eta) = E_{\text{pow}}(X) \mathcal{H}(2d/X)$.

Lemma A.6. *The function $\mathcal{H}$ is positive and is smooth from the right all the way to $Z = 0$. Every fixed derivative stays bounded on each bounded nonnegative interval. The field*

$$K(r, t) = c_\infty s^{-A} \mathcal{H}(2\tau/s), \quad s = r^2/2, \quad (\text{A.33})$$

does not depend on z . It satisfies $\partial_t K = (\partial_{rr} + r^{-1}\partial_r - r^{-2})K$, and $K_r < 0$. Its profile is $E_{\text{pow}}(X) \mathcal{H}(2d/X)$. This profile is smooth, and all of its η -derivatives extend continuously from the interior to $\eta = \pm 1$.

Proof. For the rising factorial $(b)_m = b(b+1) \cdots (b+m-1)$, with $(b)_0 = 1$, differentiation under the integral, justified by a dominating integrable function, gives

$$\mathcal{H}^{(m)}(Z) = \frac{(-1)^m (h)_m}{\Gamma(1+h)} \int_0^\infty e^{-v} v^{h+m} (1 + Zv)^{-h-m} dv, \quad (\text{A.34})$$

$$\mathcal{H}^{(m)}(0) = (-1)^m (h)_m (1+h)_m. \quad (\text{A.35})$$

Deleting the last factor leaves an integrable function that dominates the integrand, including at $Z = 0$. So the usual finite Taylor formula holds there for every fixed derivative order; a convergent Taylor series is not needed. On any fixed small nonnegative interval, it gives

$$\mathcal{H}(Z) = 1 - h(1+h)Z + O_h(Z^2), \quad |\mathcal{H}(Z) - 1| + |Z\mathcal{H}'(Z)| \leq C_h Z. \quad (\text{A.36})$$

The last constant can be chosen uniformly for $0 < h \leq h_0 < 1/2$. The reason is that $(h)_m (1+h)_m \leq C_m h$ for every fixed $m \geq 1$. Now integrate $h\partial_v [e^{-v} v^{a_K} (1 + Zv)^{-a_K}]$. This gives

$$Z^2 \mathcal{H}'' + (1 + 2a_K Z) \mathcal{H}' + a_K(a_K - 1) \mathcal{H} = 0; \quad (\text{A.37})$$

both boundary terms are zero. Differentiate (A.33) directly, using $Z = 2\tau/s$. The two sides of the heat equation become

$$\begin{aligned} \partial_t K &= -2c_\infty s^{-A-1} \mathcal{H}', \\ (\partial_{rr} + r^{-1}\partial_r - r^{-2})K &= 2c_\infty s^{-A-1} [Z^2 \mathcal{H}'' + (2A+1)Z\mathcal{H}' + (A^2 - 1/4)\mathcal{H}]. \end{aligned}$$

Since $2A+1 = 2a_K$ and $A^2 - 1/4 = a_K(a_K - 1)$, equation (A.37) proves the heat equation, including its swirl term. Next, $-Z\mathcal{H}'/\mathcal{H}$ is the average of $hZv/(1+Zv)$ with respect to the positive probability density proportional to the integrand in (A.32). So

$$0 \leq -\frac{Z\mathcal{H}'}{\mathcal{H}} < h, \quad \frac{rK_r}{2K} = -A - \frac{Z\mathcal{H}'}{\mathcal{H}} < -\frac{1}{2}.$$

The composition with $Z = 2(1 - \eta^2)/X$ only uses $Z \geq 0$. Every nonzero η -derivative is a finite sum of terms $\mathcal{H}^{(k)}(2d/X)(2/X)^k$, multiplied by polynomials in $d' = -2\eta$ and $d'' = -2$. No term divides by d . This proves smoothness on the closed interval. The integral formula is used only on its stated nonnegative domain. In particular, the large- X expansion, including its remainder after any fixed application of $((X\partial_X)^j \partial_\eta^m)$, is

$$\frac{E_{\text{heat}}(X, \eta)}{E_{\text{pow}}(X)} = \mathcal{H}(2d/X) = 1 - \frac{2h(1+h)d}{X} + O_{h,j,m}(X^{-2}). \quad (\text{A.38})$$

□

At large X , the heat profile approaches the original power law, with the derivative bounds in (A.38). Let E_{cl} stand for the reference profile built in Proposition A.4. The terminal profile will be replaced by this small change, and earlier corrections will restore the three moments it changes. The other two moments do not change, because $U = 0$ on both regions that are edited.

Proposition A.7. *Let $X_K = e^{-2}X_{tail}$. Choose a smooth step $0 \leq \chi_K(y) \leq 1$ that is zero for $y \leq .2$ and one for $y \geq .5$. Make the terminal replacement*

$$E_{cl} \mapsto E_{cl} [1 + \chi_K(y)(\mathcal{H}(2d/X) - 1)] \quad (\text{A.39})$$

Three additive E -bumps in the second reserved patch of the intermediate power-law interval can compensate for this replacement. The full edit preserves the functions M, J , the totals $S(\infty), C_p(\infty)$ from (4.15), and the renormalized angular moment $\int_0^\infty (H - H_{pow}) dX$, at every point of $[-1, 1]$. When X_R is large enough, the edit also preserves $E > 0$ and the strict admissible stress cone from that patch through $y = .5$. It leaves both the original pressure datum on the axis and the profiles before the compensation patch unchanged.

Proof. First estimate the three changes made by the heat factor. Then cancel them with three earlier angular bumps. The normalized radial moment formulas will show that these corrections keep the stress inside the admissible cone.

Put $x = X/X_K$, and let e_K be the reference amplitude at X_K . The reference tail is comparable to $e_K x^{-A}$. Equations (A.34)–(A.36), together with the chain rule, show that for every pair of fixed integers $j, m \geq 0$,

$$|(X\partial_X)^j \partial_\eta^m (E - E_{cl})| \leq C_{j,m} e_K X_K^{-1} x^{-A-1}, \quad x \geq 1.$$

The constants may depend on the schedule, which is already fixed. In particular, every derivative in this estimate is bounded at $d = 0$. Let Δ mean the change caused only by (A.39). The three differences of moments then obey

$$\left| \partial_\eta^m \Delta \int_0^\infty \frac{E^2}{2X} dX \right| \leq C_m \frac{e_K^2}{X_K} \int_1^\infty x^{-2A-2} dx, \quad (\text{A.40})$$

$$\left| \partial_\eta^m \Delta S(\infty) \right| \leq C_m e_K^2 \int_1^\infty x^{-2A-1} dx, \quad (\text{A.41})$$

$$\left| \partial_\eta^m \Delta \int_0^\infty (H - H_{pow}) dX \right| \leq C_m e_K X_K^{1/2} \int_1^\infty x^{-A-1/2} dx \leq C_{m,h} e_K X_K^{1/2}. \quad (\text{A.42})$$

Here $H = \sqrt{2X}E$ and $H_{pow} = \sqrt{2X}E_{pow}$; the pure power itself is not changed. The last integral equals $1/h$. It is finite because $h > 0$ was fixed before the large radial scale was chosen.

Once each one is divided by its natural radial scale, the three differences of moments are small. Cancel them on the second reserved patch, where the power-law profile supplies three independent weights for the moments. Let X_* be the start of this patch, and put $x_* = X/X_*$. Before the correction, the profile there is $E = e_* f(\eta) x_*^{-1/2-\lambda}$, with $U = 0$ and $f = (1 + \eta^2)^{-1} \geq 1/2$. Choose three nonnegative bumps β_i whose supports are disjoint, ordered, and contained in this patch. Add

$$\delta E = e_* \sum_{i=1}^3 c_i(\eta) \beta_i(x_*).$$

Normalize the pressure row, the S row, and the angular row, in that order, by

$$e_*^2, \quad X_* e_*^2, \quad X_*^{3/2} e_*. \quad (\text{A.43})$$

At $c = 0$, their derivatives are the integrals against the weights

$$f x_*^{-3/2-\lambda}, \quad -f x_*^{-1/2-\lambda}, \quad \sqrt{2} x_*^{1/2}.$$

The three exponents are different, so Lemma A.1 says that this matrix is invertible. Its inverse is uniform for $\eta \in [-1, 1]$. The first two rows have quadratic remainders, and the last row is linear. The ratios X_*/X_K and e_*/e_K are fixed. After dividing (A.40)–(A.42) by the scales in (A.43), the result is $O_m(X_K^{-1})$, even after any fixed number of η -derivatives. Lemma A.2 now gives a smooth local solution with $\|\partial_\eta^m c\|_\infty \leq C_m X_K^{-1}$. Choose X_R large enough that the moment equations have exactly one solution near zero correction coefficients and that the Jacobian there is invertible. Implicit differentiation then gives a finite bound for every fixed higher order of η -derivatives. Only the finitely many derivative bounds used in the cone argument have to meet specified smallness thresholds.

The three changed moments have now been restored. The functions M, J never changed, because $U = 0$ on both edited regions. It remains to check that the local changes in the profiles and the running integrals keep the stress inside the cone from the compensation patch to the terminal interval. The normalized radial moment formulas (4.16), written in X/X_R , carry the small changes in the profile and its moments over to Q_s, N_s on the fixed compact logarithmic interval from this patch to $y = .5$. More explicitly, $I = X_R^{3/2} \hat{I}$, $J = X_R^{3/2} \hat{J}$, $M = X_R \hat{M}$, and $S = X_R \hat{S}$. Substitute these into the formulas for Q_s, N_s . Every X_R -factor then cancels, including the factors inside H . This comparison costs one more η -derivative and gives $O(X_K^{-1})$ changes in the needed η -derivatives of Q_s, N_s . The positive lower bounds for E, Q_s , the slope bounds, and the admissible stress cone margins from Proposition A.4 persist for large X_R . The intermediate values of I and the transition value of Q_s may change while those cone inequalities continue to hold.

Finally, the total difference in pressure is zero. Integrating $\Pi_X = E^2/(2X)$ forward from the unchanged Π_0 still gives the pressure normalized to vanish at radial infinity. Before the compensation patch, this forward solution is unchanged. In particular, the terminal heat factor needs to be only smooth in η ; it does not change the analytic pressure datum on the axis.

□

A.7 Finding the stress from the exterior profile.

This subsection uses the corrected moments for two things. Lemma A.8 shows that the stress is zero outside the annulus, and Proposition A.10 determines the stress direction as the outer edge is approached. First, the moment identities will show that the full weighted integrals of the leading residual are zero. The stress can then be found by integrating from the current radius out to infinity. Once Proposition B.2 and Corollary B.10 have supplied the regular axis and the matching moment conditions, this formula depends only on the exterior profile.

Lemma A.8. *Suppose the leading axisymmetric field is smooth at the axis. Suppose also that it has $U = 0$ and $E = E_{\text{pow}} f_o[1 + \chi_K(\mathcal{H}(2d/X) - 1)]$ for $y \geq 0$, and that it satisfies the exact moment conditions*

$$M(\infty) = J(\infty) = S(\infty) = 0, \quad \int_0^\infty (H - H_{\text{pow}}) dX = 0, \quad \Pi(X) = -\frac{1}{2} \int_X^\infty \frac{E(x)^2}{x} dx.$$

Then the leading tangential stresses are zero for $X \geq X_b$. On $y \geq 1/2$, where $\chi_K = 1$, equation (A.46) writes them as weighted integrals of the leading residual from r to infinity, with a positive sign in front of each integral.

Proof. On the tail, the radial moment functions can be written as the following integrals from X to infinity. There $U = M = J = 0$, $W = 1$, and

$$I(X) = \frac{XH_{\text{pow}}(X)}{1-h} - \int_X^\infty (H - H_{\text{pow}}) dx, \quad S(X) = \frac{1}{2} \int_X^\infty E^2 dx,$$

$$Q_s = -1 + \frac{(1-h)I - D\eta I_\eta}{XH}, \quad N_s = \frac{4h\eta S - dS_\eta}{X} + 4A\eta\Pi - d\Pi_\eta.$$

The first formula uses $h < 1$ to integrate H_{pow} from zero. The remaining formulas follow directly from the exact moments and the integral formulas (4.16) for Q_s, N_s .

The backward stress formula is valid only if no boundary term remains, so check the boundaries next. Let $K_{\text{pow}} = c_\infty s^{-A}$. At large radius, uniformly for τ in bounded nonnegative intervals,

$$K - K_{\text{pow}} = O_h(\tau r^{-3-2h}), \quad \partial_t K = O_h(r^{-3-2h}), \quad r^2 K_r - rK = O_h(r^{-2h}). \quad (\text{A.44})$$

These bounds come from (A.36) and the corresponding bounds for any fixed derivative. In particular, at any sufficiently large fixed physical radius R_* ,

$$\int_{R_*}^{\infty} (r^2 |K - K_{\text{pow}}| + |\partial_t K|) dr \leq C_h(1 + \tau) \int_{R_*}^{\infty} r^{-1-2h} dr = \frac{C_h(1 + \tau)}{2h} R_*^{-2h}.$$

So the subtracted angular momentum converges. The same integrable upper bound works uniformly on compact time intervals, which justifies differentiating that moment with respect to time. This gives

$$\int_0^{\infty} r^2 (u_\theta - K_{\text{pow}}) dr = q^{3/2-A} \int_0^{\infty} (H - H_{\text{pow}}) dX = 0. \quad (\text{A.45})$$

At the axis, the subtracted power has weight r^{1-2h} , which is also integrable. Its time derivative is zero. The angular transport integral is the sum of the radial boundary $[r^2 u_r u_\theta]_0^\infty$ and the axial derivative of $\int r^2 u_z u_\theta dr = q^{3/2-2A} J(\infty) = 0$. Axis regularity makes the contribution at the axis zero, while $U = M = 0$ on the exterior makes the contribution at infinity zero. The latter also gives $V_0^{(0)} = 0$. By (A.44), radial viscosity integrates to $[r^2 \partial_r u_\theta - r u_\theta]_0^\infty = 0$. The leading angular residual $\mathcal{R}_\theta^{(0)}$ therefore has zero integral against $r^2 dr$.

For the axial component, $\int r u_z dr = q^{1-A} M(\infty) = 0$. The pressure formula $p(r) = -\int_r^\infty u_\theta(r')^2 dr'/r'$ gives

$$\int_0^{\infty} r(u_z^2 + p) dr = q^{1-2A} \int_0^{\infty} (U^2 - E^2/2) dX = 0.$$

The exact pressure calculation behind this identity is

$$\int_0^{\infty} r p(r) dr = - \int_0^{\infty} \frac{u_\theta(r')^2}{r'} \left(\int_0^{r'} r dr \right) dr' = -\frac{1}{2} \int_0^{\infty} r' u_\theta(r')^2 dr'.$$

The exterior bound $p = O(r^{-2-4h})$ makes the integral converge and gives $r^2 p \rightarrow 0$. The boundary terms from radial transport and viscosity in the axial equation are also zero. So $\int r \mathcal{R}_z^{(0)} dr = 0$.

Let T_θ, T_z be the physical stress components, so $T = q^{-A-1/2} \mathcal{T}_0$. Their defining integrals from the axis can now be written as

$$T_\theta(r) = \frac{1}{r^2} \int_r^\infty r'^2 \mathcal{R}_\theta^{(0)}(r') dr', \quad T_z(r) = \frac{1}{r} \int_r^\infty r' \mathcal{R}_z^{(0)}(r') dr'. \quad (\text{A.46})$$

For $X \geq X_b$, the field is $(u_r, u_\theta, u_z) = (0, K, 0)$, and its pressure does not depend on z and is normalized to vanish at radial infinity. Lemma A.6 says that both leading residuals are zero there, so the two stress components are zero for $X \geq X_b$. □

The backward integrals contain the cutoff factor, which goes to zero to every order at the outer edge. The next lemma pulls this factor away from a smooth coefficient. This makes it possible to estimate the direction of the stress even as the stress itself goes to zero.

Lemma A.9. *Let $c > 0$, let j be a fixed nonnegative integer, and let $b(u, \eta)$ be smooth for $0 \leq u \leq \delta_0$ and for η in a compact interval K . Then there is a coefficient B such that, whenever $0 < \delta \leq \delta_0$,*

$$\int_0^\delta e^{-c/u^2} u^{-j} b(u, \eta) du = \frac{1}{2} e^{-c/\delta^2} \delta^{3-j} B(\delta, \eta). \quad (\text{A.47})$$

The coefficient B is smooth on $[0, \delta_0] \times K$. At the endpoint, $B(0, \eta) = b(0, \eta)/c$, and every fixed mixed derivative is bounded there. The same result holds when η is replaced by any finite collection of smooth compact parameters.

Proof. Make the substitution $u = \delta/(1 + \delta^2 v)^{1/2}$. Its Jacobian is $-\frac{1}{2}\delta^3(1 + \delta^2 v)^{-3/2}$, and $c/u^2 = c/\delta^2 + cv$. This proves (A.47) with

$$B(\delta, \eta) = \int_0^\infty e^{-cv} (1 + \delta^2 v)^{(j-3)/2} b\left(\frac{\delta}{\sqrt{1 + \delta^2 v}}, \eta\right) dv.$$

Every fixed mixed derivative of this last integrand is bounded by e^{-cv} times a polynomial in v , with one bound for the whole closed rectangle of parameters. Differentiation under the integral is valid and proves both smoothness and the derivative bounds. Setting $\delta = 0$ inside the integral gives the stated value at the endpoint. $\square$

The stress is zero at the outer edge, so its limiting direction depends on how fast its two components go to zero relative to each other. The backward formula and the flat-integral lemma give both rates. They also give a positive leading coefficient for the angular component.

Proposition A.10. *Under the regular-axis and exact-moment assumptions of Lemma A.8, the profile on the terminal interval stays inside the admissible stress cone for $.5 \leq y < 3$. The direction of the stress extends smoothly to the outer endpoint. Through this extension, $2 - (a - 2)(T_z/T_\theta)^2$ has a uniform positive lower bound. More exactly, when $\delta = 3 - y$ is small enough,*

$$\mathcal{T}_{0,\theta} = e^{-4/\delta^2} \delta^{-3} b_\theta(\delta, \eta), \quad b_\theta(0, \eta) > 0, \quad (\text{A.48})$$

$$\mathcal{T}_{0,z} = e^{-4/\delta^2} \delta^3 b_z(\delta, \eta), \quad (\text{A.49})$$

$$\frac{\mathcal{T}_{0,z}}{\mathcal{T}_{0,\theta}} = \delta^6 \frac{b_z}{b_\theta} \longrightarrow 0. \quad (\text{A.50})$$

Both coefficients are smooth on a closed collar around the outer endpoint, and b_θ has a positive lower bound there. For every fixed mixed profile derivative ∂^I , there are finite constants C_I, N_I such that

$$|\partial^I \mathcal{T}_0| \leq C_I e^{-4/\delta^2} \delta^{-N_I}, \quad |\mathcal{T}_0| \geq c e^{-4/\delta^2} \delta^{-3}. \quad (\text{A.51})$$

These constants may depend on all the fixed choices in the profile and on I , but they do not depend on the physical scale q .

Proof. First find a positive angular stress and bound the axial stress relative to it. Then factor out the terms that become flat at the endpoint, which proves that the limiting direction is smooth.

On $y \geq 1/2$, where $\chi_K = 1$, let $f = f_o(y)$ and write $f' = \partial_y f$. Here $u_\theta = Kf$ and $u_r = u_z = 0$. Use the heat equation from Lemma A.6, together with $y_t = (qL)^{-1}$ and $y_z = -2\eta/(q^D L)$. This gives

$$\mathcal{R}_\theta^{(0)} = \frac{Kf'}{qL} - K(f_{rr} + r^{-1}f_r) - 2K_r f_r, \quad (\text{A.52})$$

$$p_z(r) = \frac{2\eta}{q^D L} \int_y^3 K(y')^2 f(y') f'(y') dy'. \quad (\text{A.53})$$

Integrate the viscous terms in (A.52) by parts inside (A.46). The exact result is

$$T_\theta = Kf_r + \frac{1}{r^2} \int_r^\infty (r'K - r'^2 K_{r'}) f_{r'} dr' + \frac{1}{qLr^2} \int_r^\infty r'^2 K f' dr'. \quad (\text{A.54})$$

All three terms are nonnegative, because $K > 0$, $K_r < 0$, and $f' \geq 0$. Along the rest of the logarithmic interval, whose length is at most 2.5, both the ratios of radii and the ratios of heat factors have upper and lower bounds.

Since $\int_y^3 f'(v) dv = \rho_o \psi_o(y)$, the last term in (A.54) gives

$$T_\theta \geq c \frac{rK}{qL} \rho_o \psi_o(y) > 0 \quad (y < 3).$$

In the same way, (A.53) and the integral formula for T_z give

$$|p_z| \leq Cq^{-D}K^2\rho_o\psi_o, \quad |T_z| \leq Crq^{-D}K^2\rho_o\psi_o, \quad \left| \frac{T_z}{T_\theta} \right| \leq Cq^{1-D}K = CE_{\text{pow}}\mathcal{H}(2d/X). \quad (\text{A.55})$$

The constants here are uniform when X_R is large enough and h is small. The last equality uses $1 - D = A$. The drop in amplitude during the exterior transition in (A.17) makes the final expression small. At the same time, $b_s = 0$, and

$$a = 2 + 2h + 2Z \frac{\mathcal{H}'}{\mathcal{H}} - 2 \frac{f'}{f} > 2 + h, \quad Z = 2d/X, \quad (\text{A.56})$$

because $f'/f < h/4$ and, once X_R is large enough, $-Z\mathcal{H}'/\mathcal{H} < h/4$. Also, $a \leq 2 + 2h$. It follows that $v_s = a$, $P_c - v_s = \mathcal{T}_{0,\theta}/F > 0$, and $J_c = \mathcal{T}_{0,z}/F$. The directional cone inequality is $(a - 2)(T_z/T_\theta)^2 < 2$. Equation (A.55) leaves a fixed positive margin in this inequality. This proves the admissible stress cone at every point before the outer endpoint $X = X_b$.

The cone inequality now holds at every interior point of the terminal interval. It remains to show that the direction of the stress extends smoothly to the endpoint. Factor out the part shared by the two stress components as they go to zero, and then show that the angular coefficient stays positive. Near $\delta = 0$, the chosen smooth step has the form

$$\psi_o = e^{-4/\delta^2} g(\delta), \quad g(\delta) = \frac{1}{e^{-(1-\delta/2)^{-2}} + e^{-4/\delta^2}}, \quad g(0) = e,$$

and so

$$f' = \rho_o e^{-4/\delta^2} \delta^{-3} [8g(\delta) + \delta^3 g'(\delta)].$$

Apply Lemma A.9 to the backward formulas for the stress.

Multiply the boundary term in (A.54) by $q^{A+1/2}$. This gives

$$q^{A+1/2} K f_r = \frac{2E_{\text{pow}}(X) \mathcal{H}(2d/X)}{\sqrt{2X}} f'.$$

Both integral terms contain f' . Equation (A.47), with $j = 3, c = 4$, pulls out a factor e^{-4/δ^2} and leaves a smooth coefficient. Compared with the boundary factor, those two terms vanish to at least order δ^3 . This proves (A.48), with

$$b_\theta(0, \eta) = \frac{16\rho_o E_{\text{pow}}(X_b)}{\sqrt{2X_b}} \mathcal{H}(2d/X_b) g(0) > 0.$$

The positive lower bound is uniform for $-1 \leq \eta \leq 1$, because $\mathcal{H}$ has a positive minimum on $[0, 2/X_b]$.

The pressure integral (A.53) contains one f' factor. The same identity first writes $p_z = e^{-4/\delta^2}$ times a smooth coefficient, apart from the factor q^{-D-2A} . The integral formula for T_z requires one more integration with smooth weights. Applying (A.47) with $j = 0$ gives exactly (A.49). In this normalization, the powers cancel because $q^{A+1/2} q^{1/2-D-2A} = q^{1-D-A} = 1$. Every coefficient left over contains only smooth functions of $X, \eta, \mathcal{H}(2d/X)$ and L^{-1} ; here $L \geq 1 - 2h > 0$. This also proves smoothness on the closed parameter ranges and shows that neither flat factor contains the physical scale.

Divide by the positive coefficient b_θ to get (A.50). The limiting direction is $(1, 0)$, where the normalized directional quadratic expression equals 2. The stress itself is zero at the endpoint. Finally, every fixed derivative of the profile factorizations adds only a finite inverse power of δ . This proves (A.51). A smooth positive factor from the other endpoint may multiply this outer weight without changing any of these local conclusions. $\square$

B Analytic profiles near the axis and how to continue them

This appendix builds the inner part of the leading profile (U, E, Π) . On this inner part, the leading residual stress $\mathcal{T}_0$ must be zero, as Proposition B.2 requires. The outer profile has already fixed the pressure datum Π_0 on the axis, in Lemma A.5. With that datum fixed, choose the remaining axis data U_*, ϕ_* and solve the nonlinear profile equations (4.13) on a short radial interval. The solution must be regular at the axis. At its outer endpoint, it must also satisfy the strict shear inequality (B.19), so that Corollary B.10 can continue it toward the outer profile. The large parameter Λ in the azimuthal datum (B.3) gives both the analytic solution and the endpoint bounds needed for this continuation, as Propositions B.2 and B.3 show. Equations (4.4)–(4.5) then show that the corresponding velocity is smooth in Cartesian coordinates across the axis.

B.1 Choosing the axis data.

The pressure datum Π_0 is already prescribed. The azimuthal datum $\phi_* = \phi(0, \eta)$ and the axial velocity U_* on the axis still have to be chosen. Choose them here so that the angular source S_q stays positive and the endpoint inequality holds throughout the parameter interval, as Proposition B.3 requires. Equation (B.2) separates the interval into two parts; the azimuthal term gives the needed lower bound on one part, and the axial term gives it on the other.

Use the datum Π_0 from Lemma A.5. It is real analytic near $[-1, 1]$, it is even, and it satisfies

$$\Pi_0 \leq -cP_*^2 f^2, \quad \eta \Pi_{0\eta} > 0 \quad (\eta \neq 0), \quad f = (1 + \eta^2)^{-1}.$$

Every schedule parameter, including $0 < h \leq 10^{-2}$, is already fixed. Choose $0 < j_0 \leq .05$ small, and set

$$\begin{aligned} U_* &= 4\eta + j_0, & H_* &= D\eta + dU_*, \\ W_* &= 1 - dU_{*\eta} - 2D\eta U_*, & Z_* &= -A(1 - 2\eta U_*)U_* - H_* U_{*\eta} - d\Pi_{0\eta} + 4A\eta \Pi_0. \end{aligned} \quad (\text{B.1})$$

The function H_* has exactly one zero $\eta_0 \in (-1, 0)$. To see why, note that on $(-1, 1)$ the function $H_*/d = D\eta/d + 4\eta + j_0$ is strictly increasing from $-\infty$ to $+\infty$. Its zero satisfies $|\eta_0| \asymp j_0$. Also,

$$-W_* = 3 - 8h\eta^2 + (1 - 2h)j_0\eta > 2.8, \quad Z_*(\eta_0) \geq c j_0 P_*^2 > 0.$$

For the second inequality, $4A\eta_0 \Pi_0(\eta_0)$ is positive and at least $c j_0 P_*^2$. The term $-d\Pi_{0\eta}(\eta_0)$ is nonnegative, the H_* term is zero, and the term that remains is $O(j_0)$. The choice of P_* , which is already large, absorbs this last term.

It is now possible to choose $\delta_* > 0$ first, so that the set $\{|Z_*| \leq \delta_*\}$ stays away from a neighborhood of η_0 . Then choose $\sigma_* > 0$ small enough that

$$\chi = \frac{H_*^2}{H_*^2 + \sigma_*^2} > .99 \quad \text{where } |Z_*| \leq \delta_*. \quad (\text{B.2})$$

Make both choices on a slightly larger compact interval $I \supset [-1, 1]$. Then choose a bounded, simply connected complex neighborhood Ω of I . On the closure of this neighborhood, neither L nor $H_*^2 + \sigma_*^2$ may vanish.

The function Π_0 is holomorphic. For $\Lambda \geq 1$, define

$$\zeta_* = -\frac{LH_*}{H_*^2 + \sigma_*^2}, \quad \tilde{\zeta}_0 = \Lambda \zeta_*, \quad \phi_* = \exp\left(\Lambda \int_0^\eta \zeta_*(w) dw\right). \quad (\text{B.3})$$

The integral is well defined throughout Ω . On the real interval, $\phi_* > 0$. The amplitude C is chosen after Λ .

B.2 A space for analytic coefficients.

The profile equations (4.13) use parameter derivatives as well as radial derivatives. Their highest radial derivatives are multiplied by the radial variable. At $X = 0$ that multiplier vanishes, so the equations cannot be solved there for their highest radial derivatives in the usual way. The vanishing is only linear, and the smooth power-series coefficients can still be solved one at a time by the radial inverse in (B.5). That is what regular singular point means here. This subsection sets up a weighted coefficient space $\mathcal{B}_\rho$ for the integral form of those equations. Its norm is given in (B.4). In this space, increasing the radial degree makes up for taking a parameter derivative. Lemma B.1 gives the bounds needed for multiplication and radial inversion, the step that solves for a function in the radial variable. Those bounds also cover the mixed nonlinear terms used in the contraction argument for Proposition B.2.

For the analytic construction at the axis, use the scalar radial variable $Y = \Lambda X$. Fix a small radius $\rho > 0$. If $F(Y, \eta) = \sum_{\alpha \geq 0} F_\alpha(\eta) Y^\alpha$, define

$$\mathfrak{a}_{\alpha\beta} = \frac{20^{-\alpha} \rho^{-\beta} \beta! \binom{\alpha+\beta}{\beta}}{(\alpha+1)^2 (\beta+1)^2}, \quad \|F\|_\rho = \sup_{\alpha, \beta \geq 0} \sup_{\eta \in I} \frac{|\partial_\eta^\beta F_\alpha(\eta)|}{\mathfrak{a}_{\alpha\beta}}. \quad (\text{B.4})$$

Let $\mathcal{B}_\rho$ be the set of smooth coefficient sequences whose norm is finite. This space is complete. In a Cauchy sequence, every coefficient and every order of its derivative converges uniformly. Uniform convergence of one derivative after another then shows that these limits are the derivatives of the limiting coefficient. In the rescaled variable Y , the logarithmic radial derivative is $\mathcal{D}_X = X\partial_X = Y\partial_Y$. Define $\mathcal{I}F = \int_0^Y F(Y', \eta) dY'$. Rescaling by $Y = \Lambda X$ does not change radial averaging, so $\mathcal{A}_X(F) = Y^{-1} \mathcal{I}F$.

Lemma B.1. *Multiplication is bounded on $\mathcal{B}_\rho$. For $\nu = 1, 2$, let $\mathcal{J}_\nu F$ be the solution G of $Y G_{YY} + \nu G_Y = F$ that extends smoothly to $Y = 0$ and satisfies $G(0) = 0$. Its coefficients are*

$$(\mathcal{J}_\nu F)_{\alpha+1} = \frac{F_\alpha}{(\alpha+1)(\alpha+\nu)}, \quad (\mathcal{J}_\nu F)_0 = 0. \quad (\text{B.5})$$

The operator $\mathcal{J}_\nu$ is bounded on $\mathcal{B}_\rho$. The same is true of radial averaging, multiplication by Y , and the operators $\mathcal{I}$ and $\partial_\eta \mathcal{I}$. Also,

$$\|\mathcal{J}_\nu[(\partial_\eta F)(\mathcal{D}_X G)]\|_\rho \leq C_\rho \|F\|_\rho \|G\|_\rho. \quad (\text{B.6})$$

The same estimate still holds if either derivative is left out, or if F is replaced by $\mathcal{A}_X(F)$. Extra factors without derivatives may also be inserted, with one more algebra constant for each extra factor.

Proof. The elementary convolution estimate

$$\sum_{i=0}^N \frac{(N+1)^2}{(i+1)^2 (N-i+1)^2} \leq 8 \sum_{i=1}^{\infty} i^{-2} =: C_{\text{sq}} \quad (\text{B.7})$$

follows by splitting the sum at $N/2$.

In the Leibniz formula for a product coefficient, the binomial coefficient from differentiation cancels the factorials in the weights. The binomial ratio that remains is at most one, because

$$\binom{i+k}{k} \binom{\alpha-i+\beta-k}{\beta-k} \leq \binom{\alpha+\beta}{\beta}. \quad (\text{B.8})$$

The reason is that the left side counts some, but not necessarily all, of the β -element subsets of a set split into blocks of sizes $i+k$ and $\alpha-i+\beta-k$. Apply (B.7) in both indices. This gives the algebra constant C_{sq}^2 .

The two weight ratios needed for integration are

$$\frac{\mathfrak{a}_{\alpha\beta}}{\mathfrak{a}_{\alpha+1,\beta}} = 20 \frac{(\alpha+2)^2}{(\alpha+1)^2} \frac{\alpha+1}{\alpha+\beta+1} \leq 80, \quad (\text{B.9})$$

$$\frac{\mathfrak{a}_{i,k+1}}{\mathfrak{a}_{i+1,k}} = \frac{20}{\rho} (i+1) \frac{(i+2)^2(k+1)^2}{(i+1)^2(k+2)^2} \leq \frac{80}{\rho} (i+1). \quad (\text{B.10})$$

The first ratio proves the stated bounds for $\mathcal{J}_\nu$, $\mathcal{I}$, and multiplication by Y . Averaging divides the term of degree α by $\alpha+1$. The second ratio proves the bound for $\partial_\eta \mathcal{I}$, because its integration divisor $i+1$ cancels the matching factor.

For the mixed derivative bound (B.6), take the coefficient of degree $N = \alpha+1$ on the left. Assign the extra degree to the factor with the parameter derivative, and write $I_1 = i+1$ and $J_1 = \alpha-i$. After applying (B.10) and dividing by $\mathfrak{a}_{N\beta}$, the β th derivative is bounded by

$$\frac{80}{\rho} \|F\|_\rho \|G\|_\rho \sum_{i=0}^{\alpha} \sum_{k=0}^{\beta} \frac{I_1 J_1}{N(\alpha+\nu)} \frac{(N+1)^2}{(I_1+1)^2 (J_1+1)^2} \frac{(\beta+1)^2}{(k+1)^2 (\beta-k+1)^2}.$$

Use (B.8) with radial indices I_1, J_1 to drop the binomial ratio. The first fraction is at most one, and (B.7) bounds both sums. If F is averaged, its divisor $i+1$ cancels I_1 , and the fraction that remains is again at most one. If the logarithmic radial derivative is left out, replace J_1 by one. A logarithmic radial derivative with no η -derivative is controlled directly by (B.9) and (B.5). For extra factors, assign the indices in the same way, leave those extra factors undifferentiated, and repeat the convolution estimate. This covers the endpoint indices and every β . $\square$

B.3 The nonlinear analytic profile at the axis.

This subsection solves the equations (4.13) for the axis data chosen in Section B.1, with the leading residual stress set to zero. Rescale the radius by $Y = \Lambda X$. The solution will be shown to approach an explicit profile as Λ grows, with the error stated in (B.13). The scalar comparison function f_0 below controls the normalized azimuthal profile ϕ/ϕ_* through $f_0(Y\chi)$. The bound (B.11) will show that it stays positive across the radial interval being used. That makes division by ϕ valid in the profile equations.

The analytic scalar comparison function is

$$f_0(z) = \sum_{\alpha \geq 0} \frac{(-z/2)^\alpha}{\alpha! (\alpha+1)!}.$$

For $0 \leq z \leq 4.1$, set $t = z/2$. After the cubic term, the terms in the alternating tail get smaller in absolute value. It follows that

$$f_0(z) \geq 1 - \frac{t}{2} + \frac{t^2}{12} - \frac{t^3}{144} \geq \frac{305719}{1152000} > .265, \quad f_0(z) \leq 1. \quad (\text{B.11})$$

The cubic decreases throughout $[0, 2.05]$, so its endpoint gives the value just stated. For the upper bound, group the alternating tail after the constant term. Starting with the linear term, its terms decrease in absolute value.

Proposition B.2. *Take the axis data from Section B.1. There is a Λ_0 . For every $\Lambda \geq \Lambda_0$, there is an amplitude threshold $C_0(\Lambda)$ such that every $C \geq C_0(\Lambda)$ gives an analytic axis profile whose leading residual stress vanishes on $0 \leq Y = \Lambda X \leq 4.1$. The profile has the form*

$$\phi = \phi_* \Phi, \quad U = U_* + \Lambda^{-1} u, \quad \Pi = \Pi_0 + \Lambda^{-1} \mathcal{I}(g^2 \Phi^2), \quad g = \phi_*/C, \quad (\text{B.12})$$

where $\Phi(0, \eta) = 1$ and $u(0, \eta) = 0$. The profile is analytic in Y and η on the same neighborhood of $[0, 4.1] \times I$. For every fixed $r, s \geq 0$,

$$\sup_{[0, 4.1] \times I} \left| \partial_Y^r \partial_\eta^s \left(\Phi - f_0(Y\chi), u + \frac{YZ_*}{2L} \right) \right| \leq \frac{C_{r,s}}{\Lambda}. \quad (\text{B.13})$$

For all sufficiently large Λ and all $C \geq C_0(\Lambda)$, the constants and a smaller common parameter neighborhood do not depend on either large parameter. Written with derivatives in the original radial variable, the bound is $C_{r,s}\Lambda^{r-1}$.

Proof. Rewrite the equations using the rescaled radius. The nonlinear remainders then come with a factor Λ^{-1} . Applying the radial inverse operators turns the problem into a contraction around the explicit comparison profile. In the notation of the stress equations, set

$$B = -2D_\eta \mathcal{A}_X(u) - d\partial_\eta \mathcal{A}_X(u), \quad W = W_* + \Lambda^{-1}B, \quad H_c = H_* + \Lambda^{-1}du, \quad p = \mathcal{I}(g^2\Phi^2). \quad (\text{B.14})$$

The equations (4.13) with zero leading residual stress, together with radial pressure balance, are

$$-2L(X\phi_{XX} + 2\phi_X)/\phi = S_q, \quad -2L(XU_{XX} + U_X) = S_n, \quad \Pi_X = \phi^2/C^2. \quad (\text{B.15})$$

Substitute (B.12). This gives

$$\begin{aligned} 2(Y\Phi_{YY} + 2\Phi_Y) &= -\chi\Phi + \Lambda^{-1}R_1, \\ 2(Yu_{YY} + u_Y) &= -Z_*/L + \Lambda^{-1}R_2, \end{aligned}$$

where the remainders are exactly

$$\begin{aligned} LR_1 &= [W + h(1 - 2\eta U) + du\zeta_*]\Phi + W\mathcal{D}_X\Phi + H_c\Phi_\eta, \\ LR_2 &= [A(1 - 4\eta U_*) + dU_{*\eta}]u - 2A_\eta\Lambda^{-1}u^2 + W\mathcal{D}_Xu + H_*u_\eta + d\Lambda^{-1}uu_\eta \\ &\quad - 4A_\eta p + d\partial_\eta p - 2\eta\mathcal{D}_Xp. \end{aligned}$$

The leading angular term has this sign because $H_*\tilde{\xi}_0/(\Lambda L) = -\chi$. To get the expanded axial remainder, subtract the constant value $-Z_*/L$ from $L^{-1}\{A(1 - 2\eta U) + W\mathcal{D}_Xu + H_cU_\eta - 4A_\eta\Pi + d\Pi_\eta - 2\eta\mathcal{D}_X\Pi\}$, and multiply the result by Λ .

The radial inverse bounds require control of the differentiated products in R_1 and R_2 . The only products that contain both a parameter derivative outside an integral and a logarithmic radial derivative are

$$(\partial_\eta \mathcal{A}_X(u))\mathcal{D}_X\Phi, \quad (\partial_\eta \mathcal{A}_X(u))\mathcal{D}_Xu.$$

The two derivatives fall on different factors. Also, $\partial_\eta p = \partial_\eta \mathcal{I}(g^2\Phi^2)$ is bounded by Lemma B.1, and $\mathcal{D}_Xp = Yg^2\Phi^2$. It follows that the maps $(\Phi, u) \mapsto \mathcal{J}_\nu R_i$ are bounded and locally Lipschitz on fixed balls in $\mathcal{B}_\rho^2$. The displayed algebra bounds give this quantitatively. To estimate the difference of two monomials, replace their factors one at a time and subtract at each step.

These coefficient bounds are uniform in the large parameters. The functions $\zeta_*, \chi, U_*, Z_*/L$ are fixed and holomorphic on Ω . Choose a smaller neighborhood, and take ρ strictly smaller than its distance from the boundary of Ω . After choosing Λ , require

$$C \geq \sup_{\eta \in \Omega} |\phi_*(\eta)|. \quad (\text{B.16})$$

Then $|g| \leq 1$ on Ω . Cauchy's inequality now bounds its derivatives on the smaller neighborhood, uniformly in both large parameters. Choosing ρ strictly smaller absorbs the factor $(\beta + 1)^2$ in (B.4).

The bound on g must hold on a complex neighborhood, because ϕ_* itself can grow exponentially with Λ .

The order-one angular term still has to be inverted without assuming that its coefficient is small. Let M_χ be a bound for multiplication by χ on $\mathcal{B}_\rho$, and set $T = \mathcal{J}_2\chi/2$, with multiplication done before integration. If F has no radial coefficient below degree b , then (B.9) gives

$$\|\mathcal{J}_2 F\|_\rho \leq \frac{80}{(b+1)(b+2)} \|F\|_\rho, \quad \|T^k\| \leq \frac{(40M_\chi)^k}{k!(k+1)!}.$$

Multiplication by χ keeps every coefficient below degree b equal to zero, while $\mathcal{J}_2$ raises the lowest possible nonzero degree by one. Parameter differentiation does not create any lower-degree coefficients. The estimate therefore includes parameter derivatives of every order in these iterated products. The series $\sum_{k \geq 0} (-T)^k$ converges absolutely in operator norm, and it is a two-sided inverse of $1 + T$.

The unperturbed solution is $\Phi^0 = (1 + T)^{-1}1 = f_0(Y\chi)$ and $u^0 = -YZ_*/(2L)$. Solve the equations with the map

$$(\Phi, u) \mapsto \left(\Phi^0 + \frac{(1+T)^{-1}\mathcal{J}_2 R_1}{2\Lambda}, u^0 + \frac{\mathcal{J}_1 R_2}{2\Lambda} \right).$$

Take a fixed ball around (Φ^0, u^0) . For every sufficiently large Λ , the uniform bounds and Lipschitz constants above make the map send this ball into itself, and make it a strict contraction there. Its fixed point is at most C/Λ away from the center in the norm. Choose $C_0(\Lambda)$ large enough to include (B.16). This keeps the parameters in the order stated in the proposition.

For $R < 20$,

$$\sum_{\alpha \geq 0} \binom{\alpha + \beta}{\beta} (R/20)^\alpha = (1 - R/20)^{-\beta-1}.$$

Choose $4.1 < R < 20$. The coefficient norm then gives a single positive parameter radius that works for all the functions involved. Any sufficiently small number below $\rho(1 - R/20)$ can be used for this radius on $|Y| \leq R$. Cauchy estimates on a smaller Y disk give (B.13), including every fixed order of radial derivative. The derivatives in the original variable satisfy $\partial_X^r = \Lambda^r \partial_Y^r$. Because the data are real and the fixed point is unique, the solution is real on the real rectangle. Analyticity in X makes the scalar profiles smooth at the axis. Along with $E = \sqrt{2X}\phi/C$, Equations (4.4) to (4.5) then give a smooth Cartesian field. The lower bound (B.11) gives $\Phi > 0$ when Λ is large, and hence $\phi > 0$. Division by ϕ in the first equation of (B.15) is therefore valid throughout the real rectangle. □

B.4 Positive sources and the alternative at the outer endpoint.

The analytic solution must satisfy the specific bounds needed to start its continuation toward the outer profile. The approximation (B.13) gives both a positive angular source S_q and the strict shear inequality (B.19) at the end of the inner interval. The same estimates will bound the normalized profiles and their parameter derivatives for the joining construction in Proposition B.3 and Lemma B.4.

Proposition B.3. *Increase Λ , and then increase $C_0(\Lambda)$, if needed. The normalized azimuthal profile satisfies $\Phi \geq c_0 > 0$ on $[0, 4.1] \times [-1, 1]$. On the same rectangle, its angular source satisfies*

$$S_q \geq 2.5 + .95\Lambda L\chi. \tag{B.17}$$

Let $p_1 = p_{s,1} = XQ_s/L$, $p_2 = p_{s,2} = XN_s/(LE)$, and $n_s = N_s/L$. On $0 < X \leq 4.1/\Lambda$,

$$p_1/X \geq c_1 > 0, \quad 0 < p_1 \leq C_1, \quad n_s = -2U_X = Z_*/L + O(\Lambda^{-1}). \tag{B.18}$$

At $X_0 = 4/\Lambda$, there is a fixed $c_{\text{ex}} > 0$ such that

$$p_1 + \frac{p_2^2}{p_1} > 2 + c_{\text{ex}}. \quad (\text{B.19})$$

The quantities $\Phi, \log \Phi, u, p_1$, and n_s have bounded derivatives at every fixed parameter order. Those bounds are uniform in all later choices $C \geq C_0(\Lambda)$. Constants for derivatives in the original radial variable X may depend on Λ .

Proof. The scalar bound (B.11) and the approximation (B.13) give $\Phi \geq c_0 > 0$ when Λ is large. On a smaller common parameter neighborhood, Φ is also nonzero. This follows from compactness and the uniform bound on its first derivative. It follows in turn that all parameter derivatives of $\log \Phi$ are uniformly bounded.

The source bound uses the logarithmic derivatives of Φ . Since $|H_* \chi'| \leq 2\|H'_*\|_{\infty} \chi$ and f_0 stays away from zero, $\mathcal{D}_X \log f_0(Y\chi) = O(\chi)$ and $H_* \partial_\eta \log f_0(Y\chi) = O(\chi)$. The approximation and positivity then give the following bound for the full solution.

$$|\mathcal{D}_X \log \Phi| + |H_* \partial_\eta \log \Phi| \leq C\chi + C/\Lambda. \quad (\text{B.20})$$

Combine this with (B.14) and (B.3). The result is

$$\begin{aligned} -H_c \tilde{\xi}_0 &= \Lambda L\chi + O_{\sigma_*}(\sqrt{\chi}), \\ -H_c(\log \phi)_\eta &\geq \Lambda L\chi - C\chi - C_{\sigma_*} \sqrt{\chi} - C/\Lambda \geq .95\Lambda L\chi - C_{\sigma_*}/\Lambda. \end{aligned} \quad (\text{B.21})$$

For the first error, use $|H_*|/(H_*^2 + \sigma_*^2) \leq \sigma_*^{-1} \sqrt{\chi}$. For the second inequality, increase Λ until the term $C\chi$ is absorbed. Then apply Young's inequality to the square-root term, using a total of $.05\Lambda L\chi$.

For the source itself, use $l = 1 + \mathcal{D}_X \log \Phi$ and the formula $S_q = -Wl - h(1 - 2\eta U) - H_c(\log E)_\eta$. The conditions $j_0 \leq .05$ and (B.13) show, after increasing Λ , that $|U| \leq 4.1$. Hence $|h(1 - 2\eta U)| \leq 10h$. The bounds above and $W - W_* = O(\Lambda^{-1})$ now give

$$S_q \geq \Lambda L\chi - W_* - 10h - C\chi - C_{\sigma_*} \sqrt{\chi} - C/\Lambda.$$

Because $h \leq 10^{-2}$ and $-W_* > 2.8$, absorbing the terms in the same way proves (B.17) once Λ is large. In particular, the argument never divides by χ near a zero of H_* .

The function Q_s extends smoothly to $X = 0$, and it has the positive integral formula

$$Q_s(X, \eta) = \frac{\int_0^X X' \Phi(\Lambda X', \eta) S_q(X', \eta) dX'}{X^2 \Phi(\Lambda X, \eta)}.$$

Uniform upper and lower bounds for Φ , together with (B.17), give $p_1/X \geq c_1$. Integrate the equations with zero leading residual stress from $X = 0$, and use regularity at the axis. This gives

$$p_1 = a = -2Y\Phi_Y/\Phi, \quad n_s = -2u_Y.$$

The upper bound for p_1 and the approximation for n_s now follow from (B.13). The same identities give the stated fixed bounds for the η -derivatives.

It remains to prove the endpoint inequality. The two parameter regions chosen in (B.2) let either the azimuthal term or the axial term do the job. At $Y = 4$, first take the case $\chi > .99$. Then $1.98 < t = 2\chi \leq 2$. The alternating series for $f_0(z) + zf'_0(z)$ gives

$$f_0(z) + zf'_0(z) \leq 1 - t + \frac{t^2}{4} - \frac{t^3}{36} + \frac{t^4}{576} < -.18.$$

This polynomial decreases on $[1.98, 2]$. At $t = 1.98$, its value is $-75535511/400000000 < -.188$. The decreasing alternating remainder has the needed negative sign. By (B.11),

$-2zf'_0/f_0 = 2 - 2(f_0 + zf'_0)/f_0 > 2.36$. Equation (B.13) then gives $p_1 > 2.3$ at the outer endpoint when Λ is large.

In the other region, $\chi \leq .99$, equation (B.2) gives $|Z_*| > \delta_*$. Then (B.18) gives $|n_s| \geq \delta_*/2$ after increasing Λ . At the outer endpoint,

$$|p_2| = C\sqrt{2/\Lambda} \frac{|n_s|}{\phi_*\Phi}.$$

Now Λ is fixed. The denominator has a finite upper bound that is uniform for all later large choices of C , while $p_1 \leq C_1$. Increase $C_0(\Lambda)$ until $p_2^2/p_1 > 2.3$ throughout this region. Both cases give (B.19). One possible choice is $c_{\text{ex}} = .2$. □

B.5 A reference continuation with $p_{s,1} + p_{s,2}^2/p_{s,1} > 2$.

The next step continues the analytic axis profile from Proposition B.2 toward the outer profile from Section A. This continuation has to introduce a nonzero leading residual stress $\mathcal{T}_0$ that satisfies the cone conditions, while keeping the pressure datum Π_0 from Lemma A.5. Start by constructing a reference continuation ϕ_r, U_r . The logarithmic radial derivatives of ϕ and U are gradually taken to zero, as specified in (B.22). Lemma B.4 bounds the integrated residual coefficients of this reference continuation. Those bounds will control the later change in shear in Proposition B.5.

As in Proposition B.3, p_1, p_2 are the components of p_s , and $n_s = N_s/L$. Recall that $\mathcal{D}_X = X\partial_X$. In a logarithmic radial coordinate, this is the same as ∂_y . A bound in C_η^k concerns only parameter derivatives, uniformly for $-1 \leq \eta \leq 1$.

Set $X_0 = 4/\Lambda$, $X_b = 100$, and $X_i = 110$. Choose a fixed $\bar{t} > 0$ such that $4e^{2\bar{t}} < 4.1$. Use the smooth step σ from (A.5). For $0 < t_1 \leq \bar{t}$, set $y = \log(X/X_0)$. The reference profile agrees with the analytic axis profile when $y \leq t_1$. When $t_1 < y < 2t_1$, prescribe

$$\begin{aligned} \partial_y \log \phi_r &= [1 - \sigma((y - t_1)/t_1)] \mathcal{D}_X \log \phi_{\text{nat}}(X, \eta), \\ \partial_y U_r &= [1 - \sigma((y - t_1)/t_1)] \mathcal{D}_X U_{\text{nat}}(X, \eta), \end{aligned} \tag{B.22}$$

and then keep both fields constant in the logarithmic radial coordinate. Compute the reference pressure and the functions Q_s, N_s by integrating radially from the axis, using the datum from Lemma A.5.

Lemma B.4. *After fixing the parameters of the outer and axis profiles, first take Λ large enough, then take C large enough, and finally take t_1 small enough. The following bounds then hold on $[X_0, X_i] \times [-1, 1]$.*

$$S_{q,r} \geq .94\Lambda\chi + 2.4, \quad l_r \leq 1, \quad v_r := p_{1,r} + \frac{p_{2,r}^2}{p_{1,r}} > 2 + c. \tag{B.23}$$

Here $p_{1,r} > 0$, and $p_{1,r} > 3$ whenever $X_b \leq X \leq X_i$. For each fixed k , the parameter norms of CE_r , its reciprocal, $p_{1,r}$, and $n_{s,r}$ on $[X_0, X_i]$ are bounded independently of all sufficiently large C and all $0 < t_1 \leq \bar{t}$.

Proof. Propositions B.2 and B.3 show that the analytic axis profile has a positive source and satisfies a strict inequality at its outer endpoint. Use the gradient bounds (B.20) and (B.21), together with the source bound (B.17), to carry these properties through the cutoff. Integrating (B.22) shows that U_r and its radial average stay within $O(\Lambda^{-1}) + o_{t_1}(1)$ of U_* in every required parameter norm. This uses the fact that radial averaging is a contraction.

$$\sup_{0 < X \leq X_i} \|\mathcal{A}_X(U_r) - U_*\|_{C_\eta^k} \leq \sup_{0 \leq X \leq X_i} \|U_r - U_*\|_{C_\eta^k}.$$

This estimate does not depend on the length of the interval where the reference profile is constant. Integrating the same slopes also bounds $\log \phi_r - \log \phi_*$ in every parameter norm,

uniformly in C, t_1 . Its real value stays bounded above and below. It follows that $CE_r = \sqrt{2X}\phi_r$ and its reciprocal have the stated bounds away from $X = 0$. Also,

$$\Pi_r(X) - \Pi_0 = C^{-2} \int_0^X \phi_r(X', \eta)^2 dX', \quad \sup_{X \leq X_i} \|\Pi_r - \Pi_0\|_{C_\eta^k} \leq C_k C^{-2}. \quad (\text{B.24})$$

The regular axis factor makes this integral finite at zero.

For the sign of the source, the leading gradient term is $-H_* \tilde{\xi}_0 = L\Lambda\chi$. The bound $|\tilde{\xi}_0| \leq L\Lambda\sigma_*^{-1}\sqrt{\chi}$ shows that replacing H_* by $H_{c,r}$ creates an error bounded by $C_{\sigma_*}\sqrt{\chi} + o_{t_1}(1)$ after Λ, C have been fixed. The logarithmic slope of the axis profile, together with the remaining gradient, contributes $C\chi + C/\Lambda$. Absorb these errors into a small fraction of $L\Lambda\chi$, using $C_{\sigma_*}\sqrt{\chi} \leq \varepsilon\Lambda L\chi + C_{\varepsilon,\sigma_*}/\Lambda$. The constant term keeps the room provided by $-W_* > 2.8$, which gives the first inequality in (B.23). The axis profile satisfies $p_1 = a > 0$, so the logarithmic ϕ slope is nonpositive. Cutting this slope off at zero gives $l_r \leq 1$.

The axial source satisfies $S_{n,r} = Z_* + O(\Lambda^{-1}) + o_{t_1}(1) + O(C^{-2})$. On the short cutoff, $\mathcal{D}_X U_{\text{nat}} = O(\Lambda^{-1})$. On the whole finite interval, (B.24) controls the pressure and the parameter derivative of the pressure that is needed here. The equations for $p_{1,r}, n_{s,r}$ are

$$\mathcal{D}_X p_{1,r} = \frac{XS_{q,r}}{L} - l_r p_{1,r}, \quad \mathcal{D}_X n_{s,r} + n_{s,r} = \frac{S_{n,r}}{L}. \quad (\text{B.25})$$

The regular initial values and bounded coefficients give the stated parameter bounds. In the region $\chi \leq .99$, the axis alternative $|Z_*| > \delta_*$ keeps the sign fixed and keeps $|n_{s,r}|$ above a positive lower bound.

The first equation in (B.25) preserves positivity and gives both comparisons below.

$$\begin{aligned} p_{1,r}(y) &\geq p_{1,r}(0)e^{-y} + 1.88\chi(e^y - e^{-y}), \\ p_{1,r}(X) &\geq p_{1,r}(X_0)\frac{X_0}{X} + 1.2\left(X - \frac{X_0^2}{X}\right). \end{aligned}$$

In the region $\chi > .99$, the first right-hand side stays above 2.3. This is because its initial value can be replaced by 2.3, and $1.88\chi > 2.3/2$. In the other region, $p_{2,r} = Xn_{s,r}/E_r$ grows uniformly in absolute value with C , while $p_{1,r}$ stays positive and bounded. This proves $v_r > 2 + c$. The second comparison gives $p_{1,r} > 3$ from $X = 100$ onward. □

B.6 Continuing to a nonzero leading residual stress.

This subsection uses the reference bounds from Lemma B.4 to reduce the shear s , while keeping the integrated residual coefficients p_s close to their reference values as stated in (B.28). The stress $\mathcal{T}_0 = F(p_s - s)$ from (4.11) must turn on smoothly at the end of the axis region. It must satisfy the admissible stress cone on an initial collar next to that region, and it must keep satisfying the relaxed cone condition throughout the rest of the continuation. Proposition B.5 does this.

Proposition B.5. *The analytic axis profile can be continued to X_i while keeping E positive and leaving the profile unchanged for $X \leq X_0$. Its stress factors as $\mathcal{T}_0 = e_a B_0$. The scalar factor e_a vanishes to infinite order at X_0 , while the smooth vector coefficient B_0 is nonzero there. The admissible stress cone holds on an initial collar, and the strict relaxed cone condition holds on the rest of $(X_0, X_i]$. At X_i , $a = .8$, $\mathcal{D}_X U = 0$, and $p_1 > 2$.*

Proof. First reduce the reference shear on a short interval. During that reduction, p_s changes by a small error that is kept under control. The residual stress then becomes nonzero and satisfies the cone conditions just stated. Next, show that the stress has a factor that vanishes to infinite order, and finish the continuation with small shear.

Fix a sufficiently large finite C . Choose $0 < t_* \leq \bar{t}$ small enough that the bounds in Lemma B.4 hold uniformly for every reference width $0 < t_1 \leq t_*$. The parameter bounds in that lemma, along with its positive lower comparison for $p_{1,r}$, give one finite upper bound $V_{\max}$ for v_r across this whole family of reference profiles. This bound may depend on C , but it does not depend on t_1 . Choose $0 < \kappa_0 < 1/2$ later from this bound. On $0 < y < t_1$, set

$$e_a = (1 - \kappa_0)\sigma(y/t_1), \quad \kappa = 1 - e_a, \quad a = \kappa p_{1,r}, \quad \mathcal{D}_X U = -\frac{\kappa X n_{s,r}}{2}, \quad \partial_y \log \phi = -\frac{\kappa p_{1,r}}{2}. \quad (\text{B.26})$$

On this interval, the reference profile agrees with the analytic axis profile. Subtracting the reference equations gives the exact identities

$$\partial_y(\log \phi - \log \phi_r) = \frac{e_a p_{1,r}}{2}, \quad \partial_y(U - U_r) = \frac{e_a X n_{s,r}}{2}. \quad (\text{B.27})$$

Since e_a is increasing, every fixed parameter derivative of these field differences is $O(ye_a)$. Constants in the activation comparisons may depend on the already fixed C, Λ , but they stay uniform as t_1 and κ_0 decrease within their allowed ranges. The bounds for the reference family control the integrated coefficients independently of both widths. Apply the radial moment identities (4.16), using one more parameter derivative. They give

$$p_s - p_{s,r} = O(ye_a), \quad E_r/E = 1 + O(ye_a). \quad (\text{B.28})$$

This comparison holds for the field values and their parameter derivatives. Radial derivatives of the narrow cutoffs may be large.

The actual shear is $\kappa(p_{1,r}, p_{2,r}E_r/E)$. The common factor κ cancels from the ratio $t_s = -b_s/a$ before that ratio is estimated. The definitions of P_c, J_c, v_s give

$$t_s = \frac{p_{2,r}E_r}{p_{1,r}E}, \quad v_s = \kappa v_r + O(ye_a), \quad P_c = v_r + O(ye_a), \quad J_c = O(ye_a). \quad (\text{B.29})$$

No inverse power of κ_0 appears in these error constants. If t_1 is small enough, then $P_c - v_s \geq ce_a$, $P_c > 2 + c$, and

$$(v_s - 2)_+ J_c^2 < 2(P_c - v_s)^2 \quad (0 < y \leq t_1).$$

The ratio of the left side of this inequality to $(P_c - v_s)^2$ is at most Cy^2 . When $v_s > 2$, the quadratic test in Lemma 4.5 proves the admissible stress cone condition. When $v_s \leq 2$, the inequality $P_c > 2$ directly gives the strict relaxed condition. Since $v_r(0) > 2 + c$, continuity gives a first collar of positive width where $v_s > 2 + c/2$. For the final fixed parameters, the collar is uniform in η .

The factor $e_a = e^{-t_1^2/y^2} g_a(y)$ is flat at zero, meaning that it vanishes there to every order, and g_a is smooth and positive. To show that division by this flat factor still gives a smooth function, apply Lemma A.9 with $c = t_1^2$, $j = 0$, and coefficient $g_a b$. For every smooth coefficient b , the lemma gives

$$\frac{1}{e_a(y)} \int_0^y e_a(u) b(u, \eta) du = y^3 B(y, \eta), \quad B \in C^\infty.$$

Dividing by the positive smooth function $g_a(y)$ keeps the result smooth and keeps every fixed order of mixed derivatives bounded. The differences in (B.27) lie in $e_a y^3 C^\infty$. Products, smooth compositions, and reciprocals of fields that never vanish stay in this class. The moment and pressure differences vanish before X_0 . One more integration puts them in $e_a y^6 C^\infty$. Substituting into (4.16) then proves (B.28) after division by e_a , for every fixed mixed derivative. It also gives the exact factorization

$$\mathcal{T}_0 = e_a B_0(y, \eta), \quad B_0(0, \eta) = F(X_0, \eta) p_{s,r}(X_0, \eta) \neq 0, \quad B_0 \in C^\infty. \quad (\text{B.30})$$

At the edge, the limiting shear is $p_{s,r}$. This gives $B_0 \cdot (-t_s, 1) = 0$ and $B_0 \cdot (1, t_s) > 0$ there. At the endpoint, $B_0 \cdot (1, t_s) > 0$ and $2(B_0 \cdot (1, t_s))^2 - (v_s - 2)(B_0 \cdot (-t_s, 1))^2 > 0$. Both quantities stay above positive lower bounds on a smaller collar. In particular, for every fixed radial and parameter multi-index I ,

$$|\mathcal{T}_0| \geq ce^{-t_1^2/y^2}, \quad |\partial^I \mathcal{T}_0| \leq C_I e^{-t_1^2/y^2} y^{-N_I}. \quad (\text{B.31})$$

The derivatives in (B.31) are taken in (y, η) . Any fixed number of derivatives in X has the same form because $X_0 > 0$. The nonzero vector $B_0(0, \eta)$ gives the limiting direction of the stress.

Continue the profile to X_i while keeping the relaxed cone condition. After the first transition, keep the prescription (B.26) with $\kappa = \kappa_0$ through the reference cutoff and the interval of constant slope. Once C, Λ are fixed, integration over $[X_0, X_b]$ and (4.16) show that the field values, their logarithms, and the required derivatives differ by $O(t_1 + \kappa_0)$ when those derivatives are taken with respect to η . Choose κ_0 small compared with $V_{\max}$ and the uniform comparison constants for the reference family. Then choose t_1 small. This order does not require fixing a reference width before κ_0 . Formula (B.29) now gives $P_c > 2 + c/2$ and $v_s < 1$. The strict relaxed cone condition follows no matter how large J_c is.

At X_b , multiply the prescribed value of $\mathcal{D}_X U$ by a smooth factor β that decreases from one to zero over a short logarithmic interval. During this transition, $P_c = p_{1,r} + \beta p_{2,r}^2 / p_{1,r} + o(1) > 2$, while $v_s < 1$. Next, use a convex interpolation to move a from $\kappa_0 p_{1,r}$ to $.8$, keeping $\mathcal{D}_X U = 0$. First arrange that $\kappa_0 \sup p_{1,r} < .8$. During this second transition, $v_s = a \leq .8$. Also, $P_c = p_1 > 2$, by continuity from the positive margin at X_b . Both transitions end strictly before X_i .

On the last interval of constant slope, $a = .8$ and $l = .6$. The same absorption of the gradient used in Lemma B.4 gives $S_q > 1$. Its constant term is $.6(-W_*) > 1.68$, while its leading gradient is $L\Lambda\chi$. Suppose that $p_1 = 2$ crossed downward. At such a crossing, $\mathcal{D}_X p_1 = XS_q/L - .6p_1 > X - 1.2 > 0$, which is impossible. This finishes the continuation. $\square$

The continuation is still analytic in η on the first collar. Reserve a smaller rectangle inside that collar. Later changes will then leave both this regularity and the factorization of the inner stress untouched.

Corollary B.6. *There is a $t_c > 0$ such that $t_c < t_1$ and $4e^{t_c} < 4.1$. The functions $E/\sqrt{2X}$, U , V_0/X , Π are defined on $[0, X_0 e^{t_c}] \times [-1, 1]$, and they are smooth in (X, η) through $X = 0$, and they are analytic in η on one complex neighborhood. Every fixed radial derivative of these four functions is analytic on that same parameter neighborhood and is bounded on a smaller one. On $X_0 < X \leq X_0 e^{t_c}$, the admissible stress cone condition and the factorization (B.30) hold. Every later change to the profile is supported strictly to the right of $X_0 e^{t_c}$.*

Proof. Choose a compact complex neighborhood inside the one from Proposition B.2. Make it small enough that the positive function Φ in the analytic axis profile has no zeros. The formulas for Q_s, N_s , computed from the reference profile, and (B.26) use only analytic parameter coefficients, their parameter derivatives, and forward integrals, multiplied by cutoffs that depend only on y . Their denominators do not vanish on a smaller neighborhood. Exponentiation gives a nonzero ϕ , so one neighborhood works across the whole fixed first collar. Radial differentiation changes the bounds for the cutoffs, but it does not change the parameter domain.

Compactness and Proposition B.5 give an inner collar where the admissible stress cone condition holds. Choose t_c strictly inside this collar, and keep the resulting rectangle free of all later changes. Neither t_c nor the radial derivative bounds are claimed to be uniform as C grows and the transition widths shrink. The radial moment and pressure integrals run forward from the axis, so changes supported later leave this rectangle unchanged. $\square$

B.7 Choosing a radial transition length before the amplitude.

This subsection constructs the remaining transition (B.34) to $E = P_* f(X/X_R)^{1/10}$, where $f = (1 + \eta^2)^{-1}$. The radial moment functions are corrected in the next subsection, in Proposition B.8. First, Lemma B.7 and (B.33) give a bound for the logarithmic radial length T_{sh} that does not depend on the amplitude C chosen later. Increasing that amplitude can then move the matching radius X_R outward without making the transition longer. This makes the normalized discrepancies in the inner moments small, as shown in (B.38) and (B.39). The estimates below concern parameter derivatives. Bounds for derivatives in X do not have to be uniform in the radial transition widths. The number $\kappa_0 \in (0, 1/2)$ in these bounds is the parameter that controls the shear reduction in (B.26).

Lemma B.7. *Fix the outer-profile data, the axis data, Λ , and a finite parameter order k . Set $\ell_i = \log(\text{CE}(X_i, \eta))$ and $G_i = U(X_i, \eta)$. Let ω_{fin} be the sum of the logarithmic radial lengths of the final axial cutoff and the interpolation of a to $.8$ in the proof of Proposition B.5. The continuations built there satisfy*

$$\begin{aligned} \|\ell_i\|_{C_\eta^k} &\leq \mathcal{B}_k, \\ \|G_i - U_*\|_{C_\eta^k} &\leq \frac{C_k^{\text{nat}}}{\Lambda} + C_k^{\text{join}}(\Lambda)(t_1 + \kappa_0 + \omega_{\text{fin}}). \end{aligned} \quad (\text{B.32})$$

The constant C_k^{nat} does not depend on either Λ or any sufficiently large C , once the earlier axis data have been fixed. The bounds $\mathcal{B}_k$ and $C_k^{\text{join}}(\Lambda)$ may depend on Λ , but they do not depend on any sufficiently large C or on any smaller radial transition widths.

Proof. Write $L_0 = \log(X_i/X_0)$. Let M_k, N_k be the amplitude-independent bounds for $p_{1,r}, n_{s,r}$ from Lemma B.4. Before the last interpolation of a to $.8$, the actual logarithmic slope is $-\kappa p_{1,r}/2$, where $0 < \kappa \leq 1$. During that interpolation, the slope changes by a convex interpolation to $-.4$, and it stays constant after that. It follows that

$$\|\log \phi(X_i)\|_{C_\eta^k} \leq \|\log \phi_* + \log \Phi(4, \cdot)\|_{C_\eta^k} + \frac{1}{2}(M_k + .8)L_0.$$

Add $\frac{1}{2} \log(2X_i)$ and use (B.13). This proves the first claim. The first axial transition changes U by at most $X_0 e^{\bar{t}} N_k t_1 / 2$. The rest changes it by at most $X_i N_k \kappa_0 / 2$, with the same bound for the transition that takes this term to zero. The approximation error for the analytic axis profile is $C_k^{\text{nat}} / \Lambda$. Equation (B.13) shows that its constant does not depend on Λ . These estimates prove (B.32). Only integrals of bounded cutoff values were used, so no inverse power of a cutoff width appears. $\square$

These bounds make it possible to choose the transition length without using the final amplitude. Choose

$$T_{\text{sh}} \geq 20 \|\sigma'\|_\infty (\mathcal{B}_0 + \|\log f\|_\infty). \quad (\text{B.33})$$

For $y_i = \log(X/X_i)$, prescribe

$$\log E = -\log C + \frac{y_i}{10} + (1 - \sigma(y_i/T_{\text{sh}}))\ell_i + \sigma(y_i/T_{\text{sh}}) \log f, \quad U = G_i. \quad (\text{B.34})$$

Then $l = .6 + T_{\text{sh}}^{-1} \sigma'(y_i/T_{\text{sh}})(\log f - \ell_i)$ lies in $[\cdot 55, \cdot 65]$, so $a \in [\cdot 7, \cdot 9]$ and $b_s = 0$.

Throughout the transition (B.34), the source still satisfies $S_q > 1$. To check its sign, make all the changes before the final $.8$ constant-slope interval small in the required C_η^k norms. The slope on that constant-slope interval does not depend on the parameter. It follows that $\ell'_i = \tilde{\xi}_0 + (\log \Phi(4, \cdot))_\eta +$

$o(1)$ and $G_i - U_* = O(\Lambda^{-1}) + o(1)$. The old gradient satisfies

$$\begin{aligned} -H_c \ell'_i &\geq L\Lambda\chi - C\chi - C_{\sigma_*}\sqrt{\chi} - C/\Lambda - o(1), \\ -H_c(\log f)' &= \frac{2\eta H_c}{1+\eta^2} \geq -Cj_0^2 - C/\Lambda - o(1). \end{aligned}$$

For the second line, use $\eta H_* = (D + 4d)\eta^2 + dj_0\eta$ and complete the square. For the first line, use (B.20), which gives $H_*\partial_\eta \log \Phi = O(\chi) + O(\Lambda^{-1})$. Multiplying the $O(\Lambda^{-1})$ error in the field by $\tilde{\xi}_0$ creates an error bounded by $C_{\sigma_*}\sqrt{\chi}$, and this error is absorbed in the same way as before. Equation (B.34) takes a convex combination of the two gradients. Since $-W$ stays close to $-W_* > 2.8$, while $l \geq .55$ and h, j_0 are small, it follows that $S_q > 1$.

The inequality $\mathcal{D}_X p_1 = XS_q/L - lp_1 > 0$ holds at the barrier $p_1 = 2$, so a downward crossing is impossible. This preserves $p_1 > 2$. The whole transition, and the later interval of constant slope before restoration, therefore continue to satisfy the strict inequalities that define the relaxed cone condition.

B.8 Matching the five radial moment functions exactly.

The azimuthal profile E now has the required reference form (A.7). Equation (B.32), together with (B.1), says that the axial profile $U = G_i$ is close to its reference value 4η . This subsection matches the five cumulative radial integrals M, I, J, S, C_p from (4.15), so that the outer profile can be attached. First choose the radial scale X_R to make their normalized differences small. Then cancel those differences with five fixed bumps while keeping the relaxed cone inequalities, as stated in Proposition B.8. Once the common pressure datum Π_0 and all five moments match exactly, Lemma 4.4 shows that every later outer field stays the same.

Set

$$X_R = X_i(CP_*)^{10}, \quad x = X/X_R.$$

Also set $X_{\text{sep}} = X_i e^{T_{\text{sh}}}$, the endpoint of the transition (B.34). This endpoint is fixed before the amplitude is chosen.

Proposition B.8. *Take the amplitude C large enough, and then take the activation transitions small enough. The profile built in Proposition B.5 and continued by (B.34) can then be continued to $\log x = -5$ while keeping the strict relaxed cone condition. At that point, its fields, its pressure Π , and all five radial moment functions M, I, J, S, C_p agree exactly with those of the reference inner profile (A.7). The profile can then follow the reference outer radial profile without changing either its prescribed pressure datum or any later value of Q_s, N_s .*

Proof. First normalize the radial moment map so that its inverse bounds do not depend on the large radius chosen later. Then show that the difference built up by the inner endpoint is small in these normalized units, and cancel it with five fixed bumps.

Introduce $\hat{H} = \sqrt{2x}E$, and define the normalized radial moment functions by

$$\begin{aligned} M &= X_R \hat{M}, & I &= X_R^{3/2} \hat{I}, & J &= X_R^{3/2} \hat{J}, & S &= X_R \hat{S}, \\ C_p &= \int_0^x \frac{E(\xi, \eta)^2}{2\xi} d\xi, & \Pi &= \Pi_0 + C_p. \end{aligned}$$

Here $\hat{M}, \hat{I}, \hat{J}, \hat{S}$ are the integrals of $U, \hat{H}, U\hat{H}, U^2 - E^2/2$, in that order, with respect to x . Substitute these definitions into (4.16). The result is

$$\begin{aligned} W &= 1 - 2D\eta\hat{M}/x - d\hat{M}_\eta/x, \\ Q_s &= -W + \frac{(1-h)\hat{I} - D\eta\hat{I}_\eta - d\hat{J}_\eta + 2(h-D)\eta\hat{J}}{x\hat{H}}, \\ N_s &= -WU + \frac{D(\hat{M} - \eta\hat{M}_\eta) + 4h\eta\hat{S} - d\hat{S}_\eta}{x} + 4A\eta\Pi - d\Pi_\eta. \end{aligned} \tag{B.35}$$

In these formulas, X_R no longer appears in Q_s, N_s . On the fixed interval $\mathcal{R} = [e^{-8}, e^{-5}]$, the ideal functions $E_0 = P_* f x^{1/10}$ and $Q_{s,0}$ have positive minimum values, and the ideal $N_{s,0}$ is bounded. These facts follow from the explicit formula (A.25). On this interval, the map (B.35) is Lipschitz in every chosen finite parameter norm. Its inputs need one more parameter derivative, and its constants depend only on the outer-profile data.

The same statements hold for the fixed restoration and moment-correction operations. To state the allowed error in the cone inequalities, set $Q_{\min} = \min_{\mathcal{R} \times [-1,1]} Q_{s,0} > 0$, $e_* = \min E_0 > 0$, and $w_* = 1 + \max |N_{s,0}/(E_0 Q_{s,0})|$. A tolerance depending only on the data of the outer profile can make the following bounds hold throughout both operations.

$$Q_s \geq Q_{\min}/2, \quad E \geq e_*/2, \quad |a - .8| \leq .1, \quad \left| \frac{N_s}{E Q_s} \right| \leq w_*, \quad |b_s| \leq \frac{.1}{1 + w_*}. \quad (\text{B.36})$$

These bounds give $v_s \leq .9 + .01/.7 < 1$. Set $G = Q_s - b_s N_s / (aE)$. Then

$$G \geq \frac{6}{7} Q_s \geq \frac{3Q_{\min}}{7}, \quad P_c = \frac{X_R x}{L} G. \quad (\text{B.37})$$

The inequality $P_c > 2$ follows by increasing X_R after the tolerance has been fixed. The tolerance for the moment correction does not have to shrink as X_R grows.

It remains to show that this fixed tolerance can be reached. In the normalized coordinate, the endpoint X_{sep} is

$$x_{\text{sep}} = X_{\text{sep}} / X_R = e^{T_{\text{sh}}} / (C P_*)^{10}. \quad (\text{B.38})$$

On $[0, X_{\text{sep}}]$, the bounds already proved for $\log(CE(X_i, \cdot))$ and $U(X_i, \cdot)$, together with (B.34), bound U and CE in every required parameter norm. Near zero, $CE = O(\sqrt{X})$. The parts of the cumulative integrals that come from $0 \leq x \leq x_{\text{sep}}$ therefore satisfy

$$\begin{aligned} \|\hat{M}\|_{C_\eta^k} + \|\hat{S}\|_{C_\eta^k} &\leq C_k x_{\text{sep}}, \\ \|\hat{I}\|_{C_\eta^k} + \|\hat{J}\|_{C_\eta^k} &\leq C_k C^{-1} x_{\text{sep}}^{3/2}, \\ \|C_p\|_{C_\eta^k} &\leq C_k C^{-2} \quad (x = x_{\text{sep}}). \end{aligned} \quad (\text{B.39})$$

The matching integrals for the ideal profile have positive weights, and they also go to zero as positive powers of x_{sep} . At x_{sep} , the pressure increment of the reference profile is $(5/2)P_*^2 f^2 x_{\text{sep}}^{1/5} = O(C^{-2})$. The constants in all these vanishing errors may depend on the axis and shape parameters, which have already been fixed.

After X_{sep} , equation (B.34) gives exactly the ideal angular profile because $C^{-1}f(X/X_i)^{1/10} = P_* f x^{1/10}$. Choose the amplitude large enough that $x_{\text{sep}} < e^{-8}$. On $-8 < \log x < -7$, bring G_i smoothly back to 4η , while keeping E equal to its ideal value. The difference $G_i - 4\eta$ can be made as small as needed by first choosing j_0 , then choosing Λ , and finally choosing the small transitions in (B.32).

Integrating over the rest of the fixed interval gives bounds depending only on the outer-profile data, because all the weights involved are integrable positive powers at zero. At the point where the moments are corrected, the five normalized moment differences are therefore at most $C_k \|G_i - 4\eta\|_{C_\eta^{k+1}} + o_C(1)$. The same estimate controls the field values, their logarithmic radial derivatives, and Q_s, N_s on the fixed restoration patch.

On $-6 < \log x < -5$, add two fixed radial bumps to U and three fixed radial bumps to E . At the ideal profile, replace the J row by $J - 4\eta I$, and replace the S row by $S - 8\eta M$. Apart from fixed nonzero normalizing factors, the derivative of the normalized moment map has the following two blocks.

$$U \text{ weights : } 1, f x^{3/5}; \quad E \text{ weights : } x^{1/2}, f x^{1/10}, f x^{-9/10}.$$

Corollary A.3 shows that this matrix has an inverse everywhere on the compact parameter interval, with one uniform bound on that inverse. The nonlinear terms are exactly quadratic. Lemma A.2 therefore solves the five equations with smooth coefficients whose prescribed finite set of η -derivatives satisfy the required smallness bounds. Every other fixed higher derivative also exists and has its own finite bound. To prove (B.36), first fix the radius of the small-discrepancy neighborhood using the outer profile data, and then use that radius in the estimates above. The bumps sit strictly inside their patch, so the fields equal the ideal near the patch's right endpoint. All five rows use the same Π_0 , and (B.35) then gives them the same pressure and the same functions Q_s, N_s . Lemma 4.4 carries this equality along the outer radial profile. $\square$

B.9 The order of the choices and the completed connection.

This subsection puts all the parameter choices in order and states the completed connection. The matching tolerance in (B.36) depends only on the outer data. Lemma B.7 bounds the radial transition length T_{sh} in (B.33) without using the final amplitude C . Because of these two facts, the amplitude can be chosen to reduce the moment discrepancies before the last transition widths are fixed. The full order is shown in (B.40).

Remark B.9. Fix any finite list of η -derivative orders whose bounds are needed to preserve the profile inequalities and solve the finite moment equations. Include the extra parameter derivative in (B.35). The choices made above and in Section A can then be made in this order.

$$\begin{aligned} M_d, T_d, P_*, \lambda, h &\longrightarrow \text{tolerance for matching moments, } j_0 \longrightarrow \delta_*, \sigma_*, \Lambda \\ &\longrightarrow (\mathcal{B}_k), T_{\text{sh}} \longrightarrow C, X_R \longrightarrow \kappa_0, t_1, \text{ widths of final transitions.} \end{aligned} \quad (\text{B.40})$$

First choose the outer profile so that $C_{\text{pre}}\sqrt{\lambda}(1 + \log(1/\lambda))$ is small. Then choose $h \ll \lambda$ and $h \ll e^{-T_d}$, after C_{pre} has already been fixed. The exterior-profile data set the tolerance for matching the moments, as (B.37) shows. The choices of j_0 and Λ make the two contributions j_0 and $C_{\text{nat}}^{\text{nat}}/\Lambda$ to the axial matching error small enough. Lemma B.7 fixes T_{sh} before the final amplitude is chosen. Then choose the amplitude to ensure $p_{1,r} + p_{2,r}^2/p_{1,r} > 2 + c$, make the outer profile obey the cone inequalities at large radius, put $X_{\text{sep}}/X_R < e^{-8}$, and give the inner-interval contributions in (B.39) the required bounds. The last narrow transitions give the activation comparisons and control the errors that remain at $X = X_i$, without losing the bound already proved. Choose the radial frequency used to realize the admissible stress cone only after all these finite choices have been made. Only the prescribed finite set of η -derivative bounds needs to be small. Once the choices are made, derivatives of every other fixed order still have finite bounds, though radial cutoff derivatives may contain inverse powers of the final widths.

Corollary B.10. *The analytic axis profile from Proposition B.2 has a smooth continuation with $E > 0$ for $X > 0$. At $\log(X/X_R) = -5$, it matches the reference outer radial profile, including the exact fields, pressure, and radial moment functions. Its leading residual stress is zero through X_0 . This is followed by an inner collar where the profile remains analytic in η , obeys the admissible stress cone condition, and has the flat factor in (B.30). From there to the matching point, it obeys the strict inequalities of the relaxed cone condition. Its finite parameters can be chosen in the order shown in (B.40).*

Proof. Combine Propositions B.5 and B.8, the fact that $p_1 > 2$ remains true along the profile (B.34), and Corollary B.6. $\square$

C Realizing the admissible stress cone

Corollary B.10 joins the inner and outer profiles E, U, Π with the required pressure datum and radial moments. One condition is still missing on part of the joining interval.

There, the shear $(a, -b_s)$ obeys only the relaxed cone condition (4.21). The wave construction also needs the inequality $v_s > 2$ from Lemma 4.5. This appendix makes that inequality hold by changing the shear rapidly in the radial direction, with the change kept away from the axis and the exterior.

The shear is determined by radial derivatives in (4.11), while the cumulative moments $\mathbf{m} = (M, I, J, S, C_p)$ are determined by profile values in (4.15). That difference makes it possible to change the shear by order one while changing the profiles and their moments only a little. First prescribe a periodic family of admissible shears $(a_L, -b_L)$ whose mean is the original shear, as in Lemma C.1. Then integrate that family at high radial frequency. A correction on the next reserved interval listed after (A.9) restores all five moments exactly, and this keeps the exterior fields unchanged, as Proposition C.2 proves.

Take the profile supplied by Corollary B.10, Proposition A.4, and Proposition A.7. Proposition B.8 matches the axis profile's radial moments to the reference outer profile's moments. That match supplies the regular axis and exact moments required in Lemma A.8. Together, these results fix the input profile E, U, Π used below. This profile and all its parameters stay fixed while the shear is modulated and the moments are corrected in this appendix. Its leading residual stress $\mathcal{T}_0$ is nonzero exactly on (X_a, X_b) , whose endpoints do not depend on η . The profile obeys the strict inequalities of the relaxed cone condition throughout that interval. It obeys the full admissible stress cone condition on an inner collar and again from the intermediate power-law interval onward, by Propositions A.4, A.7 and A.10. The two edge factorizations are (B.30) and Equations (A.48) to (A.49). Choose a compact interval $\mathcal{I} = [X_-, X_+] \Subset (X_a, X_b)$ that contains every point where the admissible stress cone condition may fail. It starts inside the first collar where that condition holds and ends in the intermediate power-law interval before its first reserved patch. Shorter collars at X_a and X_b remain unchanged. Also fix the rectangle near the axis from Corollary B.6, where $E/\sqrt{2X}, U, V_0/X, \Pi$ are smooth in X, η and analytic in η on the same complex neighborhood. Choose the rectangle's radial endpoint inside the shorter inner collar and before X_- . Call this endpoint X_{an} .

C.1 A loop with the prescribed mean.

At every profile point, this subsection constructs a family $(a_L, -b_L)$ of shear vectors while keeping the integrated residual coefficient p_s fixed, as stated in Lemma C.1. The family's average must be $(a, -b_s)$. Its difference from that original shear then has mean zero, so it has a periodic antiderivative, a periodic function whose derivative is that difference. Proposition C.2 integrates this difference into a small change of the profiles. The family must change smoothly with (X, η) and must still equal the original shear near the radial boundaries, where nothing needs to change.

Lemma C.1. *Let a, b_s, p_s be smooth on $\mathcal{I} \times [-1, 1]$, with $a > 0$, and set $t_s = -b_s/a$, $v_s = a(1 + t_s^2)$. Suppose the strict inequalities of the relaxed cone condition in (4.21) hold everywhere, and suppose the admissible stress cone holds near both radial boundary components. Then there is a smooth loop $(a_L, -b_L)(X, \eta, \varphi)$, with period one in φ and $a_L > 0$, whose mean is the original shear.*

$$\int_0^1 (a_L, -b_L) d\varphi = (a, -b_s). \quad (\text{C.1})$$

Let Ψ be the vector of cone gaps in (4.35). There are constants $\kappa_L, \delta_\partial > 0$, depending on the fixed data $a, b_s, p_s, \mathcal{I}$, for which every cone gap has the following uniform lower bound.

$$\begin{aligned} \Psi_j(a_L(X, \eta, \varphi), b_L(X, \eta, \varphi), p_s(X, \eta)) &\geq \kappa_L, \\ (X, \eta, \varphi) &\in \mathcal{I} \times [-1, 1] \times (\mathbb{R}/\mathbb{Z}), \quad j = 1, 2, 3, 4. \end{aligned} \quad (\text{C.2})$$

On the radial boundary collars, the loop is exactly the original shear.

$$\begin{aligned} (a_L, -b_L)(X, \eta, \varphi) &= (a, -b_s)(X, \eta), \\ X \in \mathcal{I}, \quad \text{dist}(X, \partial\mathcal{I}) &< \delta_\partial, \quad (\eta, \varphi) \in [-1, 1] \times (\mathbb{R}/\mathbb{Z}). \end{aligned} \quad (\text{C.3})$$

Every smoothness statement here includes the endpoints of the parameter interval $-1 \leq \eta \leq 1$.

Proof. Write each shear vector as a positive multiple of $(1, t)$. Vary the ratio t while keeping its mean equal to t_s , choose a scalar $v > 2$, and look for admissible vectors of the form $v(1, t)/(1 + t^2)$. The variance of t sets v . Reparametrizing the loop then gives these vectors the required mean $(a, -b_s)$.

Use an auxiliary angle θ' , and write $\langle f \rangle_{\theta'} = (2\pi)^{-1} \int_0^{2\pi} f(\theta') d\theta'$ for its average. This is not the physical angular average or the auxiliary-torus average. Put $P_c(t) = p_{s,1} + p_{s,2}t$ and $J_c(t) = p_{s,2} - p_{s,1}t$. Choose $0 < d_0 < \frac{1}{2} \min(P_c(t_s) - 2)$, and define

$$M_e(z) = \langle e^{z \sin \theta'} \rangle_{\theta'}, \quad (\text{C.4})$$

$$t(\theta'; \mu) = t_s + d_0 \frac{e^{\mu p_{s,2} \sin \theta'} / M_e(\mu p_{s,2}) - 1}{p_{s,2}}, \quad \mu \geq 0. \quad (\text{C.5})$$

The formula looks singular when $p_{s,2} = 0$, but that singularity is removable. For a smooth numerator $G(p)$ with $G(0) = 0$, the identity $G(p)/p = \int_0^1 G'(up) du$ removes the division by zero. Applying it to (C.5) shows that the family is smooth jointly in all its variables and gives the value $t = t_s + d_0 \mu \sin \theta'$ when $p_{s,2} = 0$. The family also obeys

$$\begin{aligned} \langle t \rangle_{\theta'} &= t_s, \quad P_c(t) \geq P_c(t_s) - d_0 > 2, \\ V(\mu, p_{s,2}) &:= \langle (t - t_s)^2 \rangle_{\theta'} = \frac{d_0^2}{p_{s,2}^2} \left[\frac{M_e(2\mu p_{s,2})}{M_e(\mu p_{s,2})^2} - 1 \right]. \end{aligned}$$

When $p_{s,2} = 0$, the last expression is $d_0^2 \mu^2 / 2$.

A prescribed finite variance can be reached uniformly over the compact range of $p_{s,2}$, including the point $p_{s,2} = 0$. To see this, let $g = \log M_e$. Its second derivative is the variance of $\sin \theta'$ under the strictly positive tilted probability density. This density weights each angle by the exponential in (C.4) and divides by the average of those weights. Therefore $g'' > 0$. For $p \neq 0$ and $\mu > 0$,

$$\partial_\mu \{g(2\mu p) - 2g(\mu p)\} = 2p \{g'(2\mu p) - g'(\mu p)\} > 0. \quad (\text{C.6})$$

So V rises strictly as $\mu > 0$ rises, including when $p = 0$. Near the maximum of $\sin \theta'$, matching quadratics bound the function above and below. Away from that maximum, the function stays below it by a uniform gap. Together these facts give $M_e(z) \asymp e^{|z|} / \sqrt{|z|}$ for $|z| \geq 1$. As a result, $M_e(2z)/M_e(z)^2 \asymp \sqrt{|z|}$, and $V(\mu, p) \rightarrow \infty$ for every fixed p , including zero. The range of $p_{s,2}$ is compact. Continuity, strict increase, and a finite cover of that range therefore give one finite $\mu_{\max}$ such that

$$V(\mu_{\max}, p_{s,2}) > 3 / \min a \quad \text{throughout } \mathcal{I} \times [-1, 1]. \quad (\text{C.7})$$

This makes it possible to choose the variance of t everywhere on the compact parameter set. Next, bound the allowed value of v so that every resulting shear vector stays inside the cone. For $0 \leq \mu \leq \mu_{\max}$, the functions $t(\theta'; \mu)$ form a compact family with $P_c(t) > 2$. The upper admissible bound $\mathcal{U}$ for v in (4.21) is the smaller root in v of $2(P_c - v)^2 = (v - 2)J_c^2$. This root changes continuously across the family and stays strictly above two. Choose $0 < \delta_L < 1$ small enough that

$$\mathcal{U}(P_c(t), J_c(t)) > 2 + \delta_L \quad (0 \leq \mu \leq \mu_{\max}). \quad (\text{C.8})$$

Make δ_L smaller again, if needed, so that it lies below the positive minimum of $v_s - 2$ on smaller neighborhoods of the radial boundaries. Choose δ_L only after $\mu_{\max}$ has been fixed.

Only the region where v_s is near or below two needs nonzero variance. Leave the admissible boundary data unchanged. Let ζ_L be a smooth nonnegative function of v_s , no larger than one, equal to one when $v_s \leq 2 + \delta_L/8$, and equal to zero when $v_s \geq 2 + \delta_L/4$. Set

$$v_* = 2 + \delta_L/2, \quad \rho = \zeta_L(v_s)^2(v_* - v_s), \quad v = v_s + \rho, \quad (\text{C.9})$$

where ρ is zero outside the support of the cutoff. Its nonnegative square root is smooth. Wherever the cutoff can be nonzero, $v_* - v_s \geq \delta_L/4$, so that square root is $\zeta_L(v_s)\sqrt{v_* - v_s}$, extended by zero. Also $0 \leq \rho < 3$. Solve

$$V(\mu, p_{s,2}) = \rho/a, \quad 0 \leq \mu < \mu_{\max}. \quad (\text{C.10})$$

Equation (C.7) and strict monotonicity give existence and uniqueness. Smooth dependence still has to be checked where $\rho = 0$. The expansion of (C.4), uniform for bounded p , is

$$V(\mu, p) = \frac{1}{2}d_0^2\mu^2 + O(\mu^4 p^2).$$

The signed square root of V is smooth and odd near $\mu = 0$, and its derivative there is $d_0/\sqrt{2}$. Apply the implicit function theorem to this square root and to the smooth right side $\sqrt{\rho}/\sqrt{a}$. This proves smoothness through $\rho = 0$. Away from zero, smoothness follows from (C.6).

The definition of ρ also gives

$$v = (1 - \zeta_L(v_s)^2)v_s + \zeta_L(v_s)^2v_*.$$

Since $0 \leq \zeta_L \leq 1$, the three cutoff regimes are

$$\begin{aligned} v &= v_* && \text{if } v_s \leq 2 + \delta_L/8, \\ 2 < v_s \leq v \leq v_* && \text{if } 2 + \delta_L/8 < v_s < 2 + \delta_L/4, \\ v &= v_s > 2 && \text{if } v_s \geq 2 + \delta_L/4. \end{aligned}$$

Thus $v > 2$ everywhere, and $v \leq v_*$ wherever the cutoff is nonzero. Wherever the cutoff is zero, $\mu = 0$ and $v = v_s > 2$, so the given data a, b_s, p_s obey the strict inequalities of the relaxed cone condition. Equation (C.8) now shows that every proposed state satisfies $2 < v < \mathcal{U}(P_c(t), J_c(t))$.

The vectors now lie in the admissible cone. Their average still depends on how long the loop spends at each value of t . To give both shear components their prescribed averages, reparametrize θ' by a lifted angle φ , starting from $\theta' = 0$, according to

$$\frac{d\varphi}{d\theta'} = \frac{a(1+t^2)}{2\pi v}, \quad (a_L, -b_L) = \frac{v(1, t)}{1+t^2}.$$

Because $a\langle 1+t^2 \rangle_{\theta'} = v_s + aV = v$, the lift obeys $\varphi(\theta' + 2\pi) = \varphi(\theta') + 1$. Its derivative has a positive lower bound across the compact family. It therefore gives a circle diffeomorphism, a smooth one-to-one change of angle whose inverse also depends smoothly on the parameters. Changing variables gives

$$\int_0^1 a_L d\varphi = a, \quad \int_0^1 (-b_L) d\varphi = a\langle t \rangle_{\theta'} = at_s = -b_s.$$

The loop obeys $-b_L/a_L = t$ and $a_L(1+t^2) = v$, so the scalar inequalities just proved give the admissible stress cone inequalities. On the boundary neighborhoods the cutoff vanishes, so the loop equals the original shear $(a, -b_s)$. Compactness gives the uniform margins in (C.2) for $a_L, v-2$, and the two strict inequalities of the quadratic cone test. The fixed boundary neighborhoods give (C.3). □

C.2 Radial modulation and exact restoration of the radial moment functions.

Lemma C.1 built the loop while keeping p_s fixed. This subsection inserts that loop into the profiles E, U and controls the resulting change in p_s , which is determined by their radial moments through (4.16). Evaluate periodic antiderivatives at $N \log X$ and divide them by N . Then the profile values change by $O(N^{-1})$, while the shear changes by order one. The proof of Proposition C.2 proves these estimates, bounds the resulting moment errors by $O(N^{-1})$, and removes the errors on the reserved correction interval. The comparison with the fixed input profile E, U, Π in the next proposition does not say that the radial derivatives are small.

Proposition C.2. *Start with the fixed input profile E, U, Π chosen at the beginning of this appendix. A sufficiently large finite integer N , together with a correction supported in the first reserved patch in the intermediate power-law interval, gives smooth profiles $\tilde{E}, \tilde{U}, \tilde{\Pi}$ with $\tilde{E} > 0$ for $X > 0$. These profiles obey the admissible stress cone everywhere in (X_a, X_b) . Both endpoint collars, the rectangle near the axis described above, and the two patches reserved for higher-order and mean corrections stay unchanged. Define $\tilde{m}, \tilde{Q}_s, \tilde{N}_s$ from these profiles by (4.15) and (4.16). Then*

$$(\tilde{m}, \tilde{\Pi}, \tilde{Q}_s, \tilde{N}_s) = (m, \Pi, Q_s, N_s) \quad \text{beyond the first correction patch.}$$

Before that point, the profile values and any fixed number of η -derivatives differ from those of the input profile by $O(N^{-1})$.

Proof. The loop has the same prescribed averages as the original shear, so their differences have mean zero in the periodic variable. Let $\mathcal{A}, \mathcal{B}$ be the unique zero-mean periodic antiderivatives

$$\partial_\varphi \mathcal{A} = -\frac{1}{2}(a_L - a), \quad \partial_\varphi \mathcal{B} = \frac{1}{2}E(b_L - b_s). \quad (\text{C.11})$$

They exist because of (C.1), are smooth, and are identically zero on the boundary neighborhoods of $\mathcal{I}$. Extend them by zero in the radial direction and set

$$E_N = E \exp\left(\frac{\mathcal{A}(X, \eta, N \log X)}{N}\right), \quad U_N = U + \frac{\mathcal{B}(X, \eta, N \log X)}{N}. \quad (\text{C.12})$$

This extension joins smoothly to the input profiles E, U . In the next identity, $\mathcal{D}_X = X\partial_X$ differentiates a function of (X, η, φ) while φ is held fixed. Every resulting term is then evaluated at $\varphi = N \log X$. The exact shears of (C.12) are

$$a_N = a_L - \frac{2\mathcal{D}_X \mathcal{A}}{N}, \quad b_N = e^{-\mathcal{A}/N} \left(b_L + \frac{2\mathcal{D}_X \mathcal{B}}{NE} \right). \quad (\text{C.13})$$

After the substitution, $X\partial_X$ becomes $\mathcal{D}_X + N\partial_\varphi$. Insert (C.11) into $a_N = 1 - 2X\partial_X \log E_N$ and $b_N = 2X\partial_X U_N / E_N$. This gives both formulas, including the exponential denominator in the axial shear.

On the fixed compact radial range from X_- through the first correction interval, X, E, H, L all have positive lower bounds. For each fixed $m \geq 0$, the phase in (C.12) does not depend on η , so

$$\sup_X \sum_{j=0}^m \left(\left| \partial_\eta^j (E_N - E) \right| + \left| \partial_\eta^j (U_N - U) \right| \right) \leq C_m N^{-1}. \quad (\text{C.14})$$

The differences between the shears in (C.13) and their loop values obey the same estimate. The derivative $X\partial_X(E_N - E)$ can still have order one. More generally, for fixed $r \geq 1$ and $m \geq 0$, radially differentiating (C.12) gives the bound $C_{r,m} N^{r-1}$ for the mixed derivatives of the profile difference. Checking the cone condition for the modified profiles also requires control of p_s , not only of the shear. Its radial integral formulas use profile values and parameter derivatives. The estimates just proved are therefore enough, even though the radial derivatives do not have to be small.

Let $\mathbf{m} = (M, I, J, S, C_p)$ and $\mathbf{m}_N$ be the five radial moment functions in (4.15) for E, U and E_N, U_N , respectively. Keep the same pressure value $\Pi_0(\eta)$ at the axis, so that $\Pi_N = \Pi_0 + C_{p,N}$. Integrate (C.14) across this fixed radial range. The weights in the radial moment definitions are bounded there, so the result is

$$\sup_X \sum_{j=0}^m \left| \partial_\eta^j (\mathbf{m}_N - \mathbf{m}) \right| + \sup_X \sum_{j=0}^m \left| \partial_\eta^j (\Pi_N - \Pi) \right| \leq C_m N^{-1}. \quad (\text{C.15})$$

The formulas for Q_s, N_s in (4.16) use these radial moment functions and their first parameter derivatives, together with the profile values. Their denominators stay bounded away from zero on this range. Therefore

$$\sup_X \sum_{j=0}^m \left| \partial_\eta^j (p_{s,N} - p_s) \right| \leq C_m N^{-1}. \quad (\text{C.16})$$

The estimate at order m uses the profile and radial moment estimates through order $m + 1$ in η . It does not compare radial derivatives. Lemma C.1 gives uniform positive lower bounds in the cone inequalities. Equations (C.13) and (C.16) then preserve the admissible stress cone on $\mathcal{I}$ once N is large enough. Between $\mathcal{I}$ and the first correction interval, the fields are still E, U , while $p_{s,N} - p_s$ obeys (C.16). The uniform positive lower bounds in the input profile's admissible stress cone inequalities therefore remain positive there as well.

The modified profiles now obey the cone inequalities all the way through the joining interval. Their moments still have errors of order N^{-1} , and those errors must be removed before the profile reaches the exterior. Make the correction on the first patch listed after (A.9). This patch comes before the terminal compensation patch and the two later correction patches. On it, the input profiles obey $U = 0$ and $E = K(\eta)X^{-1/2-\lambda}$, where K is smooth and has a positive lower bound. Add two fixed compact bumps to U and three to E , with coefficients that depend smoothly on η . List the five radial moment functions at the patch's right endpoint in the order $(M, J; I, S, C_p)$. When all coefficients are zero, their differential separates into two independent blocks at first order, with the following weights.

rows	weights in dX	powers of X
M, J	$1, H$	$0, -\lambda$
I, S, C_p	$\sqrt{2X}, -E, E/X$	$1/2, -1/2 - \lambda, -3/2 - \lambda$.

The exponents are distinct because $\lambda > 0$ has already been fixed. Choose bumps with ordered, disjoint supports inside each block. Lemma A.1, or its five-moment specialization in Corollary A.3, then gives an invertible matrix whose inverse is uniformly bounded on $[-1, 1]$. That bound may depend on λ and on the fixed patch scales. The two U -weights merge as $\lambda \rightarrow 0$, so no uniform bound in that limit is claimed.

The map from the correction coefficients to the changes in (M, J, I, S, C_p) is this linear map plus quadratic terms. The function J contains the product of the two field changes, while S, C_p contain their squares. By (C.15), the discrepancy accumulated before this correction is $O_m(N^{-1})$. Lemma A.2 therefore gives a smooth solution for coefficients near zero that cancels the discrepancy. For every fixed parameter order, these coefficients are $O_m(N^{-1})$. Choose N after λ , the patch scales, and every choice used in the fixed input profile. Choose it large enough that the discrepancy lies in the neighborhood where the solution is defined and that its derivatives through the finitely many η -orders needed for positivity and the cone inequalities meet their tolerances. Differentiating the implicit equation gives finite bounds for every further fixed η -derivative order. The derivatives do not all have to be small at once. The fixed bump shapes also keep the correction small in the radial derivatives used by the shear.

Integration gives the correction's partial radial moment functions the same $O_m(N^{-1})$ bounds. Equation (C.16) therefore remains valid inside the correction interval, and the admissible stress cone remains valid throughout that interval.

At the correction interval's right endpoint, X_{rep} , the corrected profiles obey

$$\tilde{m}(X_{\text{rep}}, \eta) = m(X_{\text{rep}}, \eta), \quad (\tilde{E}, \tilde{U}) = (E, U) \quad \text{for } X \geq X_{\text{rep}}. \quad (\text{C.17})$$

From this endpoint onward, equality of the five radial moment functions continues by integration. Lemma 4.4 then gives equality of the complete pressure and the functions Q_s, N_s for every $X \geq X_{\text{rep}}$, including equality of their parameter derivatives. This keeps the terminal compensation already built into the input moment vector m . As a result, the exterior moments and the pressure normalized to vanish at radial infinity are unchanged. The first collar is unchanged because both modifications are supported later, and the pressure is integrated from the unchanged axis datum. The last collar is unchanged because the radial moments have been restored exactly. The two unused correction patches still have their prescribed reference power law and $U = 0$.

Finally, fix one such finite N . Every mixed radial and parameter derivative of the resulting profiles is finite, though its bound may depend on this N . Fix this value before taking any limit in the physical scale q , and before choosing any later dyadic band or correction stage. □

C.3 The two edges and the completed profile.

In this subsection, E, U, Π mean the corrected profiles from Proposition C.2, and every related quantity is formed from those corrected profiles. Proposition C.2 has now supplied the interior cone inequalities and the exact radial moments. It remains to prove Proposition C.3 by checking the remaining assertions at the annular edges, where the stress $\mathcal{T}_0$ vanishes. Its direction n must extend smoothly, and the cone inequalities must keep a strict uniform margin. The derivatives of $\mathcal{T}_0$ must also obey the same exponential weight ζ required in Theorem 4.6. Both endpoint collars were left unchanged, so their original stress factorizations (B.30) and Equations (A.48) to (A.49) give these properties. In the weight below, $t_1 > 0$ is the fixed activation width in $\log(X/X_a)$ from Proposition 4.10.

Proposition C.3. *The profiles from Proposition C.2 have every property stated in Theorem 4.6, with the weight*

$$y_a = \log(X/X_a), \quad y_b = \log(X_b/X), \quad \delta = \min\{1, y_a, y_b\},$$

$$\zeta = \exp\left(-\frac{t_1^2}{y_a^2} - \frac{4}{y_b^2}\right) \quad (X_a < X < X_b), \quad (\text{C.18})$$

and this weight is extended by zero when $X \leq X_a$ or $X \geq X_b$.

Proof. The stress is zero up to X_a because of the analytic axis profile and Proposition B.5. It is zero beyond X_b because of Lemma A.8. Proposition C.2 leaves both regions unchanged and makes the admissible stress cone inequalities hold everywhere between them. If the stress were zero at an interior point, the identity (4.11) would force $p_s = (a, -b_s)$, and hence $P_c = v_s$. That contradicts the strict quadratic cone test. This proves the stress-support assertion in part (ii) of Theorem 4.6. The unchanged axis profile also obeys (4.13) when $X \leq X_a$.

For the cone assertions in part (iii), set $n = \mathcal{T}_0/|\mathcal{T}_0|$ on $X_a < X < X_b$. The interior bounds $F > 0$, $a > 0$, and $v_s > 2$ come from the positive profile construction and the admissible stress cone. At the inner endpoint, $a = p_{1,r} > 0$. Proposition B.5 gives $v_s > 2 + c$ there, and the analytic axis profile has $F > 0$. At the outer endpoint, the same bounds follow from (A.56), together with $b_s = 0$ and positivity of the heat factor. Continuity on the compact closed annulus gives the required positive lower bounds throughout the annulus.

The inner edge keeps the factorization $\mathcal{T}_0 = e_a B_0$ from (B.30), where $e_a = e^{-t_1^2/y_a^2} g_a(y_a)$, $g_a(0) > 0$, and $B_0(0, \eta) = F p_{s,r} \neq 0$. These facts make its direction smooth.

At the edge, that direction is a positive scalar multiple of the limiting shear $(a, -b_s) = a(1, t_s)$. In particular, $n_z - t_s n_\theta = 0$ and $n_\theta + t_s n_z > 0$ there. At the outer edge, the unchanged factors from Proposition A.10 give

$$n = \frac{(b_\theta, y_b^6 b_z)}{\sqrt{b_\theta^2 + y_b^{12} b_z^2}}, \quad b_\theta(0, \eta) > 0.$$

The outer limiting direction is therefore $(1, 0)$, and the limiting shear ratio is $t_s = 0$. Every statement here remains smooth through $\eta = \pm 1$; the heat factor is used only where its argument is nonnegative.

These edge limits make the interior cone inequalities uniform on the closed annulus. In the interior, (4.11) gives

$$n_\theta + t_s n_z = \frac{P_c - v_s}{|p_s - (a, -b_s)|} > 0, \quad n_z - t_s n_\theta = \frac{J_c}{|p_s - (a, -b_s)|}.$$

The admissible stress cone makes $2 - (v_s - 2)(n_z - t_s n_\theta)^2 / (n_\theta + t_s n_z)^2$ positive in the interior. This expression and its positive denominator both extend continuously to the two edges. The factorizations just proved make the expression equal to two at either edge. The expression and the denominator each have a positive minimum on the compact annulus. Those two minima give one $0 < \kappa < 2$ for which

$$n_\theta + t_s n_z \geq \kappa, \quad (v_s - 2)(n_z - t_s n_\theta)^2 \leq (2 - \kappa)(n_\theta + t_s n_z)^2.$$

These bounds, together with the endpoint directions and the positive lower bounds above, prove part (iii) of Theorem 4.6.

For part (iv), only the following weighted estimates remain to be proved.

$$|\mathcal{T}_0| \geq c\zeta, \quad |\partial^I \mathcal{T}_0| \leq C_I \zeta \delta^{-m_I} \quad (X_a < X < X_b, -1 \leq \eta \leq 1) \quad (\text{C.19})$$

They must hold for every fixed mixed profile derivative ∂^I , with finite constants that do not depend on the physical q .

On an inner collar, ζ differs from $e^{-t_1^2/y_a^2}$ only by a smooth positive factor bounded above and below. The bounds in (B.31) therefore give (C.19) there. On an outer collar, it differs from e^{-4/y_b^2} in the same way. The angular factor $e^{-4/y_b^2} y_b^{-3} b_\theta$, with b_θ bounded below, gives the lower bound, while (A.51) gives the upper bound for every fixed derivative. On the compact interval left between the two collars, both ζ and $|\mathcal{T}_0|$ have positive minima, and every fixed derivative is bounded. Putting these three regions together proves (C.19). The exponential factors also show that setting the stress equal to zero across both edges gives a smooth extension. The definition (C.18) makes ζ positive in the interior and flat at both edges. This completes part (iv) of Theorem 4.6.

For the regularity in part (i), Proposition B.2 gives the smooth axis profiles with $F > 0$, and Corollary B.6 keeps the common analytic rectangle $[0, X_{\text{an}}] \times [-1, 1]$. The factors give Cartesian smoothness across the axis through Equations (4.4) to (4.5). The later modifications in Proposition C.2 are smooth and preserve positivity, with finite mixed derivative bounds on every fixed radial interval. The inner directional margin is part (iii), proved above. The heat profile is smooth through $\eta = \pm 1$ by Lemma A.6; its factor does not need to be analytic at zero. The pressure normalization (4.25) comes from Lemma A.5 and Propositions A.7 and B.8, and (C.17) keeps it unchanged. This completes part (i). Differentiating the pressure formula gives $\Pi_X = E^2/(2X) = F^2$, including its smooth extension at $X = 0$. Proposition 4.2 gives the exact tangential residual identities. Together with the support and axis checks above, they complete part (ii).

For part (v), Propositions A.7 and B.8 give the exact exterior moments (4.28), and (C.17) keeps them unchanged. The same restoration keeps the heat exterior (4.29); Lemma A.6 proves its exact heat evolution and smoothness. For the radius in this assertion, take $X_v = X_{\text{end}} \in (X_a, X_b)$,

the pulse-end radius defined before (A.10). Beyond it, $U = M = 0$ by Proposition A.4 and (C.17). Therefore $\mathcal{A}_X(U) = M/X = 0$, and (4.7) gives $V_0 = 0$. This proves the remaining assertion in part (v).

Finally, the third and fourth reserved intervals listed after (A.9) are exactly the intervals $\mathcal{I}_{\text{pos}}, \mathcal{I}_{\text{mean}}$ in part (vi). Both lie in (X_a, X_v) , and their order in the list means that $\sup \mathcal{I}_{\text{pos}} < \inf \mathcal{I}_{\text{mean}}$. Proposition C.2 leaves both intervals unchanged. On them, the input profiles obey $U = 0$ and $E = K(\eta)X^{-1/2-\lambda}$, where $K(\eta) = c_{\text{patch}}(1 + \eta^2)^{-1}$ for a fixed positive constant c_{patch} , by Proposition A.4. They therefore still satisfy (4.30), which completes part (vi). Proposition C.2 fixed the finite frequency N before choosing the physical scale q and every later band and correction stage. The constants therefore have the independence stated at the end of Theorem 4.6. □

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